

Analysis of Financial Time Series

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Analysis of Financial Time Series

Third Edition

RUEY S. TSAY

The University of Chicago
Booth School of Business
Chicago, IL



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To Teresa and my father, and in memory of my mother

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Preface

As many countries struggle to recover from the recent global financial crisis, one thing clear is that we do not want to suffer another crisis like this in the future. We must study the past in order to prevent future financial crisis. Financial data of the past few years thus become important in empirical study. The primary objective of the revision is to update the data used and to reanalyze the examples so that one can better understand the properties of asset returns. At the same time, we also witness many new developments in financial econometrics and financial software packages. In particular, the Rmetrics now has many packages for analyzing financial time series. The second goal of the revision is to include R commands and demonstrations, making it possible and easier for readers to reproduce the results shown in the book.

Collapses of big financial institutions during the crisis show that extreme events occur in clusters; they are not independent. To deal with dependence in extremes, I include the extremal index in Chapter 7 and discuss its impact on value at risk. I also rewrite Chapter 7 to make it easier to understand and more complete. It now contains the expected shortfall, or conditional value at risk, for measuring financial risk.

Substantial efforts are made to draw a balance between the length and coverage of the book. I do not include credit risk or operational risk in this revision for three reasons. First, effective methods for assessing credit risk require further study. Second, the data are not widely available. Third, the length of the book is approaching my limit.

A brief summary of the added material in the third edition is:

1. To update the data used throughout the book.
2. To provide R commands and demonstrations. In some cases, R programs are given.
3. To reanalyze many examples with updated observations.
4. To introduce skew distributions for volatility modeling in Chapter 3.
5. To investigate properties of recent high-frequency trading data and to add applications of nonlinear duration models in Chapter 5.

6. To provide a unified approach to value at risk (VaR) via loss function, to discuss expected shortfall (ES), or equivalently the conditional value at risk (CVaR), and to introduce extremal index for dependence data in Chapter 7.
7. To discuss application of cointegration to pairs trading in Chapter 8.
8. To study applications of dynamic correlation models in Chapter 10.

I benefit greatly from constructive comments of many readers of the second edition, including students, colleagues, and friends. I am indebted to them all. In particular, I like to express my sincere thanks to Spencer Graves for creating the FinTS package for R and Tom Doan of ESTIMA and Eugene Gath for careful reading of the text. I also thank Kam Hamidieh for suggestions concerning new topics for the revision. I also like to thank colleagues at Wiley, especially Jackie Palmieri and Stephen Quigley, for their support. As always, the revision would not be possible without the constant encouragement and unconditional love of my wife and children. They are my motivation and source of energy. Part of my research is supported by the Booth School of Business, University of Chicago.

Finally, the website for the book is:

<http://faculty.chicagobooth.edu/ruey.tsay/teaching/fts3>.

Ruey S. Tsay

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Preface to the Second Edition

The subject of *financial time series analysis* has attracted substantial attention in recent years, especially with the 2003 Nobel awards to Professors Robert Engle and Clive Granger. At the same time, the field of financial econometrics has undergone various new developments, especially in high-frequency finance, stochastic volatility, and software availability. There is a need to make the material more complete and accessible for advanced undergraduate and graduate students, practitioners, and researchers. The main goals in preparing this second edition have been to bring the book up to date both in new developments and empirical analysis, and to enlarge the core material of the book by including consistent covariance estimation under heteroscedasticity and serial correlation, alternative approaches to volatility modeling, financial factor models, state-space models, Kalman filtering, and estimation of stochastic diffusion models.

The book therefore has been extended to 12 chapters and substantially revised to include S-Plus commands and illustrations. Many empirical demonstrations and exercises are updated so that they include the most recent data.

The two new chapters are Chapter 9, Principal Component Analysis and Factor Models, and Chapter 11, State-Space Models and Kalman Filter. The factor models discussed include macroeconomic, fundamental, and statistical factor models. They are simple and powerful tools for analyzing high-dimensional financial data such as portfolio returns. Empirical examples are used to demonstrate the applications. The state-space model and Kalman filter are added to demonstrate their applicability in finance and ease in computation. They are used in Chapter 12 to estimate stochastic volatility models under the general Markov chain Monte Carlo (MCMC) framework. The estimation also uses the technique of forward filtering and backward sampling to gain computational efficiency.

A brief summary of the added material in the second edition is:

1. To update the data used throughout the book.
2. To provide S-Plus commands and demonstrations.

3. To consider unit-root tests and methods for consistent estimation of the covariance matrix in the presence of conditional heteroscedasticity and serial correlation in Chapter 2.
4. To describe alternative approaches to volatility modeling, including use of high-frequency transactions data and daily high and low prices of an asset in Chapter 3.
5. To give more applications of nonlinear models and methods in Chapter 4.
6. To introduce additional concepts and applications of value at risk in Chapter 7.
7. To discuss cointegrated vector AR models in Chapter 8.
8. To cover various multivariate volatility models in Chapter 10.
9. To add an effective MCMC method for estimating stochastic volatility models in Chapter 12.

The revision benefits greatly from constructive comments of colleagues, friends, and many readers of the first edition. I am indebted to them all. In particular, I thank J. C. Artigas, Spencer Graves, Chung-Ming Kuan, Henry Lin, Daniel Peña, Jeff Russell, Michael Steele, George Tiao, Mark Wohar, Eric Zivot, and students of my MBA classes on financial time series for their comments and discussions and Rosalyn Farkas for editorial assistance. I also thank my wife and children for their unconditional support and encouragements. Part of my research in financial econometrics is supported by the National Science Foundation, the High-Frequency Finance Project of the Institute of Economics, Academia Sinica, and the Graduate School of Business, University of Chicago.

Finally, the website for the book is:

gsbwww.uchicago.edu/fac/ruey.tsay/teaching/fts2.

Ruey S. Tsay

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Preface to the First Edition

This book grew out of an MBA course in analysis of financial time series that I have been teaching at the University of Chicago since 1999. It also covers materials of Ph.D. courses in time series analysis that I taught over the years. It is an introductory book intended to provide a comprehensive and systematic account of financial econometric models and their application to modeling and prediction of financial time series data. The goals are to learn basic characteristics of financial data, understand the application of financial econometric models, and gain experience in analyzing financial time series.

The book will be useful as a text of time series analysis for MBA students with finance concentration or senior undergraduate and graduate students in business, economics, mathematics, and statistics who are interested in financial econometrics. The book is also a useful reference for researchers and practitioners in business, finance, and insurance facing value at risk calculation, volatility modeling, and analysis of serially correlated data.

The distinctive features of this book include the combination of recent developments in financial econometrics in the econometric and statistical literature. The developments discussed include the timely topics of value at risk (VaR), high-frequency data analysis, and Markov chain Monte Carlo (MCMC) methods. In particular, the book covers some recent results that are yet to appear in academic journals; see Chapter 6 on derivative pricing using jump diffusion with closed-form formulas, Chapter 7 on value at risk calculation using extreme value theory based on a nonhomogeneous two-dimensional Poisson process, and Chapter 9 on multivariate volatility models with time-varying correlations. MCMC methods are introduced because they are powerful and widely applicable in financial econometrics. These methods will be used extensively in the future.

Another distinctive feature of this book is the emphasis on real examples and data analysis. Real financial data are used throughout the book to demonstrate applications of the models and methods discussed. The analysis is carried out by using several computer packages; the SCA (the Scientific Computing Associates) for building linear time series models, the RATS (regression analysis for time series)

for estimating volatility models, and the S-Plus for implementing neural networks and obtaining postscript plots. Some commands required to run these packages are given in appendixes of appropriate chapters. In particular, complicated RATS programs used to estimate multivariate volatility models are shown in Appendix A of Chapter 9. Some Fortran programs written by myself and others are used to price simple options, estimate extreme value models, calculate VaR, and carry out Bayesian analysis. Some data sets and programs are accessible from the World Wide Web at <http://www.gsb.uchicago.edu/fac/ruey.tsay/teaching/fts>.

The book begins with some basic characteristics of financial time series data in Chapter 1. The other chapters are divided into three parts. The first part, consisting of Chapters 2 to 7, focuses on analysis and application of univariate financial time series. The second part of the book covers Chapters 8 and 9 and is concerned with the return series of multiple assets. The final part of the book is Chapter 10, which introduces Bayesian inference in finance via MCMC methods.

A knowledge of basic statistical concepts is needed to fully understand the book. Throughout the chapters, I have provided a brief review of the necessary statistical concepts when they first appear. Even so, a prerequisite in statistics or business statistics that includes probability distributions and linear regression analysis is highly recommended. A knowledge of finance will be helpful in understanding the applications discussed throughout the book. However, readers with advanced background in econometrics and statistics can find interesting and challenging topics in many areas of the book.

An MBA course may consist of Chapters 2 and 3 as a core component, followed by some nonlinear methods (e.g., the neural network of Chapter 4 and the applications discussed in Chapters 5–7 and 10). Readers who are interested in Bayesian inference may start with the first five sections of Chapter 10.

Research in financial time series evolves rapidly and new results continue to appear regularly. Although I have attempted to provide broad coverage, there are many subjects that I do not cover or can only mention in passing.

I sincerely thank my teacher and dear friend, George C. Tiao, for his guidance, encouragement, and deep conviction regarding statistical applications over the years. I am grateful to Steve Quigley, Heather Haselkorn, Leslie Galen, Danielle LaCouriere, and Amy Hendrickson for making the publication of this book possible, to Richard Smith for sending me the estimation program of extreme value theory, to Bonnie K. Ray for helpful comments on several chapters, to Steve Kou for sending me his preprint on jump diffusion models, to Robert E. McCulloch for many years of collaboration on MCMC methods, to many students in my courses on analysis of financial time series for their feedback and inputs, and to Jeffrey Russell and Michael Zhang for insightful discussions concerning analysis of high-frequency financial data. To all these wonderful people I owe a deep sense of gratitude. I am also grateful for the support of the Graduate School of Business, University of Chicago and the National Science Foundation. Finally, my heartfelt

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R. S. T.

Chicago, Illinois

CHAPTER 1

Financial Time Series and Their Characteristics

Financial time series analysis is concerned with the theory and practice of asset valuation over time. It is a highly empirical discipline, but like other scientific fields theory forms the foundation for making inference. There is, however, a key feature that distinguishes financial time series analysis from other time series analysis. Both financial theory and its empirical time series contain an element of uncertainty. For example, there are various definitions of asset volatility, and for a stock return series, the volatility is not directly observable. As a result of the added uncertainty, statistical theory and methods play an important role in financial time series analysis.

The objective of this book is to provide some knowledge of financial time series, introduce some statistical tools useful for analyzing these series, and gain experience in financial applications of various econometric methods. We begin with the basic concepts of asset returns and a brief introduction to the processes to be discussed throughout the book. Chapter 2 reviews basic concepts of linear time series analysis such as stationarity and autocorrelation function, introduces simple linear models for handling serial dependence of the series, and discusses regression models with time series errors, seasonality, unit-root nonstationarity, and long-memory processes. The chapter also provides methods for consistent estimation of the covariance matrix in the presence of conditional heteroscedasticity and serial correlations. Chapter 3 focuses on modeling conditional heteroscedasticity (i.e., the conditional variance of an asset return). It discusses various econometric models developed recently to describe the evolution of volatility of an asset return over time. The chapter also discusses alternative methods to volatility modeling, including use of high-frequency transactions data and daily high and low prices of an asset. In Chapter 4, we address nonlinearity in financial time series, introduce test statistics that can discriminate nonlinear series from linear ones, and discuss several nonlinear models. The chapter also introduces nonparametric estimation

methods and neural networks and shows various applications of nonlinear models in finance. Chapter 5 is concerned with analysis of high-frequency financial data, the effects of market microstructure, and some applications of high-frequency finance. It shows that nonsynchronous trading and bid–ask bounce can introduce serial correlations in a stock return. It also studies the dynamic of time duration between trades and some econometric models for analyzing transactions data. In Chapter 6, we introduce continuous-time diffusion models and Ito’s lemma. Black–Scholes option pricing formulas are derived, and a simple jump diffusion model is used to capture some characteristics commonly observed in options markets. Chapter 7 discusses extreme value theory, heavy-tailed distributions, and their application to financial risk management. In particular, it discusses various methods for calculating value at risk and expected shortfall of a financial position. Chapter 8 focuses on multivariate time series analysis and simple multivariate models with emphasis on the lead–lag relationship between time series. The chapter also introduces cointegration, some cointegration tests, and threshold cointegration and applies the concept of cointegration to investigate arbitrage opportunity in financial markets, including pairs trading. Chapter 9 discusses ways to simplify the dynamic structure of a multivariate series and methods to reduce the dimension. It introduces and demonstrates three types of factor model to analyze returns of multiple assets. In Chapter 10, we introduce multivariate volatility models, including those with time-varying correlations, and discuss methods that can be used to reparameterize a conditional covariance matrix to satisfy the positiveness constraint and reduce the complexity in volatility modeling. Chapter 11 introduces state-space models and the Kalman filter and discusses the relationship between state-space models and other econometric models discussed in the book. It also gives several examples of financial applications. Finally, in Chapter 12, we introduce some Markov chain Monte Carlo (MCMC) methods developed in the statistical literature and apply these methods to various financial research problems, such as the estimation of stochastic volatility and Markov switching models.

The book places great emphasis on application and empirical data analysis. Every chapter contains real examples and, in many occasions, empirical characteristics of financial time series are used to motivate the development of econometric models. Computer programs and commands used in data analysis are provided when needed. In some cases, the programs are given in an appendix. Many real data sets are also used in the exercises of each chapter.

1.1 ASSET RETURNS

Most financial studies involve returns, instead of prices, of assets. Campbell, Lo, and MacKinlay (1997) give two main reasons for using returns. First, for average investors, return of an asset is a complete and scale-free summary of the investment opportunity. Second, return series are easier to handle than price series because the former have more attractive statistical properties. There are, however, several definitions of an asset return.

Let P_t be the price of an asset at time index t . We discuss some definitions of returns that are used throughout the book. Assume for the moment that the asset pays no dividends.

One-Period Simple Return

Holding the asset for one period from date $t - 1$ to date t would result in a *simple gross return*:

$$1 + R_t = \frac{P_t}{P_{t-1}} \quad \text{or} \quad P_t = P_{t-1}(1 + R_t). \quad (1.1)$$

The corresponding one-period *simple net return* or *simple return* is

$$R_t = \frac{P_t}{P_{t-1}} - 1 = \frac{P_t - P_{t-1}}{P_{t-1}}. \quad (1.2)$$

Multiperiod Simple Return

Holding the asset for k periods between dates $t - k$ and t gives a k -period simple gross return:

$$\begin{aligned} 1 + R_t[k] &= \frac{P_t}{P_{t-k}} = \frac{P_t}{P_{t-1}} \times \frac{P_{t-1}}{P_{t-2}} \times \cdots \times \frac{P_{t-k+1}}{P_{t-k}} \\ &= (1 + R_t)(1 + R_{t-1}) \cdots (1 + R_{t-k+1}) \\ &= \prod_{j=0}^{k-1} (1 + R_{t-j}). \end{aligned}$$

Thus, the k -period simple gross return is just the product of the k one-period simple gross returns involved. This is called a compound return. The k -period simple net return is $R_t[k] = (P_t - P_{t-k})/P_{t-k}$.

In practice, the actual time interval is important in discussing and comparing returns (e.g., monthly return or annual return). If the time interval is not given, then it is implicitly assumed to be one year. If the asset was held for k years, then the annualized (average) return is defined as

$$\text{Annualized } \{R_t[k]\} = \left[\prod_{j=0}^{k-1} (1 + R_{t-j}) \right]^{1/k} - 1.$$

This is a geometric mean of the k one-period simple gross returns involved and can be computed by

$$\text{Annualized } \{R_t[k]\} = \exp \left[\frac{1}{k} \sum_{j=0}^{k-1} \ln(1 + R_{t-j}) \right] - 1,$$

where $\exp(x)$ denotes the exponential function and $\ln(x)$ is the natural logarithm of the positive number x . Because it is easier to compute arithmetic average than geometric mean and the one-period returns tend to be small, one can use a first-order Taylor expansion to approximate the annualized return and obtain

$$\text{Annualized } \{R_t[k]\} \approx \frac{1}{k} \sum_{j=0}^{k-1} R_{t-j}. \quad (1.3)$$

Accuracy of the approximation in Eq. (1.3) may not be sufficient in some applications, however.

Continuous Compounding

Before introducing continuously compounded return, we discuss the effect of compounding. Assume that the interest rate of a bank deposit is 10% per annum and the initial deposit is \$1.00. If the bank pays interest once a year, then the net value of the deposit becomes $\$1(1 + 0.1) = \1.1 one year later. If the bank pays interest semiannually, the 6-month interest rate is $10\%/2 = 5\%$ and the net value is $\$1(1 + 0.1/2)^2 = \1.1025 after the first year. In general, if the bank pays interest m times a year, then the interest rate for each payment is $10\%/m$ and the net value of the deposit becomes $\$1(1 + 0.1/m)^m$ one year later. Table 1.1 gives the results for some commonly used time intervals on a deposit of \$1.00 with interest rate of 10% per annum. In particular, the net value approaches \$1.1052, which is obtained by $\exp(0.1)$ and referred to as the result of continuous compounding. The effect of compounding is clearly seen.

In general, the net asset value A of continuous compounding is

$$A = C \exp(r \times n), \quad (1.4)$$

where r is the interest rate per annum, C is the initial capital, and n is the number of years. From Eq. (1.4), we have

$$C = A \exp(-r \times n), \quad (1.5)$$

TABLE 1.1 Illustration of Effects of Compounding: Time Interval Is 1 Year and Interest Rate Is 10% per Annum

Type	Number of Payments	Interest Rate per Period	Net Value
Annual	1	0.1	\$1.10000
Semiannual	2	0.05	\$1.10250
Quarterly	4	0.025	\$1.10381
Monthly	12	0.0083	\$1.10471
Weekly	52	0.1/52	\$1.10506
Daily	365	0.1/365	\$1.10516
Continuously	∞		\$1.10517

which is referred to as the *present value* of an asset that is worth A dollars n years from now, assuming that the continuously compounded interest rate is r per annum.

Continuously Compounded Return

The natural logarithm of the simple gross return of an asset is called the continuously compounded return or *log return*:

$$r_t = \ln(1 + R_t) = \ln \frac{P_t}{P_{t-1}} = p_t - p_{t-1}, \quad (1.6)$$

where $p_t = \ln(P_t)$. Continuously compounded returns r_t enjoy some advantages over the simple net returns R_t . First, consider multiperiod returns. We have

$$\begin{aligned} r_t[k] &= \ln(1 + R_t[k]) = \ln[(1 + R_t)(1 + R_{t-1}) \cdots (1 + R_{t-k+1})] \\ &= \ln(1 + R_t) + \ln(1 + R_{t-1}) + \cdots + \ln(1 + R_{t-k+1}) \\ &= r_t + r_{t-1} + \cdots + r_{t-k+1}. \end{aligned}$$

Thus, the continuously compounded multiperiod return is simply the sum of continuously compounded one-period returns involved. Second, statistical properties of log returns are more tractable.

Portfolio Return

The simple net return of a portfolio consisting of N assets is a weighted average of the simple net returns of the assets involved, where the weight on each asset is the percentage of the portfolio's value invested in that asset. Let p be a portfolio that places weight w_i on asset i . Then the simple return of p at time t is $R_{p,t} = \sum_{i=1}^N w_i R_{it}$, where R_{it} is the simple return of asset i .

The continuously compounded returns of a portfolio, however, do not have the above convenient property. If the simple returns R_{it} are all small in magnitude, then we have $r_{p,t} \approx \sum_{i=1}^N w_i r_{it}$, where $r_{p,t}$ is the continuously compounded return of the portfolio at time t . This approximation is often used to study portfolio returns.

Dividend Payment

If an asset pays dividends periodically, we must modify the definitions of asset returns. Let D_t be the dividend payment of an asset between dates $t-1$ and t and P_t be the price of the asset at the end of period t . Thus, dividend is not included in P_t . Then the simple net return and continuously compounded return at time t become

$$R_t = \frac{P_t + D_t}{P_{t-1}} - 1, \quad r_t = \ln(P_t + D_t) - \ln(P_{t-1}).$$

Excess Return

Excess return of an asset at time t is the difference between the asset's return and the return on some reference asset. The reference asset is often taken to be riskless

such as a short-term U.S. Treasury bill return. The simple excess return and log excess return of an asset are then defined as

$$Z_t = R_t - R_{0t}, \quad z_t = r_t - r_{0t}, \quad (1.7)$$

where R_{0t} and r_{0t} are the simple and log returns of the reference asset, respectively. In the finance literature, the excess return is thought of as the payoff on an arbitrage portfolio that goes long in an asset and short in the reference asset with no net initial investment.

Remark. A long financial position means owning the asset. A short position involves selling an asset one does not own. This is accomplished by borrowing the asset from an investor who has purchased it. At some subsequent date, the short seller is obligated to buy exactly the same number of shares borrowed to pay back the lender. Because the repayment requires equal shares rather than equal dollars, the short seller benefits from a decline in the price of the asset. If cash dividends are paid on the asset while a short position is maintained, these are paid to the buyer of the short sale. The short seller must also compensate the lender by matching the cash dividends from his own resources. In other words, the short seller is also obligated to pay cash dividends on the borrowed asset to the lender. $\square$

Summary of Relationship

The relationships between simple return R_t and continuously compounded (or log) return r_t are

$$r_t = \ln(1 + R_t), \quad R_t = e^{r_t} - 1.$$

If the returns R_t and r_t are in percentages, then

$$r_t = 100 \ln \left(1 + \frac{R_t}{100} \right), \quad R_t = 100 (e^{r_t/100} - 1).$$

Temporal aggregation of the returns produces

$$1 + R_t[k] = (1 + R_t)(1 + R_{t-1}) \cdots (1 + R_{t-k+1}),$$

$$r_t[k] = r_t + r_{t-1} + \cdots + r_{t-k+1}.$$

If the continuously compounded interest rate is r per annum, then the relationship between present and future values of an asset is

$$A = C \exp(r \times n), \quad C = A \exp(-r \times n).$$

Example 1.1. If the monthly log return of an asset is 4.46%, then the corresponding monthly simple return is $100[\exp(4.46/100) - 1] = 4.56\%$. Also, if the monthly log returns of the asset within a quarter are 4.46%, -7.34% , and 10.77% , respectively, then the quarterly log return of the asset is $(4.46 - 7.34 + 10.77)\% = 7.89\%$.

1.2 DISTRIBUTIONAL PROPERTIES OF RETURNS

To study asset returns, it is best to begin with their distributional properties. The objective here is to understand the behavior of the returns across assets and over time. Consider a collection of N assets held for T time periods, say, $t = 1, \dots, T$. For each asset i , let r_{it} be its log return at time t . The log returns under study are $\{r_{it}; i = 1, \dots, N; t = 1, \dots, T\}$. One can also consider the simple returns $\{R_{it}; i = 1, \dots, N; t = 1, \dots, T\}$ and the log excess returns $\{z_{it}; i = 1, \dots, N; t = 1, \dots, T\}$.

1.2.1 Review of Statistical Distributions and Their Moments

We briefly review some basic properties of statistical distributions and the moment equations of a random variable. Let R^k be the k -dimensional Euclidean space. A point in R^k is denoted by $\mathbf{x} \in R^k$. Consider two random vectors $\mathbf{X} = (X_1, \dots, X_k)'$ and $\mathbf{Y} = (Y_1, \dots, Y_q)'$. Let $P(\mathbf{X} \in A, \mathbf{Y} \in B)$ be the probability that $\mathbf{X}$ is in the subspace $A \subset R^k$ and $\mathbf{Y}$ is in the subspace $B \subset R^q$. For most of the cases considered in this book, both random vectors are assumed to be continuous.

Joint Distribution

The function

$$F_{X,Y}(\mathbf{x}, \mathbf{y}; \boldsymbol{\theta}) = P(\mathbf{X} \leq \mathbf{x}, \mathbf{Y} \leq \mathbf{y}; \boldsymbol{\theta}),$$

where $\mathbf{x} \in R^p$, $\mathbf{y} \in R^q$, and the inequality $\leq$ is a component-by-component operation, is a joint distribution function of $\mathbf{X}$ and $\mathbf{Y}$ with parameter $\boldsymbol{\theta}$. Behavior of $\mathbf{X}$ and $\mathbf{Y}$ is characterized by $F_{X,Y}(\mathbf{x}, \mathbf{y}; \boldsymbol{\theta})$. If the joint probability density function $f_{x,y}(\mathbf{x}, \mathbf{y}; \boldsymbol{\theta})$ of $\mathbf{X}$ and $\mathbf{Y}$ exists, then

$$F_{X,Y}(\mathbf{x}, \mathbf{y}; \boldsymbol{\theta}) = \int_{-\infty}^{\mathbf{x}} \int_{-\infty}^{\mathbf{y}} f_{x,y}(\mathbf{w}, \mathbf{z}; \boldsymbol{\theta}) d\mathbf{z} d\mathbf{w}.$$

In this case, $\mathbf{X}$ and $\mathbf{Y}$ are continuous random vectors.

Marginal Distribution

The marginal distribution of $\mathbf{X}$ is given by

$$F_X(\mathbf{x}; \boldsymbol{\theta}) = F_{X,Y}(\mathbf{x}, \infty, \dots, \infty; \boldsymbol{\theta}).$$

Thus, the marginal distribution of $\mathbf{X}$ is obtained by integrating out $\mathbf{Y}$. A similar definition applies to the marginal distribution of $\mathbf{Y}$.

If $k = 1$, X is a scalar random variable and the distribution function becomes

$$F_X(x) = P(X \leq x; \boldsymbol{\theta}),$$

which is known as the cumulative distribution function (CDF) of X . The CDF of a random variable is nondecreasing [i.e., $F_X(x_1) \leq F_X(x_2)$ if $x_1 \leq x_2$] and satisfies $F_X(-\infty) = 0$ and $F_X(\infty) = 1$. For a given probability p , the smallest real number x_p such that $p \leq F_X(x_p)$ is called the 100 p th quantile of the random variable X . More specifically,

$$x_p = \inf_x \{x | p \leq F_X(x)\}.$$

We use the CDF to compute the p value of a test statistic in the book.

Conditional Distribution

The conditional distribution of X given $Y \leq y$ is given by

$$F_{X|Y \leq y}(\mathbf{x}; \boldsymbol{\theta}) = \frac{P(X \leq \mathbf{x}, Y \leq y; \boldsymbol{\theta})}{P(Y \leq y; \boldsymbol{\theta})}.$$

If the probability density functions involved exist, then the conditional density of X given $Y = y$ is

$$f_{x|y}(\mathbf{x}; \boldsymbol{\theta}) = \frac{f_{x,y}(\mathbf{x}, y; \boldsymbol{\theta})}{f_y(y; \boldsymbol{\theta})}, \quad (1.8)$$

where the marginal density function $f_y(y; \boldsymbol{\theta})$ is obtained by

$$f_y(y; \boldsymbol{\theta}) = \int_{-\infty}^{\infty} f_{x,y}(\mathbf{x}, y; \boldsymbol{\theta}) d\mathbf{x}.$$

From Eq. (1.8), the relation among joint, marginal, and conditional distributions is

$$f_{x,y}(\mathbf{x}, y; \boldsymbol{\theta}) = f_{x|y}(\mathbf{x}; \boldsymbol{\theta}) \times f_y(y; \boldsymbol{\theta}). \quad (1.9)$$

This identity is used extensively in time series analysis (e.g., in maximum-likelihood estimation). Finally, X and Y are independent random vectors if and only if $f_{x|y}(\mathbf{x}; \boldsymbol{\theta}) = f_x(\mathbf{x}; \boldsymbol{\theta})$. In this case, $f_{x,y}(\mathbf{x}, y; \boldsymbol{\theta}) = f_x(\mathbf{x}; \boldsymbol{\theta})f_y(y; \boldsymbol{\theta})$.

Moments of a Random Variable

The ℓ th moment of a continuous random variable X is defined as

$$m'_\ell = E(X^\ell) = \int_{-\infty}^{\infty} x^\ell f(x) dx,$$

where E stands for expectation and $f(x)$ is the probability density function of X . The first moment is called the *mean* or *expectation* of X . It measures the central location of the distribution. We denote the mean of X by μ_x . The ℓ th central moment of X is defined as

$$m_\ell = E[(X - \mu_x)^\ell] = \int_{-\infty}^{\infty} (x - \mu_x)^\ell f(x) dx$$

provided that the integral exists. The second central moment, denoted by σ_x^2 , measures the variability of X and is called the *variance* of X . The positive square root, σ_x , of variance is the *standard deviation* of X . The first two moments of a random variable uniquely determine a normal distribution. For other distributions, higher order moments are also of interest.

The third central moment measures the symmetry of X with respect to its mean, whereas the fourth central moment measures the tail behavior of X . In statistics, *skewness* and *kurtosis*, which are normalized third and fourth central moments of X , are often used to summarize the extent of asymmetry and tail thickness. Specifically, the skewness and kurtosis of X are defined as

$$S(x) = E \left[\frac{(X - \mu_x)^3}{\sigma_x^3} \right], \quad K(x) = E \left[\frac{(X - \mu_x)^4}{\sigma_x^4} \right].$$

The quantity $K(x) - 3$ is called the *excess kurtosis* because $K(x) = 3$ for a normal distribution. Thus, the excess kurtosis of a normal random variable is zero. A distribution with positive excess kurtosis is said to have heavy tails, implying that the distribution puts more mass on the tails of its support than a normal distribution does. In practice, this means that a random sample from such a distribution tends to contain more extreme values. Such a distribution is said to be *leptokurtic*. On the other hand, a distribution with negative excess kurtosis has short tails (e.g., a uniform distribution over a finite interval). Such a distribution is said to be *platykurtic*.

In application, skewness and kurtosis can be estimated by their sample counterparts. Let $\{x_1, \dots, x_T\}$ be a random sample of X with T observations. The sample mean is

$$\hat{\mu}_x = \frac{1}{T} \sum_{t=1}^T x_t, \quad (1.10)$$

the sample variance is

$$\hat{\sigma}_x^2 = \frac{1}{T-1} \sum_{t=1}^T (x_t - \hat{\mu}_x)^2, \quad (1.11)$$

the sample skewness is

$$\hat{S}(x) = \frac{1}{(T-1)\hat{\sigma}_x^3} \sum_{t=1}^T (x_t - \hat{\mu}_x)^3, \quad (1.12)$$

and the sample kurtosis is

$$\hat{K}(x) = \frac{1}{(T-1)\hat{\sigma}_x^4} \sum_{t=1}^T (x_t - \hat{\mu}_x)^4. \quad (1.13)$$

Under the normality assumption, $\hat{S}(x)$ and $\hat{K}(x) - 3$ are distributed asymptotically as normal with zero mean and variances $6/T$ and $24/T$, respectively; see Snedecor and Cochran (1980, p. 78). These asymptotic properties can be used to test the normality of asset returns. Given an asset return series $\{r_1, \dots, r_T\}$, to test the skewness of the returns, we consider the null hypothesis $H_0 : S(r) = 0$ versus the alternative hypothesis $H_a : S(r) \neq 0$. The t -ratio statistic of the sample skewness in Eq. (1.12) is

$$t = \frac{\hat{S}(r)}{\sqrt{6/T}}.$$

The decision rule is as follows. Reject the null hypothesis at the α significance level, if $|t| > Z_{\alpha/2}$, where $Z_{\alpha/2}$ is the upper $100(\alpha/2)$ th quantile of the standard normal distribution. Alternatively, one can compute the p value of the test statistic t and reject H_0 if and only if the p value is less than α .

Similarly, one can test the excess kurtosis of the return series using the hypotheses $H_0 : K(r) - 3 = 0$ versus $H_a : K(r) - 3 \neq 0$. The test statistic is

$$t = \frac{\hat{K}(r) - 3}{\sqrt{24/T}},$$

which is asymptotically a standard normal random variable. The decision rule is to reject H_0 if and only if the p value of the test statistic is less than the significance level α . Jarque and Bera (1987) (JB) combine the two prior tests and use the test statistic

$$JB = \frac{\hat{S}^2(r)}{6/T} + \frac{[\hat{K}(r) - 3]^2}{24/T},$$

which is asymptotically distributed as a chi-squared random variable with 2 degrees of freedom, to test for the normality of r_t . One rejects H_0 of normality if the p value of the JB statistic is less than the significance level.

Example 1.2. Consider the daily simple returns of the International Business Machines (IBM) stock used in Table 1.2. The sample skewness and kurtosis of the returns are parts of the descriptive (or summary) statistics that can be obtained easily using various statistical software packages. Both R and S-Plus are used in the demonstration, where `d-ibm3dx7008.txt` is the data file name. Note that in R the *kurtosis* denotes excess kurtosis. From the output, the excess kurtosis is high, indicating that the daily simple returns of IBM stock have heavy tails. To test the symmetry of return distribution, we use the test statistic

$$t = \frac{0.0614}{\sqrt{6/9845}} = \frac{0.0614}{0.0247} = 2.49,$$

which gives a p value of about 0.013, indicating that the daily simple returns of IBM stock are significantly skewed to the right at the 5% level.

TABLE 1.2 Descriptive Statistics for Daily and Monthly Simple and Log Returns of Selected Indexes and Stocks^a

Security	Start	Size	Mean	Standard Deviation	Skewness	Excess Kurtosis	Minimum	Maximum
<i>Daily Simple Returns (%)</i>								
SP	70/01/02	9845	0.029	1.056	−0.73	22.81	−20.47	11.58
VW	70/01/02	9845	0.040	1.004	−0.62	18.02	−17.13	11.52
EW	70/01/02	9845	0.076	0.814	−0.77	17.08	−10.39	10.74
IBM	70/01/02	9845	0.040	1.693	0.06	9.92	−22.96	13.16
Intel	72/12/15	9096	0.108	2.891	−0.15	6.13	−29.57	26.38
3M	670/01/02	9845	0.045	1.482	−0.36	13.34	−25.98	11.54
Microsoft	86/03/14	5752	0.123	2.359	−0.13	9.92	−30.12	19.57
Citi-Grp	86/10/30	5592	0.067	2.602	1.80	55.25	−26.41	57.82
<i>Daily Log Returns (%)</i>								
SP	70/01/02	9845	0.023	1.062	−1.17	30.20	−22.90	10.96
VW	70/01/02	9845	0.035	1.008	−0.94	21.56	−18.80	10.90
EW	70/01/02	9845	0.072	0.816	−1.00	17.76	−10.97	10.20
IBM	70/01/02	9845	0.026	1.694	−0.27	12.17	−26.09	12.37
Intel	72/12/15	9096	0.066	2.905	−0.54	7.81	−35.06	23.41
3M	70/01/02	9845	0.034	1.488	−0.78	20.57	−30.08	10.92
Microsoft	86/03/14	5752	0.095	2.369	−0.63	14.23	−35.83	17.87
Citi-Grp	86/10/30	5592	0.033	2.575	0.22	33.19	−30.66	45.63
<i>Monthly Simple Returns (%)</i>								
SP	26/01	996	0.58	5.53	0.32	9.47	−29.94	42.22
VW	26/01	996	0.89	5.43	0.15	7.69	−29.01	38.37
EW	26/01	996	1.22	7.40	1.52	14.94	−31.28	66.59
IBM	26/01	996	1.35	7.15	0.44	3.43	−26.19	47.06
Intel	73/01	432	2.21	12.85	0.32	2.70	−44.87	62.50
3M	46/02	755	1.24	6.45	0.22	0.98	−27.83	25.80
Microsoft	86/04	273	2.62	11.08	0.66	1.96	−34.35	51.55
Citi-Grp	86/11	266	1.17	9.75	−0.47	1.77	−39.27	26.08
<i>Monthly Log Returns (%)</i>								
SP	26/01	996	0.43	5.54	−0.52	7.93	−35.58	35.22
VW	26/01	996	0.74	5.43	−0.58	6.85	−34.22	32.47
EW	26/01	996	0.96	7.14	0.25	8.55	−37.51	51.04
IBM	26/01	996	1.09	7.03	−0.07	2.62	−30.37	38.57
Intel	73/01	432	1.39	12.80	−0.55	3.06	−59.54	48.55
3M	46/02	755	1.03	6.37	−0.08	1.25	−32.61	22.95
Microsoft	86/04	273	2.01	10.66	0.10	1.59	−42.09	41.58
Citi-Grp	86/11	266	0.68	10.09	−1.09	3.76	−49.87	23.18

^aReturns are in percentages and the sample period ends on December 31, 2008. The statistics are defined in eqs. (1.10)–(1.13), and VW, EW and SP denote value-weighted, equal-weighted, and S&P composite index.

R Demonstration

In the following program code > is the prompt character and % denotes explanation:

```
> library(fBasics) % Load the package fBasics.
> da=read.table("d-ibm3dx7008.txt",header=T) % Load the data.
% header=T means 1st row of the data file contains
% variable names. The default is header=F, i.e., no names.

> dim(da) % Find size of the data: 9845 rows and 5 columns.
[1] 9845 5
> da[1,] % See the first row of the data
      Date      rtn    vwretd    ewretd    sprtrn % column names
1 19700102 0.000686 0.012137 0.03345 0.010211

> ibm=da[,2] % Obtain IBM simple returns
> sibm=ibm*100 % Percentage simple returns

> basicStats(sibm) % Compute the summary statistics
      sibm
nobs      9845.000000 % Sample size
NAs        0.000000 % Number of missing values
Minimum    -22.963000
Maximum     13.163600
1. Quartile -0.857100 % 25th percentile
3. Quartile  0.883300 % 75th percentile
Mean         0.040161 % Sample mean
Median        0.000000 % Sample median
Sum          395.387600 % Sum of the percentage simple returns
SE Mean       0.017058 % Standard error of the sample mean
LCL Mean      0.006724 % Lower bound of 95% conf.
                % interval for mean
UCL Mean      0.073599 % Upper bound of 95% conf.
                % interval for mean
Variance      2.864705 % Sample variance
Stdev         1.692544 % Sample standard error
Skewness      0.061399 % Sample skewness
Kurtosis      9.916359 % Sample excess kurtosis.

% Alternatively, one can use individual commands as follows:
> mean(sibm)
[1] 0.04016126
> var(sibm)
[1] 2.864705
> sqrt(var(sibm)) % Standard deviation
[1] 1.692544
> skewness(sibm)
[1] 0.06139878
attr(,"method")
```

```

[1] "moment"
> kurtosis(sibm)
[1] 9.91636
attr(,"method")
[1] "excess"

% Simple tests
> s1=skewness(sibm)
> t1=s1/sqrt(6/9845) % Compute test statistic
> t1
[1] 2.487093
> pv=2*(1-pnorm(t1)) % Compute p-value.
> pv
[1] 0.01287919

% Turn to log returns in percentages
> libm=log(ibm+1)*100
> t.test(libm) % Test mean being zero.
      One Sample t-test
data:  libm
t = 1.5126, df = 9844, p-value = 0.1304
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 -0.007641473  0.059290531
% The result shows that the hypothesis of zero expected return
% cannot be rejected at the 5% or 10% level.

> normalTest(libm,method='jb') % Normality test
Title:
  Jarque - Bera Normality Test

Test Results:
  STATISTIC:
    X-squared: 60921.9343
  P VALUE:
    Asymptotic p Value: < 2.2e-16
% The result shows the normality for log-return is rejected.

```

S-Plus Demonstration

In the following program code > is the prompt character and % marks explanation:

```

> module(finmetrics) % Load the Finmetrics module.
> da=read.table("d-ibm3dx7008.txt",header=T) % Load data.
> dim(da) % Obtain the size of the data set.
[1] 9845 5
> da[1,] % See the first row of the data
      Date      rtn   vwretd ewretd   sprtrn
1 19700102 0.000686 0.012137 0.03345 0.010211

```

```

> sibm=da[,2]*100 % Obtain percentage simple returns of
                    % IBM stock.
> summaryStats(sibm) % Obtain summary statistics

Sample Quantiles:
      min      1Q median      3Q      max
-22.96 -0.8571      0 0.8833 13.16
Sample Moments:
      mean      std skewness kurtosis
 0.04016 1.693  0.06141   12.92
Number of Observations: 9845
% simple tests
> s1=skewness(sibm) % Compute skewness
> t=s1/sqrt(6/9845) % Perform test of skewness
> t
[1] 2.487851
> pv=2*(1-pnorm(t)) % Calculate p-value.
> pv
[1] 0.01285177

> libm=log(da[,2]+1)*100 % Turn to log-return
> t.test(libm) % Test expected return being zero.
      One-sample t-Test
data:  libm
t = 1.5126, df = 9844, p-value = 0.1304
alternative hypothesis: mean is not equal to 0
95 percent confidence interval:
 -0.007641473  0.059290531

> normalTest(libm,method='jb') % Normality test
Test for Normality: Jarque-Bera
Null Hypothesis: data is normally distributed

Test Stat 60921.93
p.value      0.00
Dist. under Null: chi-square with 2 degrees of freedom
Total Observ.: 9845

```

Remark. In S-Plus, kurtosis is the regular kurtosis, not excess kurtosis. That is, S-Plus does not subtract 3 from the sample kurtosis. Also, in many cases R and S-Plus use the same commands. $\square$

1.2.2 Distributions of Returns

The most general model for the log returns $\{r_{it}; i = 1, \dots, N; t = 1, \dots, T\}$ is its joint distribution function:

$$F_r(r_{11}, \dots, r_{N1}; r_{12}, \dots, r_{N2}; \dots; r_{1T}, \dots, r_{NT}; \mathbf{Y}; \boldsymbol{\theta}), \quad (1.14)$$

where $\mathbf{Y}$ is a state vector consisting of variables that summarize the environment in which asset returns are determined and $\boldsymbol{\theta}$ is a vector of parameters that uniquely determines the distribution function $F_r(\cdot)$. The probability distribution $F_r(\cdot)$ governs the stochastic behavior of the returns r_{it} and $\mathbf{Y}$. In many financial studies, the state vector $\mathbf{Y}$ is treated as given and the main concern is the conditional distribution of $\{r_{it}\}$ given $\mathbf{Y}$. Empirical analysis of asset returns is then to estimate the unknown parameter $\boldsymbol{\theta}$ and to draw statistical inference about the behavior of $\{r_{it}\}$ given some past log returns.

The model in Eq. (1.14) is too general to be of practical value. However, it provides a general framework with respect to which an econometric model for asset returns r_{it} can be put in a proper perspective.

Some financial theories such as the capital asset pricing model (CAPM) of Sharpe (1964) focus on the joint distribution of N returns at a single time index t (i.e., the distribution of $\{r_{1t}, \dots, r_{Nt}\}$). Other theories emphasize the dynamic structure of individual asset returns (i.e., the distribution of $\{r_{i1}, \dots, r_{iT}\}$ for a given asset i). In this book, we focus on both. In the univariate analysis of Chapters 2–7, our main concern is the joint distribution of $\{r_{it}\}_{t=1}^T$ for asset i . To this end, it is useful to partition the joint distribution as

$$\begin{aligned} F(r_{i1}, \dots, r_{iT}; \boldsymbol{\theta}) &= F(r_{i1})F(r_{i2}|r_{i1}) \cdots F(r_{iT}|r_{i,T-1}, \dots, r_{i1}) \\ &= F(r_{i1}) \prod_{t=2}^T F(r_{it}|r_{i,t-1}, \dots, r_{i1}), \end{aligned} \quad (1.15)$$

where, for simplicity, the parameter $\boldsymbol{\theta}$ is omitted. This partition highlights the temporal dependencies of the log return r_{it} . The main issue then is the specification of the conditional distribution $F(r_{it}|r_{i,t-1}, \dots, r_{i1})$, in particular, how the conditional distribution evolves over time. In finance, different distributional specifications lead to different theories. For instance, one version of the random-walk hypothesis is that the conditional distribution $F(r_{it}|r_{i,t-1}, \dots, r_{i1})$ is equal to the marginal distribution $F(r_{it})$. In this case, returns are temporally independent and, hence, not predictable.

It is customary to treat asset returns as continuous random variables, especially for index returns or stock returns calculated at a low frequency, and use their probability density functions. In this case, using the identity in Eq. (1.9), we can write the partition in Eq. (1.15) as

$$f(r_{i1}, \dots, r_{iT}; \boldsymbol{\theta}) = f(r_{i1}; \boldsymbol{\theta}) \prod_{t=2}^T f(r_{it}|r_{i,t-1}, \dots, r_{i1}; \boldsymbol{\theta}). \quad (1.16)$$

For high-frequency asset returns, discreteness becomes an issue. For example, stock prices change in multiples of a tick size on the New York Stock Exchange (NYSE). The tick size was $\frac{1}{8}$ of a dollar before July 1997 and was $\frac{1}{16}$ of a dollar from July 1997 to January 2001. Therefore, the tick-by-tick return of an individual stock listed

on the NYSE is not continuous. We discuss high-frequency stock price changes and time durations between price changes later in Chapter 5.

Remark. On August 28, 2000, the NYSE began a pilot program with 7 stocks priced in decimals and the American Stock Exchange (AMEX) began a pilot program with 6 stocks and two options classes. The NYSE added 57 stocks and 94 stocks to the program on September 25 and December 4, 2000, respectively. All NYSE and AMEX stocks started trading in decimals on January 29, 2001. $\square$

Equation (1.16) suggests that conditional distributions are more relevant than marginal distributions in studying asset returns. However, the marginal distributions may still be of some interest. In particular, it is easier to estimate marginal distributions than conditional distributions using past returns. In addition, in some cases, asset returns have weak empirical serial correlations, and, hence, their marginal distributions are close to their conditional distributions.

Several statistical distributions have been proposed in the literature for the marginal distributions of asset returns, including normal distribution, lognormal distribution, stable distribution, and scale mixture of normal distributions. We briefly discuss these distributions.

Normal Distribution

A traditional assumption made in financial study is that the simple returns $\{R_{it}|t = 1, \dots, T\}$ are independently and identically distributed as normal with fixed mean and variance. This assumption makes statistical properties of asset returns tractable. But it encounters several difficulties. First, the lower bound of a simple return is -1 . Yet the normal distribution may assume any value in the real line and, hence, has no lower bound. Second, if R_{it} is normally distributed, then the multiperiod simple return $R_{it}[k]$ is not normally distributed because it is a product of one-period returns. Third, the normality assumption is not supported by many empirical asset returns, which tend to have a positive excess kurtosis.

Lognormal Distribution

Another commonly used assumption is that the log returns r_t of an asset are independent and identically distributed (iid) as normal with mean μ and variance σ^2 . The simple returns are then iid lognormal random variables with mean and variance given by

$$E(R_t) = \exp\left(\mu + \frac{\sigma^2}{2}\right) - 1, \quad \text{Var}(R_t) = \exp(2\mu + \sigma^2)[\exp(\sigma^2) - 1]. \quad (1.17)$$

These two equations are useful in studying asset returns (e.g., in forecasting using models built for log returns). Alternatively, let m_1 and m_2 be the mean and variance of the simple return R_t , which is lognormally distributed. Then the mean and

variance of the corresponding log return r_t are

$$E(r_t) = \ln \left[\frac{m_1 + 1}{\sqrt{1 + m_2/(1 + m_1)^2}} \right], \quad \text{Var}(r_t) = \ln \left[1 + \frac{m_2}{(1 + m_1)^2} \right].$$

Because the sum of a finite number of iid normal random variables is normal, $r_t[k]$ is also normally distributed under the normal assumption for $\{r_t\}$. In addition, there is no lower bound for r_t , and the lower bound for R_t is satisfied using $1 + R_t = \exp(r_t)$. However, the lognormal assumption is not consistent with all the properties of historical stock returns. In particular, many stock returns exhibit a positive excess kurtosis.

Stable Distribution

The stable distributions are a natural generalization of normal in that they are stable under addition, which meets the need of continuously compounded returns r_t . Furthermore, stable distributions are capable of capturing excess kurtosis shown by historical stock returns. However, nonnormal stable distributions do not have a finite variance, which is in conflict with most finance theories. In addition, statistical modeling using nonnormal stable distributions is difficult. An example of nonnormal stable distributions is the Cauchy distribution, which is symmetric with respect to its median but has infinite variance.

Scale Mixture of Normal Distributions

Recent studies of stock returns tend to use scale mixture or finite mixture of normal distributions. Under the assumption of scale mixture of normal distributions, the log return r_t is normally distributed with mean μ and variance σ^2 [i.e., $r_t \sim N(\mu, \sigma^2)$]. However, σ^2 is a random variable that follows a positive distribution (e.g., σ^{-2} follows a gamma distribution). An example of finite mixture of normal distributions is

$$r_t \sim (1 - X)N(\mu, \sigma_1^2) + XN(\mu, \sigma_2^2),$$

where X is a Bernoulli random variable such that $P(X = 1) = \alpha$ and $P(X = 0) = 1 - \alpha$ with $0 < \alpha < 1$, σ_1^2 is small, and σ_2^2 is relatively large. For instance, with $\alpha = 0.05$, the finite mixture says that 95% of the returns follow $N(\mu, \sigma_1^2)$ and 5% follow $N(\mu, \sigma_2^2)$. The large value of σ_2^2 enables the mixture to put more mass at the tails of its distribution. The low percentage of returns that are from $N(\mu, \sigma_2^2)$ says that the majority of the returns follow a simple normal distribution. Advantages of mixtures of normal include that they maintain the tractability of normal, have finite higher order moments, and can capture the excess kurtosis. Yet it is hard to estimate the mixture parameters (e.g., the α in the finite-mixture case).

Figure 1.1 shows the probability density functions of a finite mixture of normal, Cauchy, and standard normal random variable. The finite mixture of normal is $(1 - X)N(0, 1) + X \times N(0, 16)$ with X being Bernoulli such that $P(X = 1) =$

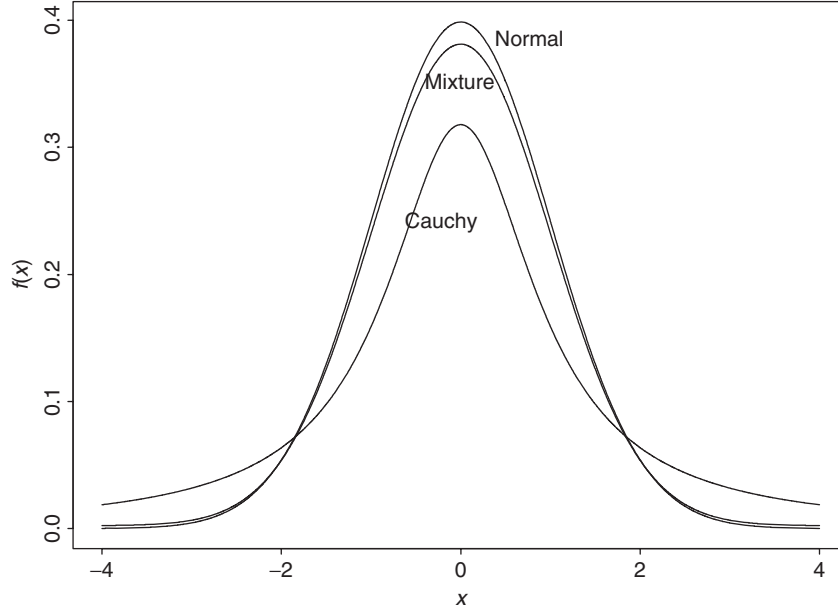


Figure 1.1 Comparison of finite mixture, stable, and standard normal density functions.

0.05, and the density function of Cauchy is

$$f(x) = \frac{1}{\pi(1+x^2)}, \quad -\infty < x < \infty.$$

It is seen that the Cauchy distribution has fatter tails than the finite mixture of normal, which, in turn, has fatter tails than the standard normal.

1.2.3 Multivariate Returns

Let $\mathbf{r}_t = (r_{1t}, \dots, r_{Nt})'$ be the log returns of N assets at time t . The multivariate analyses of Chapters 8 and 10 are concerned with the joint distribution of $\{\mathbf{r}_t\}_{t=1}^T$. This joint distribution can be partitioned in the same way as that of Eq. (1.15). The analysis is then focused on the specification of the conditional distribution function $F(\mathbf{r}_t | \mathbf{r}_{t-1}, \dots, \mathbf{r}_1, \boldsymbol{\theta})$. In particular, how the conditional expectation and conditional covariance matrix of $\mathbf{r}_t$ evolve over time constitute the main subjects of Chapters 8 and 10.

The mean vector and covariance matrix of a random vector $\mathbf{X} = (X_1, \dots, X_p)$ are defined as

$$\begin{aligned} E(\mathbf{X}) &= \boldsymbol{\mu}_x = [E(X_1), \dots, E(X_p)]', \\ \text{Cov}(\mathbf{X}) &= \boldsymbol{\Sigma}_x = E[(\mathbf{X} - \boldsymbol{\mu}_x)(\mathbf{X} - \boldsymbol{\mu}_x)'], \end{aligned}$$

provided that the expectations involved exist. When the data $\{x_1, \dots, x_T\}$ of X are available, the sample mean and covariance matrix are defined as

$$\hat{\mu}_x = \frac{1}{T} \sum_{t=1}^T x_t, \quad \hat{\Sigma}_x = \frac{1}{T-1} \sum_{t=1}^T (x_t - \hat{\mu}_x)(x_t - \hat{\mu}_x)'$$

These sample statistics are consistent estimates of their theoretical counterparts provided that the covariance matrix of X exists. In the finance literature, multivariate normal distribution is often used for the log return r_t .

1.2.4 Likelihood Function of Returns

The partition of Eq. (1.15) can be used to obtain the likelihood function of the log returns $\{r_1, \dots, r_T\}$ of an asset, where for ease in notation the subscript i is omitted from the log return. If the conditional distribution $f(r_t|r_{t-1}, \dots, r_1, \theta)$ is normal with mean μ_t and variance σ_t^2 , then θ consists of the parameters in μ_t and σ_t^2 , and the likelihood function of the data is

$$f(r_1, \dots, r_T; \theta) = f(r_1; \theta) \prod_{t=2}^T \frac{1}{\sqrt{2\pi}\sigma_t} \exp\left[-\frac{(r_t - \mu_t)^2}{2\sigma_t^2}\right], \quad (1.18)$$

where $f(r_1; \theta)$ is the marginal density function of the first observation r_1 . The value of θ that maximizes this likelihood function is the maximum-likelihood estimate (MLE) of θ . Since the log function is monotone, the MLE can be obtained by maximizing the log-likelihood function,

$$\ln f(r_1, \dots, r_T; \theta) = \ln f(r_1; \theta) - \frac{1}{2} \sum_{t=2}^T \left[\ln(2\pi) + \ln(\sigma_t^2) + \frac{(r_t - \mu_t)^2}{\sigma_t^2} \right],$$

which is easier to handle in practice. The log-likelihood function of the data can be obtained in a similar manner if the conditional distribution $f(r_t|r_{t-1}, \dots, r_1; \theta)$ is not normal.

1.2.5 Empirical Properties of Returns

The data used in this section are obtained from the Center for Research in Security Prices (CRSP) of the University of Chicago. Dividend payments, if any, are included in the returns. Figure 1.2 shows the time plots of monthly simple returns and log returns of IBM stock from January 1926 to December 2008. A *time plot* shows the data against the time index. The upper plot is for the simple returns. Figure 1.3 shows the same plots for the monthly returns of value-weighted market index. As expected, the plots show that the basic patterns of simple and log returns are similar.

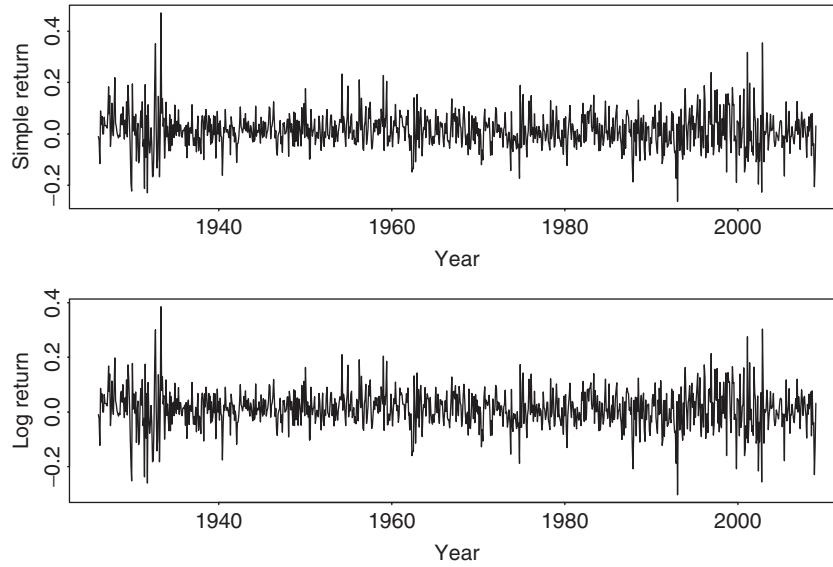


Figure 1.2 Time plots of monthly returns of IBM stock from January 1926 to December 2008. Upper panel is for simple returns, and lower panel is for log returns.

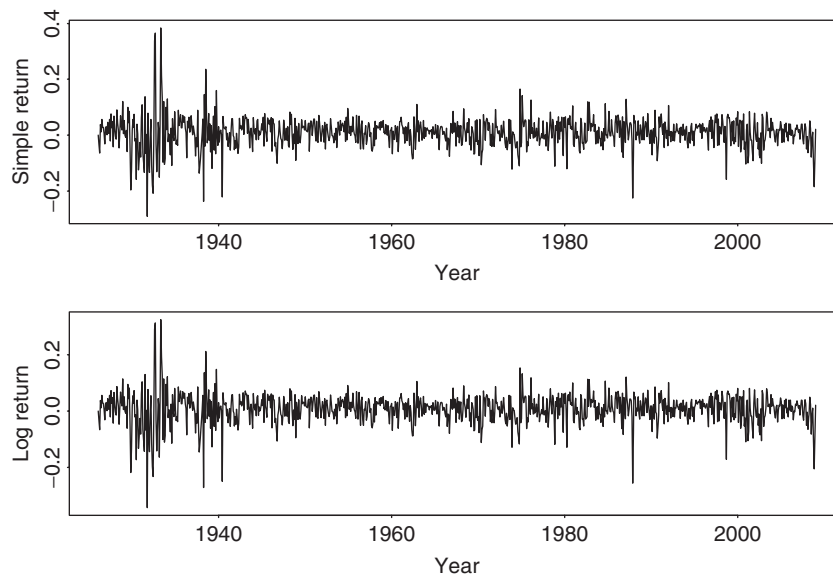


Figure 1.3 Time plots of monthly returns of value-weighted index from January 1926 to December 2008. Upper panel is for simple returns, and lower panel is for log returns.

Table 1.2 provides some descriptive statistics of simple and log returns for selected U.S. market indexes and individual stocks. The returns are for daily and monthly sample intervals and are in percentages. The data spans and sample sizes are also given in Table 1.2. From the table, we make the following observations. (a) Daily returns of the market indexes and individual stocks tend to have high excess kurtoses. For monthly series, the returns of market indexes have higher excess kurtoses than individual stocks. (b) The mean of a daily return series is close to zero, whereas that of a monthly return series is slightly larger. (c) Monthly returns have higher standard deviations than daily returns. (d) Among the daily returns, market indexes have smaller standard deviations than individual stocks. This is in agreement with common sense. (e) The skewness is not a serious problem for both daily and monthly returns. (f) The descriptive statistics show that the difference between simple and log returns is not substantial.

Figure 1.4 shows the empirical density functions of monthly simple and log returns of IBM stock from 1926 to 2008. Also shown, by a dashed line, in each graph is the normal probability density function evaluated by using the sample mean and standard deviation of IBM returns given in Table 1.2. The plots indicate that the normality assumption is questionable for monthly IBM stock returns. The empirical density function has a higher peak around its mean, but fatter tails than that of the corresponding normal distribution. In other words, the empirical density

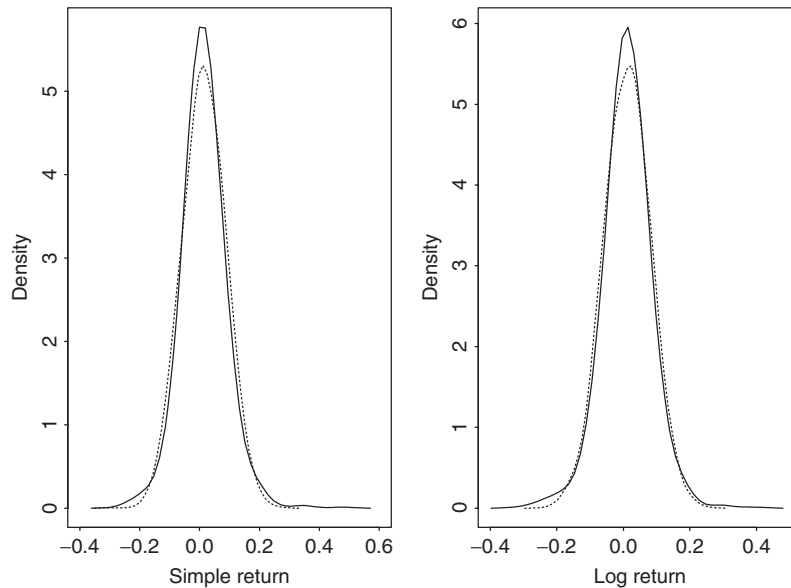


Figure 1.4 Comparison of empirical and normal densities for monthly simple and log returns of IBM stock. Sample period is from January 1926 to December 2008. Left plot is for simple returns and right plot for log returns. Normal density, shown by the dashed line, uses sample mean and standard deviation given in Table 1.2.

function is taller and skinnier, but with a wider support than the corresponding normal density.

1.3 PROCESSES CONSIDERED

Besides the return series, we also consider the volatility process and the behavior of extreme returns of an asset. The volatility process is concerned with the evolution of conditional variance of the return over time. This is a topic of interest because, as shown in Figures 1.2 and 1.3, the variabilities of returns vary over time and appear in clusters. In application, volatility plays an important role in pricing options and risk management. By extremes of a return series, we mean the large positive or negative returns. Table 1.2 shows that the minimum and maximum of a return series can be substantial. The negative extreme returns are important in risk management, whereas positive extreme returns are critical to holding a short position. We study properties and applications of extreme returns, such as the frequency of occurrence, the size of an extreme, and the impacts of economic variables on the extremes, in Chapter 7.

Other financial time series considered in the book include interest rates, exchange rates, bond yields, and quarterly earning per share of a company. Figure 1.5 shows the time plots of two U.S. monthly interest rates. They are the 10-year and 1-year

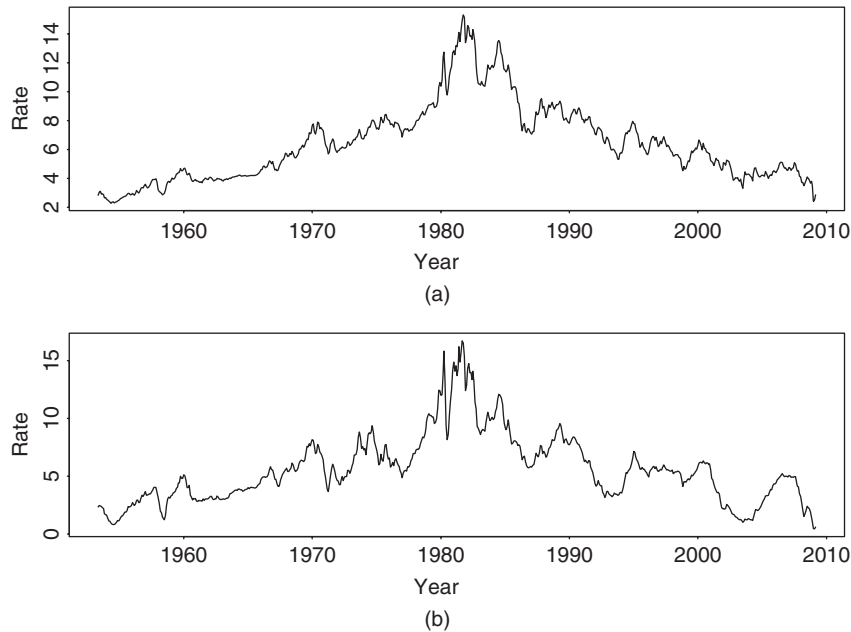


Figure 1.5 Time plots of monthly U.S. interest rates from April 1953 to February 2009: (a) 10-year Treasury constant maturity rate and (b) 1-year maturity rate.

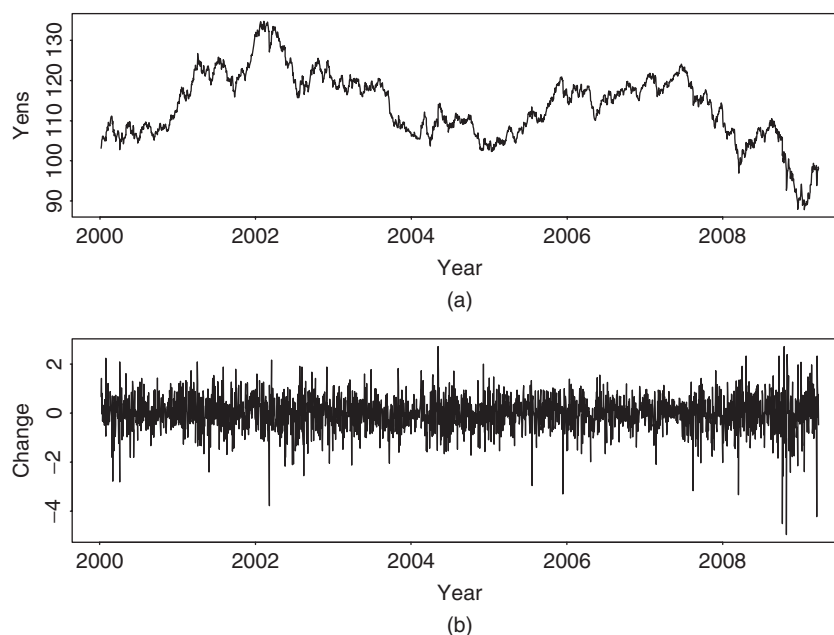


Figure 1.6 Time plot of daily exchange rate between U.S. dollar and Japanese yen from January 4, 2000, to March 27, 2009: (a) exchange rate and (b) changes in exchange rate.

Treasury constant maturity rates from April 1954 to February 2009. As expected, the two interest rates moved in unison, but the 1-year rates appear to be more volatile. Figure 1.6 shows the daily exchange rate between the U.S. dollar and the Japanese yen from January 4, 2000, to March 27, 2009. From the plot, the exchange rate encountered occasional big changes in the sampling period. Table 1.3 provides some descriptive statistics for selected U.S. financial time series. The monthly bond returns obtained from CRSP are Fama bond portfolio returns from January 1952 to December 2008. The interest rates are obtained from the Federal Reserve Bank of St. Louis. The weekly 3-month Treasury bill rate started on January 8, 1954, and the 6-month rate started on December 12, 1958. Both series ended on March 27, 2009. For the interest rate series, the sample means are proportional to the time to maturity, but the sample standard deviations are inversely proportional to the time to maturity. For the bond returns, the sample standard deviations are positively related to the time to maturity, whereas the sample means remain stable for all maturities. Most of the series considered have positive excess kurtoses.

With respect to the empirical characteristics of returns shown in Table 1.2, Chapters 2–4 focus on the first four moments of a return series and Chapter 7 on the behavior of minimum and maximum returns. Chapters 8 and 10 are concerned with moments of and the relationships between multiple asset returns, and Chapter 5 addresses properties of asset returns when the time interval is small. An introduction to mathematical finance is given in Chapter 6.

TABLE 1.3 Descriptive Statistics of Selected U.S. Financial Time Series^a

Maturity	Mean	Standard Deviation	Skewness	Excess Kurtosis	Minimum	Maximum
<i>Monthly Bond Returns: Jan. 1952 to Dec. 2008, $T = 684$</i>						
1–12 months	0.45	0.35	2.47	13.14	−0.40	3.52
12–24 months	0.49	0.67	1.88	15.44	−2.94	6.85
24–36 months	0.52	0.98	1.37	12.92	−4.90	9.33
48–60 months	0.53	1.40	0.60	4.83	−5.78	10.06
61–120 months	0.55	1.69	0.65	4.79	−7.35	10.92
<i>Monthly Treasury Rates: April 1953 to February 2009, $T = 671$</i>						
1 year	5.59	2.98	1.02	1.32	0.44	16.72
3 years	5.98	2.85	0.95	0.95	1.07	16.22
5 years	6.19	2.77	0.97	0.82	1.52	15.93
10 years	6.40	2.69	0.95	0.61	2.29	15.32
<i>Weekly Treasury Bill Rates: End on March 27, 2009.</i>						
3 months	5.07	2.82	1.08	1.80	0.02	16.76
6 months	5.52	2.73	0.99	1.53	0.20	15.76

^aThe data are in percentages. The weekly 3-month Treasury bill rate started from January 8, 1954, and the 6-month rate started from December 12, 1958. The sample sizes for Treasury bill rates are 2882 and 2625, respectively. Data sources are given in the text.

APPENDIX: R PACKAGES

R is a free software available from <http://www.r-project.org>. One can click *CRAN* on its Web page to select a nearby *CRAN Mirror* to download and install the software and selected packages. For financial time series analysis, the *Rmetrics* of Diethelm Wuertz and his associates have produced many useful packages, including *fBasics*, *timeSeries*, *fGarch*, etc. We use many functions of these packages in this book. Further information concerning installing R and the commands used can be found either on the Web page of this book or on the author's teaching Web page.

R and S-Plus are objective-oriented software. They enable users to create many objects. For instance, one can use the command `ts` to create a time series object. Treating time series data as a time series object in R has some advantages, but it requires some learning to get used to it. It is, however, not necessary to create a time series object in R to perform the analyses discussed in this book. As an illustration, consider the monthly simple returns to the General Motors stock from January 1975 to December 2008; see Exercise 1.2. The data have 408 observations. The following R commands are used to illustrate the points:

```
> da=read.table("m-gm3dx7508.txt",header=T) % Load data
> gm=da[,2] % Column 2 contains GM stock returns
> gm1=ts(gm,frequency=12,start=c(1975,1))
% Creates a ts object.
> par(mfcol=c(2,1)) % Put two plots on a page.
```

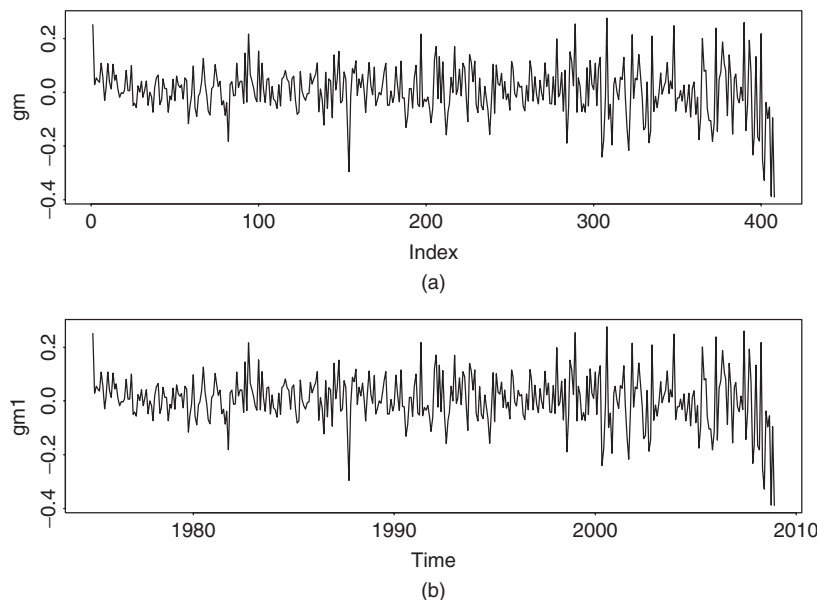


Figure 1.7 Time plots of monthly simple returns to General Motors stock from January 1975 to December 2008: (a) and (b) are without and with time series object, respectively.

```
> plot(gm, type='l')
> plot(gm1, type='l')
> acf(gm, lag=24)
> acf(gm1, lag=24)
```

In the `ts` command, `frequency = 12` says that the time unit is *year* and there are 12 equally spaced observations in each time unit, and `start = c(1975,1)` means the starting time is January 1975. Frequency and start are the two basic arguments needed in R to create a time series object. For further details, please use `help(ts)` in R to obtain details of the command. Here `gm1` is a time series object in R, but `gm` is not. Figures 1.7 and 1.8 show, respectively, the time plot and autocorrelation function (ACF) of the returns of GM stock. In each figure, the upper plot is produced without using time series object, whereas the lower plot is produced by a time series object. The upper and lower plots are identical except for the horizontal label. For the time plot, the time series object uses calendar time to label the *x* axis, which is preferred. On the other hand, for the ACF plot, the time series object uses fractions of time unit in the label, not the commonly used time lags.

EXERCISES

- 1.1. Consider the daily stock returns of American Express (AXP), Caterpillar (CAT), and Starbucks (SBUX) from January 1999 to December 2008. The

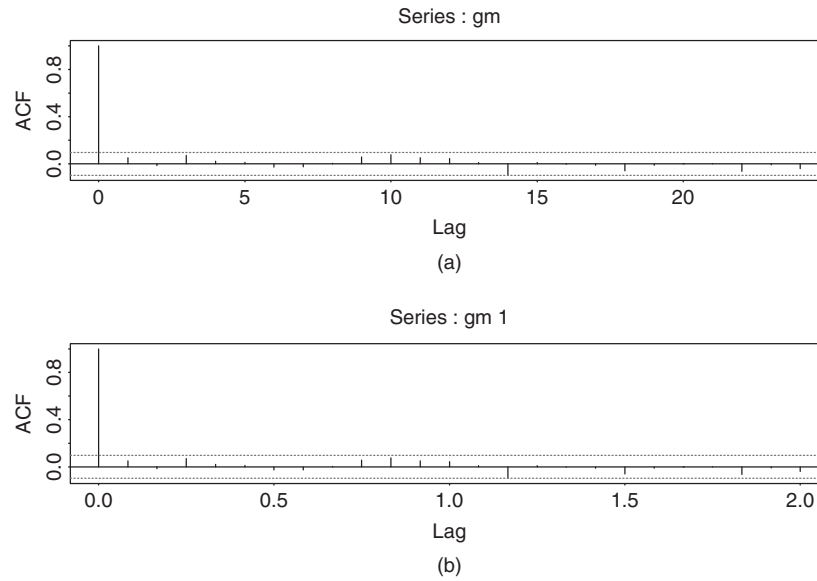


Figure 1.8 Sample ACFs of the monthly simple returns to General Motors stock from January 1975 to December 2008: (a) and (b) are without and with time series object, respectively.

data are simple returns given in the file `d-3stocks9908.txt` (date, axp, cat, sbux).

- (a) Express the simple returns in percentages. Compute the sample mean, standard deviation, skewness, excess kurtosis, minimum, and maximum of the percentage simple returns.
 - (b) Transform the simple returns to log returns.
 - (c) Express the log returns in percentages. Compute the sample mean, standard deviation, skewness, excess kurtosis, minimum, and maximum of the percentage log returns.
 - (d) Test the null hypothesis that the mean of the log returns of each stock is zero. That is, perform three separate tests. Use 5% significance level to draw your conclusion.
- 1.2. Answer the same questions as in Exercise 1.1 but using monthly stock returns for General Motors (GM), CRSP value-weighted index (VW), CRSP equal-weighted index (EW), and S&P composite index from January 1975 to December 2008. The returns of the indexes include dividend distributions. Data file is `m-gm3dx7508.txt` (date, gm, vw, ew, sp).
 - 1.3. Consider the monthly stock returns of S&P composite index from January 1975 to December 2008 in Exercise 1.2. Answer the following questions:
 - (a) What is the average annual log return over the data span?

- (b) Assume that there were no transaction costs. If one invested \$1.00 on the S&P composite index at the beginning of 1975, what was the value of the investment at the end of 2008?
- 1.4. Consider the daily log returns of American Express stock from January 1999 to December 2008 as in Exercise 1.1. Use the 5% significance level to perform the following tests: (a) Test the null hypothesis that the skewness measure of the returns is zero. (b) Test the null hypothesis that the excess kurtosis of the returns is zero.
- 1.5. Daily foreign exchange rates (spot rates) can be obtained from the Federal Reserve Bank in Chicago. The data are the noon buying rates in New York City certified by the Federal Reserve Bank of New York. Consider the exchange rates between the U.S. dollar and the Canadian dollar, euro, U.K. pound, and the Japanese yen from January 4, 2000, to March 27, 2009. The data are also on the Web. (a) Compute the daily log return of each exchange rate. (b) Compute the sample mean, standard deviation, skewness, excess kurtosis, minimum, and maximum of the log returns of each exchange rate. (c) Discuss the empirical characteristics of the log returns of exchange rates. (d) Obtain a density plot of the daily long returns of dollar–euro exchange rate.

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CHAPTER 2

Linear Time Series Analysis and Its Applications

In this chapter, we discuss basic theories of linear time series analysis, introduce some simple econometric models useful for analyzing financial data, and apply the models to financial time series such as asset returns. Discussions of the concepts are brief with emphasis on those relevant to financial applications. Understanding the simple time series models introduced here will go a long way to better appreciate the more sophisticated financial econometric models of the later chapters. There are many time series textbooks available. For basic concepts of linear time series analysis, see Box, Jenkins, and Reinsel (1994, Chapters 2 and 3) and Brockwell and Davis (1996, Chapters 1–3).

Treating an asset return (e.g., log return r_t of a stock) as a collection of random variables over time, we have a time series $\{r_t\}$. Linear time series analysis provides a natural framework to study the dynamic structure of such a series. The theories of linear time series discussed include stationarity, dynamic dependence, autocorrelation function, modeling, and forecasting. The econometric models introduced include (a) simple autoregressive (AR) models, (b) simple moving-average (MA) models, (b) mixed autoregressive moving-average (ARMA) models, (c) seasonal models, (d) unit-root nonstationarity, (e) regression models with time series errors, and (f) fractionally differenced models for long-range dependence. For an asset return r_t , simple models attempt to capture the linear relationship between r_t and information available prior to time t . The information may contain the historical values of r_t and the random vector Y in Eq. (1.14), which describes the economic environment under which the asset price is determined. As such, correlation plays an important role in understanding these models. In particular, correlations between the variable of interest and its past values become the focus of linear time series analysis. These correlations are referred to as *serial correlations* or *autocorrelations*. They are the basic tool for studying a stationary time series.

2.1 STATIONARITY

The foundation of time series analysis is stationarity. A time series $\{r_t\}$ is said to be *strictly stationary* if the joint distribution of $(r_{t_1}, \dots, r_{t_k})$ is identical to that of $(r_{t_1+t}, \dots, r_{t_k+t})$ for all t , where k is an arbitrary positive integer and $(t_1, \dots, t_k)$ is a collection of k positive integers. In other words, strict stationarity requires that the joint distribution of $(r_{t_1}, \dots, r_{t_k})$ is invariant under time shift. This is a very strong condition that is hard to verify empirically. A weaker version of stationarity is often assumed. A time series $\{r_t\}$ is *weakly stationary* if both the mean of r_t and the covariance between r_t and $r_{t-\ell}$ are time invariant, where ℓ is an arbitrary integer. More specifically, $\{r_t\}$ is weakly stationary if (a) $E(r_t) = \mu$, which is a constant, and (b) $\text{Cov}(r_t, r_{t-\ell}) = \gamma_\ell$, which only depends on ℓ . In practice, suppose that we have observed T data points $\{r_t | t = 1, \dots, T\}$. The weak stationarity implies that the time plot of the data would show that the T values fluctuate with constant variation around a fixed level. In applications, weak stationarity enables one to make inference concerning future observations (e.g., prediction).

Implicitly, in the condition of weak stationarity, we assume that the first two moments of r_t are finite. From the definitions, if r_t is strictly stationary and its first two moments are finite, then r_t is also weakly stationary. The converse is not true in general. However, if the time series r_t is normally distributed, then weak stationarity is equivalent to strict stationarity. In this book, we are mainly concerned with weakly stationary series.

The covariance $\gamma_\ell = \text{Cov}(r_t, r_{t-\ell})$ is called the lag- ℓ autocovariance of r_t . It has two important properties: (a) $\gamma_0 = \text{Var}(r_t)$ and (b) $\gamma_{-\ell} = \gamma_\ell$. The second property holds because $\text{Cov}(r_t, r_{t-(-\ell)}) = \text{Cov}(r_{t-(-\ell)}, r_t) = \text{Cov}(r_{t+\ell}, r_t) = \text{Cov}(r_{t_1}, r_{t_1-\ell})$, where $t_1 = t + \ell$.

In the finance literature, it is common to assume that an asset return series is weakly stationary. This assumption can be checked empirically provided that a sufficient number of historical returns are available. For example, one can divide the data into subsamples and check the consistency of the results obtained across the subsamples.

2.2 CORRELATION AND AUTOCORRELATION FUNCTION

The correlation coefficient between two random variables X and Y is defined as

$$\rho_{x,y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{E[(X - \mu_x)(Y - \mu_y)]}{\sqrt{E(X - \mu_x)^2 E(Y - \mu_y)^2}},$$

where μ_x and μ_y are the mean of X and Y , respectively, and it is assumed that the variances exist. This coefficient measures the strength of linear dependence between X and Y , and it can be shown that $-1 \leq \rho_{x,y} \leq 1$ and $\rho_{x,y} = \rho_{y,x}$. The two random variables are uncorrelated if $\rho_{x,y} = 0$. In addition, if both X and Y are normal random variables, then $\rho_{x,y} = 0$ if and only if X and Y are independent. When the

sample $\{(x_t, y_t)\}_{t=1}^T$ is available, the correlation can be consistently estimated by its sample counterpart

$$\hat{\rho}_{x,y} = \frac{\sum_{t=1}^T (x_t - \bar{x})(y_t - \bar{y})}{\sqrt{\sum_{t=1}^T (x_t - \bar{x})^2 \sum_{t=1}^T (y_t - \bar{y})^2}},$$

where $\bar{x} = \sum_{t=1}^T x_t / T$ and $\bar{y} = \sum_{t=1}^T y_t / T$ are the sample mean of X and Y , respectively.

Autocorrelation Function (ACF)

Consider a weakly stationary return series r_t . When the linear dependence between r_t and its past values r_{t-i} is of interest, the concept of correlation is generalized to autocorrelation. The correlation coefficient between r_t and $r_{t-\ell}$ is called the lag- ℓ *autocorrelation* of r_t and is commonly denoted by ρ_ℓ , which under the weak stationarity assumption is a function of ℓ only. Specifically, we define

$$\rho_\ell = \frac{\text{Cov}(r_t, r_{t-\ell})}{\sqrt{\text{Var}(r_t)\text{Var}(r_{t-\ell})}} = \frac{\text{Cov}(r_t, r_{t-\ell})}{\text{Var}(r_t)} = \frac{\gamma_\ell}{\gamma_0}, \quad (2.1)$$

where the property $\text{Var}(r_t) = \text{Var}(r_{t-\ell})$ for a weakly stationary series is used. From the definition, we have $\rho_0 = 1$, $\rho_\ell = \rho_{-\ell}$, and $-1 \leq \rho_\ell \leq 1$. In addition, a weakly stationary series r_t is not serially correlated if and only if $\rho_\ell = 0$ for all $\ell > 0$.

For a given sample of returns $\{r_t\}_{t=1}^T$, let $\bar{r}$ be the sample mean (i.e., $\bar{r} = \sum_{t=1}^T r_t / T$). Then the lag-1 sample autocorrelation of r_t is

$$\hat{\rho}_1 = \frac{\sum_{t=2}^T (r_t - \bar{r})(r_{t-1} - \bar{r})}{\sum_{t=1}^T (r_t - \bar{r})^2}.$$

Under some general conditions, $\hat{\rho}_1$ is a consistent estimate of ρ_1 . For example, if $\{r_t\}$ is an independent and identically distributed (iid) sequence and $E(r_t^2) < \infty$, then $\hat{\rho}_1$ is asymptotically normal with mean zero and variance $1/T$; see Brockwell and Davis (1991, Theorem 7.2.2). This result can be used in practice to test the null hypothesis $H_0 : \rho_1 = 0$ versus the alternative hypothesis $H_a : \rho_1 \neq 0$. The test statistic is the usual t ratio, which is $\sqrt{T}\hat{\rho}_1$ and follows asymptotically the standard normal distribution. The null hypothesis H_0 is rejected if the t ratio is large in magnitude or, equivalently, the p value of the t ratio is small, say less than 0.05. In general, the lag- ℓ sample autocorrelation of r_t is defined as

$$\hat{\rho}_\ell = \frac{\sum_{t=\ell+1}^T (r_t - \bar{r})(r_{t-\ell} - \bar{r})}{\sum_{t=1}^T (r_t - \bar{r})^2}, \quad 0 \leq \ell < T - 1. \quad (2.2)$$

If $\{r_t\}$ is an iid sequence satisfying $E(r_t^2) < \infty$, then $\hat{\rho}_\ell$ is asymptotically normal with mean zero and variance $1/T$ for any fixed positive integer ℓ . More generally, if r_t is a weakly stationary time series satisfying $r_t = \mu + \sum_{i=0}^q \psi_i a_{t-i}$, where

$\psi_0 = 1$ and $\{a_j\}$ is a sequence of iid random variables with mean zero, then $\hat{\rho}_\ell$ is asymptotically normal with mean zero and variance $(1 + 2 \sum_{i=1}^q \rho_i^2)/T$ for $\ell > q$. This is referred to as Bartlett's formula in the time series literature; see Box, Jenkins, and Reinsel (1994). For more information about the asymptotic distribution of sample autocorrelations, see Fuller (1976, Chapter 6) and Brockwell and Davis (1991, Chapter 7).

Testing Individual ACF

For a given positive integer ℓ , the previous result can be used to test $H_0 : \rho_\ell = 0$ vs. $H_a : \rho_\ell \neq 0$. The test statistic is

$$t \text{ ratio} = \frac{\hat{\rho}_\ell}{\sqrt{(1 + 2 \sum_{i=1}^{\ell-1} \hat{\rho}_i^2)/T}}.$$

If $\{r_t\}$ is a stationary Gaussian series satisfying $\rho_j = 0$ for $j > \ell$, the t ratio is asymptotically distributed as a standard normal random variable. Hence, the decision rule of the test is to reject H_0 if $|t \text{ ratio}| > Z_{\alpha/2}$, where $Z_{\alpha/2}$ is the $100(1 - \alpha/2)$ th percentile of the standard normal distribution. For simplicity, many software packages use $1/T$ as the asymptotic variance of $\hat{\rho}_\ell$ for all $\ell \neq 0$. They essentially assume that the underlying time series is an iid sequence.

In finite samples, $\hat{\rho}_\ell$ is a biased estimator of ρ_ℓ . The bias is in the order of $1/T$, which can be substantial when the sample size T is small. In most financial applications, T is relatively large so that the bias is not serious.

Portmanteau Test

Financial applications often require to test jointly that several autocorrelations of r_t are zero. Box and Pierce (1970) propose the Portmanteau statistic

$$Q^*(m) = T \sum_{\ell=1}^m \hat{\rho}_\ell^2$$

as a test statistic for the null hypothesis $H_0 : \rho_1 = \dots = \rho_m = 0$ against the alternative hypothesis $H_a : \rho_i \neq 0$ for some $i \in \{1, \dots, m\}$. Under the assumption that $\{r_t\}$ is an iid sequence with certain moment conditions, $Q^*(m)$ is asymptotically a chi-squared random variable with m degrees of freedom.

Ljung and Box (1978) modify the $Q^*(m)$ statistic as below to increase the power of the test in finite samples,

$$Q(m) = T(T+2) \sum_{\ell=1}^m \frac{\hat{\rho}_\ell^2}{T-\ell}. \quad (2.3)$$

The decision rule is to reject H_0 if $Q(m) > \chi_\alpha^2$, where χ_α^2 denotes the $100(1 - \alpha)$ th percentile of a chi-squared distribution with m degrees of freedom. Most software

packages will provide the p value of $Q(m)$. The decision rule is then to reject H_0 if the p value is less than or equal to α , the significance level.

In practice, the choice of m may affect the performance of the $Q(m)$ statistic. Several values of m are often used. Simulation studies suggest that the choice of $m \approx \ln(T)$ provides better power performance. This general rule needs modification in analysis of seasonal time series for which autocorrelations with lags at multiples of the seasonality are more important.

The statistics $\hat{\rho}_1, \hat{\rho}_2, \dots$ defined in Eq. (2.2) is called the *sample autocorrelation function* (ACF) of r_t . It plays an important role in linear time series analysis. As a matter of fact, a linear time series model can be characterized by its ACF, and linear time series modeling makes use of the sample ACF to capture the linear dynamic of the data. Figure 2.1 shows the sample autocorrelation functions of monthly simple and log returns of IBM stock from January 1926 to December 2008. The two sample ACFs are very close to each other, and they suggest that the serial correlations of monthly IBM stock returns are very small, if any. The sample ACFs are all within their two standard error limits, indicating that they are not significantly different from zero at the 5% level. In addition, for the simple returns, the Ljung–Box statistics give $Q(5) = 3.37$ and $Q(10) = 13.99$, which correspond to p values of 0.64 and 0.17, respectively, based on chi-squared

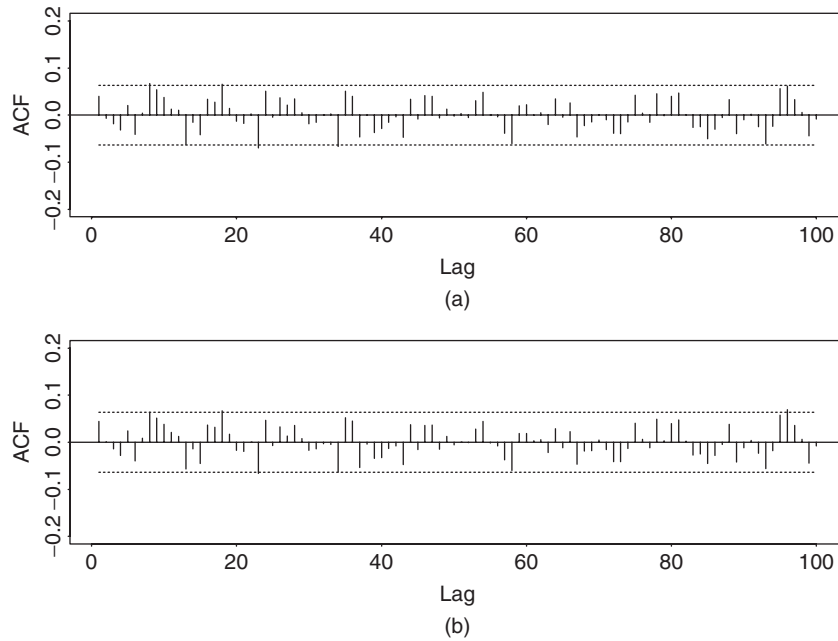


Figure 2.1 Sample autocorrelation functions of monthly (a) simple returns and (b) log returns of IBM stock from January 1926 to December 2008. In each plot, two horizontal dashed lines denote two standard error limits of sample ACF.

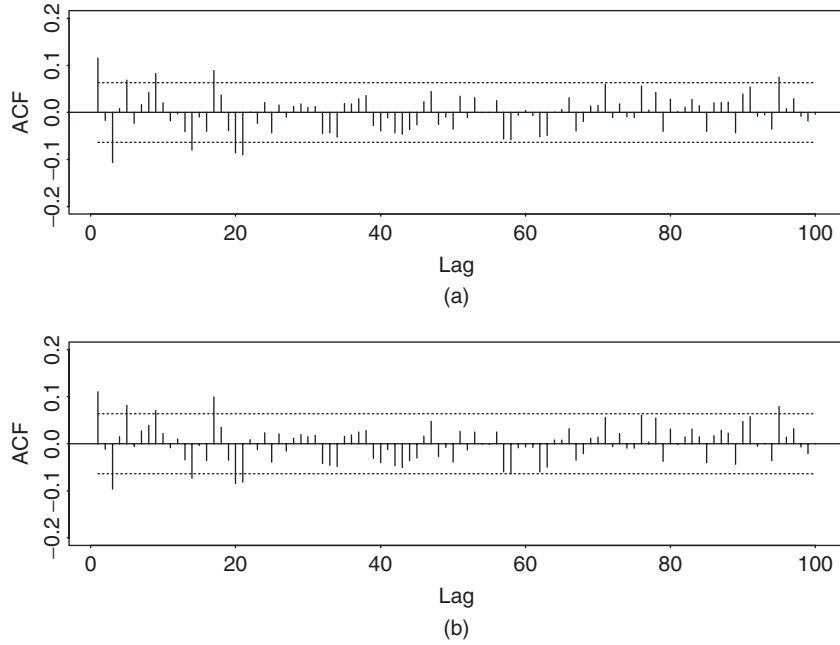


Figure 2.2 Sample autocorrelation functions of monthly (a) simple returns and (b) log returns of value-weighted index of U.S. markets from January 1926 to December 2008. In each plot, two horizontal dashed lines denote two standard error limits of sample ACF.

distributions with 5 and 10 degrees of freedom. For the log returns, we have $Q(5) = 3.52$ and $Q(10) = 13.39$ with p values 0.62 and 0.20, respectively. The joint tests confirm that monthly IBM stock returns have no significant serial correlations. Figure 2.2 shows the same for the monthly returns of the value-weighted index from the Center for Research in Security Prices (CRSP), at the University of Chicago. There are some significant serial correlations at the 5% level for both return series. The Ljung–Box statistics give $Q(5) = 29.71$ and $Q(10) = 39.55$ for the simple returns and $Q(5) = 28.38$ and $Q(10) = 36.16$ for the log returns. The p values of these four test statistics are all less than 0.0001, suggesting that monthly returns of the value-weighted index are serially correlated. Thus, the monthly market index return seems to have stronger serial dependence than individual stock returns.

In the finance literature, a version of the capital asset pricing model (CAPM) theory is that the return $\{r_t\}$ of an asset is not predictable and should have no autocorrelations. Testing for zero autocorrelations has been used as a tool to check the efficient market assumption. However, the way by which stock prices are determined and index returns are calculated might introduce autocorrelations in the observed return series. This is particularly so in analysis of high-frequency financial data. We discuss some of these issues, such as bid–ask bounce and nonsynchronous trading, in Chapter 5.

R Demonstration

The following output has been edited and % denotes explanation:

```
> da=read.table("m-ibm3dx2608.txt",header=T) % Load data
> da[1,] % Check the 1st row of the data
      date      rtn      vwrtm      ewrtm      sprtm
1 19260130 -0.010381 0.000724 0.023174 0.022472
> sibm=da[,2] % Get the IBM simple returns
> Box.test(sibm,lag=5,type='Ljung') % Ljung-Box statistic Q(5)
      Box-Ljung test
```

```
data: sibm
X-squared = 3.3682, df = 5, p-value = 0.6434
```

```
> libm=log(sibm+1) % Log IBM returns
> Box.test(libm,lag=5,type='Ljung')
```

```
      Box-Ljung test
```

```
data: libm
X-squared = 3.5236, df = 5, p-value = 0.6198
```

S-Plus Demonstration

Output edited.

```
> module(finmetrics)
> da=read.table("m-ibm3dx2608.txt",header=T) % Load data
> da[1,] % Check the 1st row of the data
      date      rtn      vwrtm      ewrtm      sprtm
1 19260130 -0.010381 0.000724 0.023174 0.022472
> sibm=da[,2] % Get IBM simple returns
> autocorTest(sibm,lag=5) % Ljung-Box Q(5) test
```

```
Test for Autocorrelation: Ljung-Box
Null Hypothesis: no autocorrelation
Test Statistics:
Test Stat 3.3682
p.value 0.6434
```

```
Dist. under Null: chi-square with 5 degrees of freedom
Total Observ.: 996
> libm=log(sibm+1) % IBM log returns
> autocorTest(libm,lag=5)
```

```
Test for Autocorrelation: Ljung-Box
Null Hypothesis: no autocorrelation
Test Statistics:
Test Stat 3.5236
p.value 0.6198
```

2.3 WHITE NOISE AND LINEAR TIME SERIES

White Noise

A time series r_t is called a white noise if $\{r_t\}$ is a sequence of independent and identically distributed random variables with finite mean and variance. In particular, if r_t is normally distributed with mean zero and variance σ^2 , the series is called a Gaussian white noise. For a white noise series, all the ACFs are zero. In practice, if all sample ACFs are close to zero, then the series is a white noise series. Based on Figures 2.1 and 2.2, the monthly returns of IBM stock are close to white noise, whereas those of the value-weighted index are not.

The behavior of sample autocorrelations of the value-weighted index returns indicates that for some asset returns it is necessary to model the serial dependence before further analysis can be made. In what follows, we discuss some simple time series models that are useful in modeling the dynamic structure of a time series. The concepts presented are also useful later in modeling volatility of asset returns.

Linear Time Series

A time series r_t is said to be linear if it can be written as

$$r_t = \mu + \sum_{i=0}^{\infty} \psi_i a_{t-i}, \quad (2.4)$$

where μ is the mean of r_t , $\psi_0 = 1$, and $\{a_t\}$ is a sequence of iid random variables with mean zero and a well-defined distribution (i.e., $\{a_t\}$ is a white noise series). It will be seen later that a_t denotes the new information at time t of the time series and is often referred to as the *innovation* or *shock* at time t . In this book, we are mainly concerned with the case where a_t is a continuous random variable. Not all financial time series are linear, however. We study nonlinearity and nonlinear models in Chapter 4.

For a linear time series in Eq. (2.4), the dynamic structure of r_t is governed by the coefficients ψ_i , which are called the ψ *weights* of r_t in the time series literature. If r_t is weakly stationary, we can obtain its mean and variance easily by using the independence of $\{a_t\}$ as

$$E(r_t) = \mu, \quad \text{Var}(r_t) = \sigma_a^2 \sum_{i=0}^{\infty} \psi_i^2, \quad (2.5)$$

where σ_a^2 is the variance of a_t . Because $\text{Var}(r_t) < \infty$, $\{\psi_i^2\}$ must be a convergent sequence, that is, $\psi_i^2 \rightarrow 0$ as $i \rightarrow \infty$. Consequently, for a stationary series, impact of the remote shock a_{t-i} on the return r_t vanishes as i increases.

The lag- ℓ autocovariance of r_t is

$$\gamma_\ell = \text{Cov}(r_t, r_{t-\ell}) = E \left[\left(\sum_{i=0}^{\infty} \psi_i a_{t-i} \right) \left(\sum_{j=0}^{\infty} \psi_j a_{t-\ell-j} \right) \right]$$

$$\begin{aligned}
&= E \left(\sum_{i,j=0}^{\infty} \psi_i \psi_j a_{t-i} a_{t-\ell-j} \right) \\
&= \sum_{j=0}^{\infty} \psi_{j+\ell} \psi_j E(a_{t-\ell-j}^2) = \sigma_a^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+\ell}.
\end{aligned} \tag{2.6}$$

Consequently, the ψ weights are related to the autocorrelations of r_t as follows:

$$\rho_\ell = \frac{\gamma_\ell}{\gamma_0} = \frac{\sum_{i=0}^{\infty} \psi_i \psi_{i+\ell}}{1 + \sum_{i=1}^{\infty} \psi_i^2}, \quad \ell \geq 0, \tag{2.7}$$

where $\psi_0 = 1$. Linear time series models are econometric and statistical models used to describe the pattern of the ψ weights of r_t . For a weakly stationary time series, $\psi_i \rightarrow 0$ as $i \rightarrow \infty$ and, hence, ρ_ℓ converges to zero as ℓ increases. For asset returns, this means that, as expected, the linear dependence of current return r_t on the remote past return $r_{t-\ell}$ diminishes for large ℓ .

2.4 SIMPLE AR MODELS

The fact that the monthly return r_t of CRSP value-weighted index has a statistically significant lag-1 autocorrelation indicates that the lagged return r_{t-1} might be useful in predicting r_t . A simple model that makes use of such predictive power is

$$r_t = \phi_0 + \phi_1 r_{t-1} + a_t, \tag{2.8}$$

where $\{a_t\}$ is assumed to be a white noise series with mean zero and variance σ_a^2 . This model is in the same form as the well-known simple linear regression model in which r_t is the dependent variable and r_{t-1} is the explanatory variable. In the time series literature, model (2.8) is referred to as an autoregressive (AR) model of order 1 or simply an AR(1) model. This simple model is also widely used in stochastic volatility modeling when r_t is replaced by its log volatility; see Chapters 3 and 12.

The AR(1) model in Eq. (2.8) has several properties similar to those of the simple linear regression model. However, there are some significant differences between the two models, which we discuss later. Here it suffices to note that an AR(1) model implies that, conditional on the past return r_{t-1} , we have

$$E(r_t | r_{t-1}) = \phi_0 + \phi_1 r_{t-1}, \quad \text{Var}(r_t | r_{t-1}) = \text{Var}(a_t) = \sigma_a^2.$$

That is, given the past return r_{t-1} , the current return is centered around $\phi_0 + \phi_1 r_{t-1}$ with standard deviation σ_a . This is a Markov property such that conditional on r_{t-1} , the return r_t is not correlated with r_{t-i} for $i > 1$. Obviously, there are situations in which r_{t-1} alone cannot determine the conditional expectation of r_t and a more

flexible model must be sought. A straightforward generalization of the AR(1) model is the AR(p) model:

$$r_t = \phi_0 + \phi_1 r_{t-1} + \cdots + \phi_p r_{t-p} + a_t, \quad (2.9)$$

where p is a nonnegative integer and $\{a_t\}$ is defined in Eq. (2.8). This model says that the past p variables r_{t-i} ($i = 1, \dots, p$) jointly determine the conditional expectation of r_t given the past data. The AR(p) model is in the same form as a multiple linear regression model with lagged values serving as the explanatory variables.

2.4.1 Properties of AR Models

For effective use of AR models, it pays to study their basic properties. We discuss properties of AR(1) and AR(2) models in detail and give the results for the general AR(p) model.

AR(1) Model

We begin with the sufficient and necessary condition for weak stationarity of the AR(1) model in Eq. (2.8). Assuming that the series is weakly stationary, we have $E(r_t) = \mu$, $\text{Var}(r_t) = \gamma_0$, and $\text{Cov}(r_t, r_{t-j}) = \gamma_j$, where μ and γ_0 are constant and γ_j is a function of j , not t . We can easily obtain the mean, variance, and autocorrelations of the series as follows. Taking the expectation of Eq. (2.8) and because $E(a_t) = 0$, we obtain

$$E(r_t) = \phi_0 + \phi_1 E(r_{t-1}).$$

Under the stationarity condition, $E(r_t) = E(r_{t-1}) = \mu$ and hence

$$\mu = \phi_0 + \phi_1 \mu \quad \text{or} \quad E(r_t) = \mu = \frac{\phi_0}{1 - \phi_1}.$$

This result has two implications for r_t . First, the mean of r_t exists if $\phi_1 \neq 1$. Second, the mean of r_t is zero if and only if $\phi_0 = 0$. Thus, for a stationary AR(1) process, the constant term ϕ_0 is related to the mean of r_t via $\phi_0 = (1 - \phi_1)\mu$ and $\phi_0 = 0$ implies that $E(r_t) = 0$.

Next, using $\phi_0 = (1 - \phi_1)\mu$, the AR(1) model can be rewritten as

$$r_t - \mu = \phi_1(r_{t-1} - \mu) + a_t. \quad (2.10)$$

By repeated substitutions, the prior equation implies that

$$\begin{aligned} r_t - \mu &= a_t + \phi_1 a_{t-1} + \phi_1^2 a_{t-2} + \cdots \\ &= \sum_{i=0}^{\infty} \phi_1^i a_{t-i}. \end{aligned} \quad (2.11)$$

This equation expresses an AR(1) model in the form of Eq. (2.4) with $\psi_i = \phi_1^i$. Thus, $r_t - \mu$ is a linear function of a_{t-i} for $i \geq 0$. Using this property and the independence of the series $\{a_t\}$, we obtain $E[(r_t - \mu)a_{t+1}] = 0$. By the stationarity assumption, we have $\text{Cov}(r_{t-1}, a_t) = E[(r_{t-1} - \mu)a_t] = 0$. This latter result can also be seen from the fact that r_{t-1} occurred before time t and a_t does not depend on any past information. Taking the square, then the expectation of Eq. (2.10), we obtain

$$\text{Var}(r_t) = \phi_1^2 \text{Var}(r_{t-1}) + \sigma_a^2,$$

where σ_a^2 is the variance of a_t , and we make use of the fact that the covariance between r_{t-1} and a_t is zero. Under the stationarity assumption, $\text{Var}(r_t) = \text{Var}(r_{t-1})$, so that

$$\text{Var}(r_t) = \frac{\sigma_a^2}{1 - \phi_1^2}$$

provided that $\phi_1^2 < 1$. The requirement of $\phi_1^2 < 1$ results from the fact that the variance of a random variable is bounded and nonnegative. Consequently, the weak stationarity of an AR(1) model implies that $-1 < \phi_1 < 1$, that is, $|\phi_1| < 1$. Yet if $|\phi_1| < 1$, then by Eq. (2.11) and the independence of the $\{a_t\}$ series, we can show that the mean and variance of r_t are finite and time invariant; see Eq. (2.5). In addition, by Eq. (2.6), all the autocovariances of r_t are finite. Therefore, the AR(1) model is weakly stationary. In summary, the necessary and sufficient condition for the AR(1) model in Eq. (2.8) to be weakly stationary is $|\phi_1| < 1$.

Using $\phi_0 = (1 - \phi_1)\mu$, one can rewrite a stationary AR(1) model as

$$r_t = (1 - \phi_1)\mu + \phi_1 r_{t-1} + a_t.$$

This model is often used in the finance literature with ϕ_1 measuring the persistence of the dynamic dependence of an AR(1) time series.

Autocorrelation Function of an AR(1) Model

Multiplying Eq. (2.10) by a_t , using the independence between a_t and r_{t-1} , and taking expectation, we obtain

$$E[a_t(r_t - \mu)] = \phi_1 E[a_t(r_{t-1} - \mu)] + E(a_t^2) = E(a_t^2) = \sigma_a^2,$$

where σ_a^2 is the variance of a_t . Multiplying Eq. (2.10) by $r_{t-\ell} - \mu$, taking expectation, and using the prior result, we have

$$\gamma_\ell = \begin{cases} \phi_1 \gamma_1 + \sigma_a^2 & \text{if } \ell = 0 \\ \phi_1 \gamma_{\ell-1} & \text{if } \ell > 0, \end{cases}$$

where we use $\gamma_\ell = \gamma_{-\ell}$. Consequently, for a weakly stationary AR(1) model in Eq. (2.8), we have

$$\text{Var}(r_t) = \gamma_0 = \frac{\sigma^2}{1 - \phi_1^2} \quad \text{and} \quad \gamma_\ell = \phi_1 \gamma_{\ell-1}, \quad \text{for } \ell > 0.$$

From the latter equation, the ACF of r_t satisfies

$$\rho_\ell = \phi_1 \rho_{\ell-1}, \quad \text{for } \ell > 0.$$

Because $\rho_0 = 1$, we have $\rho_\ell = \phi_1^\ell$. This result says that the ACF of a weakly stationary AR(1) series decays exponentially with rate ϕ_1 and starting value $\rho_0 = 1$. For a positive ϕ_1 , the plot of ACF of an AR(1) model shows a nice exponential decay. For a negative ϕ_1 , the plot consists of two alternating exponential decays with rate ϕ_1^2 . Figure 2.3 shows the ACF of two AR(1) models with $\phi_1 = 0.8$ and $\phi_1 = -0.8$.

AR(2) Model

An AR(2) model assumes the form

$$r_t = \phi_0 + \phi_1 r_{t-1} + \phi_2 r_{t-2} + a_t. \quad (2.12)$$

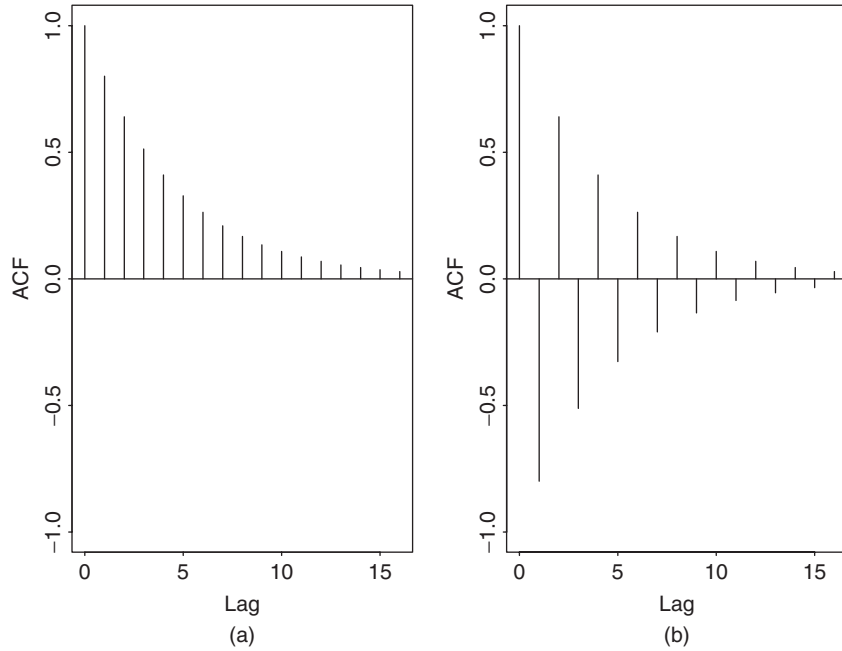


Figure 2.3 Autocorrelation function of an AR(1) model: (a) for $\phi_1 = 0.8$ and (b) for $\phi_1 = -0.8$.

Using the same technique as that of the AR(1) case, we obtain

$$E(r_t) = \mu = \frac{\phi_0}{1 - \phi_1 - \phi_2}$$

provided that $\phi_1 + \phi_2 \neq 1$. Using $\phi_0 = (1 - \phi_1 - \phi_2)\mu$, we can rewrite the AR(2) model as

$$(r_t - \mu) = \phi_1(r_{t-1} - \mu) + \phi_2(r_{t-2} - \mu) + a_t.$$

Multiplying the prior equation by $(r_{t-\ell} - \mu)$, we have

$$\begin{aligned} (r_{t-\ell} - \mu)(r_t - \mu) &= \phi_1(r_{t-\ell} - \mu)(r_{t-1} - \mu) \\ &\quad + \phi_2(r_{t-\ell} - \mu)(r_{t-2} - \mu) + (r_{t-\ell} - \mu)a_t. \end{aligned}$$

Taking expectation and using $E[(r_{t-\ell} - \mu)a_t] = 0$ for $\ell > 0$, we obtain

$$\gamma_\ell = \phi_1\gamma_{\ell-1} + \phi_2\gamma_{\ell-2}, \quad \text{for } \ell > 0.$$

This result is referred to as the *moment equation* of a stationary AR(2) model. Dividing the above equation by γ_0 , we have the property

$$\rho_\ell = \phi_1\rho_{\ell-1} + \phi_2\rho_{\ell-2}, \quad \text{for } \ell > 0, \quad (2.13)$$

for the ACF of r_t . In particular, the lag-1 ACF satisfies

$$\rho_1 = \phi_1\rho_0 + \phi_2\rho_{-1} = \phi_1 + \phi_2\rho_1.$$

Therefore, for a stationary AR(2) series r_t , we have $\rho_0 = 1$,

$$\begin{aligned} \rho_1 &= \frac{\phi_1}{1 - \phi_2} \\ \rho_\ell &= \phi_1\rho_{\ell-1} + \phi_2\rho_{\ell-2}, \quad \ell \geq 2. \end{aligned}$$

The result of Eq. (2.13) says that the ACF of a stationary AR(2) series satisfies the second-order difference equation

$$(1 - \phi_1 B - \phi_2 B^2)\rho_\ell = 0,$$

where B is called the *back-shift* operator such that $B\rho_\ell = \rho_{\ell-1}$. This difference equation determines the properties of the ACF of a stationary AR(2) time series. It also determines the behavior of the forecasts of r_t . In the time series literature, some people use the notation L instead of B for the back-shift operator. Here L stands for *lag* operator. For instance, $Lr_t = r_{t-1}$ and $L\psi_k = \psi_{k-1}$.

Corresponding to the prior difference equation, there is a second-order polynomial equation:

$$1 - \phi_1 x - \phi_2 x^2 = 0. \quad (2.14)$$

Solutions of this equation are

$$x = \frac{\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{-2\phi_2}.$$

In the time series literature, inverses of the two solutions are referred to as the *characteristic roots* of the AR(2) model. Denote the two characteristic roots by ω_1 and ω_2 . If both ω_i are real valued, then the second-order difference equation of the model can be factored as $(1 - \omega_1 B)(1 - \omega_2 B)$ and the AR(2) model can be regarded as an AR(1) model operates on top of another AR(1) model. The ACF of r_t is then a mixture of two exponential decays. If $\phi_1^2 + 4\phi_2 < 0$, then ω_1 and ω_2 are complex numbers (called a *complex-conjugate pair*), and the plot of ACF of r_t would show a picture of damping sine and cosine waves. In business and economic applications, complex characteristic roots are important. They give rise to the behavior of business cycles. It is then common for economic time series models to have complex-valued characteristic roots. For an AR(2) model in Eq. (2.12) with a pair of complex characteristic roots, the *average* length of the stochastic cycles is

$$k = \frac{2\pi}{\cos^{-1}[\phi_1/(2\sqrt{-\phi_2})]},$$

where the cosine inverse is stated in radians. If one writes the complex solutions as $a \pm bi$, where $i = \sqrt{-1}$, then we have $\phi_1 = 2a$, $\phi_2 = -(a^2 + b^2)$, and

$$k = \frac{2\pi}{\cos^{-1}(a/\sqrt{a^2 + b^2})},$$

where $\sqrt{a^2 + b^2}$ is the absolute value of $a \pm bi$. See Example 2.1 for an illustration.

Figure 2.4 shows the ACF of four stationary AR(2) models. Part (b) is the ACF of the AR(2) model $(1 - 0.6B + 0.4B^2)r_t = a_t$. Because $\phi_1^2 + 4\phi_2 = 0.36 + 4 \times (-0.4) = -1.24 < 0$, this particular AR(2) model contains two complex characteristic roots, and hence its ACF exhibits damping sine and cosine waves. The other three AR(2) models have real-valued characteristic roots. Their ACFs decay exponentially.

Example 2.1. As an illustration, consider the quarterly growth rate of U.S. real gross national product (GNP), seasonally adjusted, from the second quarter of 1947 to the first quarter of 1991. This series shown in Figure 2.5 is also used in Chapter 4 as an example of nonlinear economic time series. Here we simply employ an AR(3) model for the data. Denoting the growth rate by r_t , we can use

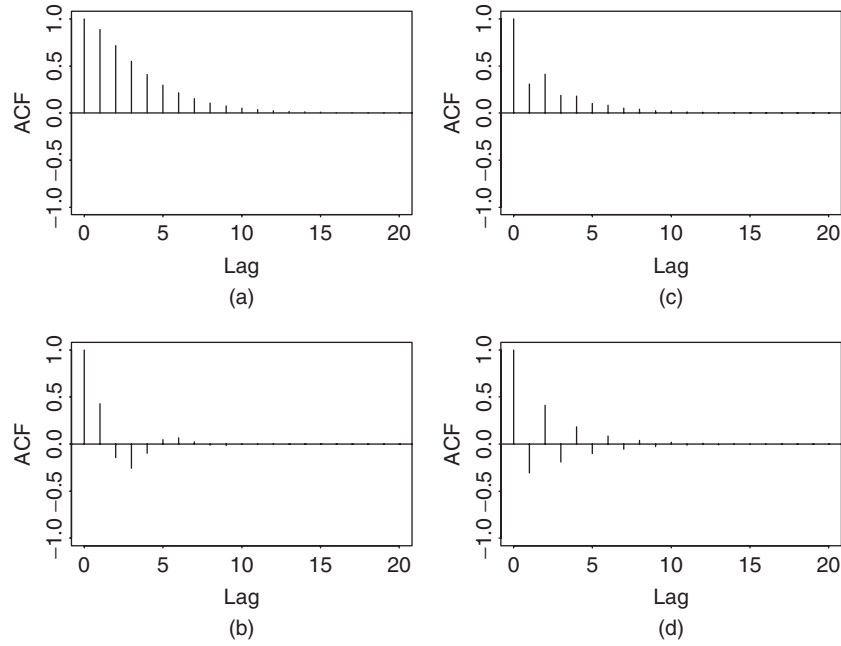


Figure 2.4 Autocorrelation function of an AR(2) model: (a) $\phi_1 = 1.2$ and $\phi_2 = -0.35$, (b) $\phi_1 = 0.6$ and $\phi_2 = -0.4$, (c) $\phi_1 = 0.2$ and $\phi_2 = 0.35$, and (d) $\phi_1 = -0.2$ and $\phi_2 = 0.35$.

the model building procedure of the next subsection to estimate the model. The fitted model is

$$r_t = 0.0047 + 0.348r_{t-1} + 0.179r_{t-2} - 0.142r_{t-3} + a_t, \quad \hat{\sigma}_a = 0.0097. \quad (2.15)$$

Rewriting the model as

$$r_t - 0.348r_{t-1} - 0.179r_{t-2} + 0.142r_{t-3} = 0.0047 + a_t,$$

we obtain a corresponding third-order difference equation

$$1 - 0.348B - 0.179B^2 + 0.141B^3 = 0,$$

which can be factored approximately as

$$(1 + 0.521B)(1 - 0.869B + 0.274B^2) = 0.$$

The first factor $(1 + 0.521B)$ shows an exponentially decaying feature of the GNP growth rate. Focusing on the second-order factor $1 - 0.869B - (-0.274)B^2 = 0$, we have $\phi_1^2 + 4\phi_2 = 0.869^2 + 4(-0.274) = -0.341 < 0$. Therefore, the second factor of the AR(3) model confirms the existence of stochastic business cycles

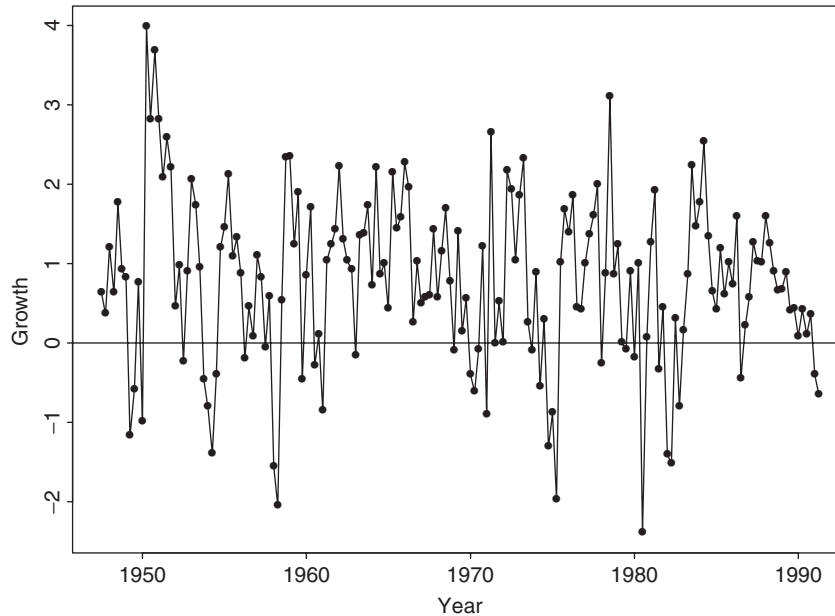


Figure 2.5 Time plot of growth rate of U.S. quarterly real GNP from 1947.II to 1991.I. Data are seasonally adjusted and in percentages.

in the quarterly growth rate of U.S. real GNP. This is reasonable as the U.S. economy went through expansion and contraction periods. The average length of the stochastic cycles is approximately

$$k = \frac{2(3.14159)}{\cos^{-1}[\phi_1/(2\sqrt{-\phi_2})]} = 10.62 \text{ quarters,}$$

which is about 3 years. If one uses a nonlinear model to separate U.S. economy into “expansion” and “contraction” periods, the data show that the average duration of contraction periods is about three quarters and that of expansion periods is about 3 years; see the analysis in Chapter 4. The average duration of 10.62 quarters is a compromise between the two separate durations. The periodic feature obtained here is common among growth rates of national economies. For example, similar features can be found for many OECD (Organization for Economic Cooperation and Development) countries.

R Demonstration

The R demonstration for Example 2.1, where % denotes explanation, follows:

```
> gnp=scan(file='dgnp82.txt') % Load data
% To create a time-series object
> gnp1=ts(gnp,frequency=4,start=c(1947,2))
> plot(gnp1)
```



```

> points(gnp1,pch='*')

> m1=ar(gnp,method='mle') % Find the AR order
> m1$order      % An AR(3) is selected based on AIC
[1] 3
> m2=arima(gnp,order=c(3,0,0)) % Estimation
> m2
Call:
arima(x = gnp, order = c(3, 0, 0))

Coefficients:
          ar1          ar2          ar3      intercept
      0.3480    0.1793   -0.1423         0.0077
s.e.   0.0745    0.0778    0.0745         0.0012

sigma^2 estimated as 9.427e-05: log likelihood=565.84,
      aic=-1121.68

% In R, ``intercept`` denotes the mean of the series.
% Therefore, the constant term is obtained below:
> (1-.348-.1793+.1423)*0.0077
[1] 0.0047355
> sqrt(m2$sigma2) % Residual standard error
[1] 0.009709322

> p1=c(1,-m2$coef[1:3]) % Characteristic equation
> roots=polyroot(p1) % Find solutions
> roots
[1] 1.590253+1.063882i -1.920152+0.000000i 1.590253-1.063882i
> Mod(roots) % Compute the absolute values of the solutions
[1] 1.913308 1.920152 1.913308
% To compute average length of business cycles:
> k=2*pi/acos(1.590253/1.913308)
> k
[1] 10.65638

```

Stationarity

The stationarity condition of an AR(2) time series is that the absolute values of its two characteristic roots are less than 1, that is, its two characteristic roots are less than 1 in modulus. Equivalently, the two solutions of the characteristic equation are greater than 1 in modulus. Under such a condition, the recursive equation in (2.13) ensures that the ACF of the model converges to 0 as the lag ℓ increases. This convergence property is a necessary condition for a stationary time series. In fact, the condition also applies to the AR(1) model where the polynomial equation is $1 - \phi_1 x = 0$. The characteristic root is $w = 1/x = \phi_1$, which must be less than 1 in modulus for r_t to be stationary. As shown before, $\rho_\ell = \phi_1^\ell$ for a stationary AR(1) model. The condition implies that $\rho_\ell \rightarrow 0$ as $\ell \rightarrow \infty$.

AR(p) Model

The results of the AR(1) and AR(2) models can readily be generalized to the general AR(p) model in Eq. (2.9). The mean of a stationary series is

$$E(r_t) = \frac{\phi_0}{1 - \phi_1 - \dots - \phi_p}$$

provided that the denominator is not zero. The associated characteristic equation of the model is

$$1 - \phi_1 x - \phi_2 x^2 - \dots - \phi_p x^p = 0.$$

If all the solutions of this equation are greater than 1 in modulus, then the series r_t is stationary. Again, inverses of the solutions are the *characteristic roots* of the model. Thus, stationarity requires that all characteristic roots are less than 1 in modulus. For a stationary AR(p) series, the ACF satisfies the difference equation

$$(1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p)\rho_\ell = 0, \quad \text{for } \ell > 0.$$

The plot of ACF of a stationary AR(p) model would then show a mixture of damping sine and cosine patterns and exponential decays depending on the nature of its characteristic roots.

2.4.2 Identifying AR Models in Practice

In application, the order p of an AR time series is unknown. It must be specified empirically. This is referred to as the *order determination* (or order specification) of AR models, and it has been extensively studied in the time series literature. Two general approaches are available for determining the value of p . The first approach is to use the partial autocorrelation function, and the second approach uses some information criteria.

Partial Autocorrelation Function (PACF)

The PACF of a stationary time series is a function of its ACF and is a useful tool for determining the order p of an AR model. A simple, yet effective way to introduce PACF is to consider the following AR models in consecutive orders:

$$\begin{aligned} r_t &= \phi_{0,1} + \phi_{1,1}r_{t-1} + e_{1t}, \\ r_t &= \phi_{0,2} + \phi_{1,2}r_{t-1} + \phi_{2,2}r_{t-2} + e_{2t}, \\ r_t &= \phi_{0,3} + \phi_{1,3}r_{t-1} + \phi_{2,3}r_{t-2} + \phi_{3,3}r_{t-3} + e_{3t}, \\ r_t &= \phi_{0,4} + \phi_{1,4}r_{t-1} + \phi_{2,4}r_{t-2} + \phi_{3,4}r_{t-3} + \phi_{4,4}r_{t-4} + e_{4t}, \\ &\vdots \end{aligned}$$

where $\phi_{0,j}$, $\phi_{i,j}$, and $\{e_{jt}\}$ are, respectively, the constant term, the coefficient of r_{t-i} , and the error term of an AR(j) model. These models are in the form of a multiple linear regression and can be estimated by the least-squares method. As a matter of fact, they are arranged in a sequential order that enables us to apply the idea of partial F test in multiple linear regression analysis. The estimate $\hat{\phi}_{1,1}$ of the first equation is called the lag-1 sample PACF of r_t . The estimate $\hat{\phi}_{2,2}$ of the second equation is the lag-2 sample PACF of r_t . The estimate $\hat{\phi}_{3,3}$ of the third equation is the lag-3 sample PACF of r_t , and so on.

From the definition, the lag-2 PACF $\hat{\phi}_{2,2}$ shows the added contribution of r_{t-2} to r_t over the AR(1) model $r_t = \phi_0 + \phi_1 r_{t-1} + e_{1t}$. The lag-3 PACF shows the added contribution of r_{t-3} to r_t over an AR(2) model, and so on. Therefore, for an AR(p) model, the lag- p sample PACF should not be zero, but $\hat{\phi}_{j,j}$ should be close to zero for all $j > p$. We make use of this property to determine the order p . For a stationary Gaussian AR(p) model, it can be shown that the sample PACF has the following properties:

- $\hat{\phi}_{p,p}$ converges to ϕ_p as the sample size T goes to infinity.
- $\hat{\phi}_{\ell,\ell}$ converges to zero for all $\ell > p$.
- The asymptotic variance of $\hat{\phi}_{\ell,\ell}$ is $1/T$ for $\ell > p$.

These results say that, for an AR(p) series, the sample PACF cuts off at lag p .

As an example, consider the monthly simple returns of CRSP value-weighted index from January 1926 to December 2008. Table 2.1 gives the first 12 lags of sample PACF of the series. With $T = 996$, the asymptotic standard error of the sample PACF is approximately 0.032. Therefore, using the 5% significant level, we identify an AR(3) or AR(9) model for the data (i.e., $p = 3$ or 9). If the 1% significant level is used, we specify an AR(3) model.

As another example, Figure 2.6 shows the PACF of the GNP growth rate series of Example 2.1. The two dotted lines of the plot denote the approximate two standard error limits $\pm(2/\sqrt{176})$. The plot suggests an AR(3) model for the data because the first three lags of sample PACF appear to be large.

TABLE 2.1 Sample Partial Autocorrelation Function and Some Information Criteria for the Monthly Simple Returns of CRSP Value-Weighted Index from January 1926 to December 2008

p	1	2	3	4	5	6
PACF	0.115	-0.030	-0.102	0.033	0.062	-0.050
AIC	-5.838	-5.837	-5.846	-5.845	-5.847	-5.847
BIC	-5.833	-5.827	-5.831	-5.825	-5.822	-5.818
p	7	8	9	10	11	12
PACF	0.031	0.052	0.063	0.005	-0.005	0.011
AIC	-5.846	-5.847	-5.849	-5.847	-5.845	-5.843
BIC	-5.812	-5.807	-5.805	-5.798	-5.791	-5.784

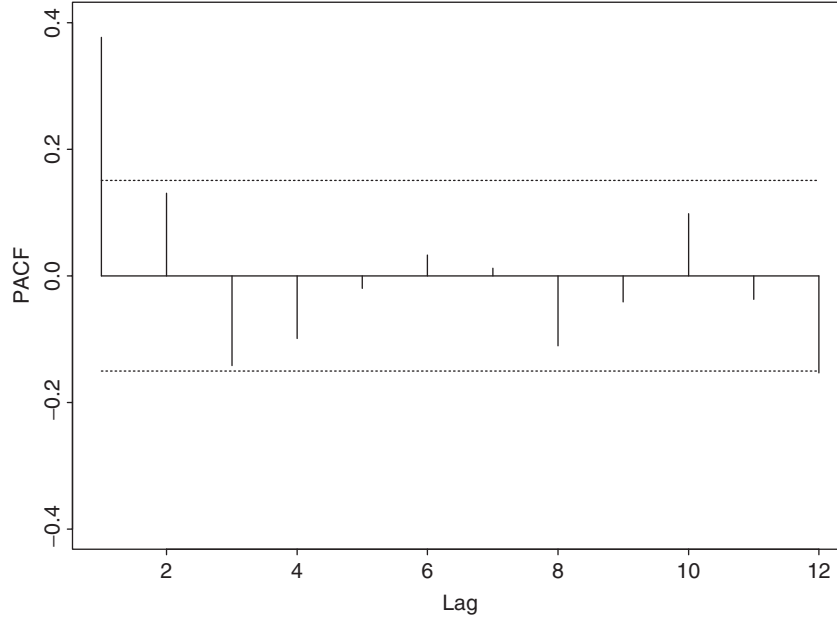


Figure 2.6 Sample partial autocorrelation function of U.S. quarterly real GNP growth rate from 1947:II to 1991:I. Dotted lines give approximate pointwise 95% confidence interval.

Information Criteria

There are several information criteria available to determine the order p of an AR process. All of them are likelihood based. For example, the well-known *Akaike information criterion* (AIC) (Akaike, 1973) is defined as

$$\text{AIC} = \frac{-2}{T} \ln(\text{likelihood}) + \frac{2}{T} \times (\text{number of parameters}), \quad (2.16)$$

where the likelihood function is evaluated at the maximum-likelihood estimates and T is the sample size. For a Gaussian $\text{AR}(\ell)$ model, AIC reduces to

$$\text{AIC}(\ell) = \ln(\tilde{\sigma}_\ell^2) + \frac{2\ell}{T}$$

where $\tilde{\sigma}_\ell^2$ is the maximum-likelihood estimate of σ_a^2 , which is the variance of a_t , and T is the sample size; see Eq. (1.18). The first term of the AIC in Eq. (2.16) measures the goodness of fit of the $\text{AR}(\ell)$ model to the data, whereas the second term is called the *penalty function* of the criterion because it penalizes a candidate model by the number of parameters used. Different penalty functions result in different information criteria.

Another commonly used criterion function is the Schwarz–Bayesian information criterion (BIC). For a Gaussian $\text{AR}(\ell)$ model, the criterion is

$$\text{BIC}(\ell) = \ln(\tilde{\sigma}_\ell^2) + \frac{\ell \ln(T)}{T}.$$

The penalty for each parameter used is 2 for AIC and $\ln(T)$ for BIC. Thus, compared with AIC, BIC tends to select a lower AR model when the sample size is moderate or large.

Selection Rule

To use AIC to select an AR model in practice, one computes $\text{AIC}(\ell)$ for $\ell = 0, \dots, P$, where p is a prespecified positive integer and selects the order k that has the minimum AIC value. The same rule applies to BIC.

Table 2.1 also gives the AIC and BIC for $p = 1, \dots, 12$. The AIC values are close to each other with minimum -5.849 occurring at $p = 9$, suggesting that an AR(9) model is preferred by the criterion. The BIC, on the other hand, attains its minimum value -5.833 at $p = 1$ with -5.831 as a close second at $p = 3$. Thus, the BIC selects an AR(1) model for the value-weighted return series. This example shows that different approaches or criteria to order determination may result in different choices of p . There is no evidence to suggest that one approach outperforms the other in a real application. Substantive information of the problem under study and simplicity are two factors that also play an important role in choosing an AR model for a given time series.

Again, consider the growth rate series of U.S. quarterly real GNP of Example 2.1. The AIC obtained from R also identifies an AR(3) model. Note that the AIC value of the `ar` command in R has been adjusted so that the minimum AIC is zero.

```
> gnp=scan(file='q-gnp4791.txt')
> ord=ar(gnp,method='mle')
> ord$aic
[1] 27.847 2.742 1.603 0.000 0.323 2.243
[7] 4.052 6.025 5.905 7.572 7.895 9.679
> ord$order
[1] 3
```

Parameter Estimation

For a specified AR(p) model in Eq. (2.9), the conditional least-squares method, which starts with the $(p + 1)$ th observation, is often used to estimate the parameters. Specifically, conditioning on the first p observations, we have

$$r_t = \phi_0 + \phi_1 r_{t-1} + \dots + \phi_p r_{t-p} + a_t, \quad t = p + 1, \dots, T,$$

which is in the form of a multiple linear regression and can be estimated by the least-squares method. Denote the estimate of ϕ_i by $\hat{\phi}_i$. The *fitted model* is

$$\hat{r}_t = \hat{\phi}_0 + \hat{\phi}_1 r_{t-1} + \dots + \hat{\phi}_p r_{t-p},$$

and the associated residual is

$$\hat{a}_t = r_t - \hat{r}_t.$$

The series $\{\hat{a}_t\}$ is called the *residual series*, from which we obtain

$$\hat{\sigma}_a^2 = \frac{\sum_{t=p+1}^T \hat{a}_t^2}{T - 2p - 1}.$$

If the conditional-likelihood method is used, the estimates of ϕ_i remain unchanged, but the estimate of σ_a^2 becomes $\tilde{\sigma}_a^2 = \hat{\sigma}_a^2 \times (T - 2p - 1)/(T - p)$. In some packages, $\tilde{\sigma}_a^2$ is defined as $\hat{\sigma}_a^2 \times (T - 2p - 1)/T$. For illustration, consider an AR(3) model for the monthly simple returns of the value-weighted index in Table 2.1. The fitted model is

$$r_t = 0.0091 + 0.116r_{t-1} - 0.019r_{t-2} - 0.104r_{t-3} + \hat{a}_t, \quad \hat{\sigma}_a = 0.054.$$

The standard errors of the coefficients are 0.002, 0.032, 0.032, and 0.032, respectively. Except for the lag-2 coefficient, all parameters are statistically significant at the 1% level.

For this example, the AR coefficients of the fitted model are small, indicating that the serial dependence of the series is weak, even though it is statistically significant at the 1% level. The significance of $\hat{\phi}_0$ of the entertained model implies that the expected mean return of the series is positive. In fact, $\hat{\mu} = 0.0091/(1 - 0.116 + 0.019 + 0.104) = 0.009$, which is small but has an important long-term implication. It implies that the long-term return of the index can be substantial. Using the multiperiod simple return defined in Chapter 1, the average annual simple gross return is $[\prod_{t=1}^{996} (1 + R_t)]^{12/996} - 1 \approx 0.093$. In other words, the monthly simple returns of the CRSP value-weighted index grew about 9.3% per annum from 1926 to 2008, supporting the common belief that equity market performs well in the long term. A one-dollar investment at the beginning of 1926 would be worth about \$1593 at the end of 2008.

```
> vw=read.table('m-ibm3dx.txt',header=T)[,3]
> t1=prod(vw+1)
> t1
[1] 1592.953
> t1^(12/996)-1
[1] 0.0929
```

Model Checking

A fitted model must be examined carefully to check for possible model inadequacy. If the model is adequate, then the residual series should behave as a white noise. The ACF and the Ljung–Box statistics in Eq. (2.3) of the residuals can be used to check the closeness of $\hat{a}_t$ to a white noise. For an AR(p) model, the Ljung–Box statistic $Q(m)$ follows asymptotically a chi-squared distribution with $m - g$ degrees of freedom, where g denotes the number of AR coefficients used in the model. The adjustment in the degrees of freedom is made based on the number of constraints added to the residuals $\hat{a}_t$ from fitting the AR(p) to an AR(0) model. If a fitted

model is found to be inadequate, it must be refined. For instance, if some of the estimated AR coefficients are not significantly different from zero, then the model should be simplified by trying to remove those insignificant parameters. If residual ACF shows additional serial correlations, then the model should be extended to take care of those correlations.

Remark. Most time series packages do not adjust the degrees of freedom when applying the Ljung–Box statistics $Q(m)$ to a residual series. This is understandable when $m \leq g$. $\square$

Consider the residual series of the fitted AR(3) model for the monthly value-weighted simple returns. We have $Q(12) = 16.35$ with a p value 0.060 based on its asymptotic chi-squared distribution with 9 degrees of freedom. Thus, the null hypothesis of no residual serial correlation in the first 12 lags is barely not rejected at the 5% level. However, since the lag-2 AR coefficient is not significant at the 5% level, one can refine the model as

$$r_t = 0.0088 + 0.114r_{t-1} - 0.106r_{t-3} + a_t, \quad \hat{\sigma}_a = 0.0536,$$

where all the estimates are now significant at the 1% level. The residual series gives $Q(12) = 16.83$ with a p value 0.078 (based on χ_{10}^2). The model is adequate in modeling the dynamic linear dependence of the data.

R Demonstration

In the following R demonstration, % denotes an explanation:

```
> vw=read.table('m-ibm3dx2608.txt',header=T)[,3]
> m3=arima(vw,order=c(3,0,0))
> m3
Call:
arima(x = vw, order = c(3, 0, 0))

Coefficients:
          ar1          ar2          ar3      intercept
      0.1158   -0.0187   -0.1042         0.0089
s.e.  0.0315    0.0317    0.0317         0.0017

sigma^2 estimated as 0.002875: log likelihood=1500.86,
aic=-2991.73

> (1-.1158+.0187+.1042)*mean(vw) % Compute
      the intercept phi(0).
[1] 0.00896761
> sqrt(m3$sigma2) % Compute standard error of residuals
[1] 0.0536189

> Box.test(m3$residuals,lag=12,type='Ljung')
```

Box-Ljung test

```
data: m3$residuals    % R uses 12 degrees of freedom

X-squared = 16.3525, df = 12, p-value = 0.1756

> pv=1-pchisq(16.35,9) % Compute p-value using 9 degrees
  of freedom
> pv
[1] 0.05992276
% To fix the AR(2) coef to zero:
> m3=arima(vw,order=c(3,0,0),fixed=c(NA,0,NA,NA))
% The subcommand 'fixed' is used to fix parameter values,
% where NA denotes estimation and 0 means fixing the
  parameter to 0.
% The ordering of the parameters can be found using m3$coef.
> m3
Call:
arima(x = vw, order = c(3, 0, 0), fixed = c(NA, 0, NA, NA))

Coefficients:
          ar1  ar2          ar3  intercept
          0.1136    0   -0.1063    0.0089
s.e.        0.0313    0    0.0315    0.0017

sigma^2 estimated as 0.002876: log likelihood=1500.69,
      aic=-2993.38
> (1-.1136+.1063)*.0089 % Compute phi(0)
[1] 0.00883503
> sqrt(m3$sigma2) % Compute residual standard error
[1] 0.05362832

> Box.test(m3$residuals,lag=12,type='Ljung')
```

Box-Ljung test

```
data: m3$residuals
X-squared = 16.8276, df = 12, p-value = 0.1562

> pv=1-pchisq(16.83,10)
> pv
[1] 0.0782113
```

S-Plus Demonstration

The following S-Plus output has been edited:

```
> vw=read.table('m-ibm3dx2608.txt',header=T)[,3]
> ar3=OLS(vw ar(3))
```



```

> summary(ar3)
Call:
OLS(formula = vw ~ ar(3))

Residuals:
    Min       1Q   Median       3Q      Max
-0.2863  -0.0263   0.0034   0.0297   0.3689

Coefficients:
            Value Std. Error t value Pr(>|t|)
(Intercept)  0.0091   0.0018    5.1653  0.0000
      lag1    0.1148   0.0316    3.6333  0.0003
      lag2   -0.0188   0.0318   -0.5894  0.5557
      lag3   -0.1043   0.0318   -3.2763  0.0011

Regression Diagnostics:
      R-Squared 0.0246
Adjusted R-Squared 0.0216
Durbin-Watson Stat 1.9913

Residual Diagnostics:
              Stat  P-Value
Jarque-Bera 1656.3928  0.0000
Ljung-Box   50.1279  0.0087

Residual standard error: 0.05375 on 989 degrees of freedom

> autocorTest(ar3$residuals,lag=12)

Test for Autocorrelation: Ljung-Box
Null Hypothesis: no autocorrelation

Test Statistics:
Test Stat 16.5668
  p.value 0.1666    % S-Plus uses 12 degrees of freedom

Dist. under Null: chi-square with 12 degrees of freedom
Total Observ.: 993
> 1-pchisq(16.57,9) % Compute p-value with 9 degrees
  of freedom
[1] 0.05589128

```

2.4.3 Goodness of Fit

A commonly used statistic to measure *goodness of fit* of a stationary model is the *R* square (R^2) defined as

$$R^2 = 1 - \frac{\text{residual sum of squares}}{\text{total sum of squares}}.$$

For a stationary AR(p) time series model with T observations $\{r_t | t = 1, \dots, T\}$, the measure becomes

$$R^2 = 1 - \frac{\sum_{t=p+1}^T \hat{a}_t^2}{\sum_{t=p+1}^T (r_t - \bar{r})^2},$$

where $\bar{r} = \sum_{t=p+1}^T r_t / (T - p)$. It is easy to show that $0 \leq R^2 \leq 1$. Typically, a larger R^2 indicates that the model provides a closer fit to the data. However, this is only true for a stationary time series. For the unit-root nonstationary series discussed later in this chapter, R^2 of an AR(1) fit converges to one when the sample size increases to infinity, regardless of the true underlying model of r_t .

For a given data set, it is well known that R^2 is a nondecreasing function of the number of parameters used. To overcome this weakness, an *adjusted* R^2 is proposed, which is defined as

$$\begin{aligned} \text{Adj} - R^2 &= 1 - \frac{\text{variance of residuals}}{\text{variance of } r_t} \\ &= 1 - \frac{\hat{\sigma}_a^2}{\hat{\sigma}_r^2}, \end{aligned}$$

where $\hat{\sigma}_r^2$ is the sample variance of r_t . This new measure takes into account the number of parameters used in the fitted model. However, it is no longer between 0 and 1.

2.4.4 Forecasting

Forecasting is an important application of time series analysis. For the AR(p) model in Eq. (2.9), suppose that we are at the time index h and are interested in forecasting $r_{h+\ell}$, where $\ell \geq 1$. The time index h is called the *forecast origin* and the positive integer ℓ is the *forecast horizon*. Let $\hat{r}_h(\ell)$ be the forecast of $r_{h+\ell}$ using the minimum squared error loss function. In other words, the forecast $\hat{r}_h(\ell)$ is chosen such that

$$E\{[r_{h+\ell} - \hat{r}_h(\ell)]^2 | F_h\} \leq \min_g E[(r_{h+\ell} - g)^2 | F_h],$$

where g is a function of the information available at time h (inclusive), that is, a function of F_h . We referred to $\hat{r}_h(\ell)$ as the ℓ -step ahead forecast of r_t at the forecast origin h . Let F_h be the collection of information available at the forecast origin h .

1-Step-Ahead Forecast

From the AR(p) model, we have

$$r_{h+1} = \phi_0 + \phi_1 r_h + \dots + \phi_p r_{h+1-p} + a_{h+1}.$$

Under the minimum squared error loss function, the point forecast of r_{h+1} given F_h is the conditional expectation

$$\hat{r}_h(1) = E(r_{h+1}|F_h) = \phi_0 + \sum_{i=1}^p \phi_i r_{h+1-i},$$

and the associated forecast error is

$$e_h(1) = r_{h+1} - \hat{r}_h(1) = a_{h+1}.$$

Consequently, the variance of the 1-step-ahead forecast error is $\text{Var}[e_h(1)] = \text{Var}(a_{h+1}) = \sigma_a^2$. If a_t is normally distributed, then a 95% 1-step-ahead interval forecast of r_{h+1} is $\hat{r}_h(1) \pm 1.96 \times \sigma_a$. For the linear model in Eq. (2.4), a_{t+1} is also the 1-step-ahead forecast error at the forecast origin t . In the econometric literature, a_{t+1} is referred to as the *shock* to the series at time $t + 1$.

In practice, estimated parameters are often used to compute point and interval forecasts. This results in a *conditional forecast* because such a forecast does not take into consideration the uncertainty in the parameter estimates. In theory, one can consider parameter uncertainty in forecasting, but it is much more involved. A natural way to consider parameter and model uncertainty in forecasting is Bayesian forecasting with Markov chain Monte Carlo (MCMC) methods. See Chapter 12 for further discussion. For simplicity, we assume that the model is given in this chapter. When the sample size used in estimation is sufficiently large, then the conditional forecast is close to the unconditional one.

2-Step-Ahead Forecast

Next consider the forecast of r_{h+2} at the forecast origin h . From the $\text{AR}(p)$ model, we have

$$r_{h+2} = \phi_0 + \phi_1 r_{h+1} + \cdots + \phi_p r_{h+2-p} + a_{h+2}.$$

Taking conditional expectation, we have

$$\hat{r}_h(2) = E(r_{h+2}|F_h) = \phi_0 + \phi_1 \hat{r}_h(1) + \phi_2 r_h + \cdots + \phi_p r_{h+2-p}$$

and the associated forecast error

$$e_h(2) = r_{h+2} - \hat{r}_h(2) = \phi_1 [r_{h+1} - \hat{r}_h(1)] + a_{h+2} = a_{h+2} + \phi_1 a_{h+1}.$$

The variance of the forecast error is $\text{Var}[e_h(2)] = (1 + \phi_1^2)\sigma_a^2$. Interval forecasts of r_{h+2} can be computed in the same way as those for r_{h+1} . It is interesting to see that $\text{Var}[e_h(2)] \geq \text{Var}[e_h(1)]$, meaning that as the forecast horizon increases the uncertainty in forecast also increases. This is in agreement with common sense

that we are more uncertain about r_{h+2} than r_{h+1} at the time index h for a linear time series.

Multistep-Ahead Forecast

In general, we have

$$r_{h+\ell} = \phi_0 + \phi_1 r_{h+\ell-1} + \cdots + \phi_p r_{h+\ell-p} + a_{h+\ell}.$$

The ℓ -step-ahead forecast based on the minimum squared error loss function is the conditional expectation of $r_{h+\ell}$ given F_h , which can be obtained as

$$\hat{r}_h(\ell) = \phi_0 + \sum_{i=1}^p \phi_i \hat{r}_h(\ell - i),$$

where it is understood that $\hat{r}_h(i) = r_{h+i}$ if $i \leq 0$. This forecast can be computed recursively using forecasts $\hat{r}_h(i)$ for $i = 1, \dots, \ell - 1$. The ℓ -step-ahead forecast error is $e_h(\ell) = r_{h+\ell} - \hat{r}_h(\ell)$. It can be shown that for a stationary AR(p) model, $\hat{r}_h(\ell)$ converges to $E(r_t)$ as $\ell \rightarrow \infty$, meaning that for such a series long-term point forecast approaches its unconditional mean. This property is referred to as the *mean reversion* in the finance literature. For an AR(1) model, the speed of mean reversion is measured by the *half-life* defined as $\ell = \ln(0.5)/\ln(|\phi_1|)$. The variance of the forecast error then approaches the unconditional variance of r_t . Note that for an AR(1) model in (2.8), let $x_t = r_t - E(r_t)$ be the mean-adjusted series. It is easy to see that the ℓ -step-ahead forecast of $x_{h+\ell}$ at the forecast origin h is $\hat{x}_h(\ell) = \phi_1^\ell x_h$. The half-life is the forecast horizon such that $\hat{x}_h(\ell) = \frac{1}{2}x_h$. That is, $\phi_1^\ell = \frac{1}{2}$. Thus, $\ell = \ln(0.5)/\ln(|\phi_1|)$.

Table 2.2 contains the 1-step- to 12-step ahead forecasts and the standard errors of the associated forecast errors at the forecast origin 984 for the monthly simple return of the value-weighted index using an AR(3) model that was reestimated using the first 984 observations. The fitted model is

$$r_t = 0.0098 + 0.1024r_{t-1} - 0.0201r_{t-2} - 0.1090r_{t-3} + a_t,$$

where $\hat{\sigma}_a = 0.054$. The actual returns of 2008 are also given in Table 2.2. Because of the weak serial dependence in the series, the forecasts and standard deviations of forecast errors converge to the sample mean and standard deviation of the data quickly. For the first 984 observations, the sample mean and standard error are 0.0095 and 0.0540, respectively.

Figure 2.7 shows the corresponding out-of-sample prediction plot for the monthly simple return series of the value-weighted index. The forecast origin $t = 984$ corresponds to December 2007. The prediction plot includes the two standard error limits of the forecasts and the actual observed returns for 2008. The forecasts and actual returns are marked by $\circ$ and $\bullet$, respectively. From the plot, except for the return of October 2008, all actual returns are within the 95% prediction intervals.

TABLE 2.2 Multistep Ahead Forecasts of an AR(3) Model for Monthly Simple Returns of CRSP Value-Weighted Index

Step	1	2	3	4	5	6
Forecast	0.0076	0.0161	0.0118	0.0099	0.0089	0.0093
Std. Error	0.0534	0.0537	0.0537	0.0540	0.0540	0.0540
Actual	-0.0623	-0.0220	-0.0105	0.0511	0.0238	-0.0786
Step	7	8	9	10	11	12
Forecast	0.0095	0.0097	0.0096	0.0096	0.0096	0.0096
Std. Error	0.0540	0.0540	0.0540	0.0540	0.0540	0.0540
Actual	-0.0132	0.0110	-0.0981	-0.1847	-0.0852	0.0215

^aThe forecast origin is $h = 984$.

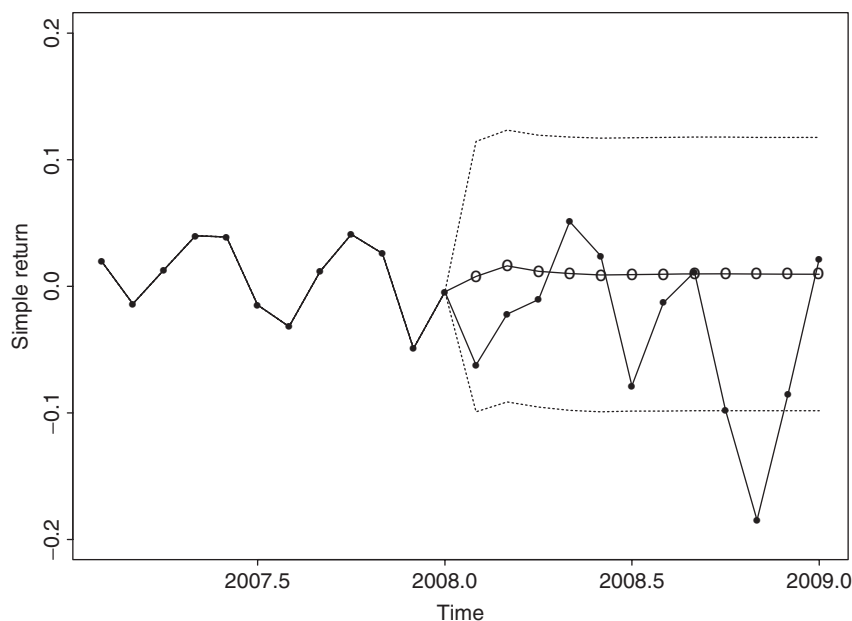


Figure 2.7 Plot of 1- to 12-step-ahead out-of-sample forecasts for monthly simple returns of CRSP value-weighted index. Forecast origin is $t = 984$, which is December 2007. Forecasts are denoted by “o” and actual observations by “•”. Two dashed lines denote two standard error limits of the forecasts.

2.5 SIMPLE MA MODELS

We now turn to another class of simple models that are also useful in modeling return series in finance. These models are the moving-average (MA) models. As is shown in Chapter 5, the bid–ask bounce in stock trading may introduce an MA(1) structure in a return series. There are several ways to introduce MA models. One approach is to treat the model as a simple extension of white noise series. Another approach is to treat the model as an infinite-order AR model with some parameter constraints. We adopt the second approach.

There is no particular reason, but simplicity, to assume a priori that the order of an AR model is finite. We may entertain, at least in theory, an AR model with infinite order as

$$r_t = \phi_0 + \phi_1 r_{t-1} + \phi_2 r_{t-2} + \cdots + a_t.$$

However, such an AR model is not realistic because it has infinite many parameters. One way to make the model practical is to assume that the coefficients ϕ_i 's satisfy some constraints so that they are determined by a finite number of parameters. A special case of this idea is

$$r_t = \phi_0 - \theta_1 r_{t-1} - \theta_1^2 r_{t-2} - \theta_1^3 r_{t-3} - \cdots + a_t, \quad (2.17)$$

where the coefficients depend on a single parameter θ_1 via $\phi_i = -\theta_1^i$ for $i \geq 1$. For the model in Eq. (2.17) to be stationary, θ_1 must be less than 1 in absolute value; otherwise, θ_1^i and the series will explode. Because $|\theta_1| < 1$, we have $\theta_1^i \rightarrow 0$ as $i \rightarrow \infty$. Thus, the contribution of r_{t-i} to r_t decays exponentially as i increases. This is reasonable as the dependence of a stationary series r_t on its lagged value r_{t-i} , if any, should decay over time.

The model in Eq. (2.17) can be rewritten in a rather compact form. To see this, rewrite the model as

$$r_t + \theta_1 r_{t-1} + \theta_1^2 r_{t-2} + \cdots = \phi_0 + a_t. \quad (2.18)$$

The model for r_{t-1} is then

$$r_{t-1} + \theta_1 r_{t-2} + \theta_1^2 r_{t-3} + \cdots = \phi_0 + a_{t-1}. \quad (2.19)$$

Multiplying Eq. (2.19) by θ_1 and subtracting the result from Eq. (2.18), we obtain

$$r_t = \phi_0(1 - \theta_1) + a_t - \theta_1 a_{t-1},$$

which says that except for the constant term r_t is a weighted average of shocks a_t and a_{t-1} . Therefore, the model is called an MA model of order 1 or MA(1) model for short. The general form of an MA(1) model is

$$r_t = c_0 + a_t - \theta_1 a_{t-1} \quad \text{or} \quad r_t = c_0 + (1 - \theta_1 B)a_t, \quad (2.20)$$

where c_0 is a constant and $\{a_t\}$ is a white noise series. Similarly, an MA(2) model is in the form

$$r_t = c_0 + a_t - \theta_1 a_{t-1} - \theta_2 a_{t-2}, \quad (2.21)$$

and an MA(q) model is

$$r_t = c_0 + a_t - \theta_1 a_{t-1} - \cdots - \theta_q a_{t-q}, \quad (2.22)$$

or $r_t = c_0 + (1 - \theta_1 B - \cdots - \theta_q B^q)a_t$, where $q > 0$.

2.5.1 Properties of MA Models

Again, we focus on the simple MA(1) and MA(2) models. The results of MA(q) models can easily be obtained by the same techniques.

Stationarity

Moving-average models are always weakly stationary because they are finite linear combinations of a white noise sequence for which the first two moments are time invariant. For example, consider the MA(1) model in Eq. (2.20). Taking expectation of the model, we have

$$E(r_t) = c_0,$$

which is time invariant. Taking the variance of Eq. (2.20), we have

$$\text{Var}(r_t) = \sigma_a^2 + \theta_1^2 \sigma_a^2 = (1 + \theta_1^2) \sigma_a^2,$$

where we use the fact that a_t and a_{t-1} are uncorrelated. Again, $\text{Var}(r_t)$ is time invariant. The prior discussion applies to general MA(q) models, and we obtain two general properties. First, the constant term of an MA model is the mean of the series [i.e., $E(r_t) = c_0$]. Second, the variance of an MA(q) model is

$$\text{Var}(r_t) = (1 + \theta_1^2 + \theta_2^2 + \cdots + \theta_q^2) \sigma_a^2.$$

Autocorrelation Function

Assume for simplicity that $c_0 = 0$ for an MA(1) model. Multiplying the model by $r_{t-\ell}$, we have

$$r_{t-\ell} r_t = r_{t-\ell} a_t - \theta_1 r_{t-\ell} a_{t-1}.$$

Taking expectation, we obtain

$$\gamma_1 = -\theta_1 \sigma_a^2 \quad \text{and} \quad \gamma_\ell = 0, \quad \text{for } \ell > 1.$$

Using the prior result and the fact that $\text{Var}(r_t) = (1 + \theta_1^2) \sigma_a^2$, we have

$$\rho_0 = 1, \quad \rho_1 = \frac{-\theta_1}{1 + \theta_1^2}, \quad \rho_\ell = 0, \quad \text{for } \ell > 1.$$

Thus, for an MA(1) model, the lag-1 ACF is not zero, but all higher order ACFs are zero. In other words, the ACF of an MA(1) model cuts off at lag 1. For the MA(2) model in Eq. (2.21), the autocorrelation coefficients are

$$\rho_1 = \frac{-\theta_1 + \theta_1 \theta_2}{1 + \theta_1^2 + \theta_2^2}, \quad \rho_2 = \frac{-\theta_2}{1 + \theta_1^2 + \theta_2^2}, \quad \rho_\ell = 0, \quad \text{for } \ell > 2.$$

Here the ACF cuts off at lag 2. This property generalizes to other MA models. For an MA(q) model, the lag- q ACF is not zero, but $\rho_\ell = 0$ for $\ell > q$. Consequently,

an $MA(q)$ series is only linearly related to its first q -lagged values and hence is a “finite-memory” model.

Invertibility

Rewriting a zero-mean $MA(1)$ model as $a_t = r_t + \theta_1 a_{t-1}$, one can use repeated substitutions to obtain

$$a_t = r_t + \theta_1 r_{t-1} + \theta_1^2 r_{t-2} + \theta_1^3 r_{t-3} + \cdots.$$

This equation expresses the current shock a_t as a linear combination of the present and past returns. Intuitively, θ_1^j should go to zero as j increases because the remote return r_{t-j} should have very little impact on the current shock, if any. Consequently, for an $MA(1)$ model to be plausible, we require $|\theta_1| < 1$. Such an $MA(1)$ model is said to be *invertible*. If $|\theta_1| = 1$, then the $MA(1)$ model is noninvertible. See Section 2.6.5 for further discussion on invertibility.

2.5.2 Identifying MA Order

The ACF is useful in identifying the order of an MA model. For a time series r_t with ACF ρ_ℓ , if $\rho_q \neq 0$, but $\rho_\ell = 0$ for $\ell > q$, then r_t follows an $MA(q)$ model.

Figure 2.8 shows the time plot of monthly simple returns of the CRSP equal-weighted index from January 1926 to December 2008 and the sample ACF of the series. The two dashed lines shown on the ACF plot denote the two standard error

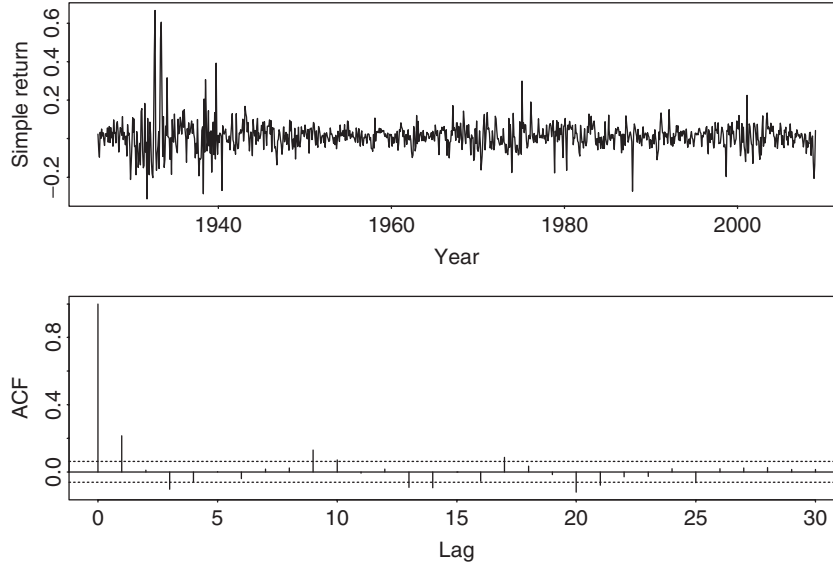


Figure 2.8 Time plot and sample autocorrelation function of monthly simple returns of CRSP equal-weighted index from January 1926 to December 2008.

limits. It is seen that the series has significant ACF at lags 1, 3, and 9. There are some marginally significant ACF at higher lags, but we do not consider them here. Based on the sample ACF, the following MA(9) model

$$r_t = c_0 + a_t - \theta_1 a_{t-1} - \theta_3 a_{t-3} - \theta_9 a_{t-9}$$

is identified for the series. Note that, unlike the sample PACF, sample ACF provides information on the nonzero MA lags of the model.

2.5.3 Estimation

Maximum-likelihood estimation is commonly used to estimate MA models. There are two approaches for evaluating the likelihood function of an MA model. The first approach assumes that the initial shocks (i.e., a_t for $t \leq 0$) are zero. As such, the shocks needed in likelihood function calculation are obtained recursively from the model, starting with $a_1 = r_1 - c_0$ and $a_2 = r_2 - c_0 + \theta_1 a_1$. This approach is referred to as the *conditional-likelihood method* and the resulting estimates the conditional maximum-likelihood estimates. The second approach treats the initial shocks a_t , $t \leq 0$, as additional parameters of the model and estimate them jointly with other parameters. This approach is referred to as the *exact-likelihood method*. The exact-likelihood estimates are preferred over the conditional ones, especially when the MA model is close to being noninvertible. The exact method, however, requires more intensive computation. If the sample size is large, then the two types of maximum-likelihood estimates are close to each other. For details of conditional- and exact-likelihood estimates of MA models, readers are referred to Box, Jenkins, and Reinsel (1994) or Chapter 8.

For illustration, consider the monthly simple return series of the CRSP equal-weighted index and the specified MA(9) model. The conditional maximum-likelihood method produces the fitted model

$$r_t = 0.012 + a_t + 0.189a_{t-1} - 0.121a_{t-3} + 0.122a_{t-9}, \quad \hat{\sigma}_a = 0.0714, \quad (2.23)$$

where standard errors of the coefficient estimates are 0.003, 0.031, 0.031, and 0.031, respectively. The Ljung–Box statistics of the residuals give $Q(12) = 17.5$ with a p value 0.041, which is based on an asymptotic chi-squared distribution with 9 degrees of freedom. The model needs some refinements in modeling the linear dynamic dependence of the data. The p value would be 0.132 if 12 degrees of freedom are used. The exact maximum-likelihood method produces the fitted model

$$r_t = 0.012 + a_t + 0.191a_{t-1} - 0.120a_{t-3} + 0.123a_{t-9}, \quad \hat{\sigma}_a = 0.0714, \quad (2.24)$$

where standard errors of the estimates are 0.003, 0.031, 0.031, and 0.031, respectively. The Ljung–Box statistics of the residuals give $Q(12) = 17.6$. The corresponding p values are 0.040 and 0.128, respectively, when the degrees of freedom

are 9 and 12. Again, this fitted model is only marginally adequate. Comparing models (2.23) and (2.24), we see that for this particular instance, the difference between the conditional- and exact-likelihood methods is negligible.

2.5.4 Forecasting Using MA Models

Forecasts of an MA model can easily be obtained. Because the model has finite memory, its point forecasts go to the mean of the series quickly. To see this, assume that the forecast origin is h and let F_h denote the information available at time h . For the 1-step-ahead forecast of an MA(1) process, the model says

$$r_{h+1} = c_0 + a_{h+1} - \theta_1 a_h.$$

Taking the conditional expectation, we have

$$\begin{aligned}\hat{r}_h(1) &= E(r_{h+1}|F_h) = c_0 - \theta_1 a_h, \\ e_h(1) &= r_{h+1} - \hat{r}_h(1) = a_{h+1}.\end{aligned}$$

The variance of the 1-step-ahead forecast error is $\text{Var}[e_h(1)] = \sigma_a^2$. In practice, the quantity a_h can be obtained in several ways. For instance, assume that $a_0 = 0$, then $a_1 = r_1 - c_0$, and we can compute a_t for $2 \leq t \leq h$ recursively by using $a_t = r_t - c_0 + \theta_1 a_{t-1}$. Alternatively, it can be computed by using the AR representation of the MA(1) model; see Section 2.6.5. Of course, a_t is the residual series of a fitted MA(1) model. Thus, a_h is readily available from the estimation.

For the 2-step-ahead forecast from the equation

$$r_{h+2} = c_0 + a_{h+2} - \theta_1 a_{h+1},$$

we have

$$\begin{aligned}\hat{r}_h(2) &= E(r_{h+2}|F_h) = c_0, \\ e_h(2) &= r_{h+2} - \hat{r}_h(2) = a_{h+2} - \theta_1 a_{h+1}.\end{aligned}$$

The variance of the forecast error is $\text{Var}[e_h(2)] = (1 + \theta_1^2)\sigma_a^2$, which is the variance of the model and is greater than or equal to that of the 1-step-ahead forecast error. The prior result shows that for an MA(1) model the 2-step-ahead forecast of the series is simply the unconditional mean of the model. This is true for any forecast origin h . More generally, $\hat{r}_h(\ell) = c_0$ for $\ell \geq 2$. In summary, for an MA(1) model, the 1-step-ahead point forecast at the forecast origin h is $c_0 - \theta_1 a_h$ and the multistep ahead forecasts are c_0 , which is the unconditional mean of the model. If we plot the forecasts $\hat{r}_h(\ell)$ versus ℓ , we see that the forecasts form a horizontal line after one step. Thus, for MA(1) models, mean reverting only takes one time period.

Similarly, for an MA(2) model, we have

$$r_{h+\ell} = c_0 + a_{h+\ell} - \theta_1 a_{h+\ell-1} - \theta_2 a_{h+\ell-2},$$

from which we obtain

$$\begin{aligned}\hat{r}_h(1) &= c_0 - \theta_1 a_h - \theta_2 a_{h-1}, \\ \hat{r}_h(2) &= c_0 - \theta_2 a_h, \\ \hat{r}_h(\ell) &= c_0, \quad \text{for } \ell > 2.\end{aligned}$$

Thus, the multistep-ahead forecasts of an MA(2) model go to the mean of the series after two steps. The variances of forecast errors go to the variance of the series after two steps. In general, for an MA(q) model, multistep-ahead forecasts go to the mean after the first q steps.

Table 2.3 gives some out-of-sample forecasts of an MA(9) model in the form of Eq. (2.24) for the monthly simple returns of the equal-weighted index at the forecast origin $h = 986$ (February 2008). The model parameters are reestimated using the first 986 observations. The sample mean and standard error of the estimation subsample are 0.0128 and 0.0736, respectively. As expected, the table shows that (a) the 10-step-ahead forecast is the sample mean, and (b) the standard deviations of the forecast errors converge to the standard deviation of the series as the forecast horizon increases. In this particular case, the point forecasts deviate substantially from the observed returns because of the worldwide financial crisis caused by the subprime mortgage problem and the collapse of Lehman Brothers.

Summary

A brief summary of AR and MA models is in order. We have discussed the following properties:

- For MA models, ACF is useful in specifying the order because ACF cuts off at lag q for an MA(q) series.
- For AR models, PACF is useful in order determination because PACF cuts off at lag p for an AR(p) process.
- An MA series is always stationary, but for an AR series to be stationary, all of its characteristic roots must be less than 1 in modulus.

TABLE 2.3 Out-of-Sample Forecast Performance of an MA(9) Model for Monthly Simple Returns of CRSP Equal-Weighted Index^a

Step	1	2	3	4	5
Forecast	0.0043	0.0136	0.0150	0.0144	0.0120
Std. Error	0.0712	0.0724	0.0729	0.0729	0.0729
Actual	-0.0260	0.0312	0.0322	-0.0871	-0.0010
Step	6	7	8	9	10
Forecast	0.0019	0.0122	0.0056	0.0085	0.0128
Std. Error	0.0729	0.0729	0.0729	0.0729	0.0734
Actual	0.0141	-0.1209	-0.2060	-0.1366	0.0431

^aThe forecast origin is February 2008 With $h = 986$. The model is estimated by the exact maximum-likelihood method.

- For a stationary series, the multistep-ahead forecasts converge to the mean of the series, and the variances of forecast errors converge to the variance of the series as the forecast horizon increases.

2.6 SIMPLE ARMA MODELS

In some applications, the AR or MA models discussed in the previous sections become cumbersome because one may need a high-order model with many parameters to adequately describe the dynamic structure of the data. To overcome this difficulty, the autoregressive moving-average (ARMA) models are introduced; see Box, Jenkins, and Reinsel (1994). Basically, an ARMA model combines the ideas of AR and MA models into a compact form so that the number of parameters used is kept small, achieving parsimony in parameterization. For the return series in finance, the chance of using ARMA models is low. However, the concept of ARMA models is highly relevant in volatility modeling. As a matter of fact, the generalized autoregressive conditional heteroscedastic (GARCH) model can be regarded as an ARMA model, albeit nonstandard, for the a_t^2 series; see Chapter 3 for details. In this section, we study the simplest ARMA(1,1) model.

A time series r_t follows an ARMA(1,1) model if it satisfies

$$r_t - \phi_1 r_{t-1} = \phi_0 + a_t - \theta_1 a_{t-1}, \quad (2.25)$$

where $\{a_t\}$ is a white noise series. The left-hand side of the Eq. (2.25) is the AR component of the model and the right-hand side gives the MA component. The constant term is ϕ_0 . For this model to be meaningful, we need $\phi_1 \neq \theta_1$; otherwise, there is a cancellation in the equation and the process reduces to a white noise series.

2.6.1 Properties of ARMA(1,1) Models

Properties of ARMA(1,1) models are generalizations of those of AR(1) models with some minor modifications to handle the impact of the MA(1) component. We start with the stationarity condition. Taking expectation of Eq. (2.25), we have

$$E(r_t) - \phi_1 E(r_{t-1}) = \phi_0 + E(a_t) - \theta_1 E(a_{t-1}).$$

Because $E(a_i) = 0$ for all i , the mean of r_t is

$$E(r_t) = \mu = \frac{\phi_0}{1 - \phi_1}$$

provided that the series is weakly stationary. This result is exactly the same as that of the AR(1) model in Eq. (2.8).

Next, assuming for simplicity that $\phi_0 = 0$, we consider the autocovariance function of r_t . First, multiplying the model by a_t and taking expectation, we have

$$E(r_t a_t) = E(a_t^2) - \theta_1 E(a_t a_{t-1}) = E(a_t^2) = \sigma_a^2. \quad (2.26)$$

Rewriting the model as

$$r_t = \phi_1 r_{t-1} + a_t - \theta_1 a_{t-1}$$

and taking the variance of the prior equation, we have

$$\text{Var}(r_t) = \phi_1^2 \text{Var}(r_{t-1}) + \sigma_a^2 + \theta_1^2 \sigma_a^2 - 2\phi_1 \theta_1 E(r_{t-1} a_{t-1}).$$

Here we make use of the fact that r_{t-1} and a_t are uncorrelated. Using Eq. (2.26), we obtain

$$\text{Var}(r_t) - \phi_1^2 \text{Var}(r_{t-1}) = (1 - 2\phi_1 \theta_1 + \theta_1^2) \sigma_a^2.$$

Therefore, if the series r_t is weakly stationary, then $\text{Var}(r_t) = \text{Var}(r_{t-1})$ and we have

$$\text{Var}(r_t) = \frac{(1 - 2\phi_1 \theta_1 + \theta_1^2) \sigma_a^2}{1 - \phi_1^2}.$$

Because the variance is positive, we need $\phi_1^2 < 1$ (i.e., $|\phi_1| < 1$). Again, this is precisely the same stationarity condition as that of the AR(1) model.

To obtain the autocovariance function of r_t , we assume $\phi_0 = 0$ and multiply the model in Eq. (2.25) by $r_{t-\ell}$ to obtain

$$r_t r_{t-\ell} - \phi_1 r_{t-1} r_{t-\ell} = a_t r_{t-\ell} - \theta_1 a_{t-1} r_{t-\ell}.$$

For $\ell = 1$, taking expectation and using Eq. (2.26) for $t - 1$, we have

$$\gamma_1 - \phi_1 \gamma_0 = -\theta_1 \sigma_a^2,$$

where $\gamma_\ell = \text{Cov}(r_t, r_{t-\ell})$. This result is different from that of the AR(1) case for which $\gamma_1 - \phi_1 \gamma_0 = 0$. However, for $\ell = 2$ and taking expectation, we have

$$\gamma_2 - \phi_1 \gamma_1 = 0,$$

which is identical to that of the AR(1) case. In fact, the same technique yields

$$\gamma_\ell - \phi_1 \gamma_{\ell-1} = 0, \quad \text{for } \ell > 1. \quad (2.27)$$

In terms of ACF, the previous results show that for a stationary ARMA(1,1) model

$$\rho_1 = \phi_1 - \frac{\theta_1 \sigma_a^2}{\gamma_0}, \quad \rho_\ell = \phi_1 \rho_{\ell-1}, \quad \text{for } \ell > 1.$$

Thus, the ACF of an ARMA(1,1) model behaves very much like that of an AR(1) model except that the exponential decay starts with lag 2. Consequently, the ACF of an ARMA(1,1) model does not cut off at any finite lag.

Turning to PACF, one can show that the PACF of an ARMA(1,1) model does not cut off at any finite lag either. It behaves very much like that of an MA(1) model except that the exponential decay starts with lag 2 instead of lag 1.

In summary, the stationarity condition of an ARMA(1,1) model is the same as that of an AR(1) model, and the ACF of an ARMA(1,1) exhibits a similar pattern like that of an AR(1) model except that the pattern starts at lag 2.

2.6.2 General ARMA Models

A general ARMA(p, q) model is in the form

$$r_t = \phi_0 + \sum_{i=1}^p \phi_i r_{t-i} + a_t - \sum_{i=1}^q \theta_i a_{t-i},$$

where $\{a_t\}$ is a white noise series and p and q are nonnegative integers. The AR and MA models are special cases of the ARMA(p, q) model. Using the back-shift operator, the model can be written as

$$(1 - \phi_1 B - \cdots - \phi_p B^p)r_t = \phi_0 + (1 - \theta_1 B - \cdots - \theta_q B^q)a_t. \quad (2.28)$$

The polynomial $1 - \phi_1 B - \cdots - \phi_p B^p$ is the AR polynomial of the model. Similarly, $1 - \theta_1 B - \cdots - \theta_q B^q$ is the MA polynomial. We require that there are no common factors between the AR and MA polynomials; otherwise the order (p, q) of the model can be reduced. Like a pure AR model, the AR polynomial introduces the characteristic equation of an ARMA model. If all of the solutions of the characteristic equation are less than 1 in absolute value, then the ARMA model is weakly stationary. In this case, the unconditional mean of the model is $E(r_t) = \phi_0/(1 - \phi_1 - \cdots - \phi_p)$.

2.6.3 Identifying ARMA Models

The ACF and PACF are not informative in determining the order of an ARMA model. Tsay and Tiao (1984) propose a new approach that uses the extended autocorrelation function (EACF) to specify the order of an ARMA process. The basic idea of EACF is relatively simple. If we can obtain a consistent estimate of the AR component of an ARMA model, then we can derive the MA component. From the derived MA series, we can use ACF to identify the order of the MA component.

The derivation of EACF is relatively involved; see Tsay and Tiao (1984) for details. Yet the function is easy to use. The output of EACF is a two-way table, where the rows correspond to AR order p and the columns to MA order q . The theoretical version of EACF for an ARMA(1,1) model is given in Table 2.4. The

TABLE 2.4 Theoretical EACF Table for an ARMA(1,1) Model, Where X Denotes Nonzero, O Denotes Zero, and * Denotes Either Zero or Nonzero^a

AR	MA							
	0	1	2	3	4	5	6	7
0	X	X	X	X	X	X	X	X
1	X	O	O	O	O	O	O	O
2	*	X	O	O	O	O	O	O
3	*	*	X	O	O	O	O	O
4	*	*	*	X	O	O	O	O
5	*	*	*	*	X	O	O	O

^aThis latter category does not play any role in identifying the order (1,1).

key feature of the table is that it contains a triangle of O with the upper left vertex located at the order (1,1). This is the characteristic we use to identify the order of an ARMA process. In general, for an ARMA(p, q) model, the triangle of O will have its upper left vertex at the (p, q) position.

For illustration, consider the monthly log stock returns of the 3M Company from February 1946 to December 2008. There are 755 observations. The return series and its sample ACF are shown in Figure 2.9. The ACF indicates that there are no significant serial correlations in the data at the 1% level. Table 2.5 shows the sample EACF and a corresponding simplified table for the series. The simplified table is constructed by using the following notation:

1. X denotes that the absolute value of the corresponding EACF is greater than or equal to $2/\sqrt{T}$, which is twice of the asymptotic standard error of the EACF.
2. O denotes that the corresponding EACF is less than $2/\sqrt{T}$ in modulus.

The simplified table exhibits a triangular pattern of O with its upper left vertex at the order (p, q) = (0, 0). A few exceptions of X appear when $q = 2, 5, 9$, and 11. However, the EACF table shows that the values of sample ACF corresponding to those X are around 0.08 or 0.09. These ACFs are only slightly greater than $2/\sqrt{755} = 0.073$. Indeed, if 1% critical value is used, those X would become O in the simplified EACF table. Consequently, the EACF suggests that the monthly log returns of 3M stock follow an ARMA(0,0) model (i.e., a white noise series). This is in agreement with the result suggested by the sample ACF in Figure 2.9.

The information criteria discussed earlier can also be used to select ARMA(p, q) models. Typically, for some prespecified positive integers P and Q , one computes AIC (or BIC) for ARMA(p, q) models, where $0 \leq p \leq P$ and $0 \leq q \leq Q$, and selects the model that gives the minimum AIC (or BIC). This approach requires maximum-likelihood estimation of many models and in some cases may encounter the difficulty of overfitting in estimation.

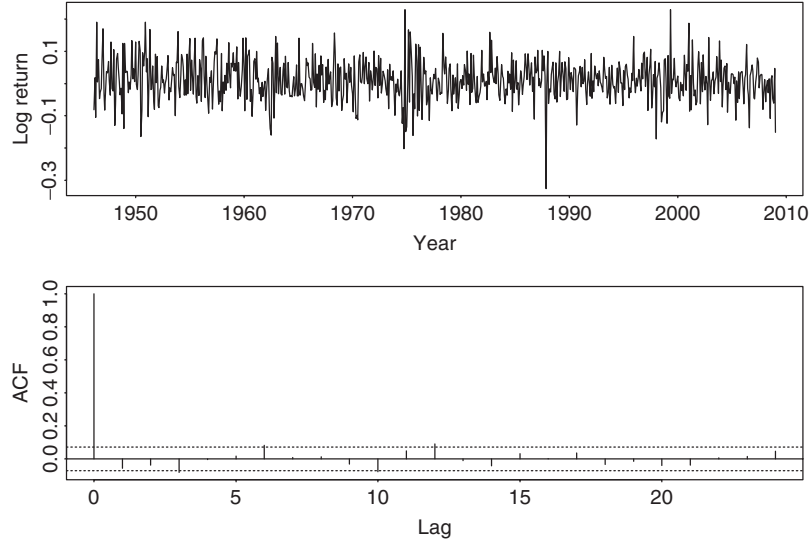


Figure 2.9 Time plot and sample autocorrelation function of monthly log stock returns of 3M Company from February 1946 to December 2008.

Once an $\text{ARMA}(p, q)$ model is specified, its parameters can be estimated by either the conditional or exact-likelihood method. In addition, the Ljung–Box statistics of the residuals can be used to check the adequacy of a fitted model. If the model is correctly specified, then $Q(m)$ follows asymptotically a chi-squared distribution with $m - g$ degrees of freedom, where g denotes the number of AR or MA coefficients fitted in the model.

2.6.4 Forecasting Using an ARMA Model

Like the behavior of ACF, forecasts of an $\text{ARMA}(p, q)$ model have similar characteristics as those of an $\text{AR}(p)$ model after adjusting for the impacts of the MA component on the lower horizon forecasts. Denote the forecast origin by h and the available information by F_h . The 1-step-ahead forecast of r_{h+1} can be easily obtained from the model as

$$\hat{r}_h(1) = E(r_{h+1}|F_h) = \phi_0 + \sum_{i=1}^p \phi_i r_{h+1-i} - \sum_{i=1}^q \theta_i a_{h+1-i},$$

and the associated forecast error is $e_h(1) = r_{h+1} - \hat{r}_h(1) = a_{h+1}$. The variance of 1-step-ahead forecast error is $\text{Var}[e_h(1)] = \sigma_a^2$. For the ℓ -step-ahead forecast, we have

$$\hat{r}_h(\ell) = E(r_{h+\ell}|F_h) = \phi_0 + \sum_{i=1}^p \phi_i \hat{r}_h(\ell - i) - \sum_{i=1}^q \theta_i a_h(\ell - i),$$

TABLE 2.5 Sample Extended Autocorrelation Function and a Simplified Table for the Monthly Log Returns of 3M Stock from February 1946 to December 2008

Sample Extended Autocorrelation Function													
MA Order: q													
p	0	1	2	3	4	5	6	7	8	9	10	11	12
0	-0.06	-0.04	-0.08	-0.00	0.02	0.08	0.01	0.01	-0.03	-0.08	0.05	0.09	-0.01
1	-0.47	0.01	-0.07	-0.02	0.00	0.08	-0.03	0.00	-0.01	-0.07	0.04	0.09	-0.02
2	-0.38	-0.35	-0.07	0.02	-0.01	0.08	0.03	0.01	0.00	-0.03	0.02	0.04	0.04
3	-0.18	0.14	0.38	-0.02	0.00	0.04	-0.02	0.02	-0.00	-0.03	0.02	0.01	0.04
4	0.42	0.03	0.45	-0.01	0.00	0.00	-0.01	0.03	0.01	0.00	0.02	-0.00	0.01
5	-0.11	0.21	0.45	0.01	0.20	-0.01	-0.00	0.04	-0.01	-0.01	0.03	0.01	0.03
6	-0.21	-0.25	0.24	0.31	0.17	-0.04	-0.00	0.04	-0.01	-0.03	0.01	0.01	0.04

Simplified EACF Table													
MA Order: q													
p	0	1	2	3	4	5	6	7	8	9	10	11	12
0	O	O	X	O	O	X	O	O	O	X	O	X	O
1	X	O	O	O	O	X	O	O	O	O	O	X	O
2	X	X	O	O	O	X	O	O	O	O	O	O	O
3	X	X	X	O	O	O	O	O	O	O	O	O	O
4	X	O	X	O	O	O	O	O	O	O	O	O	O
5	X	X	X	O	X	O	O	O	O	O	O	O	O
6	X	X	X	X	X	O	O	O	O	O	O	O	O

where it is understood that $\hat{r}_h(\ell - i) = r_{h+\ell-i}$ if $\ell - i \leq 0$ and $a_h(\ell - i) = 0$ if $\ell - i > 0$ and $a_h(\ell - i) = a_{h+\ell-i}$ if $\ell - i \leq 0$. Thus, the multistep-ahead forecasts of an ARMA model can be computed recursively. The associated forecast error is

$$e_h(\ell) = r_{h+\ell} - \hat{r}_h(\ell),$$

which can be computed easily via a formula to be given in Eq. (2.34).

2.6.5 Three Model Representations for an ARMA Model

In this section, we briefly discuss three model representations for a stationary ARMA(p, q) model. The three representations serve three different purposes. Knowing these representations can lead to a better understanding of the model. The first representation is the ARMA(p, q) model in Eq. (2.28). This representation is compact and useful in parameter estimation. It is also useful in computing recursively multistep-ahead forecasts of r_t ; see the discussion in the last section.

For the other two representations, we use long division of two polynomials. Given two polynomials $\phi(B) = 1 - \sum_{i=1}^p \phi_i B^i$ and $\theta(B) = 1 - \sum_{i=1}^q \theta_i B^i$, we can obtain, by long division, that

$$\frac{\theta(B)}{\phi(B)} = 1 + \psi_1 B + \psi_2 B^2 + \cdots \equiv \psi(B) \quad (2.29)$$

and

$$\frac{\phi(B)}{\theta(B)} = 1 - \pi_1 B - \pi_2 B^2 - \cdots \equiv \pi(B). \quad (2.30)$$

For instance, if $\phi(B) = 1 - \phi_1 B$ and $\theta(B) = 1 - \theta_1 B$, then

$$\begin{aligned} \psi(B) &= \frac{1 - \theta_1 B}{1 - \phi_1 B} = 1 + (\phi_1 - \theta_1)B + \phi_1(\phi_1 - \theta_1)B^2 + \phi_1^2(\phi_1 - \theta_1)B^3 + \cdots, \\ \pi(B) &= \frac{1 - \phi_1 B}{1 - \theta_1 B} = 1 - (\phi_1 - \theta_1)B - \theta_1(\phi_1 - \theta_1)B^2 - \theta_1^2(\phi_1 - \theta_1)B^3 - \cdots. \end{aligned}$$

From the definition, $\psi(B)\pi(B) = 1$. Making use of the fact that $Bc = c$ for any constant (because the value of a constant is time invariant), we have

$$\frac{\phi_0}{\theta(1)} = \frac{\phi_0}{1 - \theta_1 - \cdots - \theta_q} \quad \text{and} \quad \frac{\phi_0}{\phi(1)} = \frac{\phi_0}{1 - \phi_1 - \cdots - \phi_p}.$$

AR Representation

Using the result of long division in Eq. (2.30), the ARMA(p, q) model can be written as

$$r_t = \frac{\phi_0}{1 - \theta_1 - \cdots - \theta_q} + \pi_1 r_{t-1} + \pi_2 r_{t-2} + \pi_3 r_{t-3} + \cdots + a_t. \quad (2.31)$$

This representation shows the dependence of the current return r_t on the past returns r_{t-i} , where $i > 0$. The coefficients $\{\pi_i\}$ are referred to as the π weights of an ARMA model. To show that the contribution of the lagged value r_{t-i} to r_t is diminishing as i increases, the π_i coefficient should decay to zero as i increases. An ARMA(p, q) model that has this property is said to be invertible. For a pure AR model, $\theta(B) = 1$ so that $\pi(B) = \phi(B)$, which is a finite-degree polynomial. Thus, $\pi_i = 0$ for $i > p$, and the model is invertible. For other ARMA models, a sufficient condition for invertibility is that all the zeros of the polynomial $\theta(B)$ are greater than unity in modulus. For example, consider the MA(1) model $r_t = (1 - \theta_1 B)a_t$. The zero of the first-order polynomial $1 - \theta_1 B$ is $B = 1/\theta_1$. Therefore, an MA(1) model is invertible if $|1/\theta_1| > 1$. This is equivalent to $|\theta_1| < 1$.

From the AR representation in Eq. (2.31), an invertible ARMA(p, q) series r_t is a linear combination of the current shock a_t and a weighted average of the past values. The weights decay exponentially for more remote past values.

MA Representation

Again, using the result of long division in Eq. (2.29), an ARMA(p, q) model can also be written as

$$r_t = \mu + a_t + \psi_1 a_{t-1} + \psi_2 a_{t-2} + \cdots = \mu + \psi(B)a_t, \quad (2.32)$$

where $\mu = E(r_t) = \phi_0/(1 - \phi_1 - \cdots - \phi_p)$. This representation shows explicitly the impact of the past shock a_{t-i} ($i > 0$) on the current return r_t . The coefficients $\{\psi_i\}$ are referred to as the *impulse response function* of the ARMA model. For a weakly stationary series, the ψ_i coefficients decay exponentially as i increases. This is understandable as the effect of shock a_{t-i} on the return r_t should diminish over time. Thus, for a stationary ARMA model, the shock a_{t-i} does not have a permanent impact on the series. If $\phi_0 \neq 0$, then the MA representation has a constant term, which is the mean of r_t [i.e., $\phi_0/(1 - \phi_1 - \cdots - \phi_p)$].

The MA representation in Eq. (2.32) is also useful in computing the variance of a forecast error. At the forecast origin h , we have the shocks $a_h, a_{h-1}, \dots$. Therefore, the ℓ -step-ahead point forecast is

$$\hat{r}_h(\ell) = \mu + \psi_\ell a_h + \psi_{\ell+1} a_{h-1} + \cdots, \quad (2.33)$$

and the associated forecast error is

$$e_h(\ell) = a_{h+\ell} + \psi_1 a_{h+\ell-1} + \cdots + \psi_{\ell-1} a_{h+1}.$$

Consequently, the variance of ℓ -step-ahead forecast error is

$$\text{Var}[e_h(\ell)] = (1 + \psi_1^2 + \cdots + \psi_{\ell-1}^2) \sigma_a^2, \quad (2.34)$$

which, as expected, is a nondecreasing function of the forecast horizon ℓ .

Finally, the MA representation in Eq. (2.32) provides a simple proof of mean reversion of a stationary time series. The stationarity implies that ψ_i approaches zero as $i \rightarrow \infty$. Therefore, by Eq. (2.33), we have $\hat{r}_h(\ell) \rightarrow \mu$ as $\ell \rightarrow \infty$. Because $\hat{r}_h(\ell)$ is the conditional expectation of $r_{h+\ell}$ at the forecast origin h , the result says that in the long term the return series is expected to approach its mean, that is, the series is mean reverting. Furthermore, using the MA representation in Eq. (2.32), we have $\text{Var}(r_t) = (1 + \sum_{i=1}^{\infty} \psi_i^2) \sigma_a^2$. Consequently, by Eq. (2.34), we have $\text{Var}[e_h(\ell)] \rightarrow \text{Var}(r_t)$ as $\ell \rightarrow \infty$. The speed by which $\hat{r}_h(\ell)$ approaches μ determines the speed of mean reverting.

2.7 UNIT-ROOT NONSTATIONARITY

So far we have focused on return series that are stationary. In some studies, interest rates, foreign exchange rates, or the price series of an asset are of interest. These series tend to be nonstationary. For a price series, the nonstationarity is mainly

due to the fact that there is no fixed level for the price. In the time series literature, such a nonstationary series is called unit-root nonstationary time series. The best known example of unit-root nonstationary time series is the random-walk model.

2.7.1 Random Walk

A time series $\{p_t\}$ is a random walk if it satisfies

$$p_t = p_{t-1} + a_t, \quad (2.35)$$

where p_0 is a real number denoting the starting value of the process and $\{a_t\}$ is a white noise series. If p_t is the log price of a particular stock at date t , then p_0 could be the log price of the stock at its initial public offering (IPO) (i.e., the logged IPO price). If a_t has a symmetric distribution around zero, then conditional on p_{t-1} , p_t has a 50–50 chance to go up or down, implying that p_t would go up or down at random. If we treat the random-walk model as a special AR(1) model, then the coefficient of p_{t-1} is unity, which does not satisfy the weak stationarity condition of an AR(1) model. A random-walk series is, therefore, not weakly stationary, and we call it a unit-root nonstationary time series.

The random-walk model has widely been considered as a statistical model for the movement of logged stock prices. Under such a model, the stock price is not predictable or mean reverting. To see this, the 1-step-ahead forecast of model (2.35) at the forecast origin h is

$$\hat{p}_h(1) = E(p_{h+1}|p_h, p_{h-1}, \dots) = p_h,$$

which is the log price of the stock at the forecast origin. Such a forecast has no practical value. The 2-step-ahead forecast is

$$\begin{aligned} \hat{p}_h(2) &= E(p_{h+2}|p_h, p_{h-1}, \dots) = E(p_{h+1} + a_{h+2}|p_h, p_{h-1}, \dots) \\ &= E(p_{h+1}|p_h, p_{h-1}, \dots) = \hat{p}_h(1) = p_h, \end{aligned}$$

which again is the log price at the forecast origin. In fact, for any forecast horizon $\ell > 0$, we have

$$\hat{p}_h(\ell) = p_h.$$

Thus, for all forecast horizons, point forecasts of a random-walk model are simply the value of the series at the forecast origin. Therefore, the process is not mean reverting.

The MA representation of the random-walk model in Eq. (2.35) is

$$p_t = a_t + a_{t-1} + a_{t-2} + \dots$$

This representation has several important practical implications. First, the ℓ -step-ahead forecast error is

$$e_h(\ell) = a_{h+\ell} + \cdots + a_{h+1},$$

so that $\text{Var}[e_h(\ell)] = \ell\sigma_a^2$, which diverges to infinity as $\ell \rightarrow \infty$. The length of an interval forecast of $p_{h+\ell}$ will approach infinity as the forecast horizon-increases. This result says that the usefulness of point forecast $\hat{p}_h(\ell)$ diminishes as ℓ increases, which again implies that the model is not predictable. Second, the unconditional variance of p_t is unbounded because $\text{Var}[e_h(\ell)]$ approaches infinity as ℓ increases. Theoretically, this means that p_t can assume any real value for a sufficiently large t . For the log price p_t of an individual stock, this is plausible. Yet for market indexes, negative log price is very rare if it happens at all. In this sense, the adequacy of a random-walk model for market indexes is questionable. Third, from the representation, $\psi_i = 1$ for all i . Thus, the impact of any past shock a_{t-i} on p_t does not decay over time. Consequently, the series has a strong memory as it remembers all of the past shocks. In economics, the shocks are said to have a permanent effect on the series. The strong memory of a unit-root time series can be seen from the sample ACF of the observed series. The sample ACFs are all approaching 1 as the sample size increases.

2.7.2 Random Walk with Drift

As shown by empirical examples considered so far, the log return series of a market index tends to have a small and positive mean. This implies that the model for the log price is

$$p_t = \mu + p_{t-1} + a_t, \quad (2.36)$$

where $\mu = E(p_t - p_{t-1})$ and $\{a_t\}$ is a zero-mean white noise series. The constant term μ of model (2.36) is very important in financial study. It represents the time trend of the log price p_t and is often referred to as the *drift* of the model. To see this, assume that the initial log price is p_0 . Then we have

$$\begin{aligned} p_1 &= \mu + p_0 + a_1, \\ p_2 &= \mu + p_1 + a_2 = 2\mu + p_0 + a_2 + a_1, \\ &\vdots \\ p_t &= t\mu + p_0 + a_t + a_{t-1} + \cdots + a_1. \end{aligned}$$

The last equation shows that the log price consists of a time trend $t\mu$ and a pure random-walk process $\sum_{i=1}^t a_i$. Because $\text{Var}(\sum_{i=1}^t a_i) = t\sigma_a^2$, where σ_a^2 is the variance of a_t , the conditional standard deviation of p_t is $\sqrt{t}\sigma_a$, which grows at a slower rate than the conditional expectation of p_t . Therefore, if we graph p_t against

the time index t , we have a time trend with slope μ . A positive slope μ implies that the log price eventually goes to infinity. In contrast, a negative μ implies that the log price would converge to $-\infty$ as t increases. Based on the above discussion, it is then not surprising to see that the log return series of the CRSP value- and equal-weighted indexes have a small, but statistically significant, positive mean.

To illustrate the effect of the drift parameter on the price series, we consider the monthly log stock returns of the 3M Company from February 1946 to December 2008. As shown by the sample EACF in Table 2.5, the series has no significant serial correlation. The series thus follows the simple model

$$r_t = 0.0103 + a_t, \quad \hat{\sigma}_a = 0.0637, \quad (2.37)$$

where 0.0103 is the sample mean of r_t and has a standard error 0.0023. The mean of the monthly log returns of 3M stock is, therefore, significantly different from zero at the 1% level. As a matter of fact, the one-sample test of zero mean shows a t ratio of 4.44 with a p value close to 0. We use the log return series to construct two log price series, namely

$$p_t = \sum_{i=1}^t r_i \quad \text{and} \quad p_t^* = \sum_{i=1}^t a_i,$$

where a_i is the mean-corrected log return in Eq. (2.37) (i.e., $a_i = r_i - 0.0103$). The p_t is the log price of 3M stock, assuming that the initial log price is zero (i.e., the log price of January 1946 was zero). The p_t^* is the corresponding log price if the mean of log returns was zero. Figure 2.10 shows the time plots of p_t and p_t^* as well as a straight line $y_t = 0.0103 \times t + 1946$, where t is the time sequence of the returns and 1946 is the starting year of the stock. From the plots, the importance of the constant 0.0103 in Eq. (2.37) is evident. In addition, as expected, it represents the slope of the upward trend of p_t .

Interpretation of the Constant Term

From the previous discussions, it is important to understand the meaning of a constant term in a time series model. First, for an $MA(q)$ model in Eq. (2.22), the constant term is simply the mean of the series. Second, for a stationary $AR(p)$ model in Eq. (2.9) or $ARMA(p, q)$ model in Eq. (2.28), the constant term is related to the mean via $\mu = \phi_0 / (1 - \phi_1 - \cdots - \phi_p)$. Third, for a random walk with drift, the constant term becomes the time slope of the series. These different interpretations for the constant term in a time series model clearly highlight the difference between dynamic and usual linear regression models.

Another important difference between dynamic and regression models is shown by an $AR(1)$ model and a simple linear regression model,

$$r_t = \phi_0 + \phi_1 r_{t-1} + a_t \quad \text{and} \quad y_t = \beta_0 + \beta_1 x_t + a_t.$$

For the $AR(1)$ model to be meaningful, the coefficient ϕ_1 must satisfy $|\phi_1| \leq 1$. However, the coefficient β_1 can assume any fixed real number.

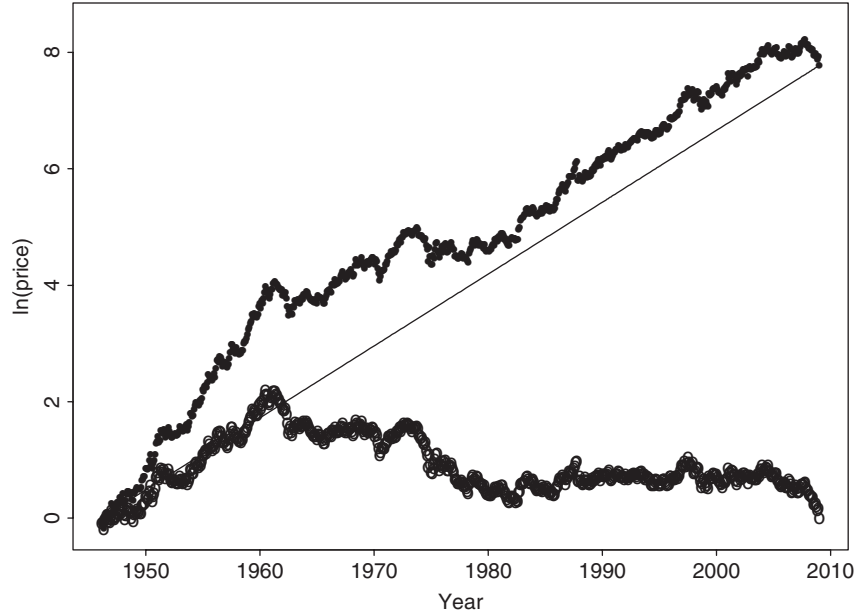


Figure 2.10 Time plots of log prices for 3M stock from February 1946 to December 2008, assuming that log price of January 1946 was zero. The “o” line is for log price without time trend. Straight line is $y_t = 0.0103 \times t + 1946$.

2.7.3 Trend-Stationary Time Series

A closely related model that exhibits linear trend is the trend-stationary time series model,

$$p_t = \beta_0 + \beta_1 t + r_t,$$

where r_t is a stationary time series, for example, a stationary $AR(p)$ series. Here p_t grows linearly in time with rate β_1 and hence can exhibit behavior similar to that of a random-walk model with drift. However, there is a major difference between the two models. To see this, suppose that p_0 is fixed. The random-walk model with drift assumes the mean $E(p_t) = p_0 + \mu t$ and variance $\text{Var}(p_t) = t\sigma_a^2$, both of them are time dependent. On the other hand, the trend-stationary model assumes the mean $E(p_t) = \beta_0 + \beta_1 t$, which depends on time, and variance $\text{Var}(p_t) = \text{Var}(r_t)$, which is finite and time invariant. The trend-stationary series can be transformed into a stationary one by removing the time trend via a simple linear regression analysis. For analysis of trend-stationary time series, see the method of Section 2.9.

2.7.4 General Unit-Root Nonstationary Models

Consider an ARMA model. If one extends the model by allowing the AR polynomial to have 1 as a characteristic root, then the model becomes the well-known

autoregressive integrated moving-average (ARIMA) model. An ARIMA model is said to be unit-root nonstationary because its AR polynomial has a unit root. Like a random-walk model, an ARIMA model has strong memory because the ψ_i coefficients in its MA representation do not decay over time to zero, implying that the past shock a_{t-i} of the model has a permanent effect on the series. A conventional approach for handling unit-root nonstationarity is to use *differencing*.

Differencing

A time series y_t is said to be an $\text{ARIMA}(p, 1, q)$ process if the change series $c_t = y_t - y_{t-1} = (1 - B)y_t$ follows a stationary and invertible $\text{ARMA}(p, q)$ model. In finance, price series are commonly believed to be nonstationary, but the log return series, $r_t = \ln(P_t) - \ln(P_{t-1})$, is stationary. In this case, the log price series is unit-root nonstationary and hence can be treated as an ARIMA process. The idea of transforming a nonstationary series into a stationary one by considering its change series is called *differencing* in the time series literature. More formally, $c_t = y_t - y_{t-1}$ is referred to as the first differenced series of y_t . In some scientific fields, a time series y_t may contain multiple unit roots and needs to be differenced multiple times to become stationary. For example, if both y_t and its first differenced series $c_t = y_t - y_{t-1}$ are unit-root nonstationary, but $s_t = c_t - c_{t-1} = y_t - 2y_{t-1} + y_{t-2}$ is weakly stationary, then y_t has double unit roots, and s_t is the second differenced series of y_t . In addition, if s_t follows an $\text{ARMA}(p, q)$ model, then y_t is an $\text{ARIMA}(p, 2, q)$ process. For such a time series, if s_t has a nonzero mean, then y_t has a quadratic time function and the quadratic time coefficient is related to the mean of s_t . The seasonally adjusted series of U.S. quarterly gross domestic product implicit price deflator might have double unit roots. However, the mean of the second differenced series is not significantly different from zero; see the Exercises of this chapter. Box, Jenkins, and Reinsel (1994) discuss many properties of general ARIMA models.

2.7.5 Unit-Root Test

To test whether the log price p_t of an asset follows a random walk or a random walk with drift, we employ the models

$$p_t = \phi_1 p_{t-1} + e_t \quad (2.38)$$

$$p_t = \phi_0 + \phi_1 p_{t-1} + e_t, \quad (2.39)$$

where e_t denotes the error term, and consider the null hypothesis $H_0 : \phi_1 = 1$ versus the alternative hypothesis $H_a : \phi_1 < 1$. This is the well-known unit-root testing problem; see Dickey and Fuller (1979). A convenient test statistic is the t ratio of the least-squares (LS) estimate of ϕ_1 under the null hypothesis. For Eq. (2.38), the LS method gives

$$\hat{\phi}_1 = \frac{\sum_{t=1}^T p_{t-1} p_t}{\sum_{t=1}^T p_{t-1}^2}, \quad \hat{\sigma}_e^2 = \frac{\sum_{t=1}^T (p_t - \hat{\phi}_1 p_{t-1})^2}{T - 1},$$

where $p_0 = 0$ and T is the sample size. The t ratio is

$$\text{DF} \equiv t \text{ ratio} = \frac{\hat{\phi}_1 - 1}{\text{std}(\hat{\phi}_1)} = \frac{\sum_{t=1}^T p_{t-1} e_t}{\hat{\sigma}_e \sqrt{\sum_{t=1}^T p_{t-1}^2}},$$

which is commonly referred to as the Dickey–Fuller (DF) test. If $\{e_t\}$ is a white noise series with finite moments of order slightly greater than 2, then the DF statistic converges to a function of the standard Brownian motion as $T \rightarrow \infty$; see Chan and Wei (1988) and Phillips (1987) for more information. If ϕ_0 is zero but Eq. (2.39) is employed anyway, then the resulting t ratio for testing $\phi_1 = 1$ will converge to another nonstandard asymptotic distribution. In either case, simulation is used to obtain critical values of the test statistics; see Fuller (1976, Chapter 8) for selected critical values. Yet if $\phi_0 \neq 0$ and Eq. (2.39) is used, then the t ratio for testing $\phi_1 = 1$ is asymptotically normal. However, large sample sizes are needed for the asymptotic normal distribution to hold. Standard Brownian motion is introduced in Chapter 6.

For many economic time series, ARIMA(p, d, q) models might be more appropriate than the simple model in Eq. (2.39). In the econometric literature, AR(p) models are often used. Denote the series by x_t . To verify the existence of a unit root in an AR(p) process, one may perform the test $H_0 : \beta = 1$ vs. $H_a : \beta < 1$ using the regression

$$x_t = c_t + \beta x_{t-1} + \sum_{i=1}^{p-1} \phi_i \Delta x_{t-i} + e_t, \quad (2.40)$$

where c_t is a deterministic function of the time index t and $\Delta x_j = x_j - x_{j-1}$ is the differenced series of x_t . In practice, c_t can be zero or a constant or $c_t = \omega_0 + \omega_1 t$. The t ratio of $\hat{\beta} - 1$,

$$\text{ADF-test} = \frac{\hat{\beta} - 1}{\text{std}(\hat{\beta})},$$

where $\hat{\beta}$ denotes the least-squares estimate of β , is the well-known *augmented Dickey–Fuller* (ADF) unit-root test. Note that because of the first differencing, Eq. (2.40) is equivalent to an AR(p) model with deterministic function c_t . Equation (2.40) can also be rewritten as

$$\Delta x_t = c_t + \beta_c x_{t-1} + \sum_{i=1}^{p-1} \phi_i \Delta x_{t-i} + e_t,$$

where $\beta_c = \beta - 1$. One can then test the equivalent hypothesis $H_0 : \beta_c = 0$ vs. $H_a : \beta_c < 0$.

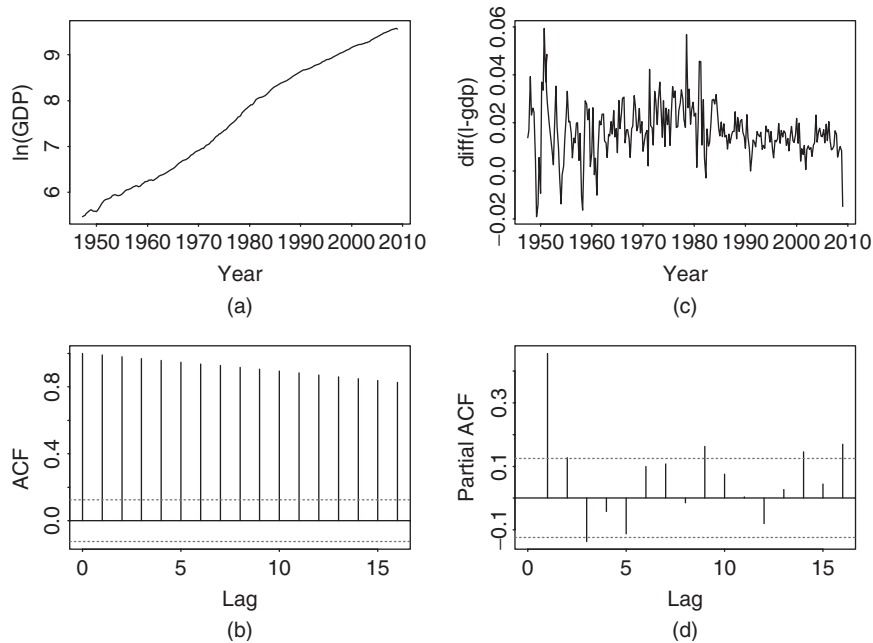


Figure 2.11 Log series of U.S. quarterly GDP from 1947.I to 2008.IV: (a) time plot of logged GDP series, (b) sample ACF of log GDP data, (c) time plot of first differenced series, and (d) sample PACF of differenced series.

Example 2.2. Consider the log series of U.S. quarterly GDP from 1947.I to 2008.IV. The series exhibits an upward trend, showing the growth of the U.S. economy, and has high sample serial correlations; see the lower left panel of Figure 2.11. The first differenced series, representing the growth rate of U.S. GDP and also shown in Figure 2.11, seems to vary around a fixed mean level, even though the variability appears to be smaller in recent years. To confirm the observed phenomenon, we apply the ADF unit-root test to the log series. Based on the sample PACF of the differenced series shown in Figure 2.11, we choose $p = 10$. Other values of p are also used, but they do not alter the conclusion of the test. With $p = 10$, the ADF test statistic is -1.701 with a p value 0.4297 , indicating that the unit-root hypothesis cannot be rejected. From the attached S-Plus output, $\hat{\beta} = 1 + \hat{\beta}_c = 1 - 0.0008 = 0.9992$.

R Demonstration

```
> library(fUnitRoots)
> da=read.table("q-gdp4708.txt",header=T)
> gdp=log(da[,4])
> m1=ar(diff(gdp),method='mle')
> m1$order
[1] 10
> adfTest(gdp,lags=10,type=c("c"))
```

Title:
Augmented Dickey-Fuller Test

Test Results:
PARAMETER:
Lag Order: 10
STATISTIC:
Dickey-Fuller: -1.6109
P VALUE: 0.4569

S-Plus Demonstration

The following output has been edited:

```
> adft=unitroot(gdp,trend='c',method='adf',lags=10)
> summary(adft)
```

```
Test for Unit Root: Augmented DF Test
Null Hypothesis: there is a unit root
Type of Test: t-test
Test Statistic: -1.701
P-value: 0.4297
```

```
Coefficients:
      Value Std. Error t value Pr(>|t|)
lag1 -0.0008  0.0005   -1.7006  0.0904
lag2  0.3799  0.0659    5.7637  0.0000
lag3  0.1883  0.0696    2.7047  0.0074
...
lag10 0.1784  0.0637    2.8023  0.0055
constant 0.0134  0.0045    2.9636  0.0034
```

```
Regression Diagnostics:
R-Squared 0.2877
Adjusted R-Squared 0.2564
Durbin-Watson Stat 1.9940
```

```
Residual standard error: 0.009318 on 234 degrees of freedom
```

As another example, consider the log series of the S&P 500 index from January 3, 1950, to April 16, 2008, for 14,462 observations. The series is shown in Figure 2.12. Testing for a unit root in the index is relevant if one wishes to verify empirically that the Index follows a random walk with drift. To this end, we use $c_t = \omega_0 + \omega_1 t$ in applying the ADF test. Furthermore, we choose $p = 15$ based on the sample PACF of the first differenced series. The resulting test statistic is -1.998 with a p value 0.602 . Thus, the unit-root hypothesis cannot be rejected at any reasonable significance level. The constant term is statistically significant, whereas the estimate of the time trend is not at the usual 5% level. The latter is

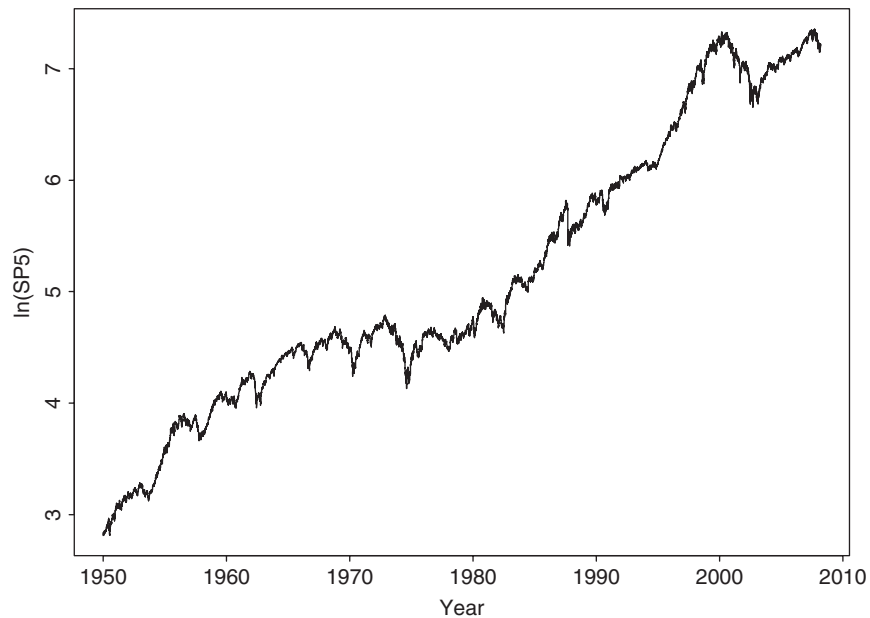


Figure 2.12 Time plot of logarithm of daily S&P 500 index from January 3, 1950, to April 16, 2008.

significant at the 10% level, however. In summary, for the period from January 1950 to April 2008, the log series of the S&P 500 index contains a unit root and a positive drift, but there is no strong evidence of a time trend.

R Demonstration

```
> library(fUnitRoots)
> da=read.table("d-sp55008.txt",header=T)
> sp5=log(da[,7])
> m2=ar(diff(sp5),method='mle')
> m2$order
[1] 2
> adfTest(sp5,lags=2,type="ct")
```

```
Title:
Augmented Dickey-Fuller Test
```

```
Test Results:
PARAMETER:
  Lag Order: 2
STATISTIC:
  Dickey-Fuller: -2.0179
P VALUE: 0.5708
```

```
> adfTest(sp5,lags=15,type="ct")
```

Title:
Augmented Dickey-Fuller Test

Test Results:
PARAMETER:
Lag Order: 15
STATISTIC:
Dickey-Fuller: -1.9946
P VALUE: 0.5807

S-Plus Demonstration

The following output has been edited:

```
> adft=unitroot(sp5,method='adf',trend='ct',lags=15)
> summary(adft)
```

Test for Unit Root: Augmented DF Test

Null Hypothesis: there is a unit root
Type of Test: t-test
Test Statistic: -1.998
P-value: 0.602

Coefficients:

	Value	Std. Error	t value	Pr(> t)
lag1	-0.0005	0.0003	-1.9977	0.0458
lag2	0.0722	0.0083	8.7374	0.0000
lag3	-0.0386	0.0083	-4.6532	0.0000
lag4	-0.0071	0.0083	-0.8548	0.3927
...				
lag15	0.0133	0.0083	1.6122	0.1069
constant	0.0019	0.0008	2.3907	0.0168
time	0.0020	0.0011	1.8507	0.0642

Regression Diagnostics:
R-Squared 0.0081
Adjusted R-Squared 0.0070
Durbin-Watson Stat 1.9995

Residual standard error: 0.008981 on 14643 degrees of freedom

2.8 SEASONAL MODELS

Some financial time series such as quarterly earnings per share of a company exhibits certain cyclical or periodic behavior. Such a time series is called a *seasonal time series*. Figure 2.13(a) shows the time plot of quarterly earnings per share of Johnson & Johnson from the first quarter of 1960 to the last quarter of 1980.

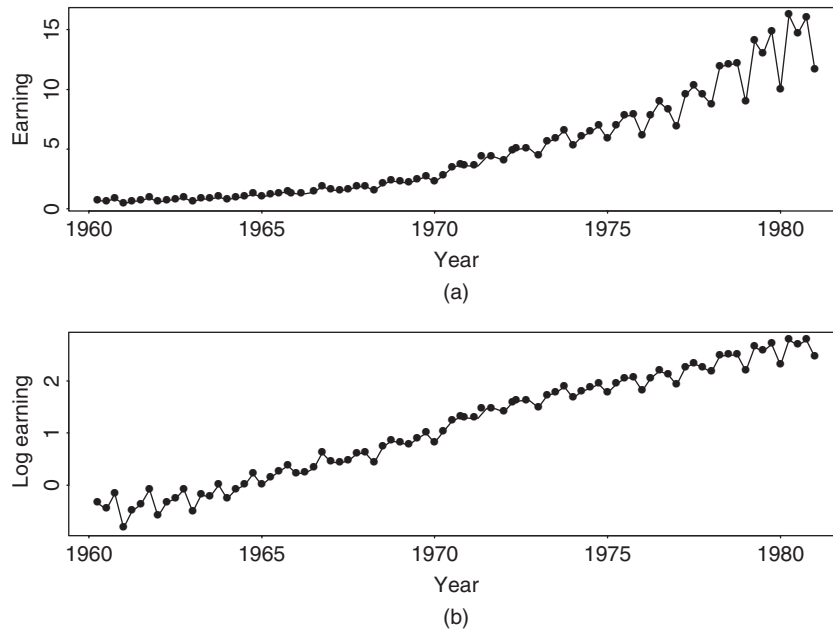


Figure 2.13 Time plots of quarterly earnings per share of Johnson & Johnson from 1960 to 1980: (a) observed earnings and (b) log earnings.

The data obtained from Shumway and Stoffer (2000) possess some special characteristics. In particular, the earnings grew exponentially during the sample period and had a strong seasonality. Furthermore, the variability of earnings increased over time. The cyclical pattern repeats itself every year so that the periodicity of the series is 4. If monthly data are considered (e.g., monthly sales of Wal-Mart stores), then the periodicity is 12. Seasonal time series models are also useful in pricing weather-related derivatives and energy futures because most environmental time series exhibit strong seasonal behavior.

Analysis of seasonal time series has a long history. In some applications, seasonality is of secondary importance and is removed from the data, resulting in a seasonally adjusted time series that is then used to make inference. The procedure to remove seasonality from a time series is referred to as *seasonal adjustment*. Most economic data published by the U.S. government are seasonally adjusted (e.g., the growth rate of gross domestic product and the unemployment rate). In other applications such as forecasting, seasonality is as important as other characteristics of the data and must be handled accordingly. Because forecasting is a major objective of financial time series analysis, we focus on the latter approach and discuss some econometric models that are useful in modeling seasonal time series.

2.8.1 Seasonal Differencing

Figure 2.13(b) shows the time plot of log earnings per share of Johnson & Johnson. We took the log transformation for two reasons. First, it is used to handle the

exponential growth of the series. Indeed, the plot confirms that the growth is linear in the log scale. Second, the transformation is used to stabilize the variability of the series. Again, the increasing pattern in variability of Figure 2.13(a) disappears in the new plot. Log transformation is commonly used in analysis of financial and economic time series. In this particular instance, all earnings are positive so that no adjustment is needed before taking the transformation. In some cases, one may need to add a positive constant to every data point before taking the transformation.

Denote the log earnings by x_t . The upper left panel of Figure 2.14 shows the sample ACF of x_t , which indicates that the quarterly log earnings per share has strong serial correlations. A conventional method to handle such strong serial correlations is to consider the first differenced series of x_t [i.e., $\Delta x_t = x_t - x_{t-1} = (1 - B)x_t$]. The lower left plot of Figure 2.14 gives the sample ACF of Δx_t . The ACF is strong when the lag is a multiple of periodicity 4. This is a well-documented behavior of sample ACF of a seasonal time series. Following the procedure of Box, Jenkins, and Reinsel (1994, Chapter 9), we take another difference of the data, that is,

$$\Delta_4(\Delta x_t) = (1 - B^4)\Delta x_t = \Delta x_t - \Delta x_{t-4} = x_t - x_{t-1} - x_{t-4} + x_{t-5}.$$

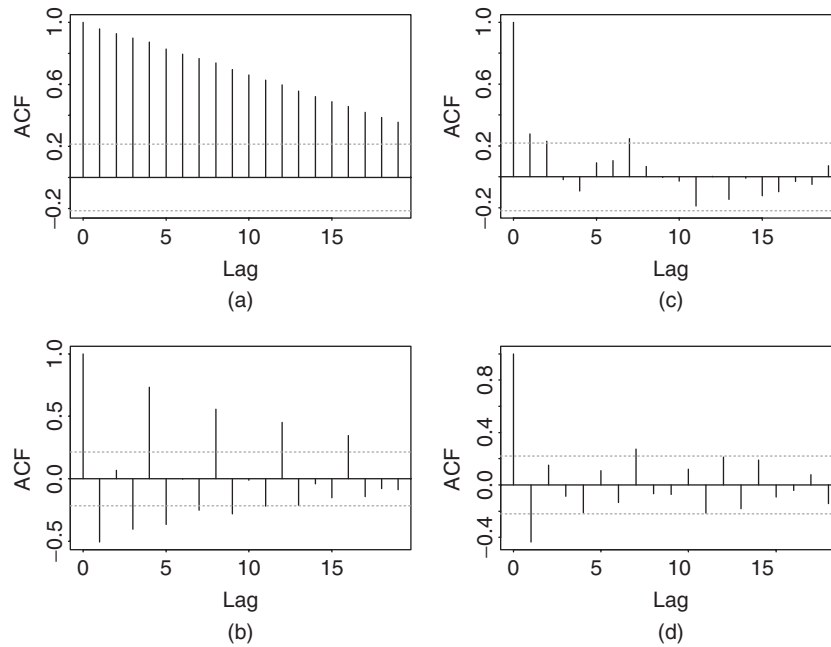


Figure 2.14 Sample ACF of log series of quarterly earnings per share of Johnson & Johnson from 1960 to 1980. (a) log earnings, (b) first differenced series, (c) seasonally differenced series, and (d) series with regular and seasonal differencing.

The operation $\Delta_4 = (1 - B^4)$ is called a *seasonal differencing*. In general, for a seasonal time series y_t with periodicity s , seasonal differencing means

$$\Delta_s y_t = y_t - y_{t-s} = (1 - B^s)y_t.$$

The conventional difference $\Delta y_t = y_t - y_{t-1} = (1 - B)y_t$ is referred to as the *regular differencing*. The lower right plot of Figure 2.14 shows the sample ACF of $\Delta_4 \Delta x_t$, which has a significant negative ACF at lag 1 and a marginal negative correlation at lag 4. For completeness, Figure 2.14 also gives the sample ACF of the seasonally differenced series $\Delta_4 x_t$.

2.8.2 Multiplicative Seasonal Models

The behavior of the sample ACF of $(1 - B^4)(1 - B)x_t$ in Figure 2.14 is common among seasonal time series. It led to the development of the following special seasonal time series model:

$$(1 - B^s)(1 - B)x_t = (1 - \theta B)(1 - \Theta B^s)a_t, \quad (2.41)$$

where s is the periodicity of the series, a_t is a white noise series, $|\theta| < 1$, and $|\Theta| < 1$. This model is referred to as the *airline model* in the literature; see Box, Jenkins, and Reinsel (1994, Chapter 9). It has been found to be widely applicable in modeling seasonal time series. The AR part of the model simply consists of the regular and seasonal differences, whereas the MA part involves two parameters. Focusing on the MA part (i.e., on the model),

$$w_t = (1 - \theta B)(1 - \Theta B^s)a_t = a_t - \theta a_{t-1} - \Theta a_{t-s} + \theta \Theta a_{t-s-1},$$

where $w_t = (1 - B^s)(1 - B)x_t$ and $s > 1$. It is easy to obtain that $E(w_t) = 0$ and

$$\begin{aligned} \text{Var}(w_t) &= (1 + \theta^2)(1 + \Theta^2)\sigma_a^2, \\ \text{Cov}(w_t, w_{t-1}) &= -\theta(1 + \Theta^2)\sigma_a^2, \\ \text{Cov}(w_t, w_{t-s+1}) &= \theta\Theta\sigma_a^2, \\ \text{Cov}(w_t, w_{t-s}) &= -\Theta(1 + \theta^2)\sigma_a^2, \\ \text{Cov}(w_t, w_{t-s-1}) &= \theta\Theta\sigma_a^2, \\ \text{Cov}(w_t, w_{t-\ell}) &= 0, \quad \text{for } \ell \neq 0, 1, s-1, s, s+1. \end{aligned}$$

Consequently, the ACF of the w_t series is given by

$$\rho_1 = \frac{-\theta}{1 + \theta^2}, \quad \rho_s = \frac{-\Theta}{1 + \Theta^2}, \quad \rho_{s-1} = \rho_{s+1} = \rho_1 \rho_s = \frac{\theta \Theta}{(1 + \theta^2)(1 + \Theta^2)},$$

and $\rho_\ell = 0$ for $\ell > 0$ and $\ell \neq 1, s-1, s, s+1$. For example, if w_t is a quarterly time series, then $s = 4$ and for $\ell > 0$, the ACF ρ_ℓ is nonzero at lags 1, 3, 4, and 5 only.

It is interesting to compare the prior ACF with those of the MA(1) model $y_t = (1 - \theta B)a_t$ and the MA(s) model $z_t = (1 - \Theta B^s)a_t$. The ACF of y_t and z_t series are

$$\begin{aligned} \rho_1(y) &= \frac{-\theta}{1 + \theta^2} & \text{and} & & \rho_\ell(y) &= 0, & \ell > 1, \\ \rho_s(z) &= \frac{-\Theta}{1 + \Theta^2} & \text{and} & & \rho_\ell(z) &= 0, & \ell > 0, \neq s. \end{aligned}$$

We see that (i) $\rho_1 = \rho_1(y)$, (ii) $\rho_s = \rho_s(z)$, and (iii) $\rho_{s-1} = \rho_{s+1} = \rho_1(y) \times \rho_s(z)$. Therefore, the ACF of w_t at lags $(s-1)$ and $(s+1)$ can be regarded as the *interaction* between lag-1 and lag- s serial dependence, and the model of w_t is called a *multiplicative* seasonal MA model. In practice, a multiplicative seasonal model says that the dynamics of the regular and seasonal components of the series are approximately orthogonal.

The model

$$w_t = (1 - \theta B - \Theta B^s)a_t, \quad (2.42)$$

where $|\theta| < 1$ and $|\Theta| < 1$, is a nonmultiplicative seasonal MA model. It is easy to see that for the model in Eq. (2.42), $\rho_{s+1} = 0$. A multiplicative model is more parsimonious than the corresponding nonmultiplicative model because both models use the same number of parameters, but the multiplicative model has more nonzero ACFs.

Example 2.3. In this example we apply the airline model to the log series of quarterly earnings per share of Johnson & Johnson from 1960 to 1980. Based on the exact-likelihood method, the fitted model is

$$(1 - B)(1 - B^4)x_t = (1 - 0.678B)(1 - 0.314B^4)a_t, \quad \hat{\sigma}_a = 0.089,$$

where standard errors of the two MA parameters are 0.080 and 0.101, respectively. The Ljung-Box statistics of the residuals show $Q(12) = 10.0$ with a p value of 0.44. The model appears to be adequate.

To illustrate the forecasting performance of the prior seasonal model, we reestimate the model using the first 76 observations and reserve the last 8 data points for forecasting evaluation. We compute 1-step- to 8-step-ahead forecasts and their standard errors of the fitted model at the forecast origin $h = 76$. An antilog transformation is taken to obtain forecasts of earnings per share using the relationship between normal and lognormal distributions given in Chapter 1. Figure 2.15 shows the forecast performance of the model, where the observed data are in solid line, point forecasts are shown by dots, and the dashed lines show 95% interval forecasts. The forecasts show a strong seasonal pattern and are close to the observed



Figure 2.15 Out-of-sample point and interval forecasts for quarterly earnings of Johnson & Johnson. Forecast origin is fourth quarter of 1978. In plot, solid line shows actual observations, dots represent point forecasts, and dashed lines show 95% interval forecasts.

data. Finally, for an alternative approach to modeling the quarterly earnings data, see Example 11.3.

When the seasonal pattern of a time series is stable over time (e.g., close to a deterministic function), dummy variables may be used to handle the seasonality. This approach is taken by some analysts. However, deterministic seasonality is a special case of the multiplicative seasonal model discussed before. Specifically, if $\Theta = 1$, then model (2.41) contains a deterministic seasonal component. Consequently, the same forecasts are obtained by using either dummy variables or a multiplicative seasonal model when the seasonal pattern is deterministic. Yet use of dummy variables can lead to inferior forecasts if the seasonal pattern is not deterministic. In practice, we recommend that the exact-likelihood method should be used to estimate a multiplicative seasonal model, especially when the sample size is small or when there is the possibility of having a deterministic seasonal component.

Example 2.4. To demonstrate deterministic seasonal behavior, consider the monthly simple returns of the CRSP Decile 1 Index from January 1970 to December 2008 for 468 observations. The series is shown in Figure 2.16(a), and the time plot does not show any clear pattern of seasonality. However, the sample ACF of the return series shown in Figure 2.16(b) contains significant lags at 12, 24, and 36 as well as lag 1. If seasonal ARMA models are entertained, a model in the form

$$(1 - \phi_1 B)(1 - \phi_{12} B^{12})R_t = (1 - \theta_{12} B^{12})a_t$$

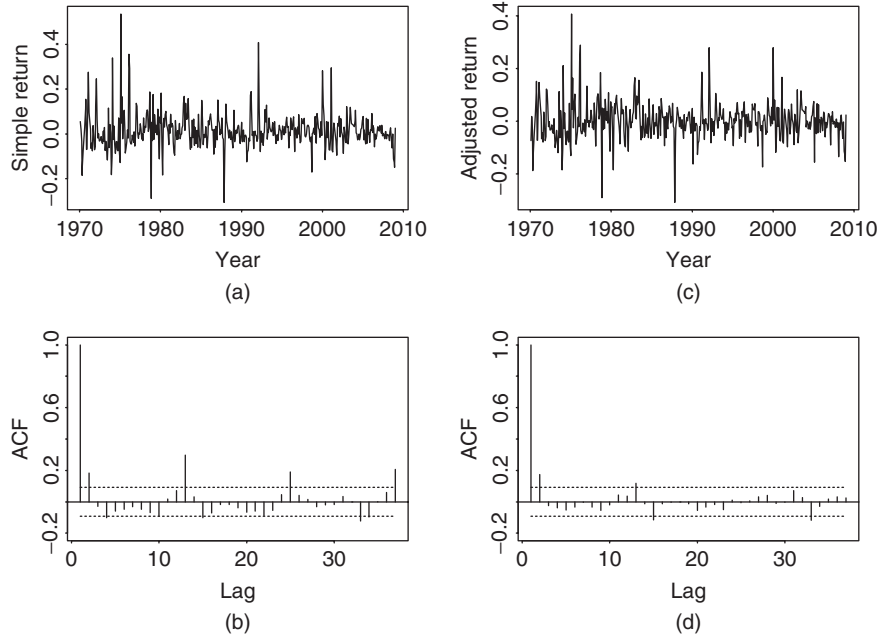


Figure 2.16 Monthly simple returns of CRSP Decile 1 index from January 1970 to December 2008: (a) time plot of the simple returns, (b) sample ACF of simple returns, (c) time plot of simple returns after adjusting for January effect, and (d) sample ACF of adjusted simple returns.

is identified, where R_t denotes the monthly simple return. Using the conditional-likelihood method, the fitted model is

$$(1 - 0.18B)(1 - 0.87B^{12})R_t = (1 - 0.74B^{12})a_t, \quad \tilde{\sigma}_a = 0.069.$$

See the attached SCA (Scientific Computing Associates) output below. The estimates of the seasonal AR and MA coefficients are of similar magnitude. If the exact-likelihood method is used, we have

$$(1 - 0.188B)(1 - 0.951B^{12})R_t = (1 - 0.997B^{12})a_t, \quad \tilde{\sigma}_a = 0.063.$$

The cancellation between seasonal AR and MA factors is clearly seen. This highlights the usefulness of using the exact-likelihood method and, the estimation result suggests that the seasonal behavior might be deterministic. To further confirm this assertion, we define the dummy variable for January, that is,

$$\text{Jan}_t = \begin{cases} 1 & \text{if } t \text{ is January,} \\ 0 & \text{otherwise,} \end{cases}$$

and employ the simple linear regression

$$R_t = \beta_0 + \beta_1 \text{Jan}_t + e_t.$$

The fitted model is $R_t = 0.0029 + 0.1253\text{Jan}_t + e_t$, where the standard errors of the estimates are 0.0033 and 0.0115, respectively. The right panel of Figure 2.16 shows the time plot and sample ACF of the residual series of the prior simple linear regression. From the sample ACF, serial correlations at lags 12, 24, and 36 largely disappear, suggesting that the seasonal pattern of the Decile 1 returns has been successfully removed by the January dummy variable. Consequently, the seasonal behavior in the monthly simple return of Decile 1 is mainly due to the *January effect*.

R Demonstration

The following output has been edited and % denotes explanation:

```
> da=read.table("m-deciles08.txt",header=T)
> d1=da[,2]
> jan=rep(c(1,rep(0,11)),39) % Create January dummy.
> m1=lm(d1 ~ jan)
> summary(m1)
Call:
lm(formula = d1 ~ jan)

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.002864   0.003333    0.859   0.391
jan           0.125251   0.011546   10.848 <2e-16 ***
---

Residual standard error: 0.06904 on 466 degrees of freedom
Multiple R-squared:  0.2016,    Adjusted R-squared:  0.1999

> m2=arima(d1,order=c(1,0,0),seasonal=list(order=c(1,0,1),
+ period=12))
> m2
Coefficients:
          ar1      sar1      sma1  intercept
          0.1769   0.9882  -0.9144          0.0118
s.e. 0.0456   0.0093   0.0335          0.0129

sigma^2 estimated as 0.004717: log likelihood=584.07,
      aic=-1158.14
> tsdiag(m2,gof=36) % plot not shown.

> m2=arima(d1,order=c(1,0,0),seasonal=list(order=c(1,0,1),
+ period=12),include.mean=F)
> m2
Call:
arima(x=d1,order=c(1,0,0),seasonal=list(order=c(1,0,1),
      period=12),include.mean = F)
```

Coefficients:

```

      ar1    sar1    sma1
      0.1787 0.9886 -0.9127 % Slightly differ from those of SCA.
s.e. 0.0456 0.0089 0.0335

```

```

sigma^2 estimated as 0.00472: log likelihood=583.68,
aic=-1159.36

```

SCA Demonstration

The following output has been edited:

```

input date,dec1,d2,d9,d10. file 'm-deciles08.txt'.
--
tsm m1. model (1)(12)dec1=(12)noise.
--
estim m1. hold resi(r1).      % Conditional MLE estimation

SUMMARY FOR UNIVARIATE TIME SERIES MODEL -- M1
-----
VAR  TYPE OF  ORIGINAL  DIFFERENCING
    VARIABLE OR CENTERED

DEC1 RANDOM    ORIGINAL    NONE
-----
PAR.  VAR.  NUM./  FACTOR  ORDER  CONS-  VALUE  STD  T
LABEL NAME  DENOM.          TRRAINT  ERROR  VALUE
1      D1    MA      1      12     NONE   .7388  .0488 15.14
2      D1    AR      1       1     NONE   .1765  .0447  3.95
3      D1    AR      2      12     NONE   .8698  .0295 29.49

EFFECTIVE NUMBER OF OBSERVATIONS . . . . . 455
R-SQUARE . . . . . 0.199
RESIDUAL STANDARD ERROR. . . . . 0.689906E-01
RESIDUAL STANDARD ERROR. . . . . 0.705662E-01
--
estim m1. method exact. hold resi(r1) % Exact MLE estimation

SUMMARY FOR UNIVARIATE TIME SERIES MODEL -- M1
-----
VAR.  TYPE OF  ORIGINAL  DIFFERENCING
    VAR.    OR CENTERED

DEC1 RANDOM    ORIGINAL    NONE
-----
PAR.  VARI.  NUM./  FACTOR  ORDER  CONS-  VALUE  STD  T
LABEL NAME  DENOM.          TRRAINT  ERROR  VALUE
1      D1    MA      1      12     NONE   .9968  .0150 66.31

```

2	D1	AR	1	1	NONE	.1884	.0448	4.21
3	D1	AR	2	12	NONE	.9505	.0070	135.46

EFFECTIVE NUMBER OF OBSERVATIONS	. .	455
R-SQUARE		0.328
RESIDUAL STANDARD ERROR		0.631807E-01

2.9 REGRESSION MODELS WITH TIME SERIES ERRORS

In many applications, the relationship between two time series is of major interest. An obvious example is the *market model* in finance that relates the excess return of an individual stock to that of a market index. The term structure of interest rates is another example in which the time evolution of the relationship between interest rates with different maturities is investigated. These examples lead naturally to the consideration of a linear regression in the form

$$y_t = \alpha + \beta x_t + e_t, \quad (2.43)$$

where y_t and x_t are two time series and e_t denotes the error term. The least-squares (LS) method is often used to estimate model (2.43). If $\{e_t\}$ is a white noise series, then the LS method produces consistent estimates. In practice, however, it is common to see that the error term e_t is serially correlated. In this case, we have a regression model with time series errors, and the LS estimates of α and β may not be consistent.

A regression model with time series errors is widely applicable in economics and finance, but it is one of the most commonly misused econometric models because the serial dependence in e_t is often overlooked. It pays to study the model carefully.

We introduce the model by considering the relationship between two U.S. weekly interest rate series:

1. r_{1t} : the 1-year Treasury constant maturity rate
2. r_{3t} : the 3-year Treasury constant maturity rate

Both series have 2467 observations from January 5, 1962, to April 10, 2009, and are measured in percentages. The series are obtained from the Federal Reserve Bank of St Louis. Strictly speaking, we should model the two interest series jointly using multivariate time series analysis in Chapter 8. However, for simplicity, we focus here on regression-type analysis and ignore the issue of simultaneity.

Figure 2.17 shows the time plots of the two interest rates with a solid line denoting the 1-year rate and a dashed line the 3-year rate. Figure 2.18(a) plots r_{1t} versus r_{3t} , indicating that, as expected, the two interest rates are highly correlated. A naive way to describe the relationship between the two interest rates is to use

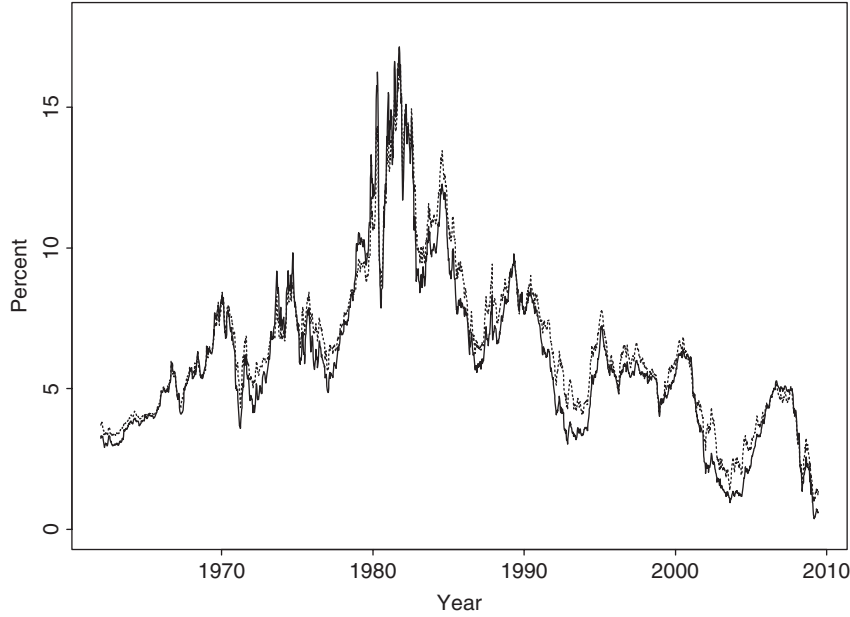


Figure 2.17 Time plots of U.S. weekly interest rates (in percentages) from January 5, 1962, to April 10, 2009. Solid line is Treasury 1-year constant maturity rate and dashed line Treasury 3-year constant maturity rate.

the simple model $r_{3t} = \alpha + \beta r_{1t} + e_t$. This results in a fitted model

$$r_{3t} = 0.832 + 0.930r_{1t} + e_t, \quad \hat{\sigma}_e = 0.523 \quad (2.44)$$

with $R^2 = 96.5\%$, where the standard errors of the two coefficients are 0.024 and 0.004, respectively. Model (2.44) confirms the high correlation between the two interest rates. However, the model is seriously inadequate, as shown by Figure 2.19, which gives the time plot and ACF of its residuals. In particular, the sample ACF of the residuals is highly significant and decays slowly, showing the pattern of a unit-root nonstationary time series. The behavior of the residuals suggests that marked differences exist between the two interest rates. Using the modern econometric terminology, if one assumes that the two interest rate series are unit-root nonstationary, then the behavior of the residuals of Eq. (2.44) indicates that the two interest rates are not *cointegrated*; see Chapter 8 for discussion of cointegration. In other words, the data fail to support the hypothesis that there exists a long-term equilibrium between the two interest rates. In some sense, this is not surprising because the pattern of “inverted yield curve” did occur during the data span. By inverted yield curve we mean the situation under which interest rates are inversely related to their time to maturities.

The unit-root behavior of both interest rates and the residuals of Eq. (2.44) leads to the consideration of the change series of interest rates. Let

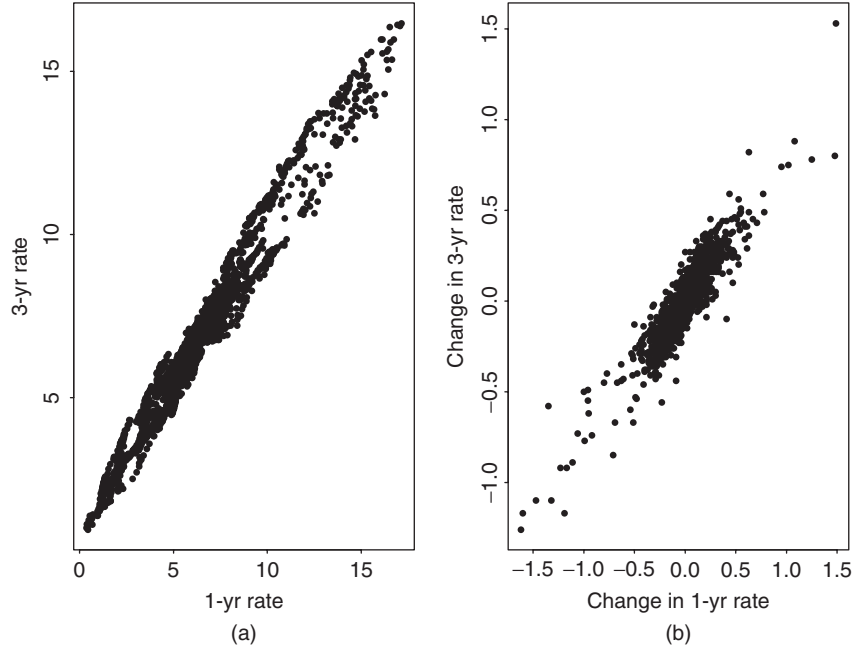


Figure 2.18 Scatterplots of U.S. weekly interest rates from January 5, 1962, to April 10, 2009: (a) 3-year rate vs. 1-year rate and (b) changes in 3-year rate vs. changes in 1-year rate.

1. $c_{1t} = r_{1t} - r_{1,t-1} = (1 - B)r_{1t}$ for $t \geq 2$: changes in the 1-year interest rate
2. $c_{3t} = r_{3t} - r_{3,t-1} = (1 - B)r_{3t}$ for $t \geq 2$: changes in the 3-year interest rate

and consider the linear regression $c_{3t} = \beta c_{1t} + e_t$. Figure 2.20 shows time plots of the two change series, whereas Figure 2.18(b) provides a scatterplot between them. The change series remain highly correlated with a fitted linear regression model given by

$$c_{3t} = 0.792c_{1t} + e_t, \quad \hat{\sigma}_e = 0.0690, \quad (2.45)$$

with $R^2 = 82.5\%$. The standard error of the coefficient is 0.0073. This model further confirms the strong linear dependence between interest rates. Figure 2.21 shows the time plot and sample ACF of the residuals of Eq. (2.45). Once again, the ACF shows some significant serial correlations in the residuals, but magnitudes of the correlations are much smaller. This weak serial dependence in the residuals can be modeled by using the simple time series models discussed in the previous sections, and we have a linear regression with time series errors.

The main objective of this section is to discuss a simple approach for building a linear regression model with time series errors. The approach is straightforward. We employ a simple time series model discussed in this chapter for the residual series

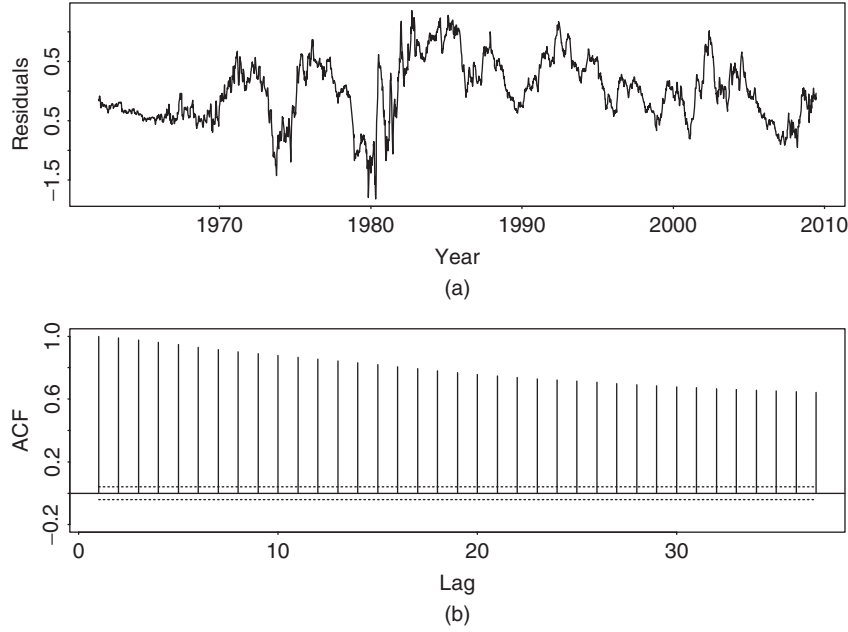


Figure 2.19 Residual series of linear regression (2.44) for two U.S. weekly interest rates: (a) time plot and (b) sample ACF.

and estimate the whole model jointly. For illustration, consider the simple linear regression in Eq. (2.45). Because residuals of the model are serially correlated, we shall identify a simple ARMA model for the residuals. From the sample ACF of the residuals shown in Figure 2.21, we specify an MA(1) model for the residuals and modify the linear regression model to

$$c_{3t} = \beta c_{1t} + e_t, \quad e_t = a_t - \theta_1 a_{t-1}, \quad (2.46)$$

where $\{a_t\}$ is assumed to be a white noise series. In other words, we simply use an MA(1) model, without the constant term, to capture the serial dependence in the error term of Eq. (2.45). The resulting model is a simple example of linear regression with time series errors. In practice, more elaborated time series models can be added to a linear regression equation to form a general regression model with time series errors.

Estimating a regression model with time series errors was not easy before the advent of modern computers. Special methods such as the Cochrane–Orcutt estimator have been proposed to handle the serial dependence in the residuals; see Greene (2003, p. 273). By now, the estimation is as easy as that of other time series models. If the time series model used is stationary and invertible, then one can estimate the model jointly via the maximum-likelihood method. This is the approach we take by using either the SCA or R package. R and S-Plus demonstrations are given

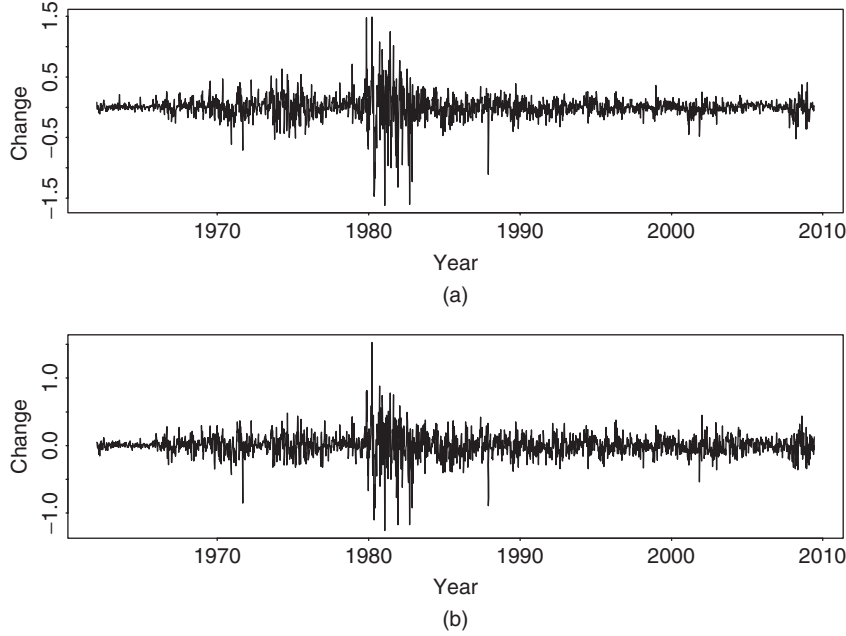


Figure 2.20 Time plots of change series of U.S. weekly interest rates from January 12, 1962, to April 10, 2009: (a) changes in Treasury 1-year constant maturity rate and (b) changes in Treasury 3-year constant maturity rate.

later. For the U.S. weekly interest rate data, the fitted version of model (2.46) is

$$c_{3t} = 0.794c_{1t} + e_t, \quad e_t = a_t + 0.1823a_{t-1}, \quad \hat{\sigma}_a = 0.0678, \quad (2.47)$$

with $R^2 = 83.1\%$. The standard errors of the parameters are 0.0075 and 0.0196, respectively. The model no longer has a significant lag-1 residual ACF, even though some minor residual serial correlations remain at lags 4, 6, and 7. The incremental improvement of adding additional MA parameters at lags 4, 6, and 7 to the residual equation is small and the result is not reported here.

Comparing the models in Eqs. (2.44), (2.45), and (2.47), we make the following observations. First, the high R^2 96.5% and coefficient 0.930 of model (2.44) are misleading because the residuals of the model show strong serial correlations. Second, for the change series, R^2 and the coefficient of c_{1t} of models (2.45) and (2.47) are close. In this particular instance, adding the MA(1) model to the change series only provides a marginal improvement. This is not surprising because the estimated MA coefficient is small numerically, even though it is statistically highly significant. Third, the analysis demonstrates that it is important to check residual serial dependence in linear regression analysis.

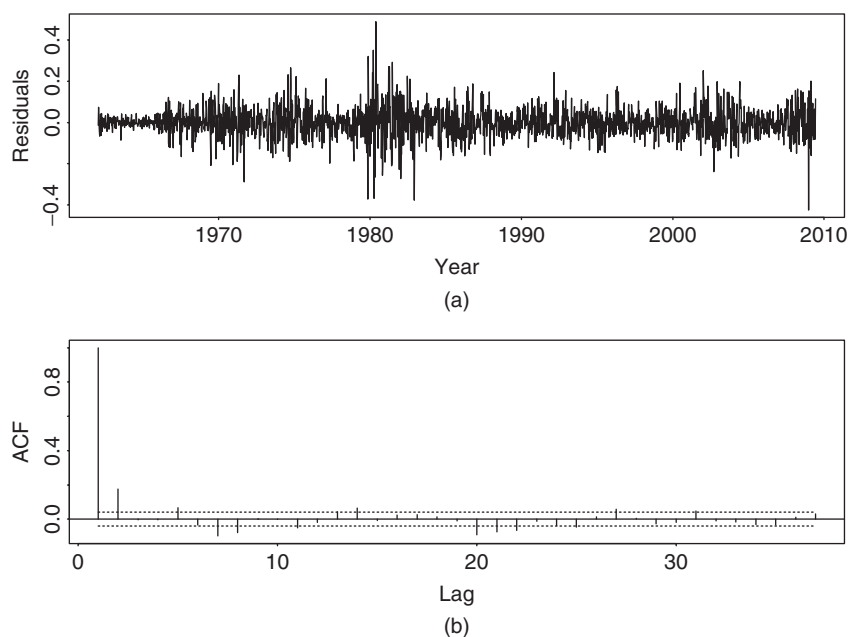


Figure 2.21 Residual series of linear regression (2.45) for two change series of U.S. weekly interest rates: (a) time plot and (b) sample ACF.

From Eq. (2.47), the model shows that the two weekly interest rate series are related as

$$r_{3t} = r_{3,t-1} + 0.794(r_{1t} - r_{1,t-1}) + a_t + 0.182a_{t-1}.$$

The interest rates are concurrently and serially correlated.

R Demonstration

The following output has been edited.

```
> r1=read.table("w-gs1yr.txt",header=T)[,4]
> r3=read.table("w-gs3yr.txt",header=T)[,4]
> m1=lm(r3 ~ r1)
> summary(m1)
Call:
lm(formula = r3 ~ r1)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept )  0.83214    0.02417   34.43  <2e-16 ***
r1             0.92955    0.00357  260.40  <2e-16 ***
```

```

---
Residual standard error: 0.5228 on 2465 degrees of freedom
Multiple R-squared: 0.9649,    Adjusted R-squared: 0.9649

> plot(m1$residuals,type='l')
> acf(m1$residuals,lag=36)
> c1=diff(r1)
> c3=diff(r3)
> m2=lm(c3 ~ -1+c1)
> summary(m2)
Call:
lm(formula = c3 ~ -1 + c1)
Coefficients:
      Estimate Std. Error t value Pr(>|t|)
c1  0.791935    0.007337   107.9  <2e-16 ***
---
Residual standard error: 0.06896 on 2465 degrees of freedom
Multiple R-squared: 0.8253,    Adjusted R-squared: 0.8253

> acf(m2$residuals,lag=36)

> m3=arima(c3,order=c(0,0,1),xreg=c1,include.mean=F)
> m3
Call:
arima(x = c3, order = c(0, 0, 1), xreg = c1, include.mean = F)
Coefficients:
          ma1          c1
          0.1823    0.7936
s.e.    0.0196    0.0075

sigma^2 estimated as 0.0046: log likelihood=3136.62,
aic=-6267.23
>
> rsq=(sum(c3^2)-sum(m3$residuals^2))/sum(c3^2)
> rsq
[1] 0.8310077

```

Summary

We outline a general procedure for analyzing linear regression models with time series errors:

1. Fit the linear regression model and check serial correlations of the residuals.
2. If the residual series is unit-root nonstationary, take the first difference of both the dependent and explanatory variables. Go to step 1. If the residual series appears to be stationary, identify an ARMA model for the residuals and modify the linear regression model accordingly.

3. Perform a joint estimation via the maximum-likelihood method and check the fitted model for further improvement.

To check the serial correlations of residuals, we recommend that the Ljung–Box statistics be used instead of the Durbin–Watson (DW) statistic because the latter only considers the lag-1 serial correlation. There are cases in which serial dependence in residuals appears at higher order lags. This is particularly so when the time series involved exhibits some seasonal behavior.

Remark. For a residual series e_t with T observations, the Durbin–Watson statistic is

$$DW = \frac{\sum_{t=2}^T (e_t - e_{t-1})^2}{\sum_{t=1}^T e_t^2}.$$

Straightforward calculation shows that $DW \approx 2(1 - \hat{\rho}_1)$, where $\hat{\rho}_1$ is the lag-1 ACF of $\{e_t\}$. $\square$

In S-Plus, regression models with time series errors can be analyzed by the command `OLS` (ordinary least squares) if the residuals assume an AR model. Also, to identify a lagged variable, the command is `tslag`, for example, `y = tslag(r, 1)`. For the interest rate series, the relevant commands follow, where % denotes explanation of the command:

```
> r1t=read.table("w-gs1yr.txt",header=T)[,4] %load data
> r3t=read.table("w-gs3yr.txt",header=T)[,4]
> fit=OLS(r3t~r1t) % fit the first regression
> summary(fit)
> c3t=diff(r3t) % take difference
> c1t=diff(r1t)
> fit1=OLS(c3t~c1t) % fit second regression
> summary(fit1)
> fit2=OLS(c3t~c1t+tslag(c3t,1)+tslag(c1t,1), na.rm=T)
> summary(fit2)
```

See the output in the next section for more information.

2.10 CONSISTENT COVARIANCE MATRIX ESTIMATION

Consider again the regression model in Eq. (2.43). There may exist situations in which the error term e_t has serial correlations and/or conditional heteroscedasticity, but the main objective of the analysis is to make inference concerning the regression coefficients α and β . See Chapter 3 for discussion of conditional heteroscedasticity. In situations under which the OLS estimates of the coefficients remain consistent, methods are available to provide consistent estimate of the covariance matrix of

the coefficient estimates. Two such methods are widely used. The first method is called the heteroscedasticity consistent (HC) estimator; see Eicker (1967) and White (1980). The second method is called the heteroscedasticity and autocorrelation consistent (HAC) estimator; see Newey and West (1987).

For ease in discussion, we shall rewrite the regression model as

$$y_t = \mathbf{x}_t' \boldsymbol{\beta} + e_t, \quad t = 1, \dots, T, \quad (2.48)$$

where y_t is the dependent variable, $\mathbf{x}_t = (x_{1t}, \dots, x_{kt})'$ is a k -dimensional vector of explanatory variables including constant, and $\boldsymbol{\beta} = (\beta_1, \dots, \beta_k)'$ is the parameter vector. Here $\mathbf{c}'$ denotes the transpose of the vector $\mathbf{c}$. The LS estimate of $\boldsymbol{\beta}$ and the associate covariance matrix are

$$\hat{\boldsymbol{\beta}} = \left[\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right]^{-1} \sum_{t=1}^T \mathbf{x}_t y_t, \quad \text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma_e^2 \left[\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right]^{-1},$$

where σ_e^2 is the variance of e_t and is estimated by the variance of the residuals of the regression. In the presence of serial correlations or conditional heteroscedasticity, the prior covariance matrix estimator is inconsistent, often resulting in inflating the t ratios of $\hat{\boldsymbol{\beta}}$.

The estimator of White (1980) is

$$\text{Cov}(\hat{\boldsymbol{\beta}})_{\text{HC}} = \left[\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right]^{-1} \left[\sum_{t=1}^T \hat{e}_t^2 \mathbf{x}_t \mathbf{x}_t' \right] \left[\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right]^{-1}, \quad (2.49)$$

where $\hat{e}_t = y_t - \mathbf{x}_t' \hat{\boldsymbol{\beta}}$ is the residual at time t . The estimator of Newey and West (1987) is

$$\text{Cov}(\hat{\boldsymbol{\beta}})_{\text{HAC}} = \left[\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right]^{-1} \hat{\mathbf{C}}_{\text{HAC}} \left[\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right]^{-1}, \quad (2.50)$$

where

$$\hat{\mathbf{C}}_{\text{HAC}} = \sum_{t=1}^T \hat{e}_t^2 \mathbf{x}_t \mathbf{x}_t' + \sum_{j=1}^{\ell} w_j \sum_{t=j+1}^T (\mathbf{x}_t \hat{e}_t \mathbf{x}_{t-j}' + \mathbf{x}_{t-j} \hat{e}_{t-j} \mathbf{x}_t'),$$

where ℓ is a truncation parameter and w_j is a weight function such as the Bartlett weight function defined by

$$w_j = 1 - \frac{j}{\ell + 1}.$$

Other weight functions can also be used. Newey and West (1987) suggest choosing ℓ to be the integer part of $4(T/100)^{2/9}$. This estimator essentially uses a nonparametric method to estimate the covariance matrix of $\{\sum_{t=1}^T \hat{e}_t x_t\}$.

For illustration, we employ the first differenced interest rate series in Eq. (2.45). The t ratio of the coefficient of c_{1t} is 107.91 if both serial correlation and heteroscedasticity in the residuals are ignored, it becomes 48.44 when the HC estimator is used, and it reduces to 39.92 when the HAC estimator is used. The S-Plus demonstration below also uses a regression that includes lagged values $c_{1,t-1}$ and $c_{3,t-1}$ as regressors to take care of serial correlations in the residuals. One can also apply the HC or HAC estimator to the fitted model to refine the t ratios of the coefficient estimates.

S-Plus Demonstration

The following output has been edited and % denotes explanation:

```
> module(finmetrics)
> r1=read.table("w-gs1yr.txt",header=T)[,4] % Load data
> r3=read.table("w-gs3yr.txt",header=T)[,4]
> c1=diff(r1) % Take 1st difference
> c3=diff(r3)

> reg.fit=OLS(c3~c1) % Fit a simple linear regression.
> summary(reg.fit)
Call:
OLS(formula = c3 ~ c1)

Residuals:
    Min       1Q   Median       3Q      Max
-0.4246 -0.0358 -0.0012  0.0347  0.4892

Coefficients:
              Value Std. Error  t value Pr(>|t|)
(Intercept) -0.0001    0.0014  -0.0757   0.9397
          c1   0.7919    0.0073  107.9063   0.0000

Regression Diagnostics:
      R-Squared 0.8253
Adjusted R-Squared 0.8253
Durbin-Watson Stat 1.6456

Residual Diagnostics:
              Stat      P-Value
Jarque-Bera 1644.6146   0.0000
Ljung-Box   230.0477   0.0000

Residual standard error: 0.06897 on 2464 degrees of freedom

> summary(reg.fit,correction="white") % Use HC the estimator
```

Coefficients:

	Value	Std. Error	t value	Pr(> t)
(Intercept)	-0.0001	0.0014	-0.0757	0.9396
c1	0.7919	0.0163	48.4405	0.0000

```
> summary(reg.fit,correction="nw") % Use the HAC estimator
```

Coefficients:

	Value	Std. Error	t value	Pr(> t)
(Intercept)	-0.0001	0.0016	-0.0678	0.9459
c1	0.7919	0.0198	39.9223	0.0000

```
% Below, fit a regression model with time series errors.
```

```
> reg.ts=OLS(c3~c1+tslag(c3,1)+tslag(c1,1),na.rm=T)
```

```
> summary(reg.ts)
```

Call:

```
OLS(formula = c3 ~ c1 + tslag(c3, 1)+tslag(c1, 1), na.rm = T)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.4481	-0.0355	-0.0008	0.0341	0.4582

Coefficients:

	Value	Std. Error	t value	Pr(> t)
(Intercept)	-0.0001	0.0014	-0.0636	0.9493
c1	0.7971	0.0077	103.6320	0.0000
tslag(c3, 1)	0.1766	0.0198	8.9057	0.0000
tslag(c1, 1)	-0.1580	0.0174	-9.0583	0.0000

Regression Diagnostics:

R-Squared 0.8312

Adjusted R-Squared 0.8310

Durbin-Watson Stat 1.9865

Residual Diagnostics:

	Stat	P-Value
Jarque-Bera	1620.5090	0.0000
Ljung-Box	131.6048	0.0000

Residual standard error: 0.06785 on 2461 degrees of freedom

Let $\hat{\beta}_j$ be the j th element of $\hat{\beta}$. When $k > 1$, the HC variance of $\hat{\beta}_j$ in Eq. (2.49) can be obtained by using an auxiliary regression. Let $\mathbf{x}_{-j,t}$ be the $(k-1)$ -dimensional vector obtained by removing the element x_{jt} from $\mathbf{x}_t$. Consider the auxiliary regression

$$x_{jt} = \mathbf{x}'_{-j,t} \boldsymbol{\gamma} + v_t, \quad t = 1, \dots, T. \quad (2.51)$$

Let $\hat{v}_t$ be the least-squares residual of this auxiliary regression. It can be shown that

$$\text{Var}(\hat{\beta}_j)_{\text{HC}} = \frac{\sum_{t=1}^T \hat{e}_t^2 \hat{v}_t^2}{\left(\sum_{t=1}^T \hat{v}_t^2\right)^2},$$

where $\hat{e}_t$ is the residual of original regression in Eq. (2.48). The auxiliary regression is simply a step taken to achieve orthogonality between $\hat{v}_t$ and the rest of the regressors so that the formula in Eq. (2.49) can be simplified.

2.11 LONG-MEMORY MODELS

We have discussed that for a stationary time series the ACF decays exponentially to zero as lag increases. Yet for a unit-root nonstationary time series, it can be shown that the sample ACF converges to 1 for all fixed lags as the sample size increases; see Chan and Wei (1988) and Tiao and Tsay (1983). There exist some time series whose ACF decays slowly to zero at a polynomial rate as the lag increases. These processes are referred to as long-memory time series. One such example is the fractionally differenced process defined by

$$(1 - B)^d x_t = a_t, \quad -0.5 < d < 0.5, \quad (2.52)$$

where $\{a_t\}$ is a white noise series. Properties of model (2.52) have been widely studied in the literature (e.g., Hosking, 1981). We now summarize some of these properties:

1. If $d < 0.5$, then x_t is a weakly stationary process and has the infinite MA representation

$$\begin{aligned} x_t &= a_t + \sum_{i=1}^{\infty} \psi_i a_{t-i} \quad \text{with} \quad \psi_k \\ &= \frac{d(1+d) \cdots (k-1+d)}{k!} = \frac{(k+d-1)!}{k!(d-1)!}. \end{aligned}$$

2. If $d > -0.5$, then x_t is invertible and has the infinite AR representation

$$\begin{aligned} x_t &= \sum_{i=1}^{\infty} \pi_i x_{t-i} + a_t \quad \text{with} \quad \pi_k \\ &= \frac{-d(1-d) \cdots (k-1-d)}{k!} = \frac{(k-d-1)!}{k!(-d-1)!}. \end{aligned}$$

3. For $-0.5 < d < 0.5$, the ACF of x_t is

$$\rho_k = \frac{d(1+d) \cdots (k-1+d)}{(1-d)(2-d) \cdots (k-d)}, \quad k = 1, 2, \dots$$

In particular, $\rho_1 = d/(1-d)$ and

$$\rho_k \approx \frac{(-d)!}{(d-1)!} k^{2d-1} \quad \text{as } k \rightarrow \infty.$$

4. For $-0.5 < d < 0.5$, the PACF of x_t is $\phi_{k,k} = d/(k-d)$ for $k = 1, 2, \dots$
 5. For $-0.5 < d < 0.5$, the spectral density function $f(\omega)$ of x_t , which is the Fourier transform of the ACF of x_t , satisfies

$$f(\omega) \sim \omega^{-2d}, \quad \text{as } \omega \rightarrow 0, \quad (2.53)$$

where $\omega \in [0, 2\pi]$ denotes the frequency.

Of particular interest here is the behavior of ACF of x_t when $d < 0.5$. The property says that $\rho_k \sim k^{2d-1}$, which decays at a polynomial, instead of exponential, rate. For this reason, such an x_t process is called a long-memory time series. A special characteristic of the spectral density function in Eq. (2.53) is that the spectrum diverges to infinity as $\omega \rightarrow 0$. However, the spectral density function of a stationary ARMA process is bounded for all $\omega \in [0, 2\pi]$.

Earlier we used the binomial theorem for noninteger powers

$$(1-B)^d = \sum_{k=0}^{\infty} (-1)^k \binom{d}{k} B^k, \quad \binom{d}{k} = \frac{d(d-1) \cdots (d-k+1)}{k!}.$$

If the fractionally differenced series $(1-B)^d x_t$ follows an ARMA(p, q) model, then x_t is called an ARFIMA(p, d, q) process, which is a generalized ARIMA model by allowing for noninteger d .

In practice, if the sample ACF of a time series is not large in magnitude, but decays slowly, then the series may have long memory. As an illustration, Figure 2.22 shows the sample ACFs of the absolute series of daily simple returns for the CRSP value- and equal-weighted indexes from January 2, 1970, to December 31, 2008. The ACFs are relatively small in magnitude but decay very slowly; they appear to be significant at the 5% level even after 300 lags. For more information about the behavior of sample ACF of absolute return series, see Ding, Granger, and Engle (1993). For the pure fractionally differenced model in Eq. (2.52), one can estimate d using either a maximum-likelihood method or a regression method with logged periodogram at the lower frequencies. Finally, long-memory models have attracted some attention in the finance literature in part because of the work on fractional Brownian motion in the continuous-time models.

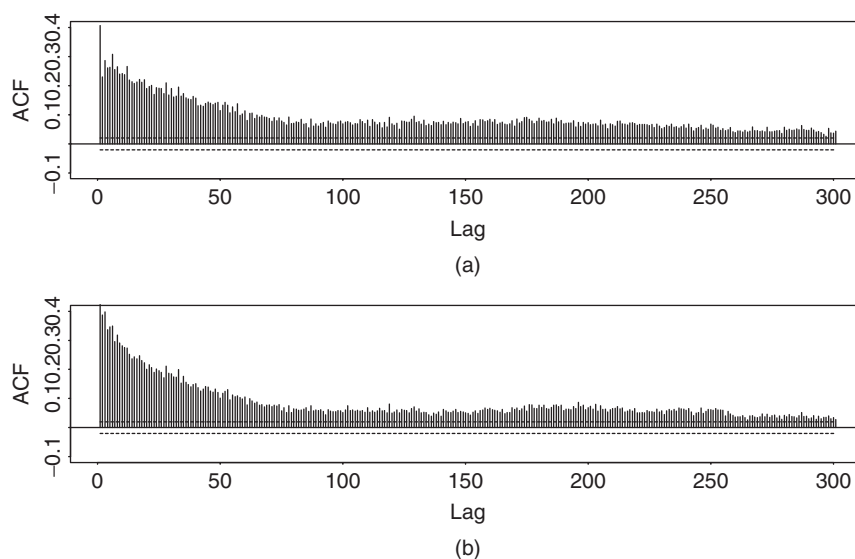


Figure 2.22 Sample autocorrelation function of absolute series of daily simple returns for CRSP value- and equal-weighted indexes: (a) value-weighted index return and (b) equal-weighted index return. Sample period is from January 2, 1970, to December 31, 2008.

APPENDIX: SOME SCA COMMANDS

In this appendix, we give the SCA commands used in Section 2.9. The 1-year maturity interest rates are in the file `w-gs1yr.txt` and the 3-year rates are in the file `w-gs3yr.txt`.

```
-- load the data into SCA, denote the data by rate1 and rate3.
input year,mom,day,rate1. file 'w-gs1yr.txt'
--
input year,mon,day,rate3. file 'w-gs3yr.txt'
-- specify a simple linear regression model.
tsm m1. model rate3=b0+(b1)rate1+noise.
-- estimate the specified model and store residual in r1.
estim m1. hold resi(r1).
-- compute 10 lags of residual acf.
acf r1. maxl 10.
-- difference the two series, denote the new series by c1t
   and c3t
diff old rate1,rate3. new c1t, c3t. compress.
-- specify a linear regression model for the differenced data
tsm m2. model c3t=h0+(h1)c1t+noise.
-- estimation
estim m2. hold resi(r2).
-- compute residual acf.
```

```

acf r2. maxl 10.
-- specify a regression model with time series errors.
tsm m3. model c3t=g0+(g1)c1t+(1)noise.
-- estimate the model using the exact likelihood method.
estim m3. method exact. hold resi(r3).
-- compute residual acf.
acf r3. maxl 10.
-- refine the model to include more MA lags.
tsm m4. model c3t=g0+(g1)c1t+(1,4,6,7)noise.
-- estimation
estim m4. method exact. hold resi(r4).
-- compute residual acf.
acf r4. maxl 10.
-- exit SCA
stop

```

EXERCISES

If not specifically specified, use 5% significance level to draw conclusions in the exercises.

- 2.1. Suppose that the simple return of a monthly bond index follows the MA(1) model

$$R_t = a_t + 0.2a_{t-1}, \quad \sigma_a = 0.025.$$

Assume that $a_{100} = 0.01$. Compute the 1-step- and 2-step-ahead forecasts of the return at the forecast origin $t = 100$. What are the standard deviations of the associated forecast errors? Also compute the lag-1 and lag-2 autocorrelations of the return series.

- 2.2. Suppose that the daily log return of a security follows the model

$$r_t = 0.01 + 0.2r_{t-2} + a_t,$$

where $\{a_t\}$ is a Gaussian white noise series with mean zero and variance 0.02. What are the mean and variance of the return series r_t ? Compute the lag-1 and lag-2 autocorrelations of r_t . Assume that $r_{100} = -0.01$, and $r_{99} = 0.02$. Compute the 1- and 2-step-ahead forecasts of the return series at the forecast origin $t = 100$. What are the associated standard deviations of the forecast errors?

- 2.3. Consider the monthly U.S. unemployment rate from January 1948 to March 2009 in the file `m-unrate.txt`. The data are seasonally adjusted and obtained from the Federal Reserve Bank of St Louis. Build a time series model for the series and use the model to forecast the unemployment rate for the April, May, June, and July of 2009. In addition, does the fitted model imply the existence of business cycles? Why? (Note that there are more than one model fits the data well. You only need an adequate model.)

- 2.4. Consider the monthly simple returns of the Decile 1, Decile 2, Decile 9, and Decile 10 of NYSE/AMEX/NASDAQ based on market capitalization. The data span is from January 1970 to December 2008, and the data are obtained from CRSP.
- (a) For the return series of Decile 2 and Decile 10, test the null hypothesis that the first 12 lags of autocorrelations are zero at the 5% level. Draw your conclusion.
 - (b) Build an ARMA model for the return series of Decile 2. Perform model checking and write down the fitted model.
 - (c) Use the fitted ARMA model to produce 1- to 12-step-ahead forecasts of the series and the associated standard errors of forecasts.
- 2.5. Consider the daily simple returns of IBM stock from 1970 to 2008 in the file `d-ibm3dx7008.txt`. Compute the first 100 lags of ACF of the absolute series of daily simple returns of IBM stock. Is there evidence of long-range dependence? Why?
- 2.6. Consider the demand of electricity of a manufacturing sector in the United States. The data are logged, denote the demand of a fixed day of each month, and are in `power6.txt`. Build a time series model for the series and use the fitted model to produce 1- to 24-step-ahead forecasts.
- 2.7. Consider the daily simple returns of IBM stock, CRSP value-weighted index, CRSP equal-weighted index, and the S&P composite index from January 1980 to December 2008. The index returns include dividend distributions. The data file is `d-ibm3dxwkdays8008.txt`, which has 12 columns. The columns are (year, month, day, IBM, VW, EW, SP, M, T, W, H, F), where M, T, W, R, and F denotes indicator variables for Monday to Friday, respectively. Use a regression model to study the effects of trading days on the equal-weighted index returns. What is the fitted model? Are the weekday effects significant in the returns at the 5% level? Use the HAC estimator of the covariance matrix to obtain the t ratio of regression estimates. Does the HAC estimator change the conclusion of weekday effects? Are there serial correlations in the regression residuals? If yes, build a regression model with time series error to study weekday effects.
- 2.8. Consider the data set of the previous question, but focus on the daily simple returns of the S&P composite index. Perform the necessary data analysis and statistical tests using the 5% significance level to answer the following questions:
- (a) Is there any weekday effect on the daily simple returns of the S&P composite index? You may employ a linear regression model to answer this question. Estimate the model, check its validity, and test the hypothesis that there is no Friday effect. Draw your conclusion.
 - (b) Check the residual serial correlations using $Q(12)$ statistic. Are there any significant serial correlations in the residuals? If yes, build a regression model with time series errors for the data.

- 2.9. Now consider similar questions of the previous exercise for the IBM stock returns.
- Is there any weekday effect on the daily simple returns of IBM stock? Estimate your model and test the hypothesis that there is no Friday effect. Draw your conclusion.
 - Are there serial correlations in the residuals? Use $Q(12)$ to perform the test. Draw your conclusion.
 - Refine the above model by using the technique of regression model with time series errors. Is there a significant weekday effect based on the refined model?
- 2.10. Consider the weekly yields of Moody's Aaa and Baa seasoned bonds from January 5, 1962, to April 10, 2009. The data are obtained from the Federal Reserve Bank of St Louis. Weekly yields are averages of daily yields. Obtain the summary statistics (sample mean, standard deviation, skewness, excess kurtosis, minimum, and maximum) of the two yield series. Are the bond yields skewed? Do they have heavy tails? Answer the questions using 5% significance level.
- 2.11. Consider the monthly Aaa bond yields of the prior problem. Build a time series model for the series.
- 2.12. Again, consider the two bond yield series, that is, Aaa and Baa. What is the relationship between the two series? To answer this question, build a time series model using yields of Aaa bonds as the dependent variable and yields of Baa bonds as independent variable.
- 2.13. Consider the monthly log returns of CRSP equal-weighted index from January 1962 to December 1999 for 456 observations. You may obtain the data from CRSP directly or from the file `m-ew6299.txt` on the Web.
- Build an AR model for the series and check the fitted model.
 - Build an MA model for the series and check the fitted model.
 - Compute 1- and 2-step-ahead forecasts of the AR and MA models built in the previous two questions.
 - Compare the fitted AR and MA models.
- 2.14. This problem is concerned with the dynamic relationship between the spot and futures prices of the S&P 500 index. The data file `sp5may.dat` has three columns: $\log(\text{futures price})$, $\log(\text{spot price})$, and $\text{cost-of-carry} (\times 100)$. The data were obtained from the Chicago Mercantile Exchange for the S&P 500 stock index in May 1993 and its June futures contract. The time interval is 1 minute (intraday). Several authors used the data to study index futures arbitrage. Here we focus on the first two columns. Let f_t and s_t be the log prices of futures and spot, respectively. Consider $y_t = f_t - f_{t-1}$ and $x_t = s_t - s_{t-1}$. Build a regression model with time series errors between $\{y_t\}$ and $\{x_t\}$, with y_t being the dependent variable.

- 2.15. The quarterly gross domestic product implicit price deflator is often used as a measure of inflation. The file `q-gdpdef.txt` contains the data for the United States from the first quarter of 1947 to the last quarter of 2008. Data format is year, month, day, and deflator. The data are seasonally adjusted and equal to 100 for year 2000. Build an ARIMA model for the series and check the validity of the fitted model. Use the fitted model to predict the inflation for each quarter of 2009. The data are obtained from the Federal Reserve Bank of St Louis.

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Conditional Heteroscedastic Models

The objective of this chapter is to study some methods and econometric models available in the literature for modeling the volatility of an asset return. The models are referred to as conditional heteroscedastic models.

Volatility is an important factor in options trading. Here volatility means the conditional standard deviation of the underlying asset return. Consider, for example, the price of a European *call option*, which is a contract giving its holder the right, but not the obligation, to buy a fixed number of shares of a specified common stock at a fixed price on a given date. The fixed price is called the *strike price* and is commonly denoted by K . The given date is called the expiration date. The important time duration here is the time to expiration (measured in years), and we denote it by ℓ . The well-known Black–Scholes option pricing formula states that the price of such a call option is

$$c_t = P_t \Phi(x) - K e^{-r\ell} \Phi(x - \sigma_t \sqrt{\ell}), \quad \text{and} \quad x = \frac{\ln(P_t/K) + r\ell}{\sigma_t \sqrt{\ell}} + \frac{1}{2} \sigma_t \sqrt{\ell}, \quad (3.1)$$

where P_t is the current price of the underlying stock, r is the continuously compounded risk-free interest rate, σ_t is the annualized conditional standard deviation of the log return of the specified stock, and $\Phi(x)$ is the cumulative distribution function of the standard normal random variable evaluated at x . A derivation of the formula is given in Chapter 6. The formula has several nice interpretations, but it suffices to say here that the conditional standard deviation of the log return of the underlying stock plays an important role. This volatility evolves over time. If the holder can exercise her right any time on or before the expiration date, then the option is called an *American call option*.

Volatility has many other financial applications. As discussed in Chapter 7, volatility modeling provides a simple approach to calculating value at risk of a financial position in risk management. It plays an important role in asset allocation

under the mean-variance framework. Furthermore, modeling the volatility of a time series can improve the efficiency in parameter estimation and the accuracy in interval forecast. Finally, the volatility index of a market has recently become a financial instrument. The VIX volatility index compiled by the Chicago Board of Option Exchange (CBOE) started to trade in futures on March 26, 2004.

The univariate volatility models discussed in this chapter include the autoregressive conditional heteroscedastic (ARCH) model of Engle (1982), the generalized ARCH (GARCH) model of Bollerslev (1986), the exponential GARCH (EGARCH) model of Nelson (1991), the threshold GARCH (TGARCH) model of Glosten, Jagannathan, and Runkle (1993) and Zakoian (1994), the conditional heteroscedastic autoregressive moving-average (CHARMA) model of Tsay (1987), the random coefficient autoregressive (RCA) model of Nicholls and Quinn (1982), and the stochastic volatility (SV) models of Melino and Turnbull (1990), Taylor (1994), Harvey, Ruiz, and Shephard (1994), and Jacquier, Polson, and Rossi (1994). We also discuss advantages and weaknesses of each volatility model and show some applications of the models. Multivariate volatility models, including those with time-varying correlations, are discussed in Chapter 10. The chapter also discusses some alternative approaches to volatility modeling in Section 3.15, including use of daily high and low prices of an asset.

3.1 CHARACTERISTICS OF VOLATILITY

A special feature of stock volatility is that it is not directly observable. For example, consider the daily log returns of IBM stock. The daily volatility is not directly observable from the return data because there is only one observation in a trading day. If intraday data of the stock, such as 10-minute returns, are available, then one can estimate the daily volatility. See Section 3.15. The accuracy of such an estimate deserves a careful study, however. For example, stock volatility consists of intraday volatility and overnight volatility with the latter denoting variation between trading days. The high-frequency intraday returns contain only very limited information about the overnight volatility. The unobservability of volatility makes it difficult to evaluate the forecasting performance of conditional heteroscedastic models. We discuss this issue later.

In options markets, if one accepts the idea that the prices are governed by an econometric model such as the Black–Scholes formula, then one can use the price to obtain the “implied” volatility. Yet this approach is often criticized for using a specific model, which is based on some assumptions that might not hold in practice. For instance, from the observed prices of a European call option, one can use the Black–Scholes formula in Eq. (3.1) to deduce the conditional standard deviation σ_t . The resulting value of σ_t is called the *implied volatility* of the underlying stock. However, this implied volatility is derived under the assumption that the price of the underlying asset follows a geometric Brownian motion. It might be different from the actual volatility. Experience shows that implied volatility of an asset return tends to be larger than that obtained by using a GARCH type of volatility model.

This might be due to the risk premium for volatility or to the way daily returns are calculated. The VIX of CBOE is an implied volatility.

Although volatility is not directly observable, it has some characteristics that are commonly seen in asset returns. First, there exist volatility clusters (i.e., volatility may be high for certain time periods and low for other periods). Second, volatility evolves over time in a continuous manner—that is, volatility jumps are rare. Third, volatility does not diverge to infinity—that is, volatility varies within some fixed range. Statistically speaking, this means that volatility is often stationary. Fourth, volatility seems to react differently to a big price increase or a big price drop, referred to as the *leverage* effect. These properties play an important role in the development of volatility models. Some volatility models were proposed specifically to correct the weaknesses of the existing ones for their inability to capture the characteristics mentioned earlier. For example, the EGARCH model was developed to capture the asymmetry in volatility induced by big “positive” and “negative” asset returns.

3.2 STRUCTURE OF A MODEL

Let r_t be the log return of an asset at time index t . The basic idea behind volatility study is that the series $\{r_t\}$ is either serially uncorrelated or with minor lower order serial correlations, but it is a dependent series. For illustration, consider the monthly log stock returns of Intel Corporation from January 1973 to December 2008 shown in Figure 3.1. Figure 3.2(a) shows the sample ACF of the log return series, which suggests no significant serial correlations except for a minor one at lag 7. Figure 3.2(c) shows the sample ACF of the absolute log returns (i.e., $|r_t|$), whereas Figure 3.2(b) shows the sample ACF of the squared log returns r_t^2 . These two plots clearly suggest that the monthly log returns are not serially independent. Combining the three plots, it seems that the log returns are indeed serially uncorrelated but dependent. Volatility models attempt to capture such dependence in the return series.

To put the volatility models in proper perspective, it is informative to consider the conditional mean and variance of r_t given F_{t-1} ; that is,

$$\mu_t = E(r_t|F_{t-1}), \quad \sigma_t^2 = \text{Var}(r_t|F_{t-1}) = E[(r_t - \mu_t)^2|F_{t-1}], \quad (3.2)$$

where F_{t-1} denotes the information set available at time $t - 1$. Typically, F_{t-1} consists of all linear functions of the past returns. As shown by the empirical examples of Chapter 2 and Figure 3.2, serial dependence of a stock return series r_t is weak if it exists at all. Therefore, the equation for μ_t in (3.2) should be simple, and we assume that r_t follows a simple time series model such as a stationary ARMA(p, q) model with some explanatory variables. In other words, we entertain the model

$$r_t = \mu_t + a_t, \quad \mu_t = \sum_{i=1}^p \phi_i y_{t-i} - \sum_{i=1}^q \theta_i a_{t-i}, \quad y_t = r_t - \phi_0 - \sum_{i=1}^k \beta_i x_{it}, \quad (3.3)$$

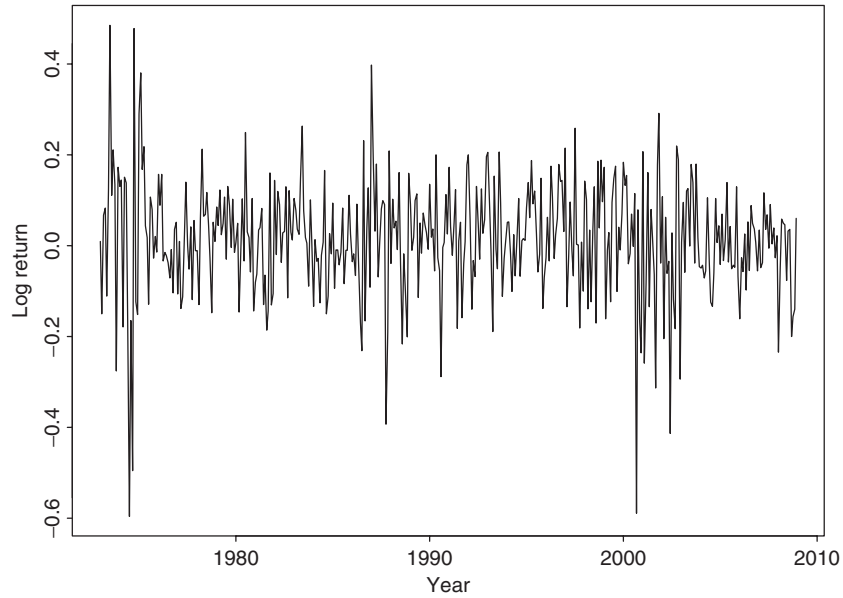


Figure 3.1 Time plot of monthly log returns of Intel stock from January 1973 to December 2008.

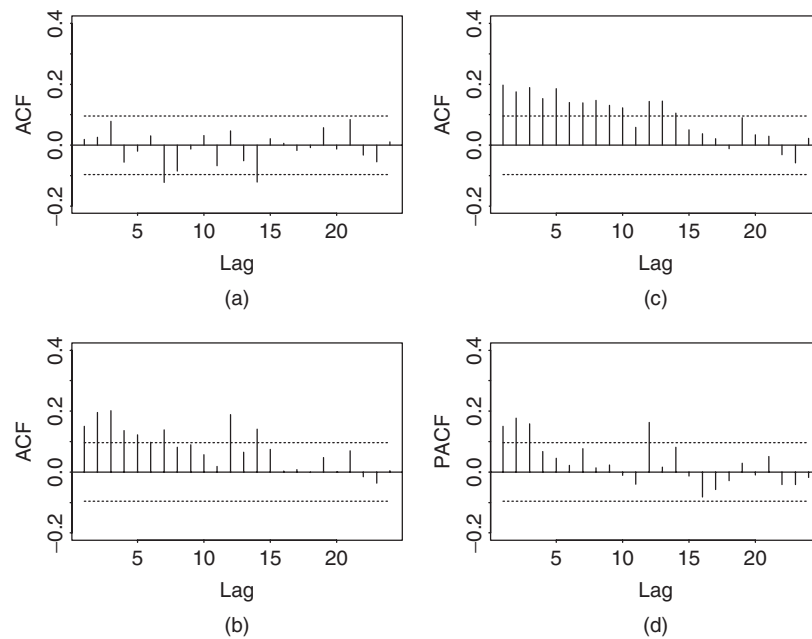


Figure 3.2 Sample ACF and PACF of various functions of monthly log stock returns of Intel Corporation from January 1973 to December 2008: (a) ACF of the log returns, (b) ACF of the squared log returns, (c) ACF of the absolute log returns, and (d) PACF of the squared log returns.

for r_t , where k , p , and q are nonnegative integers, and x_{it} are explanatory variables. Here y_t is simply a notation representing the adjusted return series after removing the effect of explanatory variables.

Model (3.3) illustrates a possible financial application of the regression model with time series errors of Chapter 2. The order (p, q) of an ARMA model may depend on the frequency of the return series. For example, daily returns of a market index often show some minor serial correlations, but monthly returns of the index may not contain any significant serial correlation. The explanatory variables x_t in Eq. (3.3) are flexible. For example, a dummy variable can be used for the Mondays to study the effect of the weekend on daily stock returns. In the capital asset pricing model (CAPM), the mean equation of r_t can be written as $r_t = \phi_0 + \beta r_{m,t} + a_t$, where $r_{m,t}$ denotes the market return.

Combining Eqs. (3.2) and (3.3), we have

$$\sigma_t^2 = \text{Var}(r_t | F_{t-1}) = \text{Var}(a_t | F_{t-1}). \quad (3.4)$$

The conditional heteroscedastic models of this chapter are concerned with the evolution of σ_t^2 . The manner under which σ_t^2 evolves over time distinguishes one volatility model from another.

Conditional heteroscedastic models can be classified into two general categories. Those in the first category use an exact function to govern the evolution of σ_t^2 , whereas those in the second category use a stochastic equation to describe σ_t^2 . The GARCH model belongs to the first category, whereas the stochastic volatility model is in the second category.

Throughout the book, a_t is referred to as the *shock* or *innovation* of an asset return at time t and σ_t is the positive square root of σ_t^2 . The model for μ_t in Eq. (3.3) is referred to as the *mean* equation for r_t and the model for σ_t^2 is the *volatility* equation for r_t . Therefore, modeling conditional heteroscedasticity amounts to augmenting a dynamic equation, which governs the time evolution of the conditional variance of the asset return, to a time series model.

3.3 MODEL BUILDING

Building a volatility model for an asset return series consists of four steps:

1. Specify a mean equation by testing for serial dependence in the data and, if necessary, building an econometric model (e.g., an ARMA model) for the return series to remove any linear dependence.
2. Use the residuals of the mean equation to test for ARCH effects.
3. Specify a volatility model if ARCH effects are statistically significant, and perform a joint estimation of the mean and volatility equations.
4. Check the fitted model carefully and refine it if necessary.

For most asset return series, the serial correlations are weak, if any. Thus, building a mean equation amounts to removing the sample mean from the data if the sample mean is significantly different from zero. For some daily return series, a simple AR model might be needed. In some cases, the mean equation may employ some explanatory variables such as an indicator variable for weekend or January effects.

In what follows, we use R (both with and without OX) and S-Plus in empirical illustrations. Other software packages (e.g., Eviews, SCA, and RATS) can also be used.

3.3.1 Testing for ARCH Effect

For ease in notation, let $a_t = r_t - \mu_t$ be the residuals of the mean equation. The squared series a_t^2 is then used to check for conditional heteroscedasticity, which is also known as the *ARCH* effects. Two tests are available. The first test is to apply the usual Ljung–Box statistics $Q(m)$ to the $\{a_t^2\}$ series; see McLeod and Li (1983). The null hypothesis is that the first m lags of ACF of the a_t^2 series are zero. The second test for conditional heteroscedasticity is the Lagrange multiplier test of Engle (1982). This test is equivalent to the usual F statistic for testing $\alpha_i = 0$ ($i = 1, \dots, m$) in the linear regression

$$a_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \dots + \alpha_m a_{t-m}^2 + e_t, \quad t = m+1, \dots, T,$$

where e_t denotes the error term, m is a prespecified positive integer, and T is the sample size. Specifically, the null hypothesis is $H_0 : \alpha_1 = \dots = \alpha_m = 0$. Let $SSR_0 = \sum_{t=m+1}^T (a_t^2 - \bar{a}^2)^2$, where $\bar{a}^2 = (1/T) \sum_{t=1}^T a_t^2$ is the sample mean of a_t^2 , and $SSR_1 = \sum_{t=m+1}^T \hat{e}_t^2$, where $\hat{e}_t$ is the least-squares residual of the prior linear regression. Then we have

$$F = \frac{(SSR_0 - SSR_1)/m}{SSR_1/(T - 2m - 1)},$$

which is asymptotically distributed as a chi-squared distribution with m degrees of freedom under the null hypothesis. The decision rule is to reject the null hypothesis if $F > \chi_m^2(\alpha)$, where $\chi_m^2(\alpha)$ is the upper $100(1 - \alpha)$ th percentile of χ_m^2 , or the p value of F is less than α , type-I error.

To demonstrate, we consider the monthly log stock returns of Intel Corporation from 1973 to 2008; see Example 3.1. The series does not have significant serial correlations so that it can be directly used to test for the ARCH effect. Indeed, the $Q(m)$ statistics of the return series give $Q(12) = 18.26$ with a p value of 0.11, confirming no serial correlations in the data. On the other hand, the Lagrange multiplier test shows strong ARCH effects with test statistic $F \approx 53.62$, the p value of which is close to zero. The Ljung–Box statistics of the a_t series also shows strong ARCH effects with $Q(12) = 89.85$, the p value of which is close to zero.

S-Plus Demonstration

Denote the return series by `intc`. Note that the command `archTest` applies directly to the a_t series, not to a_t^2 .

```
> da=read.table("m-intc7308.txt",header=T)
> intc=log(da[,2]+1)
> autocorTest(intc,lag=12)
Test for Autocorrelation: Ljung-Box
Null Hypothesis: no autocorrelation

Test Statistics:
Test Stat 18.2635  p.value  0.1079

Dist. under Null: chi-square with 12 degrees of freedom
Total Observ.: 432

> archTest(intc,lag=12)
Test for ARCH Effects: LM Test
Null Hypothesis: no ARCH effects

Test Statistics:
Test Stat 53.6197  p.value  0.0000

Dist. under Null: chi-square with 12 degrees of freedom
```

R Demonstration

```
> da=read.table("m-intc7308.txt",header=T)
> intc=log(da[,2]+1)
> Box.test(intc,lag=12,type='Ljung')
Box-Ljung test

data:  intc
X-squared = 18.2635, df = 12, p-value = 0.1079

> at=intc-mean(intc)
> Box.test(at^2,lag=12,type='Ljung')
Box-Ljung test

data:  at^2
X-squared = 89.8509, df = 12, p-value = 5.274e-14
```

3.4 THE ARCH MODEL

The first model that provides a systematic framework for volatility modeling is the ARCH model of Engle (1982). The basic idea of ARCH models is that (a) the shock a_t of an asset return is serially uncorrelated, but dependent, and

(b) the dependence of a_t can be described by a simple quadratic function of its lagged values. Specifically, an ARCH(m) model assumes that

$$a_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \cdots + \alpha_m a_{t-m}^2, \quad (3.5)$$

where $\{\epsilon_t\}$ is a sequence of independent and identically distributed (iid) random variables with mean zero and variance 1, $\alpha_0 > 0$, and $\alpha_i \geq 0$ for $i > 0$. The coefficients α_i must satisfy some regularity conditions to ensure that the unconditional variance of a_t is finite. In practice, ϵ_t is often assumed to follow the standard normal or a standardized Student- t or a generalized error distribution.

From the structure of the model, it is seen that large past squared shocks $\{a_{t-i}^2\}_{i=1}^m$ imply a large conditional variance σ_t^2 for the innovation a_t . Consequently, a_t tends to assume a large value (in modulus). This means that, under the ARCH framework, large shocks tend to be followed by another large shock. Here I use the word *tend* because a large variance does not necessarily produce a large realization. It only says that the probability of obtaining a large variate is greater than that of a smaller variance. This feature is similar to the volatility clusterings observed in asset returns.

The ARCH effect also occurs in other financial time series. Figure 3.3 shows the time plots of (a) the percentage changes in Deutsche mark/U.S. dollar exchange rate measured in 10-minute intervals from June 5, 1989, to June 19, 1989, for 2488 observations, and (b) the squared series of the percentage changes. Big percentage

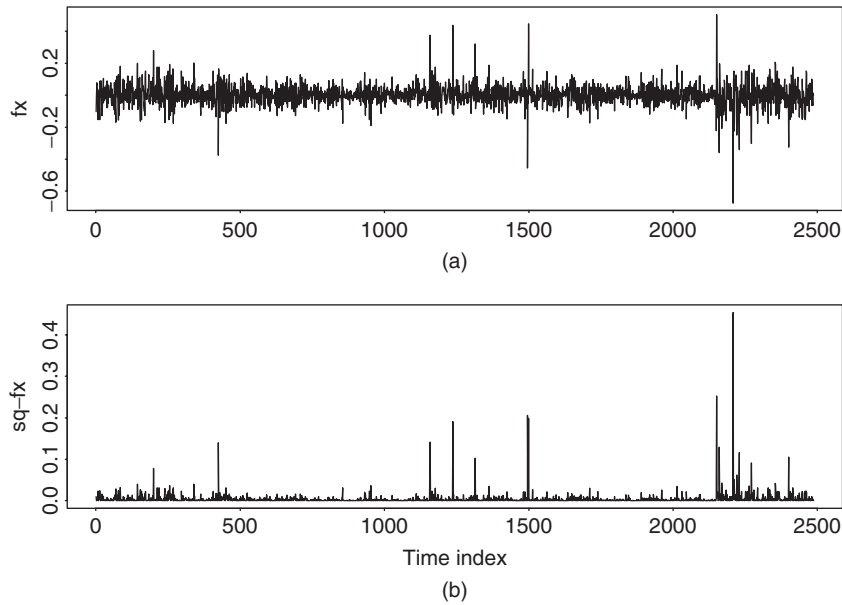


Figure 3.3 (a) Time plot of 10-minute returns of exchange rate between Deutsche mark and U.S. dollar from June 5, 1989, to June 19, 1989, and (b) the squared returns.

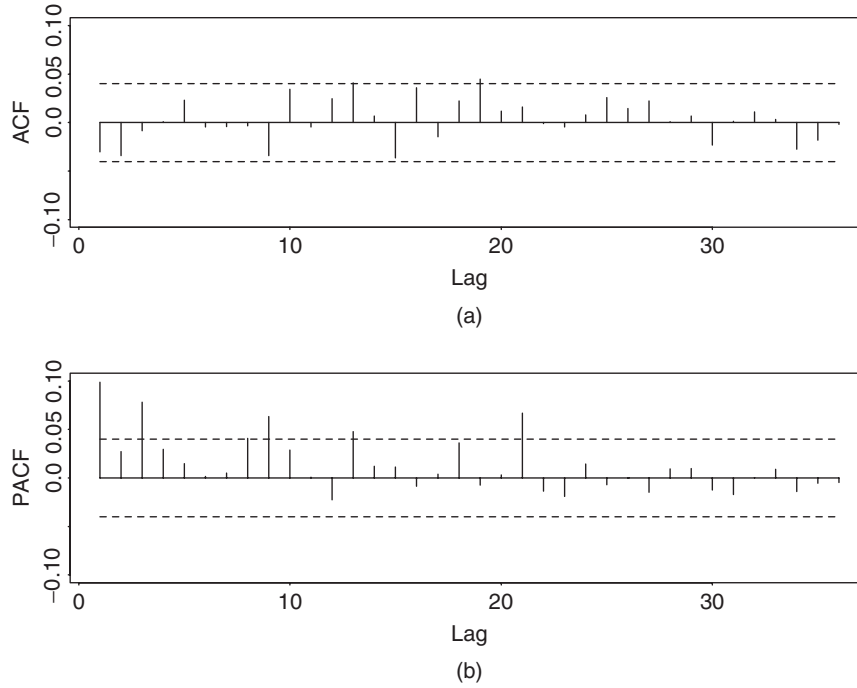


Figure 3.4 (a) Sample autocorrelation function of return series of mark/dollar exchange rate and (b) sample partial autocorrelation function of squared returns.

changes occurred occasionally, but there were certain stable periods. Figure 3.4(a) shows the sample ACF of the percentage change series. Clearly, the series has no serial correlation. Figure 3.4(b) shows the sample PACF of the squared series of percentage change. It is seen that there are some big spikes in the PACF. Such spikes suggest that the percentage changes are not serially independent and have some ARCH effects.

Remark. Some authors use h_t to denote the conditional variance in Eq. (3.5). In this case, the shock becomes $a_t = \sqrt{h_t}\epsilon_t$. $\square$

3.4.1 Properties of ARCH Models

To understand the ARCH models, it pays to carefully study the ARCH(1) model

$$a_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2,$$

where $\alpha_0 > 0$ and $\alpha_1 \geq 0$. First, the unconditional mean of a_t remains zero because

$$E(a_t) = E[E(a_t|F_{t-1})] = E[\sigma_t E(\epsilon_t)] = 0.$$

Second, the unconditional variance of a_t can be obtained as

$$\begin{aligned}\text{Var}(a_t) &= E(a_t^2) = E[E(a_t^2|F_{t-1})] \\ &= E(\alpha_0 + \alpha_1 a_{t-1}^2) = \alpha_0 + \alpha_1 E(a_{t-1}^2).\end{aligned}$$

Because a_t is a stationary process with $E(a_t) = 0$, $\text{Var}(a_t) = \text{Var}(a_{t-1}) = E(a_{t-1}^2)$. Therefore, we have $\text{Var}(a_t) = \alpha_0 + \alpha_1 \text{Var}(a_t)$ and $\text{Var}(a_t) = \alpha_0/(1 - \alpha_1)$. Since the variance of a_t must be positive, we require $0 \leq \alpha_1 < 1$. Third, in some applications, we need higher order moments of a_t to exist and, hence, α_1 must also satisfy some additional constraints. For instance, to study its tail behavior, we require that the fourth moment of a_t is finite. Under the normality assumption of ϵ_t in Eq. (3.5), we have

$$E(a_t^4|F_{t-1}) = 3[E(a_t^2|F_{t-1})]^2 = 3(\alpha_0 + \alpha_1 a_{t-1}^2)^2.$$

Therefore,

$$E(a_t^4) = E[E(a_t^4|F_{t-1})] = 3E(\alpha_0 + \alpha_1 a_{t-1}^2)^2 = 3E(\alpha_0^2 + 2\alpha_0\alpha_1 a_{t-1}^2 + \alpha_1^2 a_{t-1}^4).$$

If a_t is fourth-order stationary with $m_4 = E(a_t^4)$, then we have

$$\begin{aligned}m_4 &= 3[\alpha_0^2 + 2\alpha_0\alpha_1 \text{Var}(a_t) + \alpha_1^2 m_4] \\ &= 3\alpha_0^2 \left(1 + 2\frac{\alpha_1}{1 - \alpha_1}\right) + 3\alpha_1^2 m_4.\end{aligned}$$

Consequently,

$$m_4 = \frac{3\alpha_0^2(1 + \alpha_1)}{(1 - \alpha_1)(1 - 3\alpha_1^2)}.$$

This result has two important implications: (a) since the fourth moment of a_t is positive, we see that α_1 must also satisfy the condition $1 - 3\alpha_1^2 > 0$; that is, $0 \leq \alpha_1^2 < \frac{1}{3}$; and (b) the unconditional kurtosis of a_t is

$$\frac{E(a_t^4)}{[\text{Var}(a_t)]^2} = 3 \frac{\alpha_0^2(1 + \alpha_1)}{(1 - \alpha_1)(1 - 3\alpha_1^2)} \times \frac{(1 - \alpha_1)^2}{\alpha_0^2} = 3 \frac{1 - \alpha_1^2}{1 - 3\alpha_1^2} > 3.$$

Thus, the excess kurtosis of a_t is positive and the tail distribution of a_t is heavier than that of a normal distribution. In other words, the shock a_t of a conditional Gaussian ARCH(1) model is more likely than a Gaussian white noise series to produce “outliers.” This is in agreement with the empirical finding that “outliers” appear more often in asset returns than that implied by an iid sequence of normal random variates.

These properties continue to hold for general ARCH models, but the formulas become more complicated for higher order ARCH models. The condition $\alpha_i \geq 0$ in

Eq. (3.5) can be relaxed. It is a condition to ensure that the conditional variance σ_t^2 is positive for all t . In fact, a natural way to achieve positiveness of the conditional variance is to rewrite an ARCH(m) model as

$$a_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = \alpha_0 + A'_{m,t-1} \Omega A_{m,t-1}, \quad (3.6)$$

where $A_{m,t-1} = (a_{t-1}, \dots, a_{t-m})'$ and Ω is an $m \times m$ nonnegative definite matrix. The ARCH(m) model in Eq. (3.5) requires Ω to be diagonal. Thus, Engle's model uses a parsimonious approach to approximate a quadratic function. A simple way to achieve Eq. (3.6) is to employ a random-coefficient model for a_t ; see the CHARMA and RCA models discussed later.

3.4.2 Weaknesses of ARCH Models

The advantages of ARCH models include properties discussed in the previous section. The model also has some weaknesses:

1. The model assumes that positive and negative shocks have the same effects on volatility because it depends on the square of the previous shocks. In practice, it is well known that the price of a financial asset responds differently to positive and negative shocks.
2. The ARCH model is rather restrictive. For instance, α_1^2 of an ARCH(1) model must be in the interval $[0, \frac{1}{3}]$ if the series has a finite fourth moment. The constraint becomes complicated for higher order ARCH models. In practice, it limits the ability of ARCH models with Gaussian innovations to capture excess kurtosis.
3. The ARCH model does not provide any new insight for understanding the source of variations of a financial time series. It merely provides a mechanical way to describe the behavior of the conditional variance. It gives no indication about what causes such behavior to occur.
4. ARCH models are likely to overpredict the volatility because they respond slowly to large isolated shocks to the return series.

3.4.3 Building an ARCH Model

Among volatility models, specifying an ARCH model is relatively easy. Details are given below.

Order Determination

If an ARCH effect is found to be significant, one can use the PACF of a_t^2 to determine the ARCH order. Using PACF of a_t^2 to select the ARCH order can be justified as follows. From the model in Eq. (3.5), we have

$$\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \dots + \alpha_m a_{t-m}^2.$$

For a given sample, a_t^2 is an unbiased estimate of σ_t^2 . Therefore, we expect that a_t^2 is linearly related to $a_{t-1}^2, \dots, a_{t-m}^2$ in a manner similar to that of an autoregressive model of order m . Note that a single a_t^2 is generally not an efficient estimate of σ_t^2 , but it can serve as an approximation that could be informative in specifying the order m .

Alternatively, define $\eta_t = a_t^2 - \sigma_t^2$. It can be shown that $\{\eta_t\}$ is an uncorrelated series with mean 0. The ARCH model then becomes

$$a_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \dots + \alpha_m a_{t-m}^2 + \eta_t,$$

which is in the form of an AR(m) model for a_t^2 , except that $\{\eta_t\}$ is not an iid series. From Chapter 2, PACF of a_t^2 is a useful tool to determine the order m . Because $\{\eta_t\}$ are not identically distributed, the least-squares estimates of the prior model are consistent but not efficient. The PACF of a_t^2 may not be effective when the sample size is small.

Estimation

Several likelihood functions are commonly used in ARCH estimation, depending on the distributional assumption of ϵ_t . Under the normality assumption, the likelihood function of an ARCH(m) model is

$$\begin{aligned} f(a_1, \dots, a_T | \boldsymbol{\alpha}) &= f(a_T | F_{T-1}) f(a_{T-1} | F_{T-2}) \cdots f(a_{m+1} | F_m) f(a_1, \dots, a_m | \boldsymbol{\alpha}) \\ &= \prod_{t=m+1}^T \frac{1}{\sqrt{2\pi\sigma_t^2}} \exp\left(-\frac{a_t^2}{2\sigma_t^2}\right) \times f(a_1, \dots, a_m | \boldsymbol{\alpha}), \end{aligned}$$

where $\boldsymbol{\alpha} = (\alpha_0, \alpha_1, \dots, \alpha_m)'$ and $f(a_1, \dots, a_m | \boldsymbol{\alpha})$ is the joint probability density function of $a_1, \dots, a_m$. Since the exact form of $f(a_1, \dots, a_m | \boldsymbol{\alpha})$ is complicated, it is commonly dropped from the prior likelihood function, especially when the sample size is sufficiently large. This results in using the conditional-likelihood function

$$f(a_{m+1}, \dots, a_T | \boldsymbol{\alpha}, a_1, \dots, a_m) = \prod_{t=m+1}^T \frac{1}{\sqrt{2\pi\sigma_t^2}} \exp\left(-\frac{a_t^2}{2\sigma_t^2}\right),$$

where σ_t^2 can be evaluated recursively. We refer to estimates obtained by maximizing the prior likelihood function as the conditional maximum-likelihood estimates (MLEs) under normality.

Maximizing the conditional-likelihood function is equivalent to maximizing its logarithm, which is easier to handle. The conditional log-likelihood function is

$$\ell(a_{m+1}, \dots, a_T | \boldsymbol{\alpha}, a_1, \dots, a_m) = \sum_{t=m+1}^T \left[-\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\sigma_t^2) - \frac{1}{2} \frac{a_t^2}{\sigma_t^2} \right].$$

Since the first term $\ln(2\pi)$ does not involve any parameters, the log-likelihood function becomes

$$\ell(a_{m+1}, \dots, a_T | \boldsymbol{\alpha}, a_1, \dots, a_m) = - \sum_{t=m+1}^T \left[\frac{1}{2} \ln(\sigma_t^2) + \frac{1}{2} \frac{a_t^2}{\sigma_t^2} \right],$$

where $\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \dots + \alpha_m a_{t-m}^2$ can be evaluated recursively.

In some applications, it is more appropriate to assume that ϵ_t follows a heavy-tailed distribution such as a standardized Student- t distribution. Let x_v be a Student- t distribution with v degrees of freedom. Then $\text{Var}(x_v) = v/(v-2)$ for $v > 2$, and we use $\epsilon_t = x_v / \sqrt{v/(v-2)}$. The probability density function of ϵ_t is

$$f(\epsilon_t | v) = \frac{\Gamma[(v+1)/2]}{\Gamma(v/2)\sqrt{(v-2)\pi}} \left(1 + \frac{\epsilon_t^2}{v-2} \right)^{-(v+1)/2}, \quad v > 2, \quad (3.7)$$

where $\Gamma(x)$ is the usual gamma function (i.e., $\Gamma(x) = \int_0^\infty y^{x-1} e^{-y} dy$). Using $a_t = \sigma_t \epsilon_t$, we obtain the conditional-likelihood function of a_t as

$$f(a_{m+1}, \dots, a_T | \boldsymbol{\alpha}, A_m) = \prod_{t=m+1}^T \frac{\Gamma[(v+1)/2]}{\Gamma(v/2)\sqrt{(v-2)\pi}} \frac{1}{\sigma_t} \left[1 + \frac{a_t^2}{(v-2)\sigma_t^2} \right]^{-(v+1)/2},$$

where $v > 2$ and $A_m = (a_1, a_2, \dots, a_m)$. We refer to the estimates that maximize the prior likelihood function as the conditional MLEs under t distribution. The degrees of freedom of the t distribution can be specified a priori or estimated jointly with other parameters. A value between 4 and 8 is often used if it is prespecified.

If the degrees of freedom v of the Student- t distribution is prespecified, then the conditional log-likelihood function is

$$\ell(a_{m+1}, \dots, a_T | \boldsymbol{\alpha}, A_m) = - \sum_{t=m+1}^T \left[\frac{v+1}{2} \ln \left(1 + \frac{a_t^2}{(v-2)\sigma_t^2} \right) + \frac{1}{2} \ln(\sigma_t^2) \right]. \quad (3.8)$$

If one wishes to estimate v jointly with other parameters, then the log-likelihood function becomes

$$\begin{aligned} \ell(a_{m+1}, \dots, a_T | \boldsymbol{\alpha}, v, A_m) &= (T-m) \left\{ \ln \left[\Gamma \left(\frac{v+1}{2} \right) \right] - \ln \left[\Gamma \left(\frac{v}{2} \right) \right] \right. \\ &\quad \left. - 0.5 \ln[(v-2)\pi] \right\} + \ell(a_{m+1}, \dots, a_T | \boldsymbol{\alpha}, A_m), \end{aligned}$$

where the second term is given in Eq. (3.8).

Besides fat tails, empirical distributions of asset returns may also be skewed. To handle this additional characteristic of asset returns, the Student- t distribution

has been modified to become a skew-Student- t distribution. There are multiple versions of skew-Student- t distribution, but we shall adopt the approach of Fernández and Steel (1998), which can introduce skewness into any continuous unimodal and symmetric (with respect to 0) univariate distribution. Specifically, for the innovation ϵ_t of an ARCH process, Lambert and Laurent (2001) apply the Fernández and Steel method to the standardized Student- t distribution in Eq. (3.7) to obtain a standardized skew-Student- t distribution. The resulting probability density function is

$$g(\epsilon_t|\xi, v) = \begin{cases} \frac{2}{\xi + \frac{1}{\xi}} \varrho f[\xi(\varrho\epsilon_t + \bar{\omega})|v] & \text{if } \epsilon_t < -\bar{\omega}/\varrho, \\ \frac{2}{\xi + \frac{1}{\xi}} \varrho f[(\varrho\epsilon_t + \bar{\omega})/\xi|v] & \text{if } \epsilon_t \geq -\bar{\omega}/\varrho, \end{cases} \quad (3.9)$$

where $f(\cdot)$ is the probability density function (pdf) of the standardized Student- t distribution in Eq. (3.7), ξ is the skewness parameter, $v > 2$ is the degrees of freedom, and the parameters ϱ and $\bar{\omega}$ are given below:

$$\varpi = \frac{\Gamma[(v-1)/2]\sqrt{v-2}}{\sqrt{\pi}\Gamma(v/2)}(\xi - \frac{1}{\xi}), \quad \varrho^2 = (\xi^2 + \frac{1}{\xi^2} - 1) - \varpi^2.$$

In Eq. (3.9), ξ^2 is equal to the ratio of probability masses above and below the mode of the distribution and, hence, it is a measure of the skewness.

Finally, ϵ_t may assume a generalized error distribution (GED) with probability density function

$$f(x) = \frac{v}{\lambda 2^{(1+1/v)}\Gamma(1/v)} \exp(-\frac{1}{2}|x/\lambda|^v), \quad -\infty < x < \infty, \quad 0 < v \leq \infty, \quad (3.10)$$

where $\Gamma(\cdot)$ is the gamma function and $\lambda = [2^{(-2/v)}\Gamma(1/v)/\Gamma(3/v)]^{1/2}$. This distribution reduces to a Gaussian distribution if $v = 2$, and it has heavy tails when $v < 2$. The conditional log-likelihood function $\ell(a_{m+1}, \dots, a_T|\boldsymbol{\alpha}, \mathbf{A}_m)$ can easily be obtained.

Remark. Skew Student- t , skew normal, and skew GED distributions are available in the `fGarch` package of `Rmetrics`. The commands are `sstd`, `snorm`, and `sged`, respectively. See the R demonstration below for an example. $\square$

Model Checking

For a properly specified ARCH model, the standardized residuals

$$\tilde{a}_t = \frac{a_t}{\sigma_t}$$

form a sequence of iid random variables. Therefore, one can check the adequacy of a fitted ARCH model by examining the series $\{\tilde{a}_t\}$. In particular, the Ljung–Box

statistics of $\tilde{a}_t$ can be used to check the adequacy of the mean equation and that of $\tilde{a}_t^2$ can be used to test the validity of the volatility equation. The skewness, kurtosis, and quantile-to-quantile plot (i.e., QQ plot) of $\{\tilde{a}_t\}$ can be used to check the validity of the distribution assumption. Many residual plots are available in S-Plus for model checking.

Forecasting

Forecasts of the ARCH model in Eq. (3.5) can be obtained recursively as those of an AR model. Consider an ARCH(m) model. At the forecast origin h , the 1-step-ahead forecast of σ_{h+1}^2 is

$$\sigma_h^2(1) = \alpha_0 + \alpha_1 a_h^2 + \cdots + \alpha_m a_{h+1-m}^2.$$

The 2-step-ahead forecast is

$$\sigma_h^2(2) = \alpha_0 + \alpha_1 \sigma_h^2(1) + \alpha_2 a_h^2 + \cdots + \alpha_m a_{h+2-m}^2,$$

and the ℓ -step-ahead forecast for $\sigma_{h+\ell}^2$ is

$$\sigma_h^2(\ell) = \alpha_0 + \sum_{i=1}^m \alpha_i \sigma_h^2(\ell - i), \quad (3.11)$$

where $\sigma_h^2(\ell - i) = a_{h+\ell-i}^2$ if $\ell - i \leq 0$.

3.4.4 Some Examples

In this section, we illustrate ARCH modeling by considering two examples.

Example 3.1. We first apply the modeling procedure to build a simple ARCH model for the monthly log returns of Intel stock. The sample ACF and PACF of the squared returns in Figure 3.2 clearly show the existence of conditional heteroscedasticity. This is confirmed by the ARCH test shown in Section 3.3.1, and we proceed to identify the order of an ARCH model. The sample PACF in Figure 3.2(d) indicates that an ARCH(3) model might be appropriate. Consequently, we specify the model

$$r_t = \mu + a_t, \quad a_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \alpha_2 a_{t-2}^2 + \alpha_3 a_{t-3}^2$$

for the monthly log returns of Intel stock. Assuming that ϵ_t are iid standard normal, we obtain the fitted model

$$r_t = 0.0122 + a_t, \quad \sigma_t^2 = 0.0106 + 0.2131 a_{t-1}^2 + 0.0770 a_{t-2}^2 + 0.0599 a_{t-3}^2,$$

where the standard errors of the parameters are 0.0057, 0.0010, 0.0757, 0.0480, and 0.0688, respectively; see the output below. While the estimates meet the general requirement of an ARCH(3) model, the estimates of α_2 and α_3 appear to be statistically nonsignificant at the 5% level. Therefore, the model can be simplified.

S-Plus Demonstration

The following output has been edited and % marks explanation:

```
> module(finmetrics)
> da=read.table("m-intc7308.txt",header=T)
> intc=log(da[,2]+1)
> arch3.fit=garch(intc~1,~garch(3,0))
> summary(arch3.fit)
garch(formula.mean = intc ~ 1, formula.var = ~ garch(3, 0))

Mean Equation: structure(.Data = intc ~ 1, class = "formula")
Conditional Variance Equation:structure(.Data=~garch(3,0),...)
Conditional Distribution: gaussian
-----
Estimated Coefficients:
-----
              Value Std.Error t value Pr(>|t|)
      C 0.01216 0.0056986  2.1341 0.033402
      A 0.01058 0.0009643 10.9739 0.000000
ARCH(1) 0.21307 0.0756708  2.8157 0.005093
ARCH(2) 0.07698 0.0480170  1.6032 0.109638
ARCH(3) 0.05988 0.0688081  0.8703 0.384628
-----

> arch1=garch(intc~1,~garch(1,0))
> summary(arch1)
garch(formula.mean = intc ~ 1, formula.var = ~ garch(1, 0))

Conditional Distribution: gaussian
-----
Estimated Coefficients:
-----
              Value Std.Error t value  Pr(>|t|)
      C 0.01261 0.0052624  2.397 1.695e-02
      A 0.01113 0.0009971 11.164 0.000e+00
ARCH(1) 0.35602 0.0761267  4.677 3.912e-06
-----

AIC(3) = -570.0179, BIC(3) = -557.8126

Ljung-Box test for standardized residuals:
-----
Statistic P-value Chi^2-d.f.
    14.26  0.2844          12

Ljung-Box test for squared standardized residuals:
-----
Statistic P-value Chi^2-d.f.
    32.11 0.001329          12
> stres=arch1$residuals/arch1$sigma.t %standardized residuals
```



```

> autocorTest(stres, lag=10)
Test for Autocorrelation: Ljung-Box

Null Hypothesis: no autocorrelation
Test Statistics:
Test Stat 12.6386,  p.value  0.2446

Dist. under Null: chi-square with 10 degrees of freedom
> archTest(stres, lag=10)
Test for ARCH Effects: LM Test

Null Hypothesis: no ARCH effects
Test Statistics:
Test Stat 14.7481,  p.value  0.1415

Dist. under Null: chi-square with 10 degrees of freedom
> arch1$asymp.sd  %Obtain unconditional standard error

[1] 0.1314698
> plot(arch1)  % Obtain various plots, including the
               % fitted volatility series.

```

Dropping the two nonsignificant parameters, we obtain the model

$$r_t = 0.0126 + a_t, \quad \sigma_t^2 = 0.0111 + 0.3560a_{t-1}^2, \quad (3.12)$$

where the standard errors of the parameters are 0.0053, 0.0010, and 0.0761, respectively. All the estimates are highly significant. Figure 3.5 shows the standardized residuals $\{\tilde{a}_t\}$ and the sample ACF of some functions of $\{\tilde{a}_t\}$. The Ljung–Box statistics of standardized residuals give $Q(10) = 12.64$ with a p value of 0.24 and those of $\{\tilde{a}_t^2\}$ give $Q(10) = 14.75$ with a p value of 0.14. See the output. Consequently, the ARCH(1) model in Eq. (3.12) is adequate for describing the conditional heteroscedasticity of the data at the 5% significance level.

The ARCH(1) model in Eq. (3.12) has some interesting properties. First, the expected monthly log return for Intel stock is about 1.26%, which is remarkable, especially since the data span includes the period after the Internet bubble. Second, $\hat{a}_1^2 = 0.356^2 < \frac{1}{3}$ so that the unconditional fourth moment of the monthly log return of Intel stock exists. Third, the unconditional standard deviation of r_t is $\sqrt{0.0111/(1-0.356)} \approx 0.1315$. Finally, the ARCH(1) model can be used to predict the monthly volatility of Intel stock returns.

t Innovation

For comparison, we also fit an ARCH(1) model with Student- t innovations to the series. The resulting model is

$$r_t = 0.0169 + a_t, \quad \sigma_t^2 = 0.0120 + 0.2845a_{t-1}^2, \quad (3.13)$$

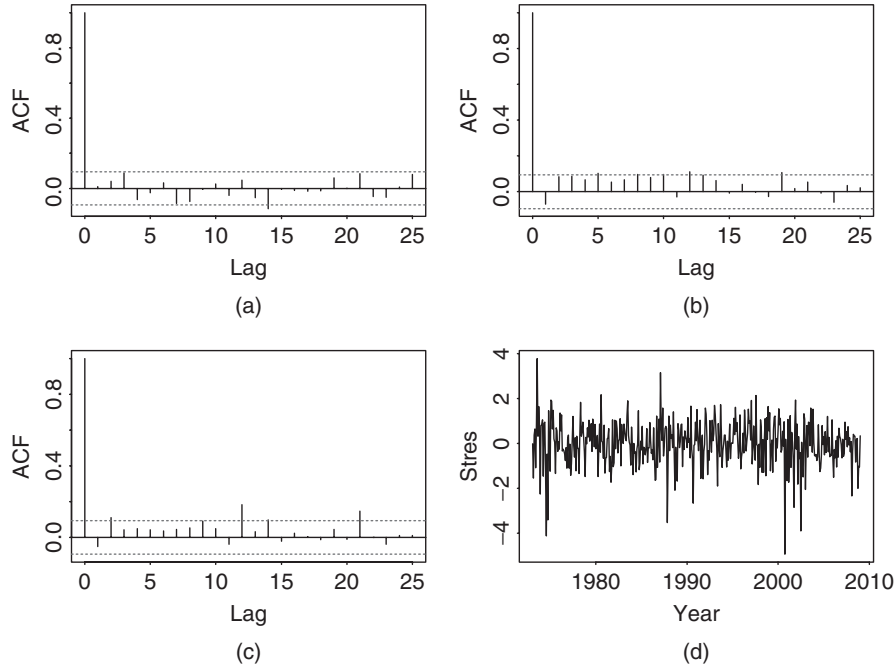


Figure 3.5 Model checking statistics of Gaussian ARCH(1) model in Eq. (3.12) for monthly log returns of Intel stock from January 1973 to December 2008: Parts (a), (b), and (c) show sample ACF of standardized residuals, their squared series, and absolute series, respectively; part (d) is time plot of standardized residuals.

where the standard errors of the parameters are 0.0053, 0.0017, and 0.1120, respectively. The estimated degrees of freedom is 6.01 with standard error 1.50. All the estimates are significant at the 5% level. The unconditional standard deviation of a_t is $\sqrt{0.0120/(1 - 0.2845)} \approx 0.1295$, which is close to that obtained under normality. The Ljung–Box statistics of the standardized residuals give $Q(12) = 14.88$ with a p value of 0.25, confirming that the mean equation is adequate. However, the Ljung–Box statistics for the squared standardized residuals show $Q(12) = 35.42$ with a p value of 0.0004. The volatility equation is inadequate at the 1% level. Further analysis shows that $Q(10) = 15.90$ with a p value of 0.10 for the squared standardized residuals. The inadequacy of the volatility equation is due to a large lag-12 ACF ($\rho_{12} = 0.188$) of the squared standardized residuals.

Comparing models (3.12) and (3.13), we see that (a) using a heavy-tailed distribution for ϵ_t reduces the ARCH coefficient, and (b) the difference between the two models is small for this particular instance. Finally, a more appropriate conditional heteroscedastic model for the monthly log returns of Intel stock is a GARCH(1,1) model, which is discussed in the next section.

S-Plus Demonstration

Note the following output with t innovations:

```
> arch1t=garch(intc~1,~garch(1,0),cond.dist="t")
> summary(arch1t)
Call:
garch(formula.mean=intc~1,formula.var=~garch(1,0),
      cond.dist="t")

Mean Equation: structure(.Data = intc ~ 1, class = "formula")
Cond. Variance Equation:structure(.Data=~ garch(1,0), ...)
Cond. Distribution:  t
  with estimated parameter 6.012769 and standard error 1.502179
-----
Estimated Coefficients:
-----
              Value Std.Error t value  Pr(>|t|)
      C 0.01688    0.005288   3.193 1.512e-03
      A 0.01195    0.001667   7.169 3.345e-12
ARCH(1) 0.28445    0.111998   2.540 1.145e-02
-----
AIC(4) = -597.3379, BIC(4) = -581.0642

Ljung-Box test for standardized residuals:
-----
Statistic P-value Chi^2-d.f.
      14.88  0.2482          12

Ljung-Box test for squared standardized residuals:
-----
Statistic    P-value Chi^2-d.f.
      35.42 0.0004014          12
```

Remark. In S-Plus, the command `garch` allows for several conditional distributions. They are specified by `cond.dist = 't'` or `'ged'`. The default is Gaussian. The R output is given below. The estimates are close to those of S-Plus. $\square$

R Demonstration

The following output uses the `fGarch` package with command `garchFit` and % denotes explanation:

```
> da=read.table("m-intc7308.txt",header=T)
> library(fGarch) % Load the package
> intc=log(da[,2]+1)
> m1=garchFit(intc~garch(1,0),data=intc,trace=F)
> summary(m1) % Obtain results
```

```

Title:
  GARCH Modelling
Call:
garchFit(formula=intc~garch(1,0), data=intc, trace=F)

Mean and Variance Equation: data ~ garch(1, 0) [data = intc]
Conditional Distribution: norm

Coefficient(s):
      mu      omega    alpha1
0.012637 0.011195 0.379492

Std. Errors:
  based on Hessian

Error Analysis:
      Estimate Std. Error t value Pr(>|t|)
mu      0.012637   0.005428   2.328 0.01990 *
omega   0.011195   0.001239   9.034 < 2e-16 ***
alpha1  0.379492   0.115534   3.285 0.00102 **
---
Log Likelihood:
 288.0589    normalized: 0.6668031

Standardised Residuals Tests:    %Model checking
                                Statistic p-Value
Jarque-Bera Test    R    Chi^2  137.919    0
Shapiro-Wilk Test   R    W      0.9679255 4.025172e-08
Ljung-Box Test      R    Q(10)  12.54002 0.2505382
Ljung-Box Test      R    Q(15)  21.33508 0.1264607
Ljung-Box Test      R    Q(20)  23.19679 0.2792354
Ljung-Box Test      R^2  Q(10)  16.0159 0.09917815
Ljung-Box Test      R^2  Q(15)  36.08022 0.001721296
Ljung-Box Test      R^2  Q(20)  37.43683 0.01036728
LM Arch Test        R    TR^2   26.57744 0.008884587

Information Criterion Statistics:
      AIC      BIC      SIC      HQIC
-1.319717 -1.291464 -1.319813 -1.308563

> predict(m1,5) % Obtain 1 to 5-step predictions
      meanForecast meanError standardDeviation
1    0.01263656 0.1278609      0.1098306
2    0.01263656 0.1278609      0.1255897
3    0.01263656 0.1278609      0.1310751
4    0.01263656 0.1278609      0.1330976
5    0.01263656 0.1278609      0.1338571

```

```

% The next command fits a GARCH(1,1) model
> m2=garchFit(intc~garch(1,1),data=intc,trace=F)
> summary(m2) % output edited.
Coefficient(s):
      mu      omega      alpha1      beta1
0.01073352  0.00095445  0.08741989  0.85118414

Error Analysis:
      Estimate Std. Error  t value Pr(>|t|)
mu      0.0107335  0.0055289   1.941  0.05222 .
omega   0.0009544  0.0003989   2.392  0.01674 *
alpha1  0.0874199  0.0269810   3.240  0.00120 **
beta1   0.8511841  0.0393702  21.620 < 2e-16 ***
---
Standardised Residuals Tests:
      Statistic p-Value
Jarque-Bera Test  R      Chi^2  165.5740  0
Shapiro-Wilk Test  R      W      0.9712087  1.626824e-07
Ljung-Box Test     R      Q(10)  8.267633  0.6027128
Ljung-Box Test     R      Q(15)  14.42612  0.4934871
Ljung-Box Test     R      Q(20)  15.13331  0.7687297
Ljung-Box Test     R^2    Q(10)  0.9891848  0.9998363
Ljung-Box Test     R^2    Q(15)  11.36596  0.7262473
Ljung-Box Test     R^2    Q(20)  12.68143  0.8906302
LM Arch Test       R      TR^2   10.70199  0.5546164

% The next command fits an ARCH(1) model with Student-t dist.
> m3=garchFit(intc~garch(1,0),data=intc,trace=F,
  cond.dist='std')
> summary(m3) % Output shortened.

Call:
garchFit(formula=intc~garch(1,0), data=intc, cond.dist="std",
  trace = F)
Mean and Variance Equation: data ~ garch(1, 0) [data = intc]
Conditional Distribution: std % Student-t distribution

Coefficient(s):
      mu      omega      alpha1      shape
0.016731  0.011939  0.285320  6.015195

Error Analysis:
      Estimate Std. Error  t value Pr(>|t|)
mu      0.016731  0.005302   3.155  0.001603 **
omega   0.011939  0.001603   7.449  9.4e-14 ***
alpha1  0.285320  0.110607   2.580  0.009892 **
shape   6.015195  1.562620   3.849  0.000118 ***
% Degrees of freedom

```

```
% The next command fits an ARCH(1) model with skew
%Student-t dist.
> m4=garchFit(intc~garch(1,0),data=intc,cond.dist='sstd',
  trace=F)
% Next, fit an ARMA(1,0)+GARCH(1,1) model with
% Gaussian noises.
> m5=garchFit(intc~arma(1,0)+garch(1,1),data=intc,trace=F)
```

R Demonstration

The following output was generated with Ox and G@RCH4.2 package and % denotes explanation:

```
> source("garchoxfit_R.txt")
% In G@RCH package, an ARCH(1) model is specified as
% GARCH(0,1).
> m1=garchOxFit(formula.mean=~arma(0,0),
  formula.var=~garch(0,1), series=intc)
% ** SPECIFICATIONS **
Dependent variable : X
Mean Equation : ARMA (0, 0) model.
No regressor in the mean
Variance Equation : GARCH (0, 1) model.
No regressor in the variance
The distribution is a Gauss distribution.

Maximum Likelihood Estimation(Std.Errors based on 2nd deriv.)
               Coefficient Std.Error t-value t-prob
Cst(M)          0.012630  0.0054130   2.333  0.0201
Cst(V)          0.011129  0.0012355   9.007  0.0000
ARCH(Alpha1)    0.387223   0.11688   3.313  0.0010

% ** TESTS **
Q-Statistics on Standardized Residuals
  Q(10)=12.4952 [0.2532785],  Q(20)=23.1210 [0.2828934]
H0: No serial correlation ==> Accept H0 when prob. is High.
-----
Q-Statistics on Squared Standardized Residuals
  --> P-values adjusted by 1 degree(s) of freedom
  Q(10)=15.7849 [0.0715122], Q( 20)=37.0238 [0.0078807]
-----
ARCH 1-10 test:   F(10,410)=   1.4423 [0.1592]
-----
% Apply Student-t distribution
> m2=garchOxFit(formula.mean=~arma(0,0),
  formula.var=~garch(0,1),
  series=intc,cond.dist="t")
% ** SPECIFICATIONS **
```

Dependent variable : X
Mean Equation : ARMA (0, 0) model.
No regressor in the mean
Variance Equation : GARCH (0, 1) model.
No regressor in the variance
The distribution is a Student distribution, with 6.02272 df.

Maximum Likelihood Estimation(Std.Errors based on 2nd deriv.)

	Coefficient	Std.Error	t-value	t-prob
Cst(M)	0.016702	0.0052934	3.155	0.0017
Cst(V)	0.011870	0.0015969	7.433	0.0000
ARCH(Alpha1)	0.292318	0.11223	2.605	0.0095
Student(DF)	6.022723	1.5663	3.845	0.0001

** TESTS **

Q-Statistics on Standardized Residuals

Q(10)=13.0837 [0.2190281], Q(20)=24.0724 [0.2392436]

Q-Statistics on Squared Standardized Residuals

--> P-values adjusted by 1 degree(s) of freedom

Q(10)=18.6982 [0.0278845], Q(20)=41.7182 [0.0019343]

Example 3.2. Consider the percentage changes of the exchange rate between mark and dollar in 10-minute intervals. The data are shown in Figure 3.3(a). As shown in Figure 3.4(a), the series has no serial correlations. However, the sample PACF of the squared series a_t^2 shows some big spikes, especially at lags 1 and 3. There are some large PACF at higher lags, but the lower order lags tend to be more important. Following the procedure discussed in the previous section, we specify an ARCH(3) model for the series. Using the conditional Gaussian likelihood function, we obtain the fitted model $r_t = 0.0018 + \sigma_t \epsilon_t$ and

$$\sigma_t^2 = 0.22 \times 10^{-2} + 0.322a_{t-1}^2 + 0.074a_{t-2}^2 + 0.093a_{t-3}^2,$$

where all the estimates in the volatility equation are statistically significant at the 5% significant level, and the standard errors of the parameters are 0.47×10^{-6} , 0.017, 0.016, and 0.014, respectively. Model checking, using the standardized residual $\tilde{a}_t$, indicates that the model is adequate.

3.5 THE GARCH MODEL

Although the ARCH model is simple, it often requires many parameters to adequately describe the volatility process of an asset return. For instance, consider the monthly excess returns of S&P 500 index of Example 3.3. An ARCH(9) model is needed for the volatility process. Some alternative model must be sought. Bollerslev (1986) proposes a useful extension known as the generalized ARCH (GARCH) model. For a log return series r_t , let $a_t = r_t - \mu_t$ be the innovation at time t . Then

a_t follows a GARCH(m, s) model if

$$a_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = \alpha_0 + \sum_{i=1}^m \alpha_i a_{t-i}^2 + \sum_{j=1}^s \beta_j \sigma_{t-j}^2, \quad (3.14)$$

where again $\{\epsilon_t\}$ is a sequence of iid random variables with mean 0 and variance 1.0, $\alpha_0 > 0$, $\alpha_i \geq 0$, $\beta_j \geq 0$, and $\sum_{i=1}^{\max(m,s)} (\alpha_i + \beta_i) < 1$. Here it is understood that $\alpha_i = 0$ for $i > m$ and $\beta_j = 0$ for $j > s$. The latter constraint on $\alpha_i + \beta_i$ implies that the unconditional variance of a_t is finite, whereas its conditional variance σ_t^2 evolves over time. As before, ϵ_t is often assumed to follow a standard normal or standardized Student- t distribution or generalized error distribution. Equation (3.14) reduces to a pure ARCH(m) model if $s = 0$. The α_i and β_j are referred to as ARCH and GARCH parameters, respectively.

To understand properties of GARCH models, it is informative to use the following representation. Let $\eta_t = a_t^2 - \sigma_t^2$ so that $\sigma_t^2 = a_t^2 - \eta_t$. By plugging $\sigma_{t-i}^2 = a_{t-i}^2 - \eta_{t-i}$ ($i = 0, \dots, s$) into Eq. (3.14), we can rewrite the GARCH model as

$$a_t^2 = \alpha_0 + \sum_{i=1}^{\max(m,s)} (\alpha_i + \beta_i) a_{t-i}^2 + \eta_t - \sum_{j=1}^s \beta_j \eta_{t-j}. \quad (3.15)$$

It is easy to check that $\{\eta_t\}$ is a martingale difference series [i.e., $E(\eta_t) = 0$ and $\text{cov}(\eta_t, \eta_{t-j}) = 0$ for $j \geq 1$]. However, $\{\eta_t\}$ in general is not an iid sequence. Equation (3.15) is an ARMA form for the squared series a_t^2 . Thus, a GARCH model can be regarded as an application of the ARMA idea to the squared series a_t^2 . Using the unconditional mean of an ARMA model, we have

$$E(a_t^2) = \frac{\alpha_0}{1 - \sum_{i=1}^{\max(m,s)} (\alpha_i + \beta_i)}$$

provided that the denominator of the prior fraction is positive.

The strengths and weaknesses of GARCH models can easily be seen by focusing on the simplest GARCH(1,1) model with

$$\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \quad 0 \leq \alpha_1, \beta_1 \leq 1, (\alpha_1 + \beta_1) < 1. \quad (3.16)$$

First, a large a_{t-1}^2 or σ_{t-1}^2 gives rise to a large σ_t^2 . This means that a large a_{t-1}^2 tends to be followed by another large a_t^2 , generating, again, the well-known behavior of volatility clustering in financial time series. Second, it can be shown that if $1 - 2\alpha_1^2 - (\alpha_1 + \beta_1)^2 > 0$, then

$$\frac{E(a_t^4)}{[E(a_t^2)]^2} = \frac{3[1 - (\alpha_1 + \beta_1)^2]}{1 - (\alpha_1 + \beta_1)^2 - 2\alpha_1^2} > 3.$$

Consequently, similar to ARCH models, the tail distribution of a GARCH(1,1) process is heavier than that of a normal distribution. Third, the model provides a simple parametric function that can be used to describe the volatility evolution.

Forecasts of a GARCH model can be obtained using methods similar to those of an ARMA model. Consider the GARCH(1,1) model in Eq. (3.16) and assume that the forecast origin is h . For 1-step-ahead forecast, we have

$$\sigma_{h+1}^2 = \alpha_0 + \alpha_1 a_h^2 + \beta_1 \sigma_h^2,$$

where a_h and σ_h^2 are known at the time index h . Therefore, the 1-step-ahead forecast is

$$\sigma_h^2(1) = \alpha_0 + \alpha_1 a_h^2 + \beta_1 \sigma_h^2.$$

For multistep-ahead forecasts, we use $a_t^2 = \sigma_t^2 \epsilon_t^2$ and rewrite the volatility equation in Eq. (3.16) as

$$\sigma_{t+1}^2 = \alpha_0 + (\alpha_1 + \beta_1) \sigma_t^2 + \alpha_1 \sigma_t^2 (\epsilon_t^2 - 1).$$

When $t = h + 1$, the equation becomes

$$\sigma_{h+2}^2 = \alpha_0 + (\alpha_1 + \beta_1) \sigma_{h+1}^2 + \alpha_1 \sigma_{h+1}^2 (\epsilon_{h+1}^2 - 1).$$

Since $E(\epsilon_{h+1}^2 - 1 | F_h) = 0$, the 2-step-ahead volatility forecast at the forecast origin h satisfies the equation

$$\sigma_h^2(2) = \alpha_0 + (\alpha_1 + \beta_1) \sigma_h^2(1).$$

In general, we have

$$\sigma_h^2(\ell) = \alpha_0 + (\alpha_1 + \beta_1) \sigma_h^2(\ell - 1), \quad \ell > 1. \quad (3.17)$$

This result is exactly the same as that of an ARMA(1,1) model with AR polynomial $1 - (\alpha_1 + \beta_1)B$. By repeated substitutions in Eq. (3.17), we obtain that the ℓ -step-ahead forecast can be written as

$$\sigma_h^2(\ell) = \frac{\alpha_0 [1 - (\alpha_1 + \beta_1)^{\ell-1}]}{1 - \alpha_1 - \beta_1} + (\alpha_1 + \beta_1)^{\ell-1} \sigma_h^2(1).$$

Therefore,

$$\sigma_h^2(\ell) \rightarrow \frac{\alpha_0}{1 - \alpha_1 - \beta_1}, \quad \text{as } \ell \rightarrow \infty$$

provided that $\alpha_1 + \beta_1 < 1$. Consequently, the multistep-ahead volatility forecasts of a GARCH(1,1) model converge to the unconditional variance of a_t as the forecast horizon increases to infinity provided that $\text{Var}(a_t)$ exists.

The literature on GARCH models is enormous; see Bollerslev, Chou, and Kroner (1992), Bollerslev, Engle, and Nelson (1994), and the references therein. The model encounters the same weaknesses as the ARCH model. For instance, it responds equally to positive and negative shocks. In addition, recent empirical studies of high-frequency financial time series indicate that the tail behavior of GARCH models remains too short even with standardized Student- t innovations. For further information about kurtosis of GARCH models, see Section 3.16.

3.5.1 An Illustrative Example

The modeling procedure of ARCH models can also be used to build a GARCH model. However, specifying the order of a GARCH model is not easy. Only lower order GARCH models are used in most applications, say, GARCH(1,1), GARCH(2,1), and GARCH(1,2) models. The conditional maximum-likelihood method continues to apply provided that the starting values of the volatility $\{\sigma_t^2\}$ are assumed to be known. Consider, for instance, a GARCH(1,1) model. If σ_1^2 is treated as fixed, then σ_t^2 can be computed recursively for a GARCH(1,1) model. In some applications, the sample variance of a_t serves as a good starting value of σ_1^2 . The fitted model can be checked by using the standardized residual $\tilde{a}_t = a_t/\sigma_t$ and its squared process.

Example 3.3. In this example, we consider the monthly excess returns of S&P 500 index starting from 1926 for 792 observations. The series is shown in Figure 3.6. Denote the excess return series by r_t . Figure 3.7 shows the sample ACF

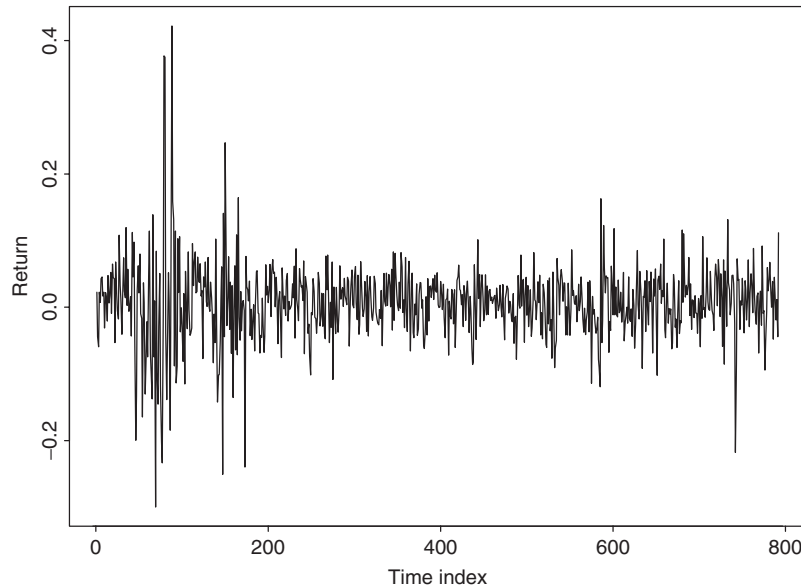


Figure 3.6 Time series plot of monthly excess returns of S&P 500 index from 1926 to 1991.

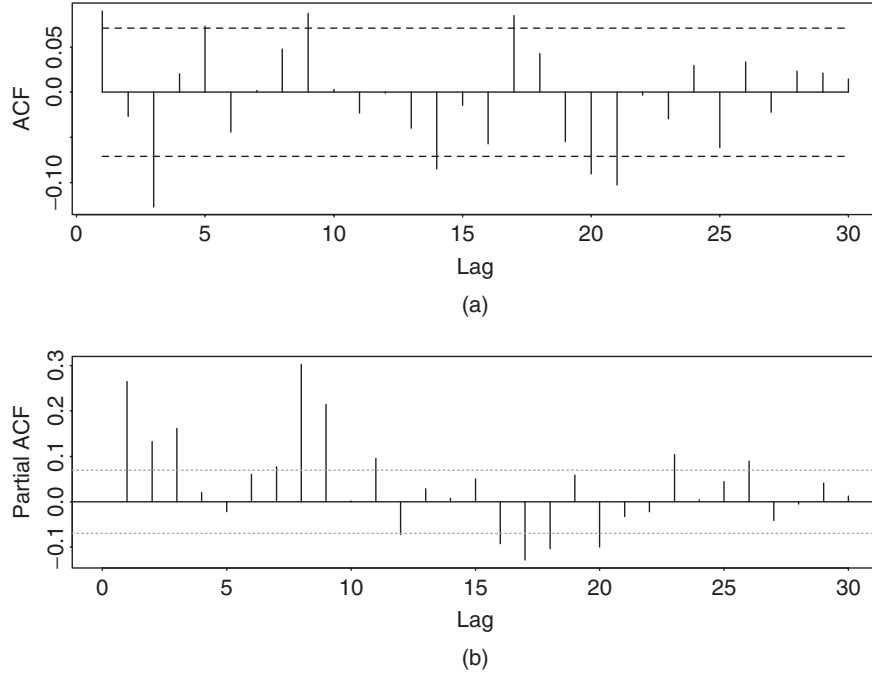


Figure 3.7 (a) Sample ACF of monthly excess returns of S&P 500 index and (b) sample PACF of squared monthly excess returns. Sample period is from 1926 to 1991.

of r_t and the sample PACF of r_t^2 . The r_t series has some serial correlations at lags 1 and 3, but the key feature is that the PACF of r_t^2 shows strong linear dependence. If an MA(3) model is entertained, we obtain

$$r_t = 0.0062 + a_t + 0.0944a_{t-1} - 0.1407a_{t-3}, \quad \hat{\sigma}_a = 0.0576$$

for the series, where all of the coefficients are significant at the 5% level. However, for simplicity, we use instead an AR(3) model

$$r_t = \phi_1 r_{t-1} + \phi_2 r_{t-2} + \phi_3 r_{t-3} + \beta_0 + a_t.$$

The fitted AR(3) model, under the normality assumption, is

$$r_t = 0.088r_{t-1} - 0.023r_{t-2} - 0.123r_{t-3} + 0.0066 + a_t, \quad \hat{\sigma}_a^2 = 0.00333. \quad (3.18)$$

For the GARCH effects, we use the GARCH(1,1) model

$$a_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = \alpha_0 + \beta_1 \sigma_{t-1}^2 + \alpha_1 a_{t-1}^2.$$

A joint estimation of the AR(3)–GARCH(1,1) model gives

$$r_t = 0.0078 + 0.032r_{t-1} - 0.029r_{t-2} - 0.008r_{t-3} + a_t,$$

$$\sigma_t^2 = 0.000084 + 0.1213a_{t-1}^2 + 0.8523\sigma_{t-1}^2.$$

From the volatility equation, the implied unconditional variance of a_t is

$$\frac{0.000084}{1 - 0.8523 - 0.1213} = 0.00317,$$

which is close to that of Eq. (3.18). However, t ratios of the parameters in the mean equation suggest that all three AR coefficients are insignificant at the 5% level. Therefore, we refine the model by dropping all AR parameters. The refined model is

$$r_t = 0.0076 + a_t, \quad \sigma_t^2 = 0.000086 + 0.1216a_{t-1}^2 + 0.8511\sigma_{t-1}^2. \quad (3.19)$$

The standard error of the constant in the mean equation is 0.0015, whereas those of the parameters in the volatility equation are 0.000024, 0.0197, and 0.0190, respectively. The unconditional variance of a_t is $0.000086/(1 - 0.8511 - 0.1216) = 0.00314$. This is a simple stationary GARCH(1,1) model. Figure 3.8 shows the estimated volatility process, σ_t , and the standardized shocks $\tilde{a}_t = a_t/\sigma_t$ for the

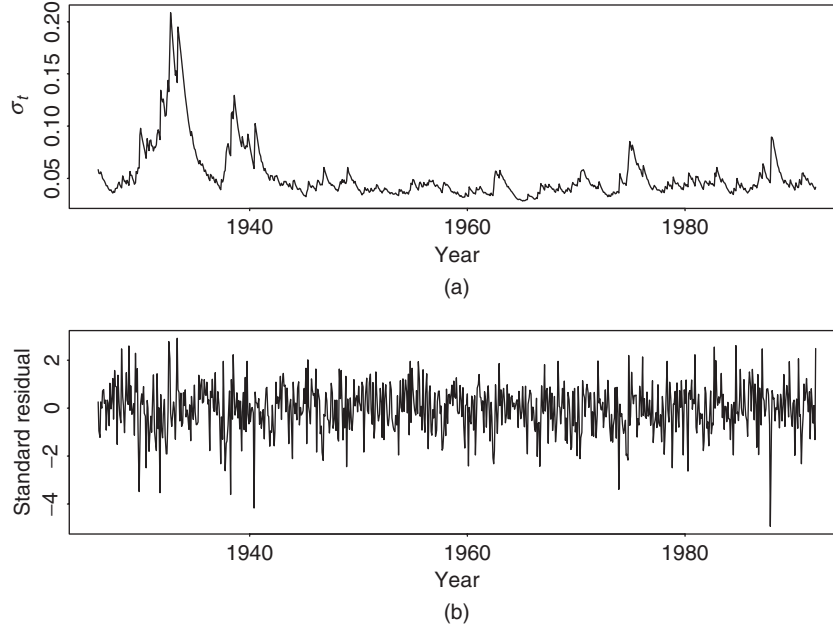


Figure 3.8 (a) Time series plot of estimated volatility (σ_t) for monthly excess returns of S&P 500 index and (b) standardized shocks of monthly excess returns of S&P 500 index. Both plots are based on GARCH(1,1) model in Eq. (3.19).

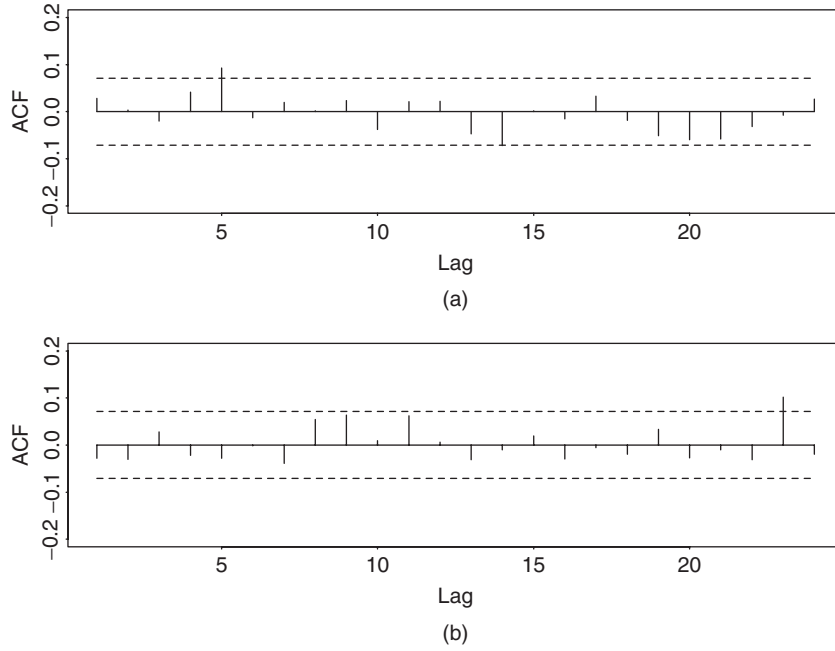


Figure 3.9 Model checking of GARCH(1,1) model in Eq. (3.19) for monthly excess returns of S&P 500 index: (a) Sample ACF of standardized residuals and (b) sample ACF of the squared standardized residuals.

GARCH(1,1) model in Eq. (3.19). The $\tilde{a}_t$ series looks like a white noise process. Figure 3.9 provides the sample ACF of the standardized residuals $\tilde{a}_t$ and the squared process $\tilde{a}_t^2$. These ACFs fail to suggest any significant serial correlations or conditional heteroscedasticity in the standardized residual series. More specifically, we have $Q(12) = 11.99(0.45)$ and $Q(24) = 28.52(0.24)$ for $\tilde{a}_t$, and $Q(12) = 13.11(0.36)$ and $Q(24) = 26.45(0.33)$ for $\tilde{a}_t^2$, where the number in parentheses is the p value of the test statistic. Thus, the model appears to be adequate in describing the linear dependence in the return and volatility series. Note that the fitted model shows $\hat{\alpha}_1 + \hat{\beta}_1 = 0.9772$, which is close to 1. This phenomenon is commonly observed in practice and it leads to imposing the constraint $\alpha_1 + \beta_1 = 1$ in a GARCH(1,1) model, resulting in an integrated GARCH (or IGARCH) model; see Section 3.6.

Finally, to forecast the volatility of monthly excess returns of the S&P 500 index, we can use the volatility equation in Eq. (3.19). For instance, at the forecast origin h , we have $\sigma_{h+1}^2 = 0.000086 + 0.1216a_h^2 + 0.8511\sigma_h^2$. The 1-step-ahead forecast is then

$$\sigma_h^2(1) = 0.000086 + 0.1216a_h^2 + 0.8511\sigma_h^2,$$

TABLE 3.1 Volatility Forecasts for Monthly Excess Returns of S&P 500 Index^a

Horizon	1	2	3	4	5	∞
Return	0.0076	0.0076	0.0076	0.0076	0.0076	0.0076
Volatility	0.0536	0.0537	0.0537	0.0538	0.0538	0.0560

^aThe forecast origin is $h = 792$, which corresponds to December 1991. Here volatility denotes conditional standard deviation.

where a_h is the residual of the mean equation at time h and σ_h is obtained from the volatility equation. The starting value σ_0^2 is fixed at either zero or the unconditional variance of a_t . For multistep-ahead forecasts, we use the recursive formula in Eq. (3.17). Table 3.1 shows some mean and volatility forecasts for the monthly excess return of the S&P 500 index with forecast origin $h = 792$ based on the GARCH(1,1) model in Eq. (3.19).

Some S-Plus Commands Used in Example 3.3.

```
> fit=garch(sp~ar(3),~garch(1,1))
> summary(fit)
> fit=garch(sp~1,~garch(1,1))
> summary(fit)
> names(fit)
[1] "residuals" "sigma.t"      "df.residual" "coef" "model"
[6] "cond.dist" "likelihood" "opt.index"   "cov"
    "prediction"
[11] "call"      "asympt.sd"   "series"
>
> stdresi=fit$residuals/fit$sigma.t
> autocorTest(stdresi,lag=24)
> autocorTest(stdresi^2,lag=24)
> predict(fit,5)
```

Note that in the prior commands the volatility series σ_t is stored in `fit$sigma.t` and the residual series of the returns in `fit$residuals`.

t Innovation

Assuming that ϵ_t follows a standardized Student- t distribution with 5 degrees of freedom, we reestimate the GARCH(1,1) model and obtain

$$r_t = 0.0085 + a_t, \quad \sigma_t^2 = 0.00012 + 0.1121a_{t-1}^2 + 0.8432\sigma_{t-1}^2, \quad (3.20)$$

where the standard errors of the parameters are 0.0015, 0.51×10^{-4} , 0.0296, and 0.0371, respectively. This model is essentially an IGARCH(1,1) model as $\hat{\alpha}_1 + \hat{\beta}_1 \approx 0.95$, which is close to 1. The Ljung–Box statistics of the standardized residuals give $Q(10) = 11.38$ with a p value of 0.33 and those of the $\{\tilde{a}_t^2\}$ series give $Q(10) = 10.48$ with a p value of 0.40. Thus, the fitted GARCH(1,1) model with Student- t distribution is adequate.

S-Plus Commands Used

```

> fit1 = garch(sp~1, ~garch(1,1), cond.dist='t', cond.par=5,
+ cond.est=F)
> summary(fit1)
> stres1=fit1$residuals/fit1$sigma.t
> autocorTest(stres1, lag=10)
> autocorTest(stres1^2, lag=10)

```

Estimation of Degrees of Freedom

If we further extend the GARCH(1,1) model by estimating the degrees of freedom of the Student-*t* distribution used, we obtain the model

$$r_t = 0.0085 + a_t, \quad \sigma_t^2 = 0.00012 + 0.1121a_{t-1}^2 + 0.8432\sigma_{t-1}^2, \quad (3.21)$$

where the estimated degrees of freedom is 7.02. Standard errors of the estimates in Eq. (3.21) are close to those in Eq. (3.20). The standard error of the estimated degrees of freedom is 1.78. Consequently, we cannot reject the hypothesis of using a standardized Student-*t* distribution with 5 degrees of freedom at the 5% significance level.

S-Plus Commands Used

```

> fit2 = garch(sp~1, ~garch(1,1), cond.dist='t')
> summary(fit2)

```

R Commands Used in Example 3.3

```

> library(fGarch)
> sp5=scan(file='sp500.txt') % Load data
> plot(sp5, type='l')
% Below, fit an AR(3)+GARCH(1,1) model.
> m1=garchFit(~arma(3,0)+garch(1,1), data=sp5, trace=F)
> summary(m1)
% Below, fit a GARCH(1,1) model with Student-t distribution.
> m2=garchFit(~garch(1,1), data=sp5, trace=F, cond.dist="std")
> summary(m2)
% Obtain standardized residuals.
> stres1=residuals(m2, standardize=T)
> plot(stres1, type='l')
> Box.test(stres1, 10, type='Ljung')
> predict(m2, 5)

```

3.5.2 Forecasting Evaluation

Since the volatility of an asset return is not directly observable, comparing the forecasting performance of different volatility models is a challenge to data analysts. In the literature, some researchers use out-of-sample forecasts and compare the

volatility forecasts $\sigma_h^2(\ell)$ with the shock $a_{h+\ell}^2$ in the forecasting sample to assess the forecasting performance of a volatility model. This approach often finds a low correlation coefficient between $a_{h+\ell}^2$ and $\sigma_h^2(\ell)$, that is, low R^2 . However, such a finding is not surprising because $a_{h+\ell}^2$ alone is not an adequate measure of the volatility at time index $h + \ell$. Consider the 1-step-ahead forecasts. From a statistical point of view, $E(a_{h+1}^2|F_h) = \sigma_{h+1}^2$ so that a_{h+1}^2 is a consistent estimate of σ_{h+1}^2 . But it is not an accurate estimate of σ_{h+1}^2 because a single observation of a random variable with a known mean value cannot provide an accurate estimate of its variance. Consequently, such an approach to evaluate forecasting performance of volatility models is strictly speaking not proper. For more information concerning forecasting evaluation of GARCH models, readers are referred to Andersen and Bollerslev (1998).

3.5.3 A Two-Pass Estimation Method

Based on Eq. (3.15), a two-pass estimation method can be used to estimate GARCH models. First, ignoring any ARCH effects, one estimates the mean equation of a return series using the methods discussed in Chapter 2 (e.g., maximum-likelihood method). Denote the residual series by a_t . Second, treating $\{a_t^2\}$ as an observed time series, one applies the maximum-likelihood method to estimate parameters of Eq. (3.15). Denote the AR and MA coefficient estimates by $\hat{\phi}_i$ and $\hat{\theta}_i$. The GARCH estimates are obtained as $\hat{\beta}_i = \hat{\theta}_i$ and $\hat{\alpha}_i = \hat{\phi}_i - \hat{\theta}_i$. Obviously, such estimates are approximations to the true parameters and their statistical properties have not been rigorously investigated. However, limited experience shows that this simple approach often provides good approximations, especially when the sample size is moderate or large. For instance, consider the monthly excess return series of the S&P 500 index of Example 3.3. Using the conditional MLE method in SCA, we obtain the model

$$r_t = 0.0061 + a_t, \quad a_t^2 = 0.00014 + 0.9583a_{t-1}^2 + \eta_t - 0.8456\eta_{t-1},$$

where all estimates are significantly different from zero at the 5% level. From the estimates, we have $\hat{\beta}_1 = 0.8456$ and $\hat{\alpha}_1 = 0.9583 - 0.8456 = 0.1127$. These approximate estimates are very close to those in Eq. (3.19) or (3.21). Furthermore, the fitted volatility series of the two-pass method is very close to that of Figure 3.8(a).

3.6 THE INTEGRATED GARCH MODEL

If the AR polynomial of the GARCH representation in Eq. (3.15) has a unit root, then we have an IGARCH model. Thus, IGARCH models are unit-root GARCH models. Similar to ARIMA models, a key feature of IGARCH models is that the impact of past squared shocks $\eta_{t-i} = a_{t-i}^2 - \sigma_{t-i}^2$ for $i > 0$ on a_t^2 is persistent.

An IGARCH(1,1) model can be written as

$$a_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = \alpha_0 + \beta_1 \sigma_{t-1}^2 + (1 - \beta_1) a_{t-1}^2,$$

where $\{\epsilon_t\}$ is defined as before and $1 > \beta_1 > 0$. For the monthly excess returns of the S&P 500 index, an estimated IGARCH(1,1) model is

$$\begin{aligned} r_t &= 0.0067 + a_t, & a_t &= \sigma_t \epsilon_t, \\ \sigma_t^2 &= 0.000119 + 0.8059 \sigma_{t-1}^2 + 0.1941 a_{t-1}^2, \end{aligned}$$

where the standard errors of the estimates in the volatility equation are 0.0017, 0.000013, and 0.0144, respectively. The parameter estimates are close to those of the GARCH(1,1) model shown before, but there is a major difference between the two models. The unconditional variance of a_t , hence that of r_t , is not defined under the above IGARCH(1,1) model. This seems hard to justify for an excess return series. From a theoretical point of view, the IGARCH phenomenon might be caused by occasional level shifts in volatility. The actual cause of persistence in volatility deserves a careful investigation.

When $\alpha_1 + \beta_1 = 1$, repeated substitutions in Eq. (3.17) give

$$\sigma_h^2(\ell) = \sigma_h^2(1) + (\ell - 1)\alpha_0, \quad \ell \geq 1, \quad (3.22)$$

where h is the forecast origin. Consequently, the effect of $\sigma_h^2(1)$ on future volatilities is also persistent, and the volatility forecasts form a straight line with slope α_0 . Nelson (1990) studies some probability properties of the volatility process σ_t^2 under an IGARCH model. The process σ_t^2 is a martingale for which some nice results are available in the literature. Under certain conditions, the volatility process is strictly stationary but not weakly stationary because it does not have the first two moments.

The case of $\alpha_0 = 0$ is of particular interest in studying the IGARCH(1,1) model. In this case, the volatility forecasts are simply $\sigma_h^2(1)$ for all forecast horizons; see Eq. (3.22). This special IGARCH(1,1) model is the volatility model used in RiskMetrics, which is an approach for calculating value at risk; see Chapter 7. The model is also an exponential smoothing model for the $\{a_t^2\}$ series. To see this, rewrite the model as

$$\begin{aligned} \sigma_t^2 &= (1 - \beta_1) a_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \\ &= (1 - \beta_1) a_{t-1}^2 + \beta_1 [(1 - \beta_1) a_{t-2}^2 + \beta_1 \sigma_{t-2}^2] \\ &= (1 - \beta_1) a_{t-1}^2 + (1 - \beta_1) \beta_1 a_{t-2}^2 + \beta_1^2 \sigma_{t-2}^2. \end{aligned}$$

By repeated substitutions, we have

$$\sigma_t^2 = (1 - \beta_1)(a_{t-1}^2 + \beta_1 a_{t-2}^2 + \beta_1^2 a_{t-3}^2 + \cdots),$$

which is the well-known exponential smoothing formation with β_1 being the discounting factor. Exponential smoothing methods can thus be used to estimate such an IGARCH(1,1) model.

3.7 THE GARCH-M MODEL

In finance, the return of a security may depend on its volatility. To model such a phenomenon, one may consider the GARCH-M model, where M stands for GARCH *in the mean*. A simple GARCH(1,1)-M model can be written as

$$\begin{aligned} r_t &= \mu + c\sigma_t^2 + a_t, & a_t &= \sigma_t\epsilon_t, \\ \sigma_t^2 &= \alpha_0 + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \end{aligned} \quad (3.23)$$

where μ and c are constants. The parameter c is called the risk premium parameter. A positive c indicates that the return is positively related to its volatility. Other specifications of risk premium have also been used in the literature, including $r_t = \mu + c\sigma_t + a_t$ and $r_t = \mu + c \ln(\sigma_t^2) + a_t$.

The formulation of the GARCH-M model in Eq. (3.23) implies that there are serial correlations in the return series r_t . These serial correlations are introduced by those in the volatility process $\{\sigma_t^2\}$. The existence of risk premium is, therefore, another reason that some historical stock returns have serial correlations.

For illustration, we consider a GARCH(1,1)-M model with Gaussian innovations for the monthly excess returns of the S&P 500 index from January 1926 to December 1991. The fitted model is

$$r_t = 0.0055 + 1.09\sigma_t^2 + a_t, \quad \sigma_t^2 = 8.76 \times 10^{-5} + 0.123a_{t-1}^2 + 0.849\sigma_{t-1}^2,$$

where the standard errors for the two parameters in the mean equation are 0.0023 and 0.818, respectively, and those for the parameters in the volatility equation are 2.51×10^{-5} , 0.0205, and 0.0196, respectively. The estimated risk premium for the index return is positive but is not statistically significant at the 5% level. Here the result is obtained using S-Plus. Other forms of GARCH-M specification in S-Plus are given in Table 3.2. The idea of risk premium applies to other GARCH models.

TABLE 3.2 GARCH-M Models Allowed in S-Plus:
Mean Equation Is $r_t = \mu + cg(\sigma_t) + a_t$

$g(\sigma_t)$	Command
σ_t^2	var.in.mean
σ_t	sd.in.mean
$\ln(\sigma_t^2)$	logvar.in.mean

S-Plus Demonstration

```
> sp.fit = garch(sp~1+var.in.mean, ~garch(1,1))
> summary(sp.fit)
```

3.8 THE EXPONENTIAL GARCH MODEL

To overcome some weaknesses of the GARCH model in handling financial time series, Nelson (1991) proposes the exponential GARCH (EGARCH) model. In particular, to allow for asymmetric effects between positive and negative asset returns, he considered the weighted innovation

$$g(\epsilon_t) = \theta\epsilon_t + \gamma[|\epsilon_t| - E(|\epsilon_t|)], \quad (3.24)$$

where θ and γ are real constants. Both ϵ_t and $|\epsilon_t| - E(|\epsilon_t|)$ are zero-mean iid sequences with continuous distributions. Therefore, $E[g(\epsilon_t)] = 0$. The asymmetry of $g(\epsilon_t)$ can easily be seen by rewriting it as

$$g(\epsilon_t) = \begin{cases} (\theta + \gamma)\epsilon_t - \gamma E(|\epsilon_t|) & \text{if } \epsilon_t \geq 0, \\ (\theta - \gamma)\epsilon_t - \gamma E(|\epsilon_t|) & \text{if } \epsilon_t < 0. \end{cases}$$

Remark. For the standard Gaussian random variable ϵ_t , $E(|\epsilon_t|) = \sqrt{2/\pi}$. For the standardized Student- t distribution in Eq. (3.7), we have

$$E(|\epsilon_t|) = \frac{2\sqrt{v-2}\Gamma[(v+1)/2]}{(v-1)\Gamma(v/2)\sqrt{\pi}}.$$

□

An EGARCH(m, s) model can be written as

$$a_t = \sigma_t \epsilon_t, \quad \ln(\sigma_t^2) = \alpha_0 + \frac{1 + \beta_1 B + \cdots + \beta_{s-1} B^{s-1}}{1 - \alpha_1 B - \cdots - \alpha_m B^m} g(\epsilon_{t-1}), \quad (3.25)$$

where α_0 is a constant, B is the back-shift (or lag) operator such that $Bg(\epsilon_t) = g(\epsilon_{t-1})$, and $1 + \beta_1 B + \cdots + \beta_{s-1} B^{s-1}$ and $1 - \alpha_1 B - \cdots - \alpha_m B^m$ are polynomials with zeros outside the unit circle and have no common factors. By outside the unit circle we mean that absolute values of the zeros are greater than 1. Again, Eq. (3.25) uses the usual ARMA parameterization to describe the evolution of the conditional variance of a_t . Based on this representation, some properties of the EGARCH model can be obtained in a similar manner as those of the GARCH model. For instance, the unconditional mean of $\ln(\sigma_t^2)$ is α_0 . However, the model differs from the GARCH model in several ways. First, it uses logged conditional variance to relax the positiveness constraint of model coefficients. Second, the use of $g(\epsilon_t)$ enables the model to respond asymmetrically to positive and negative

lagged values of a_t . Some additional properties of the EGARCH model can be found in Nelson (1991).

To better understand the EGARCH model, let us consider the simple model with order (1,1):

$$a_t = \sigma_t \epsilon_t, \quad (1 - \alpha B) \ln(\sigma_t^2) = (1 - \alpha)\alpha_0 + g(\epsilon_{t-1}), \quad (3.26)$$

where the ϵ_t are iid standard normal and the subscript of α_1 is omitted. In this case, $E(|\epsilon_t|) = \sqrt{2/\pi}$ and the model for $\ln(\sigma_t^2)$ becomes

$$(1 - \alpha B) \ln(\sigma_t^2) = \begin{cases} \alpha_* + (\gamma + \theta)\epsilon_{t-1} & \text{if } \epsilon_{t-1} \geq 0, \\ \alpha_* + (\gamma - \theta)(-\epsilon_{t-1}) & \text{if } \epsilon_{t-1} < 0, \end{cases} \quad (3.27)$$

where $\alpha_* = (1 - \alpha)\alpha_0 - \sqrt{2/\pi}\gamma$. This is a nonlinear function similar to that of the threshold autoregressive (TAR) model of Tong (1978, 1990). It suffices to say that for this simple EGARCH model the conditional variance evolves in a nonlinear manner depending on the sign of a_{t-1} . Specifically, we have

$$\sigma_t^2 = \sigma_{t-1}^{2\alpha} \exp(\alpha_*) \begin{cases} \exp\left[(\gamma + \theta)\frac{a_{t-1}}{\sigma_{t-1}}\right] & \text{if } a_{t-1} \geq 0, \\ \exp\left[(\gamma - \theta)\frac{|a_{t-1}|}{\sigma_{t-1}}\right] & \text{if } a_{t-1} < 0. \end{cases}$$

The coefficients $(\gamma + \theta)$ and $(\gamma - \theta)$ show the asymmetry in response to positive and negative a_{t-1} . The model is, therefore, nonlinear if $\theta \neq 0$. Since negative shocks tend to have larger impacts, we expect θ to be negative. For higher order EGARCH models, the nonlinearity becomes much more complicated. Cao and Tsay (1992) use nonlinear models, including EGARCH models, to obtain multistep-ahead volatility forecasts. We discuss nonlinearity in financial time series in Chapter 4.

3.8.1 Alternative Model Form

An alternative form for the EGARCH(m, s) model is

$$\ln(\sigma_t^2) = \alpha_0 + \sum_{i=1}^s \alpha_i \frac{|a_{t-i}| + \gamma_i a_{t-i}}{\sigma_{t-i}} + \sum_{j=1}^m \beta_j \ln(\sigma_{t-j}^2). \quad (3.28)$$

Here a positive a_{t-i} contributes $\alpha_i(1 + \gamma_i)|\epsilon_{t-i}|$ to the log volatility, whereas a negative a_{t-i} gives $\alpha_i(1 - \gamma_i)|\epsilon_{t-i}|$, where $\epsilon_{t-i} = a_{t-i}/\sigma_{t-i}$. The γ_i parameter thus signifies the leverage effect of a_{t-i} . Again, we expect γ_i to be negative in real applications. This is the model form used in S-Plus.

3.8.2 Illustrative Example

Nelson (1991) applies an EGARCH model to the daily excess returns of the value-weighted market index from the Center for Research in Security Prices from July 1962 to December 1987. The excess returns are obtained by removing monthly Treasury bill returns from the value-weighted index returns, assuming that the Treasury bill return was constant for each calendar day within a given month. There are 6408 observations. Denote the excess return by r_t . The model used is as follows:

$$r_t = \phi_0 + \phi_1 r_{t-1} + c\sigma_t^2 + a_t, \quad (3.29)$$

$$\ln(\sigma_t^2) = \alpha_0 + \ln(1 + wN_t) + \frac{1 + \beta B}{1 - \alpha_1 B - \alpha_2 B^2} g(\epsilon_{t-1}),$$

where σ_t^2 is the conditional variance of a_t given F_{t-1} , N_t is the number of nontrading days between trading days $t-1$ and t , α_0 and w are real parameters, $g(\epsilon_t)$ is defined in Eq. (3.24), and ϵ_t follows a generalized error distribution in Eq. (3.10). Similar to a GARCH-M model, the parameter c in Eq. (3.29) is the risk premium parameter. Table 3.3 gives the parameter estimates and their standard errors of the model. The mean equation of model (3.29) has two features that are of interest. First, it uses an AR(1) model to take care of possible serial correlation in the excess returns. Second, it uses the volatility σ_t^2 as a regressor to account for risk premium. The estimated risk premium is negative, but statistically insignificant.

3.8.3 Second Example

As another illustration, we consider the monthly log returns of IBM stock from January 1926 to December 1997 for 864 observations. An AR(1)–EGARCH(1,1) model is entertained and the fitted model is

$$r_t = 0.0105 + 0.092r_{t-1} + a_t, \quad a_t = \sigma_t \epsilon_t, \quad (3.30)$$

$$\ln(\sigma_t^2) = -5.496 + \frac{g(\epsilon_{t-1})}{1 - 0.856B},$$

$$g(\epsilon_{t-1}) = -0.0795\epsilon_{t-1} + 0.2647 \left(|\epsilon_{t-1}| - \sqrt{2/\pi} \right), \quad (3.31)$$

TABLE 3.3 Estimated AR(1)–EGARCH(2,2) Model for Daily Excess Returns of Value-Weighted CRSP Market Index: July 1962–December 1987

Parameter	α_0	w	γ	α_1	α_2	β
Estimate	-10.06	0.183	0.156	1.929	-0.929	-0.978
Error	0.346	0.028	0.013	0.015	0.015	0.006
Parameter	θ	ϕ_0	ϕ_1	c	v	
Estimate	-0.118	$3.5 \cdot 10^{-4}$	0.205	-3.361	1.576	
Error	0.009	$9.9 \cdot 10^{-5}$	0.012	2.026	0.032	

where $\{\epsilon_t\}$ is a sequence of independent standard Gaussian random variates. All parameter estimates are statistically significant at the 5% level. For model checking, the Ljung–Box statistics give $Q(10) = 6.31(0.71)$ and $Q(20) = 21.4(0.32)$ for the standardized residual process $\tilde{a}_t = a_t/\sigma_t$ and $Q(10) = 4.13(0.90)$ and $Q(20) = 15.93(0.66)$ for the squared process $\tilde{a}_t^2$, where again the number in parentheses denotes p value. Therefore, there is no serial correlation or conditional heteroscedasticity in the standardized residuals of the fitted model. The prior AR(1)–EGARCH(1,1) model is adequate.

From the estimated volatility equation in (3.31) and using $\sqrt{2/\pi} \approx 0.7979$, we obtain the volatility equation as

$$\ln(\sigma_t^2) = -1.001 + 0.856 \ln(\sigma_{t-1}^2) + \begin{cases} 0.1852\epsilon_{t-1} & \text{if } \epsilon_{t-1} \geq 0, \\ -0.3442\epsilon_{t-1} & \text{if } \epsilon_{t-1} < 0. \end{cases}$$

Taking antilog transformation, we have

$$\sigma_t^2 = \sigma_{t-1}^{2 \times 0.856} e^{-1.001} \times \begin{cases} e^{0.1852\epsilon_{t-1}} & \text{if } \epsilon_{t-1} \geq 0, \\ e^{-0.3442\epsilon_{t-1}} & \text{if } \epsilon_{t-1} < 0. \end{cases}$$

This equation highlights the asymmetric responses in volatility to the past positive and negative shocks under an EGARCH model. For example, for a standardized shock with magnitude 2 (i.e., two standard deviations), we have

$$\frac{\sigma_t^2(\epsilon_{t-1} = -2)}{\sigma_t^2(\epsilon_{t-1} = 2)} = \frac{\exp[-0.3442 \times (-2)]}{\exp(0.1852 \times 2)} = e^{0.318} = 1.374.$$

Therefore, the impact of a negative shock of size 2 standard deviations is about 37.4% higher than that of a positive shock of the same size. This example clearly demonstrates the asymmetric feature of EGARCH models. In general, the bigger the shock, the larger the difference in volatility impact.

Finally, we extend the sample period to include the log returns from 1998 to 2003 so that there are 936 observations and use S-Plus to fit an EGARCH(1,1) model. The results are given below.

S-Plus Demonstration

The following output has been edited:

```
> ibm.egarch=garch(ibmln~1,~egarch(1,1),leverage=T,
+ cond.dist='ged')

> summary(ibm.egarch)
Call:
garch(formula.mean = ibmln ~ 1, formula.var = ~ egarch(1, 1),
      leverage = T,cond.dist = "ged")

Mean Equation: ibmln ~ 1
```

```

Conditional Variance Equation: ~ egarch(1, 1)
Conditional Distribution:  ged
    with estimated parameter 1.5003 and standard error 0.09912
-----
Estimated Coefficients:
-----
              Value Std.Error t value  Pr(>|t|)
      C  0.01181   0.002012    5.870 3.033e-09
      A -0.55680   0.171602   -3.245 6.088e-04
    ARCH(1)  0.22025   0.052824    4.169 1.669e-05
  GARCH(1)  0.92910   0.026743   34.742 0.000e+00
    LEV(1) -0.26400   0.126096   -2.094 1.828e-02
-----
Ljung-Box test for standardized residuals:
-----
Statistic P-value Chi^2-d.f.
      17.87  0.1195         12

Ljung-Box test for squared standardized residuals:
-----
Statistic P-value Chi^2-d.f.
      6.723  0.8754         12

```

The fitted GARCH(1,1) model is

$$\begin{aligned}
 r_t &= 0.0118 + a_t, & a_t &= \sigma_t \epsilon_t, \\
 \ln(\sigma_t^2) &= -0.557 + 0.220 \frac{|a_{t-1}| - 0.264a_{t-1}}{\sigma_{t-1}} + 0.929 \ln(\sigma_{t-1}^2), \quad (3.32)
 \end{aligned}$$

where ϵ_t follows a GED distribution with parameter 1.5. This model is adequate and based on the Ljung–Box statistics of the standardized residual series and its squared process. As expected, the output shows that the estimated leverage effect is negative and is statistically significant at the 5% level with a t ratio of -2.094 .

3.8.4 Forecasting Using an EGARCH Model

We use the EGARCH(1,1) model to illustrate multistep-ahead forecasts of EGARCH models, assuming that the model parameters are known and the innovations are standard Gaussian. For such a model, we have

$$\begin{aligned}
 \ln(\sigma_t^2) &= (1 - \alpha_1)\alpha_0 + \alpha_1 \ln(\sigma_{t-1}^2) + g(\epsilon_{t-1}), \\
 g(\epsilon_{t-1}) &= \theta\epsilon_{t-1} + \gamma(|\epsilon_{t-1}| - \sqrt{2/\pi}).
 \end{aligned}$$

Taking exponentials, the model becomes

$$\begin{aligned}
 \sigma_t^2 &= \sigma_{t-1}^{2\alpha_1} \exp[(1 - \alpha_1)\alpha_0] \exp[g(\epsilon_{t-1})], \\
 g(\epsilon_{t-1}) &= \theta\epsilon_{t-1} + \gamma(|\epsilon_{t-1}| - \sqrt{2/\pi}). \quad (3.33)
 \end{aligned}$$

Let h be the forecast origin. For the 1-step-ahead forecast, we have

$$\sigma_{h+1}^2 = \sigma_h^{2\alpha_1} \exp[(1 - \alpha_1)\alpha_0] \exp[g(\epsilon_h)],$$

where all of the quantities on the right-hand side are known. Thus, the 1-step-ahead volatility forecast at the forecast origin h is simply $\hat{\sigma}_h^2(1) = \sigma_{h+1}^2$ given earlier. For the 2-step-ahead forecast, Eq. (3.33) gives

$$\sigma_{h+2}^2 = \sigma_{h+1}^{2\alpha_1} \exp[(1 - \alpha_1)\alpha_0] \exp[g(\epsilon_{h+1})].$$

Taking conditional expectation at time h , we have

$$\hat{\sigma}_h^2(2) = \hat{\sigma}_h^{2\alpha_1}(1) \exp[(1 - \alpha_1)\alpha_0] E_h\{\exp[g(\epsilon_{h+1})]\},$$

where E_h denotes a conditional expectation taken at the time origin h . The prior expectation can be obtained as follows:

$$\begin{aligned} E\{\exp[g(\epsilon)]\} &= \int_{-\infty}^{\infty} \exp[\theta\epsilon + \gamma(|\epsilon| - \sqrt{2/\pi})] f(\epsilon) d\epsilon \\ &= \exp\left(-\gamma\sqrt{2/\pi}\right) \left[\int_0^{\infty} e^{(\theta+\gamma)\epsilon} \frac{1}{\sqrt{2\pi}} e^{-\epsilon^2/2} d\epsilon \right. \\ &\quad \left. + \int_{-\infty}^0 e^{(\theta-\gamma)\epsilon} \frac{1}{\sqrt{2\pi}} e^{-\epsilon^2/2} d\epsilon \right] \\ &= \exp\left(-\gamma\sqrt{2/\pi}\right) \left[e^{(\theta+\gamma)^2/2} \Phi(\theta + \gamma) + e^{(\theta-\gamma)^2/2} \Phi(\gamma - \theta) \right], \end{aligned}$$

where $f(\epsilon)$ and $\Phi(x)$ are the probability density function and CDF of the standard normal distribution, respectively. Consequently, the 2-step-ahead volatility forecast is

$$\begin{aligned} \hat{\sigma}_h^2(2) &= \hat{\sigma}_h^{2\alpha_1}(1) \exp\left[(1 - \alpha_1)\alpha_0 - \gamma\sqrt{2/\pi}\right] \\ &\quad \times \left\{ \exp[(\theta + \gamma)^2/2] \Phi(\theta + \gamma) + \exp[(\theta - \gamma)^2/2] \Phi(\gamma - \theta) \right\}. \end{aligned}$$

Repeating the previous procedure, we obtain a recursive formula for a j -step-ahead forecast:

$$\begin{aligned} \hat{\sigma}_h^2(j) &= \widehat{\sigma}_h^{2\alpha_1}(j-1) \exp(\omega) \\ &\quad \times \left\{ \exp[(\theta + \gamma)^2/2] \Phi(\theta + \gamma) + \exp[(\theta - \gamma)^2/2] \Phi(\gamma - \theta) \right\}, \end{aligned}$$

where $\omega = (1 - \alpha_1)\alpha_0 - \gamma\sqrt{2/\pi}$. The values of $\Phi(\theta + \gamma)$ and $\Phi(\gamma - \theta)$ can be obtained from most statistical packages. Alternatively, accurate approximations to these values can be obtained by using the method in Appendix B of Chapter 6.

For illustration, consider the AR(1)–EGARCH(1,1) model of the previous section for the monthly log returns of IBM stock, ending December 1997. Using the fitted EGARCH(1,1) model, we can compute the volatility forecasts for the series. At the forecast origin $t = 864$, the forecasts are $\hat{\sigma}_{864}^2(1) = 6.05 \times 10^{-3}$, $\hat{\sigma}_{864}^2(2) = 5.82 \times 10^{-3}$, $\hat{\sigma}_{864}^2(3) = 5.63 \times 10^{-3}$, and $\hat{\sigma}_{864}^2(10) = 4.94 \times 10^{-3}$. These forecasts converge gradually to the sample variance 4.37×10^{-3} of the shock process a_t of Eq. (3.30).

3.9 THE THRESHOLD GARCH MODEL

Another volatility model commonly used to handle leverage effects is the threshold GARCH (or TGARCH) model; see Glosten, Jagannathan, and Runkle (1993) and Zakoian (1994). A TGARCH(m, s) model assumes the form

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^s (\alpha_i + \gamma_i N_{t-i}) a_{t-i}^2 + \sum_{j=1}^m \beta_j \sigma_{t-j}^2, \quad (3.34)$$

where N_{t-i} is an indicator for *negative* a_{t-i} , that is,

$$N_{t-i} = \begin{cases} 1 & \text{if } a_{t-i} < 0, \\ 0 & \text{if } a_{t-i} \geq 0, \end{cases}$$

and α_i , γ_i , and β_j are nonnegative parameters satisfying conditions similar to those of GARCH models. From the model, it is seen that a positive a_{t-i} contributes $\alpha_i a_{t-i}^2$ to σ_t^2 , whereas a negative a_{t-i} has a larger impact $(\alpha_i + \gamma_i) a_{t-i}^2$ with $\gamma_i > 0$. The model uses zero as its *threshold* to separate the impacts of past shocks. Other threshold values can also be used; see Chapter 4 for the general concept of threshold models. Model (3.34) is also called the GJR model because Glosten et al. (1993) proposed essentially the same model.

For illustration, consider the monthly log returns of IBM stock from 1926 to 2003. The fitted TGARCH(1,1) model with conditional GED innovations is

$$\begin{aligned} r_t &= 0.0121 + a_t, & a_t &= \sigma_t \epsilon_t, \\ \sigma_t^2 &= 3.45 \times 10^{-4} + (0.0658 + 0.0843 N_{t-1}) a_{t-1}^2 + 0.8182 \sigma_{t-1}^2, \end{aligned} \quad (3.35)$$

where the estimated parameter of the GED is 1.51 with standard error 0.099. The standard error of the parameter for the mean equation is 0.002 and the standard errors of the parameters in the volatility equation are 1.26×10^{-4} , 0.0314, 0.0395, and 0.049, respectively. To check the fitted model, we have $Q(12) = 18.34(0.106)$ for the standardized residual $\tilde{a}_t$ and $Q(12) = 5.36(0.95)$ for $\tilde{a}_t^2$. The model is adequate in modeling the first two conditional moments of the log return series. Based on the fitted model, the leverage effect is significant at the 5% level.

S-Plus Commands Used

```

> ibm.tgarch = garch(ibm.ln~1, ~tgarch(1,1), leverage=T,
+ cond.dist='ged')
> summary(ibm.tgarch)
> plot(ibm.tgarch)

```

It is interesting to compare the two models in Eqs. (3.32) and (3.35) for the monthly log returns of IBM stock. Assume that $a_{t-1} = \pm 2\sigma_{t-1}$ so that $\epsilon_{t-1} = \pm 2$. The EGARCH(1,1) model gives

$$\frac{\sigma_t^2(\epsilon_{t-1} = -2)}{\sigma_t^2(\epsilon_{t-1} = 2)} = e^{0.22 \times 2 \times 0.632} \approx 1.264.$$

On the other hand, ignoring the constant term 0.000345, the TGARCH(1,1) model gives

$$\frac{\sigma_t^2(\epsilon_{t-1} = -2)}{\sigma_t^2(\epsilon_{t-1} = 2)} \approx \frac{[(0.0658 + 0.0843)4 + 0.8182]\sigma_{t-1}^2}{(0.0658 \times 4 + 0.8182)\sigma_{t-1}^2} = 1.312.$$

The two models provide similar leverage effects.

3.10 THE CHARMA MODEL

Many other econometric models have been proposed in the literature to describe the evolution of the conditional variance σ_t^2 in Eq. (3.2). We mention the conditional heteroscedastic ARMA (CHARMA) model that uses random coefficients to produce conditional heteroscedasticity; see Tsay (1987). The CHARMA model is not the same as the ARCH model, but the two models have similar second-order conditional properties. A CHARMA model is defined as

$$r_t = \mu_t + a_t, \quad a_t = \delta_{1t}a_{t-1} + \delta_{2t}a_{t-2} + \cdots + \delta_{mt}a_{t-m} + \eta_t, \quad (3.36)$$

where $\{\eta_t\}$ is a Gaussian white noise series with mean zero and variance σ_η^2 , $\{\delta_t\} = \{(\delta_{1t}, \dots, \delta_{mt})'\}$ is a sequence of iid random vectors with mean zero and nonnegative definite covariance matrix $\mathbf{\Omega}$, and $\{\delta_t\}$ is independent of $\{\eta_t\}$. In this section, we use some basic properties of vector and matrix operations to simplify the presentation. Readers may consult Appendix A of Chapter 8 for a brief review of these properties. For $m > 0$, the model can be written as

$$a_t = \mathbf{a}_{t-1}'\boldsymbol{\delta}_t + \eta_t,$$

where $\mathbf{a}_{t-1} = (a_{t-1}, \dots, a_{t-m})'$ is a vector of lagged values of a_t and is available at time $t - 1$. The conditional variance of a_t of the CHARMA model in Eq. (3.36)

is then

$$\begin{aligned}\sigma_t^2 &= \sigma_\eta^2 + \mathbf{a}_{t-1}' \text{Cov}(\boldsymbol{\delta}_t) \mathbf{a}_{t-1} \\ &= \sigma_\eta^2 + (a_{t-1}, \dots, a_{t-m}) \boldsymbol{\Omega} (a_{t-1}, \dots, a_{t-m})'.\end{aligned}\quad (3.37)$$

Denote the (i, j) th element of $\boldsymbol{\Omega}$ by ω_{ij} . Because the matrix is symmetric, we have $\omega_{ij} = \omega_{ji}$. If $m = 1$, then Eq. (3.37) reduces to $\sigma_t^2 = \sigma_\eta^2 + \omega_{11}a_{t-1}^2$, which is an ARCH(1) model. If $m = 2$, then Eq. (3.37) reduces to

$$\sigma_t^2 = \sigma_\eta^2 + \omega_{11}a_{t-1}^2 + 2\omega_{12}a_{t-1}a_{t-2} + \omega_{22}a_{t-2}^2,$$

which differs from an ARCH(2) model by the cross-product term $a_{t-1}a_{t-2}$. In general, the conditional variance of a CHARMA(m) model is equivalent to that of an ARCH(m) model if $\boldsymbol{\Omega}$ is a diagonal matrix. Because $\boldsymbol{\Omega}$ is a covariance matrix, which is nonnegative definite, and σ_η^2 is a variance, which is positive, we have $\sigma_t^2 \geq \sigma_\eta^2 > 0$ for all t . In other words, the positiveness of σ_t^2 is automatically satisfied under a CHARMA model.

An obvious difference between ARCH and CHARMA models is that the latter use cross products of the lagged values of a_t in the volatility equation. The cross-product terms might be useful in some applications. For example, in modeling an asset return series, cross-product terms denote interactions between previous returns. It is conceivable that stock volatility may depend on such interactions. However, the number of cross-product terms increases rapidly with the order m , and some constraints are needed to keep the model simple. A possible constraint is to use a small number of cross-product terms in a CHARMA model. Another difference between the two models is that higher order properties of CHARMA models are harder to obtain than those of ARCH models because it is in general harder to handle multiple random variables.

For illustration, we employ the CHARMA model

$$r_t = \phi_0 + a_t, \quad a_t = \delta_{1t}a_{t-1} + \delta_{2t}a_{t-2} + \eta_t$$

for the monthly excess returns of the S&P 500 index used before in GARCH modeling. The fitted model is

$$r_t = 0.00635 + a_t, \quad \sigma_t^2 = 0.00179 + (a_{t-1}, a_{t-2}) \hat{\boldsymbol{\Omega}} (a_{t-1}, a_{t-2})',$$

where

$$\hat{\boldsymbol{\Omega}} = \begin{bmatrix} 0.1417(0.0333) & -0.0594(0.0365) \\ -0.0594(0.0365) & 0.3081(0.0340) \end{bmatrix},$$

where the numbers in parentheses are standard errors. The cross-product term of $\hat{\boldsymbol{\Omega}}$ has a t ratio of -1.63 , which is marginally significant at the 10% level. If we refine the model to

$$r_t = \phi_0 + a_t, \quad a_t = \delta_{1t}a_{t-1} + \delta_{2t}a_{t-2} + \delta_{3t}a_{t-3} + \eta_t,$$

but assume that δ_{3t} is uncorrelated with $(\delta_{1t}, \delta_{2t})$, then we obtain the fitted model

$$r_t = 0.0068 + a_t, \quad \sigma_t^2 = 0.00136 + (a_{t-1}, a_{t-2}, a_{t-3})\hat{\mathbf{\Omega}}(a_{t-1}, a_{t-2}, a_{t-3})',$$

where the elements of $\hat{\mathbf{\Omega}}$ and their standard errors, shown in parentheses, are

$$\hat{\mathbf{\Omega}} = \begin{bmatrix} 0.1212(0.0355) & -0.0622(0.0283) & 0 \\ -0.0622(0.0283) & 0.1913(0.0254) & 0 \\ 0 & 0 & 0.2988(0.0420) \end{bmatrix}.$$

All of the estimates are now statistically significant at the 5% level. From the model, $a_t = r_t - 0.0068$ is the deviation of the monthly excess return from its average. The fitted CHARMA model shows that there is some interaction effect between the first two lagged deviations. Indeed, the volatility equation can be written approximately as

$$\sigma_t^2 = 0.00136 + 0.12a_{t-1}^2 - 0.12a_{t-1}a_{t-2} + 0.19a_{t-2}^2 + 0.30a_{t-3}^2.$$

The conditional variance is slightly larger when $a_{t-1}a_{t-2}$ is negative.

3.10.1 Effects of Explanatory Variables

The CHARMA model can easily be generalized so that the volatility of r_t may depend on some explanatory variables. Let $\{x_{it}\}_{i=1}^m$ be m explanatory variables available at time t . Consider the model

$$r_t = \mu_t + a_t, \quad a_t = \sum_{i=1}^m \delta_{it}x_{i,t-1} + \eta_t, \quad (3.38)$$

where $\delta_t = (\delta_{1t}, \dots, \delta_{mt})'$ and η_t are random vector and variable defined in Eq. (3.36). Then the conditional variance of a_t is

$$\sigma_t^2 = \sigma_\eta^2 + (x_{1,t-1}, \dots, x_{m,t-1})\mathbf{\Omega}(x_{1,t-1}, \dots, x_{m,t-1})'.$$

In application, the explanatory variables may include some lagged values of a_t .

3.11 RANDOM COEFFICIENT AUTOREGRESSIVE MODELS

In the literature, the random coefficient autoregressive (RCA) model is introduced to account for variability among different subjects under study, similar to the panel data analysis in econometrics and the hierarchical model in statistics. We classify the RCA model as a conditional heteroscedastic model, but historically it is used to obtain a better description of the conditional mean equation of the process by

allowing for the parameters to evolve over time. A time series r_t is said to follow an RCA(p) model if it satisfies

$$r_t = \phi_0 + \sum_{i=1}^p (\phi_i + \delta_{it}) r_{t-i} + a_t, \quad (3.39)$$

where p is a positive integer, $\{\delta_t\} = \{(\delta_{1t}, \dots, \delta_{pt})'\}$ is a sequence of independent random vectors with mean zero and covariance matrix $\mathbf{\Omega}_\delta$, and $\{\delta_t\}$ is independent of $\{a_t\}$; see Nicholls and Quinn (1982) for further discussions of the model. The conditional mean and variance of the RCA model in Eq. (3.39) are

$$\begin{aligned} \mu_t &= E(r_t | F_{t-1}) = \phi_0 + \sum_{i=1}^p \phi_i r_{t-i}, \\ \sigma_t^2 &= \sigma_a^2 + (r_{t-1}, \dots, r_{t-p}) \mathbf{\Omega}_\delta (r_{t-1}, \dots, r_{t-p})', \end{aligned}$$

which is in the same form as that of a CHARMA model. However, there is a subtle difference between RCA and CHARMA models. For the RCA model, the volatility is a quadratic function of the observed lagged values r_{t-i} . Yet the volatility is a quadratic function of the lagged innovations a_{t-i} in a CHARMA model.

3.12 STOCHASTIC VOLATILITY MODEL

An alternative approach to describe the volatility evolution of a financial time series is to introduce an innovation to the conditional variance equation of a_t ; see Melino and Turnbull (1990), Taylor (1994), Harvey, Ruiz, and Shephard (1994), and Jacquier, Polson, and Rossi (1994). The resulting model is referred to as a stochastic volatility (SV) model. Similar to EGARCH models, to ensure positiveness of the conditional variance, SV models use $\ln(\sigma_t^2)$ instead of σ_t^2 . A SV model is defined as

$$a_t = \sigma_t \epsilon_t, \quad (1 - \alpha_1 B - \dots - \alpha_m B^m) \ln(\sigma_t^2) = \alpha_0 + v_t, \quad (3.40)$$

where the ϵ_t are iid $N(0, 1)$, the v_t are iid $N(0, \sigma_v^2)$, $\{\epsilon_t\}$ and $\{v_t\}$ are independent, α_0 is a constant, and all zeros of the polynomial $1 - \sum_{i=1}^m \alpha_i B^i$ are greater than 1 in modulus. Adding the innovation v_t substantially increases the flexibility of the model in describing the evolution of σ_t^2 , but it also increases the difficulty in parameter estimation. To estimate an SV model, we need a quasi-likelihood method via Kalman filtering or a Monte Carlo method. Jacquier, Polson, and Rossi (1994) provide some comparison of estimation results between quasi-likelihood and Markov chain Monte Carlo (MCMC) methods. The difficulty in estimating an SV model is understandable because for each shock a_t the model uses two innovations ϵ_t and v_t . We discuss an MCMC method to estimate SV models in Chapter 12. For more discussions on stochastic volatility models, see Taylor (1994).

The appendixes of Jacquier, Polson, and Rossi (1994) provide some properties of the SV model when $m = 1$. For instance, with $m = 1$, we have

$$\ln(\sigma_t^2) \sim N\left(\frac{\alpha_0}{1 - \alpha_1}, \frac{\sigma_v^2}{1 - \alpha_1^2}\right) \equiv N(\mu_h, \sigma_h^2),$$

and $E(a_t^2) = \exp(\mu_h + \sigma_h^2/2)$, $E(a_t^4) = 3 \exp(2\mu_h^2 + 2\sigma_h^2)$, and $\text{corr}(a_t^2, a_{t-i}^2) = [\exp(\sigma_h^2 \alpha_1^i) - 1] / [3 \exp(\sigma_h^2) - 1]$. Limited experience shows that SV models often provided improvements in model fitting, but their contributions to out-of-sample volatility forecasts received mixed results.

3.13 LONG-MEMORY STOCHASTIC VOLATILITY MODEL

More recently, the SV model is further extended to allow for long memory in volatility, using the idea of fractional difference. As stated in Chapter 2, a time series is a long-memory process if its autocorrelation function decays at a hyperbolic, instead of an exponential, rate as the lag increases. The extension to long-memory models in volatility study is motivated by the fact that the autocorrelation function of the squared or absolute-valued series of an asset return often decays slowly, even though the return series has no serial correlation; see Ding, Granger, and Engle (1993). Figure 3.10 shows the sample ACF of the daily absolute returns for IBM stock and the S&P 500 index from July 3, 1962, to December 31, 2003. These sample ACFs are positive with moderate magnitude but decay slowly.

A simple long-memory stochastic volatility (LMSV) model can be written as

$$a_t = \sigma_t \epsilon_t, \quad \sigma_t = \sigma \exp(u_t/2), \quad (1 - B)^d u_t = \eta_t, \quad (3.41)$$

where $\sigma > 0$, the ϵ_t are iid $N(0, 1)$, the η_t are iid $N(0, \sigma_\eta^2)$ and independent of ϵ_t , and $0 < d < 0.5$. The feature of long memory stems from the fractional difference $(1 - B)^d$, which implies that the ACF of u_t decays slowly at a hyperbolic, instead of an exponential, rate as the lag increases. For model (3.41), we have

$$\begin{aligned} \ln(a_t^2) &= \ln(\sigma^2) + u_t + \ln(\epsilon_t^2) \\ &= [\ln(\sigma^2) + E(\ln \epsilon_t^2)] + u_t + [\ln(\epsilon_t^2) - E(\ln \epsilon_t^2)] \\ &\equiv \mu + u_t + e_t. \end{aligned}$$

Thus, the $\ln(a_t^2)$ series is a Gaussian long-memory signal plus a non-Gaussian white noise; see Breidt, Crato, and de Lima (1998). Estimation of the LMSV model is complicated, but the fractional difference parameter d can be estimated by using either a quasi-maximum-likelihood method or a regression method. Using the log series of squared daily returns for companies in the S&P 500 index, Bollerslev and Jubinski (1999) and Ray and Tsay (2000) found that the median estimate of d is about 0.38. For applications, Ray and Tsay (2000) studied common long-memory

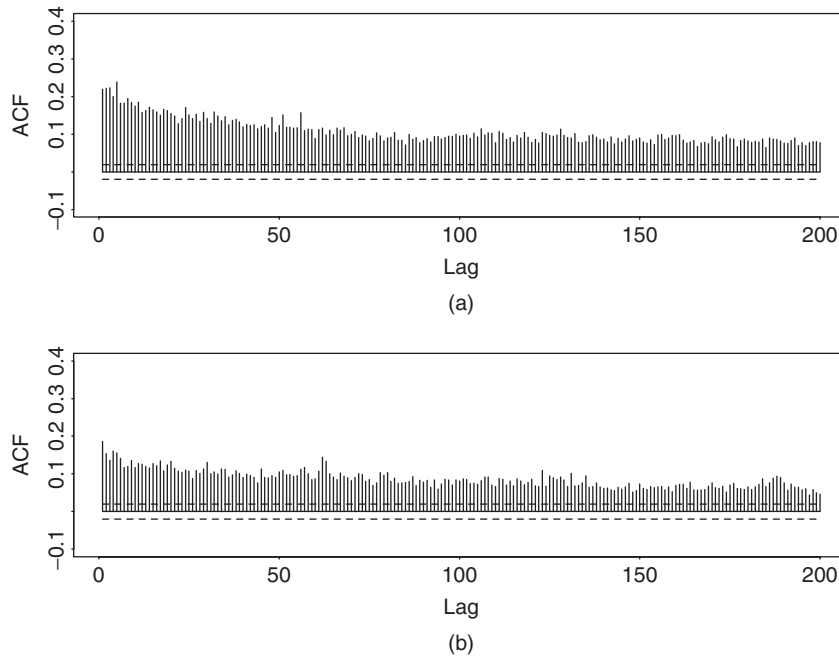


Figure 3.10 Sample ACF of daily absolute log returns for (a) S&P 500 index and (b) IBM stock for period from July 3, 1962, to December 31, 2003. Two horizontal lines denote asymptotic 5% limits.

components in daily stock volatilities of groups of companies classified by various characteristics. They found that companies in the same industrial or business sector tend to have more common long-memory components (e.g., big U.S. national banks and financial institutions).

3.14 APPLICATION

In this section, we apply the volatility models discussed in this chapter to investigate some problems of practical importance. The data used are the monthly log returns of IBM stock and the S&P 500 index from January 1926 to December 1999. There are 888 observations, and the returns are in percentages and include dividends. Figure 3.11 shows the time plots of the two return series. Note that the result of this section was obtained by the RATS program.

Example 3.4. The questions we address here are whether the daily volatility of a stock is lower in the summer and, if so, by how much. Affirmative answers to these two questions have practical implications in stock option pricing. We use the monthly log returns of IBM stock shown in Figure 3.11(a) as an illustrative example.

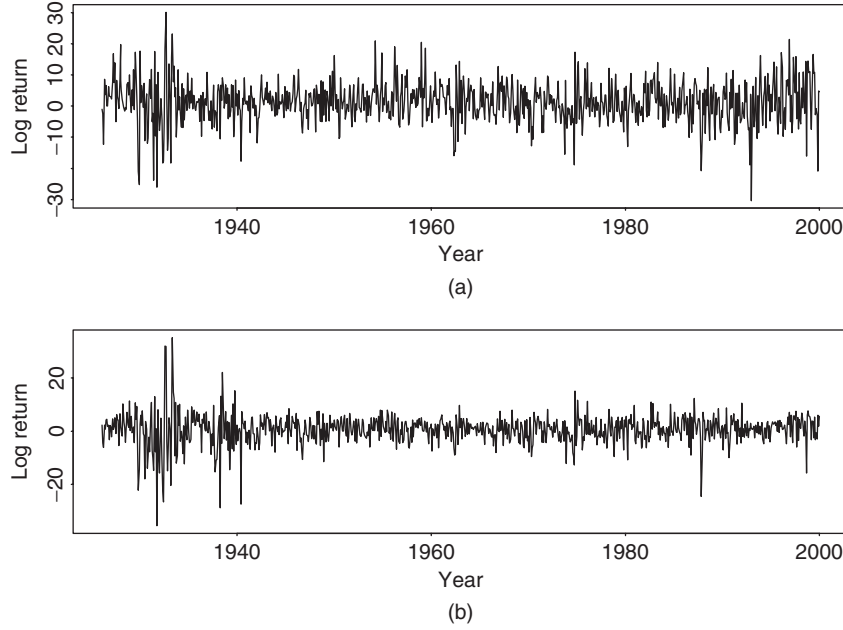


Figure 3.11 Time plots of monthly log returns for (a) IBM stock and (b) S&P 500 index. Sample period is from January 1926 to December 1999. Returns are in percentages and include dividends.

Denote the monthly log return series by r_t . If Gaussian GARCH models are entertained, we obtain the GARCH(1,1) model:

$$\begin{aligned} r_t &= 1.23 + 0.099r_{t-1} + a_t, & a_t &= \sigma_t \epsilon_t, \\ \sigma_t^2 &= 3.206 + 0.103a_{t-1}^2 + 0.825\sigma_{t-1}^2, \end{aligned} \quad (3.42)$$

for the series. The standard errors of the two parameters in the mean equation are 0.222 and 0.037, respectively, whereas those of the parameters in the volatility equation are 0.947, 0.021, and 0.037, respectively. Using the standardized residuals $\tilde{a}_t = a_t/\sigma_t$, we obtain $Q(10) = 7.82(0.553)$ and $Q(20) = 21.22(0.325)$, where the p value is in parentheses. Therefore, there are no serial correlations in the residuals of the mean equation. The Ljung–Box statistics of the $\tilde{a}_t^2$ series show $Q(10) = 2.89(0.98)$ and $Q(20) = 7.26(0.99)$, indicating that the standardized residuals have no conditional heteroscedasticity. The fitted model seems adequate. This model serves as a starting point for further study.

To study the summer effect on stock volatility of an asset, we define an indicator variable

$$u_t = \begin{cases} 1 & \text{if } t \text{ is June, July, or August,} \\ 0 & \text{otherwise,} \end{cases} \quad (3.43)$$

and modify the volatility equation to

$$\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2 + u_t(\alpha_{00} + \alpha_{10} a_{t-1}^2 + \beta_{10} \sigma_{t-1}^2).$$

This equation uses two GARCH(1,1) models to describe the volatility of a stock return; one model for the summer months and the other for the remaining months. For the monthly log returns of IBM stock, estimation results show that the estimates of α_{10} and β_{10} are statistically nonsignificant at the 10% level. Therefore, we refine the equation and obtain the model

$$\begin{aligned} r_t &= 1.21 + 0.099r_{t-1} + a_t, & a_t &= \sigma_t \epsilon_t, \\ \sigma_t^2 &= 4.539 + 0.113a_{t-1}^2 + 0.816\sigma_{t-1}^2 - 5.514u_t. \end{aligned} \quad (3.44)$$

The standard errors of the parameters in the mean equation are 0.218 and 0.037, respectively, and those of the parameters in the volatility equation are 1.071, 0.022, 0.037, and 1.900, respectively. The Ljung–Box statistics for the standardized residuals $\tilde{a}_t = a_t/\sigma_t$ show $Q(10) = 7.66(0.569)$ and $Q(20) = 21.64(0.302)$. Therefore, there are no serial correlations in the standardized residuals. The Ljung–Box statistics for $\tilde{a}_t^2$ give $Q(10) = 3.38(0.97)$ and $Q(20) = 6.82(0.99)$, indicating no conditional heteroscedasticity in the standardized residuals either. The refined model seems adequate.

Comparing the volatility models in Eqs. (3.42) and (3.44), we obtain the following conclusions. First, because the coefficient -5.514 is significantly different from zero with a p value of 0.0067, the summer effect on stock volatility is statistically significant at the 1% level. Furthermore, the negative sign of the estimate confirms that the volatility of IBM monthly log stock returns is indeed lower during the summer. Second, rewrite the volatility model in Eq. (3.44) as

$$\sigma_t^2 = \begin{cases} -0.615 + 0.113a_{t-1}^2 + 0.816\sigma_{t-1}^2 & \text{if } t \text{ is June, July, or August,} \\ 4.539 + 0.113a_{t-1}^2 + 0.816\sigma_{t-1}^2 & \text{otherwise.} \end{cases}$$

The negative constant term $-0.615 = 4.539 - 5.514$ is counterintuitive. However, since the standard errors of 4.539 and 5.514 are relatively large, the estimated difference -0.615 might not be significantly different from zero. To verify the assertion, we refit the model by imposing the constraint that the constant term of the volatility equation is zero for the summer months. This can easily be done by using the equation

$$\sigma_t^2 = \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2 + \gamma(1 - u_t).$$

The fitted model is

$$\begin{aligned} r_t &= 1.21 + 0.099r_{t-1} + a_t, & a_t &= \sigma_t \epsilon_t, \\ \sigma_t^2 &= 0.114a_{t-1}^2 + 0.811\sigma_{t-1}^2 + 4.552(1 - u_t). \end{aligned} \quad (3.45)$$

The standard errors of the parameters in the mean equation are 0.219 and 0.038, respectively, and those of the parameters in the volatility equation are 0.022, 0.034, and 1.094, respectively. The Ljung–Box statistics of the standardized residuals show $Q(10) = 7.68$ and $Q(20) = 21.67$, and those of the $\tilde{a}_t^2$ series give $Q(10) = 3.17$ and $Q(20) = 6.85$. These test statistics are close to what we had before and are not significant at the 5% level.

The volatility Eq. (3.45) can readily be used to assess the summer effect on the IBM stock volatility. For illustration, based on the model in Eq. (3.45) the medians of a_t^2 and σ_t^2 are 29.4 and 75.1, respectively, for the IBM monthly log returns in 1999. Using these values, we have $\sigma_t^2 = 0.114 \times 29.4 + 0.811 \times 75.1 = 64.3$ for the summer months and $\sigma_t^2 = 68.8$ for the other months. The ratio of the two volatilities is $64.3/68.8 \approx 93\%$. Thus, there is a 7% reduction in the volatility of the monthly log return of IBM stock in the summer months.

Example 3.5. The S&P 500 index is widely used in the derivative markets. As such, modeling its volatility is a subject of intensive study. The question we ask in this example is whether the past returns of individual components of the index contribute to the modeling of the S&P 500 index volatility in the presence of its own returns. A thorough investigation on this topic is beyond the scope of this chapter, but we use the past returns of IBM stock as explanatory variables to address the question.

The data used are shown in Figure 3.11. Denote by r_t the monthly log return series of the S&P 500 index. Using the r_t series and Gaussian GARCH models, we obtain the following special GARCH(2,1) model:

$$r_t = 0.609 + a_t, \quad a_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = 0.717 + 0.147a_{t-2}^2 + 0.839\sigma_{t-1}^2. \quad (3.46)$$

The standard error of the constant term in the mean equation is 0.138, and those of the parameters in the volatility equation are 0.214, 0.021, and 0.017, respectively. Based on the standardized residuals $\tilde{a}_t = a_t/\sigma_t$, we have $Q(10) = 11.51(0.32)$ and $Q(20) = 23.71(0.26)$, where the number in parentheses denotes the p value. For the $\tilde{a}_t^2$ series, we have $Q(10) = 9.42(0.49)$ and $Q(20) = 13.01(0.88)$. Therefore, the model seems adequate at the 5% significance level.

Next, we evaluate the contributions, if any, of using the past returns of IBM stock, which is a component of the S&P 500 index, in modeling the index volatility. As a simple illustration, we modify the volatility equation as

$$\sigma_t^2 = \alpha_0 + \alpha_2 a_{t-2}^2 + \beta_1 \sigma_{t-1}^2 + \gamma (x_{t-1} - 1.24)^2,$$

where x_t is the monthly log return of IBM stock and 1.24 is the sample mean of x_t . The fitted model for r_t becomes

$$r_t = 0.616 + a_t, \quad a_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = 1.069 + 0.148a_{t-2}^2 + 0.834\sigma_{t-1}^2 - 0.007(x_{t-1} - 1.24)^2. \quad (3.47)$$

TABLE 3.4 Fitted Volatilities for Monthly Log Returns of S&P 500 Index from July to December 1999 Using Models with and without Past Log Return of IBM Stock

Month	7/99	8/99	9/99	10/99	11/99	12/99
Model (3.46)	26.30	26.01	24.73	21.69	20.71	22.46
Model (3.47)	23.32	23.13	22.46	20.00	19.45	18.27

The standard error of the parameter in the mean equation is 0.139 and the standard errors of the parameters in the volatility equation are 0.271, 0.020, 0.018, and 0.002, respectively. For model checking, we have $Q(10) = 11.39(0.33)$ and $Q(20) = 23.63(0.26)$ for the standardized residuals $\tilde{a}_t = a_t/\sigma_t$ and $Q(10) = 9.35(0.50)$ and $Q(20) = 13.51(0.85)$ for the $\tilde{a}_t^2$ series. Therefore, the model is adequate.

Since the p value for testing $\gamma = 0$ is 0.0039, the contribution of the lag-1 IBM stock return to the S&P 500 index volatility is statistically significant at the 1% level. The negative sign is understandable because it implies that using the lag-1 past return of IBM stock reduces the volatility of the S&P 500 index return. Table 3.4 gives the fitted volatility of the S&P 500 index from July to December of 1999 using models (3.46) and (3.47). From the table, the past value of IBM log stock return indeed contributes to the modeling of the S&P 500 index volatility.

3.15 ALTERNATIVE APPROACHES

In this section, we discuss two alternative methods to volatility modeling.

3.15.1 Use of High-Frequency Data

French, Schwert, and Stambaugh (1987) consider an alternative approach for volatility estimation that uses high-frequency data to calculate volatility of low-frequency returns. In recent years, this approach has attracted substantial interest due to the availability of high-frequency financial data; see Andersen, Bollerslev, Diebold, and Labys (2001a, 2001b).

Suppose that we are interested in the monthly volatility of an asset for which daily returns are available. Let r_t^m be the monthly log return of the asset at month t . Assume that there are n trading days in month t and the daily log returns of the asset in the month are $\{r_{t,i}\}_{i=1}^n$. Using properties of log returns, we have

$$r_t^m = \sum_{i=1}^n r_{t,i}.$$

Assuming that the conditional variance and covariance exist, we have

$$\text{Var}(r_t^m | F_{t-1}) = \sum_{i=1}^n \text{Var}(r_{t,i} | F_{t-1}) + 2 \sum_{i < j} \text{Cov}[r_{t,i}, r_{t,j} | F_{t-1}], \quad (3.48)$$

where F_{t-1} denotes the information available at month $t - 1$ (inclusive). The prior equation can be simplified if additional assumptions are made. For example, if we assume that $\{r_{t,i}\}$ is a white noise series, then

$$\text{Var}(r_t^m | F_{t-1}) = n \text{Var}(r_{t,1}),$$

where $\text{Var}(r_{t,1})$ can be estimated from the daily returns $\{r_{t,i}\}_{i=1}^n$ by

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n (r_{t,i} - \bar{r}_t)^2}{n - 1},$$

where $\bar{r}_t$ is the sample mean of the daily log returns in month t [i.e., $\bar{r}_t = (\sum_{i=1}^n r_{t,i})/n$]. The estimated monthly volatility is then

$$\hat{\sigma}_m^2 = \frac{n}{n - 1} \sum_{i=1}^n (r_{t,i} - \bar{r}_t)^2. \quad (3.49)$$

If $\{r_{t,i}\}$ follows an MA(1) model, then

$$\text{Var}(r_t^m | F_{t-1}) = n \text{Var}(r_{t,1}) + 2(n - 1) \text{Cov}(r_{t,1}, r_{t,2}),$$

which can be estimated by

$$\hat{\sigma}_m^2 = \frac{n}{n - 1} \sum_{i=1}^n (r_{t,i} - \bar{r}_t)^2 + 2 \sum_{i=1}^{n-1} (r_{t,i} - \bar{r}_t)(r_{t,i+1} - \bar{r}_t). \quad (3.50)$$

The previous approach for volatility estimation is simple, but it encounters several difficulties in practice. First, the model for daily returns $\{r_{t,i}\}$ is unknown. This complicates the estimation of covariances in Eq. (3.48). Second, there are roughly 21 trading days in a month, resulting in a small sample size. The accuracy of the estimates of variance and covariance in Eq. (3.48) might be questionable. The accuracy depends on the dynamic structure of $\{r_{t,i}\}$ and their distribution. If the daily log returns have high excess kurtosis and serial correlations, then the sample estimates $\hat{\sigma}_m^2$ in Eqs. (3.49) and (3.50) may not even be consistent; see Bai, Russell, and Tiao (2004). Further research is needed to make this approach valuable.

Example 3.6. Consider the monthly volatility of the log returns of the S&P 500 index from January 1980 to December 1999. We calculate the volatility by three methods. In the first method, we use daily log returns and Eq. (3.49) (i.e., assuming that the daily log returns form a white noise series). The second method also uses daily returns but assumes an MA(1) model [i.e., using Eq. (3.50)]. The third method applies a GARCH(1,1) model to the monthly returns from January

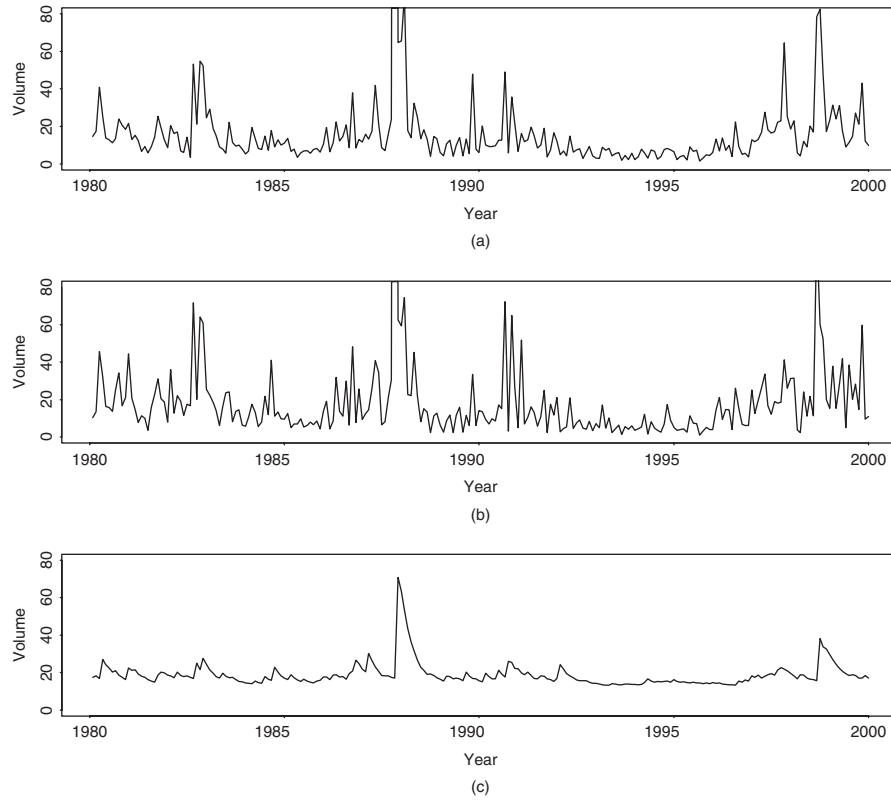


Figure 3.12 Time plots of estimated monthly volatility for log returns of S&P 500 index from January 1980 to December 1999: (a) assumes that daily log returns form a white noise series, (b) assumes that daily log returns follow an MA(1) model, and (c) uses monthly returns from January 1962 to December 1999 and a GARCH(1,1) model.

1962 to December 1999. We use a longer data span to obtain a more accurate estimate of the monthly volatility. The GARCH(1,1) model used is

$$r_t^m = 0.658 + a_t, \quad a_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = 3.349 + 0.086a_{t-1}^2 + 0.735\sigma_{t-1}^2,$$

where ϵ_t is a standard Gaussian white noise series. Figure 3.12 shows the time plots of the estimated monthly volatility. Clearly the estimated volatilities based on daily returns are much higher than those based on monthly returns and a GARCH(1,1) model. In particular, the estimated volatility for October 1987 was about 680 when daily returns are used. The plots shown were truncated to have the same scale.

In Eq. (3.49), if we further assume that the sample mean $\bar{r}_t$ is zero, then we have $\hat{\sigma}_m^2 \approx \sum_{i=1}^n r_{t,i}^2$. In this case, the cumulative sum of squares of daily log returns in a month is used as an estimate of monthly volatility. This concept has been generalized to estimate daily volatility of an asset by using intradaily log returns.

Let r_t be the daily log return of an asset. Suppose that there are n equally spaced intradaily log returns available such that $r_t = \sum_{i=1}^n r_{t,i}$. The quantity

$$RV_t = \sum_{i=1}^n r_{t,i}^2,$$

is called the *realized* volatility of r_t ; see Andersen et al. (2001a,b). Mathematically, realized volatility is a quadratic variation of r_t , and it assumes that $\{r_{t,i}\}_{i=1}^n$ forms an iid sequence with mean zero and finite variance. Limited experience indicates that $\ln(RV_t)$ often follows approximately a Gaussian ARIMA(0,1, q) model, which can be used to produce forecasts. See demonstration in Section 1.1 for further information.

Advantages of realized volatility include simplicity and making use of intradaily returns. Intuitively, one would like to use as much information as possible by choosing a large n . However, when the time interval between $r_{t,i}$ is small, the returns are subject to the effects of market microstructure, for example, bid–ask bounce, which often results in a biased estimate of the volatility. The problem of choosing an optimal time interval for constructing realized volatility has attracted much research lately. For heavily traded assets in the United States, a time interval of 4–15 minutes is often used. Another problem of using realized volatility for stock returns is that the overnight return, which is the return from the closing price of day $t - 1$ to the opening price of t , tends to be substantial. Ignoring overnight returns can seriously underestimate the volatility. On the other hand, our limited experience shows that overnight returns appear to be small for index returns or foreign exchange returns.

In a series of recent articles, Barndorff-Nielsen and Shephard (2004) have used high-frequency returns to study bi-power variations of an asset return and developed some methods to detect jumps in volatility.

3.15.2 Use of Daily Open, High, Low, and Close Prices

For many assets, daily opening, high, low, and closing prices are available. Parkinson (1980), Garman and Klass (1980), Rogers and Satchell (1991), and Yang and Zhang (2000) showed that one can use such information to improve volatility estimation. Figure 3.13 shows a time plot of price versus time for the t th trading day, assuming that time is continuous. For an asset, define the following variables:

- C_t = closing price of the t th trading day.
- O_t = opening price of the t th trading day.
- f = fraction of the day (in interval $[0,1]$) that trading is closed.
- H_t = highest price of the t th trading period.
- L_t = lowest price of the t th trading period.
- F_{t-1} = public information available at time $t - 1$.

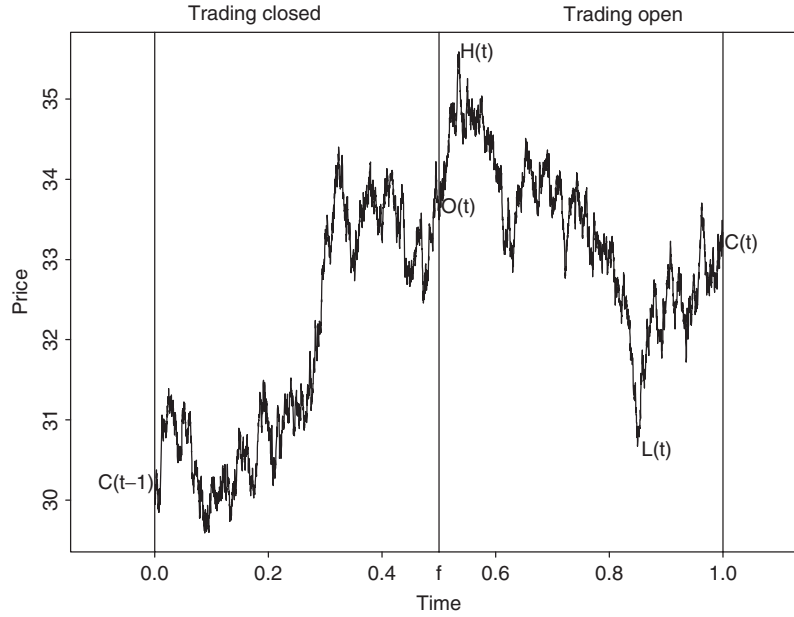


Figure 3.13 Time plot of price over time: scale for price is arbitrary.

The conventional variance (or volatility) is $\sigma_t^2 = E[(C_t - C_{t-1})^2 | F_{t-1}]$. Garman and Klass (1980) considered several estimates of σ_t^2 assuming that the price follows a simple diffusion model without drift; see Chapter 6 for more information about stochastic diffusion models. The estimators considered include:

- $\hat{\sigma}_{0,t}^2 = (C_t - C_{t-1})^2$.
- $\hat{\sigma}_{1,t}^2 = \frac{(O_t - C_{t-1})^2}{2f} + \frac{(C_t - O_t)^2}{2(1-f)}, \quad 0 < f < 1$.
- $\hat{\sigma}_{2,t}^2 = \frac{(H_t - L_t)^2}{4 \ln(2)} \approx 0.3607(H_t - L_t)^2$.
- $\hat{\sigma}_{3,t}^2 = 0.17 \frac{(O_t - C_{t-1})^2}{f} + 0.83 \frac{(H_t - L_t)^2}{(1-f)4 \ln(2)}, \quad 0 < f < 1$.
- $\hat{\sigma}_{5,t}^2 = 0.5(H_t - L_t)^2 - [2 \ln(2) - 1](C_t - O_t)^2$, which is $\approx 0.5(H_t - L_t)^2 - 0.386(C_t - O_t)^2$.
- $\hat{\sigma}_{6,t}^2 = 0.12 \frac{(O_t - C_{t-1})^2}{f} + 0.88 \frac{\hat{\sigma}_{5,t}^2}{1-f}, \quad 0 < f < 1$.

A more precise, but complicated, estimator $\hat{\sigma}_{4,t}^2$ was also considered. However, it is close to $\hat{\sigma}_{5,t}^2$. Defining the efficiency factor of a volatility estimator as

$$\text{Eff}(\hat{\sigma}_{i,t}^2) = \frac{\text{Var}(\hat{\sigma}_{0,t}^2)}{\text{Var}(\hat{\sigma}_{i,t}^2)},$$

Garman and Klass (1980) found that $\text{Eff}(\hat{\sigma}_{i,t}^2)$ is approximately 2, 5.2, 6.2, 7.4, and 8.4 for $i = 1, 2, 3, 5$, and 6, respectively, for the simple diffusion model entertained. Note that $\hat{\sigma}_{2,t}^2$ was derived by Parkinson (1980) with $f = 0$.

Turn to log returns. Define the following:

- $o_t = \ln(O_t) - \ln(C_{t-1})$, the normalized open.
- $u_t = \ln(H_t) - \ln(O_t)$, the normalized high.
- $d_t = \ln(L_t) - \ln(O_t)$, the normalized low.
- $c_t = \ln(C_t) - \ln(O_t)$, the normalized close.

Suppose that there are n days of data available and the volatility is constant over the period. Yang and Zhang (2000) recommend the estimate

$$\hat{\sigma}_{yz}^2 = \hat{\sigma}_o^2 + k\hat{\sigma}_c^2 + (1 - k)\hat{\sigma}_{rs}^2$$

as a robust estimator of the volatility, where

$$\begin{aligned}\hat{\sigma}_o^2 &= \frac{1}{n-1} \sum_{t=1}^n (o_t - \bar{o})^2 \quad \text{with} \quad \bar{o} = \frac{1}{n} \sum_{t=1}^n o_t, \\ \hat{\sigma}_c^2 &= \frac{1}{n-1} \sum_{t=1}^n (c_t - \bar{c})^2 \quad \text{with} \quad \bar{c} = \frac{1}{n} \sum_{t=1}^n c_t, \\ \hat{\sigma}_{rs}^2 &= \frac{1}{n} \sum_{t=1}^n [u_t(u_t - c_t) + d_t(d_t - c_t)], \\ k &= \frac{0.34}{1.34 + (n+1)/(n-1)}.\end{aligned}$$

The estimate $\hat{\sigma}_{rs}^2$ was proposed by Rogers and Satchell (1991), and the quantity k is chosen to minimize the variance of the estimator of $\hat{\sigma}_{yz}^2$, which is a linear combination of three estimates.

The quantity $H_t - L_t$ is called the *range* of the price in the t th day. This estimator has led to the use of range-based volatility estimates; see, for instance, Alizadeh, Brandt, and Diebold (2002). In practice, stock prices are only observed at discrete time points. As such, the observed daily high is likely lower than H_t and the observed daily low is likely higher than L_t . Consequently, the observed daily price range tends to underestimate the actual range and, hence, may lead to underestimation of volatility. This bias in volatility estimation depends on the trading frequency and tick size of the stocks. For intensively traded stocks, the bias should be negligible. For other stocks, further study is needed to better understand the performance of range-based volatility estimation.

3.16 KURTOSIS OF GARCH MODELS

Uncertainty in volatility estimation is an important issue, but it is often overlooked. To assess the variability of an estimated volatility, one must consider the kurtosis of a volatility model. In this section, we derive the excess kurtosis of a GARCH(1,1) model. The same idea applies to other GARCH models, however. The model considered is

$$a_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2,$$

where $\alpha_0 > 0$, $\alpha_1 \geq 0$, $\beta_1 \geq 0$, $\alpha_1 + \beta_1 < 1$, and $\{\epsilon_t\}$ is an iid sequence satisfying

$$E(\epsilon_t) = 0, \quad \text{Var}(\epsilon_t) = 1, \quad E(\epsilon_t^4) = K_\epsilon + 3,$$

where K_ϵ is the excess kurtosis of the innovation ϵ_t . Based on the assumption, we have the following:

- $\text{Var}(a_t) = E(\sigma_t^2) = \alpha_0 / [1 - (\alpha_1 + \beta_1)]$.
- $E(a_t^4) = (K_\epsilon + 3)E(\sigma_t^4)$ provided that $E(\sigma_t^4)$ exists.

Taking the square of the volatility model, we have

$$\sigma_t^4 = \alpha_0^2 + \alpha_1^2 a_{t-1}^4 + \beta_1^2 \sigma_{t-1}^4 + 2\alpha_0 \alpha_1 a_{t-1}^2 + 2\alpha_0 \beta_1 \sigma_{t-1}^2 + 2\alpha_1 \beta_1 \sigma_{t-1}^2 a_{t-1}^2.$$

Taking expectation of the equation and using the two properties mentioned earlier, we obtain

$$E(\sigma_t^4) = \frac{\alpha_0^2(1 + \alpha_1 + \beta_1)}{[1 - (\alpha_1 + \beta_1)][1 - \alpha_1^2(K_\epsilon + 2) - (\alpha_1 + \beta_1)^2]},$$

provided that $1 > \alpha_1 + \beta_1 \geq 0$ and $1 - \alpha_1^2(K_\epsilon + 2) - (\alpha_1 + \beta_1)^2 > 0$. The excess kurtosis of a_t , if it exists, is then

$$\begin{aligned} K_a &= \frac{E(a_t^4)}{[E(a_t^2)]^2} - 3 \\ &= \frac{(K_\epsilon + 3)[1 - (\alpha_1 + \beta_1)^2]}{1 - 2\alpha_1^2 - (\alpha_1 + \beta_1)^2 - K_\epsilon \alpha_1^2} - 3. \end{aligned}$$

This excess kurtosis can be written in an informative expression. First, consider the case that ϵ_t is normally distributed. In this case, $K_\epsilon = 0$, and some algebra shows that

$$K_a^{(g)} = \frac{6\alpha_1^2}{1 - 2\alpha_1^2 - (\alpha_1 + \beta_1)^2},$$

where the superscript (g) is used to denote Gaussian distribution. This result has two important implications: (a) the kurtosis of a_t exists if $1 - 2\alpha_1^2 - (\alpha_1 + \beta_1)^2 > 0$, and (b) if $\alpha_1 = 0$, then $K_a^{(g)} = 0$, meaning that the corresponding GARCH(1,1) model does not have heavy tails.

Second, consider the case that ϵ_t is not Gaussian. Using the prior result, we have

$$\begin{aligned} K_a &= \frac{K_\epsilon - K_\epsilon(\alpha_1 + \beta_1) + 6\alpha_1^2 + 3K_\epsilon\alpha_1^2}{1 - 2\alpha_1^2 - (\alpha_1 + \beta_1)^2 - K_\epsilon\alpha_1^2} \\ &= \frac{K_\epsilon[1 - 2\alpha_1^2 - (\alpha_1 + \beta_1)^2] + 6\alpha_1^2 + 5K_\epsilon\alpha_1^2}{1 - 2\alpha_1^2 - (\alpha_1 + \beta_1)^2 - K_\epsilon\alpha_1^2} \\ &= \frac{K_\epsilon + K_a^{(g)} + \frac{5}{6}K_\epsilon K_a^{(g)}}{1 - \frac{1}{6}K_\epsilon K_a^{(g)}}. \end{aligned}$$

This result was obtained originally by George C. Tiao; see Bai, Russell, and Tiao (2003). It holds for all GARCH models provided that the kurtosis exists. For instance, if $\beta_1 = 0$, then the model reduces to an ARCH(1) model. In this case, it is easy to verify that $K_a^{(g)} = 6\alpha_1^2/(1 - 3\alpha_1^2)$ provided that $1 > 3\alpha_1^2$ and the excess kurtosis of a_t is

$$\begin{aligned} K_a &= \frac{(K_\epsilon + 3)(1 - \alpha_1^2)}{1 - (K_\epsilon + 3)\alpha_1^2} - 3 = \frac{K_\epsilon + 2K_\epsilon\alpha_1^2 + 6\alpha_1^2}{1 - 3\alpha_1^2 - K_\epsilon\alpha_1^2} \\ &= \frac{K_\epsilon(1 - 3\alpha_1^2) + 6\alpha_1^2 + 5K_\epsilon\alpha_1^2}{1 - 3\alpha_1^2 - K_\epsilon\alpha_1^2} \\ &= \frac{K_\epsilon + K_a^{(g)} + \frac{5}{6}K_\epsilon K_a^{(g)}}{1 - \frac{1}{6}K_\epsilon K_a^{(g)}}. \end{aligned}$$

The prior result shows that for a GARCH(1,1) model the coefficient α_1 plays a critical role in determining the tail behavior of a_t . If $\alpha_1 = 0$, then $K_a^{(g)} = 0$ and $K_a = K_\epsilon$. In this case, the tail behavior of a_t is similar to that of the standardized noise ϵ_t . Yet if $\alpha_1 > 0$, then $K_a^{(g)} > 0$ and the a_t process has heavy tails.

For a (standardized) Student- t distribution with v degrees of freedom, we have $E(\epsilon_t^4) = 6/(v - 4) + 3$ if $v > 4$. Therefore, the excess kurtosis of ϵ_t is $K_\epsilon = 6/(v - 4)$ for $v > 4$. This is part of the reason that we used t_5 in the chapter when the degrees of freedom of a t -distribution are prespecified. The excess kurtosis of a_t becomes $K_a = [6 + (v + 1)K_a^{(g)}]/[v - 4 - K_a^{(g)}]$ provided that $1 - 2\alpha_1^2(v - 1)/(v - 4) - (\alpha_1 + \beta_1)^2 > 0$.

APPENDIX: SOME RATS PROGRAMS FOR ESTIMATING VOLATILITY MODELS

The data file used in the illustration is `sp500.txt`, which contains the monthly excess returns of the S&P 500 index with 792 observations. Comments in a RATS program start with `*`.

A Gaussian GARCH(1,1) Model with a Constant Mean Equation

```
all 0 792:1
open data sp500.txt
data(org=obs) / rt
*** initialize the conditional variance function
set h = 0.0
*** specify the parameters of the model
nonlin mu a0 a1 b1
*** specify the mean equation
frml at = rt(t)-mu
*** specify the volatility equation
frml gvar = a0+a1*at(t-1)**2+b1*h(t-1)
*** specify the log likelihood function
frml garchln = -0.5*log(h(t)=gvar(t))-0.5*at(t)**2/h(t)
*** sample period used in estimation
smpl 2 792
*** initial estimates
compute a0 = 0.01, a1 = 0.1, b1 = 0.5, mu = 0.1
maximize(method=bhhh,recursive,iterations=150) garchln
set fv = gvar(t)
set resid = at(t)/sqrt(fv(t))
set residsq = resid(t)*resid(t)
*** Checking standardized residuals
cor(qstats,number=20,span=10) resid
*** Checking squared standardized residuals
cor(qstats,number=20,span=10) residsq
```

A GARCH(1,1) Model with Student-t Innovation

```
all 0 792:1
open data sp500.txt
data(org=obs) / rt
set h = 0.0
nonlin mu a0 a1 b1 v
frml at = rt(t)-mu
frml gvar = a0+a1*at(t-1)**2+b1*h(t-1)
frml tt = at(t)**2/(h(t)=gvar(t))
frml tln = %LNGAMMA((v+1)/2.)-%LNGAMMA(v/2.)-0.5*log(v-2.)
frml gln = tln-((v+1)/2.)*log(1.0+tt(t)/(v-2.0))-0.5*log(h(t))
smpl 2 792
```

```

compute a0 = 0.01, a1 = 0.1, b1 = 0.5, mu = 0.1, v = 10
maximize(method=bhhh,recursive,iterations=150) gln
set fv = gvar(t)
set resid = at(t)/sqrt(fv(t))
set residsq = resid(t)*resid(t)
cor(qstats,number=20,span=10) resid
cor(qstats,number=20,span=10) residsq

```

An AR(1)–EGARCH(1,1) Model for Monthly Log Returns of IBM Stock

```

all 0 864:1
open data m-ibm.txt
data(org=obs) / rt
set h = 0.0
nonlin c0 p1 th ga a0 a1
frml at = rt(t)-c0-p1*rt(t-1)
frml epsi = at(t)/(sqrt(exp(h(t))))
frml g = th*epsi(t)+ga*(abs(epsi(t))-sqrt(2./% PI))
frml gvar = a1*h(t-1)+(1-a1)*a0+g(t-1)
frml garchln = -0.5*(h(t)=gvar(t))-0.5*epsi(t)**2
smp1 3 864
compute c0 = 0.01, p1 = 0.01, th = 0.1, ga = 0.1
compute a0 = 0.01, a1 = 0.5
maximize(method=bhhh,recursive,iterations=150) garchln
set fv = gvar(t)
set resid = epsi(t)
set residsq = resid(t)*resid(t)
cor(qstats,number=20,span=10) resid
cor(qstats,number=20,span=10) residsq

```

EXERCISES

- 3.1. Derive multistep-ahead forecasts for a GARCH(1,2) model at the forecast origin h .
- 3.2. Derive multistep-ahead forecasts for a GARCH(2,1) model at the forecast origin h .
- 3.3. Suppose that $r_1, \dots, r_n$ are observations of a return series that follows the AR(1)-GARCH(1,1) model

$$r_t = \mu + \phi_1 r_{t-1} + a_t, \quad a_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2,$$

where ϵ_t is a standard Gaussian white noise series. Derive the conditional log-likelihood function of the data.

- 3.4. In the equation in Exercise 3.3, assume that ϵ_t follows a standardized Student- t distribution with v degrees of freedom. Derive the conditional log-likelihood function of the data.

- 3.5. Consider the monthly simple returns of Intel stock from January 1973 to December 2008 in `m-intc7308.txt`. Transform the returns into log returns. Build a GARCH model for the transformed series and compute 1-step- to 5-step-ahead volatility forecasts at the forecast origin December 2008.
- 3.6. The file `m-mrk4608.txt` contains monthly simple returns of Merck stock from June 1946 to December 2008. The file has two columns denoting date and simple return. Transform the simple returns to log returns.
- Is there any evidence of serial correlations in the log returns? Use autocorrelations and 5% significance level to answer the question. If yes, remove the serial correlations.
 - Is there any evidence of ARCH effects in the log returns? Use the residual series if there are serial correlations in part (a). Use Ljung–Box statistics for the squared returns (or residuals) with 6 and 12 lags of autocorrelations and 5% significance level to answer the question.
 - Identify an ARCH model for the data and fit the identified model. Write down the fitted model.
- 3.7. The file `m-3m4608.txt` contains two columns. They are date and the monthly simple return for 3M stock. Transform the returns to log returns.
- Is there any evidence of ARCH effects in the log returns? Use Ljung–Box statistics with 6 and 12 lags of autocorrelations and 5% significance level to answer the question.
 - Use the PACF of the squared returns to identify an ARCH model. What is the fitted model?
 - There are 755 data points. Refit the model using the first 750 observations and use the fitted model to predict the volatilities for t from 751 to 755 (the forecast origin is 750).
 - Build an ARCH-M model for the log return series of 3M stock. Test the hypothesis that the risk premium is zero at the 5% significance level. Draw your conclusion.
 - Build an EGARCH model for the log return series of 3M stock using the first 750 observations. Use the fitted model to compute 1-step- to 5-step-ahead volatility forecasts at the forecast origin $h = 750$.
- 3.8. The file `m-gmsp5008.txt` contains the dates and monthly simple returns of General Motors stock and the S&P 500 index from 1950 to 2008.
- Build a GARCH model with Gaussian innovations for the log returns of GM stock. Check the model and write down the fitted model.
 - Build a GARCH-M model with Gaussian innovations for the log returns of GM stock. What is the fitted model?
 - Build a GARCH model with Student- t distribution for the log returns of GM stock, including estimation of the degrees of freedom. Write

- down the fitted model. Let v be the degrees of freedom of the Student- t distribution. Test the hypothesis $H_0 : v = 6$ versus $H_a : v \neq 6$, using the 5% significance level.
- (d) Build an EGARCH model for the log returns of GM stock. What is the fitted model?
 - (e) Obtain 1-step- to 6-step-ahead volatility forecasts for all the models obtained. Compare the forecasts.
- 3.9. Consider the monthly log returns of GM stock in `m-gmsp5008.txt`. Build an adequate TGARCH model for the series. Write down the fitted model and test for the significance of the leverage effect. Obtain 1-step- to 6-steps-ahead volatility forecasts.
- 3.10. Again, consider the returns in `m-gmsp5008.txt`.
- (a) Build a Gaussian GARCH model for the monthly log returns of the S&P 500 index. Check the model carefully.
 - (b) Is there a summer effect on the volatility of the index return? Use the GARCH model built in part (a) to answer this question.
 - (c) Are lagged returns of GM stock useful in modeling the index volatility? Again, use the GARCH model of part (a) as a baseline model for comparison.
- 3.11. The file `d-gmsp9908.txt` contains the daily simple returns of GM stock and the S&P composite index from 1999 to 2008. It has three columns denoting date, GM return, and S&P return.
- (a) Compute the daily log returns of GM stock. Is there any evidence of ARCH effects in the log returns? You may use 10 lags of the squared returns and 5% significance level to perform the test.
 - (b) Compute the PACF of the squared log returns (10 lags).
 - (c) Specify a GARCH model for the GM log return using a normal distribution for the innovations. Perform model checking and write down the fitted model.
 - (d) Find an adequate GARCH model for the series but using the generalized error distribution for the innovations. Write down the fitted model.
- 3.12. Consider the daily simple returns of the S&P composite index in the file `d-gmsp9908.txt`.
- (a) Is there any ARCH effect in the simple return series? Use 10 lags of the squared returns and 5% significance level to perform the test.
 - (b) Build an adequate GARCH model for the simple return series.
 - (c) Compute 1-step- to 4-step-ahead forecasts of the simple return and its volatility based on the fitted model.
- 3.13. Again, consider the daily simple returns of GM stock in the file `d-gmsp9908.txt`.

- (a) Find an adequate GARCH-M model for the series. Write down the fitted model.
 - (b) Find an adequate EGARCH model for the series. Is the “leverage” effect significant at the 5% level?
- 3.14. Revisit the file `d-gmsp9908.txt`. However, we shall investigate the value of using market volatility in modeling volatility of individual stocks. Convert the two simple return series into percentage log return series.
- (a) Build an AR(5)–GARCH(1,1) model with generalized error distribution for the log S&P returns. The AR(5) contains only lags 3 and 5. Denote the fitted volatility series by `spvol`.
 - (b) Estimate a GARCH(1,1) model with `spvol` as an exogenous variable to the log GM return series. Check the adequacy of the model, and write down the fitted model. In S-Plus, the command is


```
fit = garch(gm ~ 1, ~garch(1,1)+spvol, cond.dist='ged')
```
 - (c) Discuss the implication of the fitted model.
- 3.15. Again, consider the percentage daily log returns of GM stock and the S&P 500 index from 1999 to 2008 as before, but we shall investigate whether the volatility of GM stock has any contribution in modeling the S&P index volatility. Follow the steps below to perform the analysis.
- (a) Fit a GARCH(1,1) model with generalized error distribution to the percentage log returns of GM stock. Denote the fitted volatility by `gmvol`. Build an adequate GARCH model plus `gmvol` as the exogenous variable for the log S&P return series. Write down the fitted model.
 - (b) Is the volatility of GM stock returns helpful in modeling the volatility of the S&P index returns? Why?

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Multivariate Time Series Analysis and Its Applications

Economic globalization and Internet communication have accelerated the integration of world financial markets in recent years. Price movements in one market can spread easily and instantly to another market. For this reason, financial markets are more dependent on each other than ever before, and one must consider them jointly to better understand the dynamic structure of the global finance. One market may lead the other market under some circumstances, yet the relationship may be reversed under other circumstances. Consequently, knowing how the markets are interrelated is of great importance in finance. Similarly, for an investor or a financial institution holding multiple assets, the dynamic relationships between returns of the assets play an important role in decision making. In this and the next two chapters, we introduce econometric models and methods useful for studying jointly multiple return series. In the statistical literature, these models and methods belong to vector or multivariate time series analysis.

A multivariate time series consists of multiple single series referred to as *components*. As such, concepts of vector and matrix are useful in understanding multivariate time series analysis. We use boldface notation to indicate vectors and matrices. If necessary, readers may consult Appendix A of this chapter for some basic operations and properties of vectors and matrices. Appendix B provides some results of multivariate normal distribution, which is widely used in multivariate statistical analysis (e.g., Johnson and Wichern, 1998).

Let $\mathbf{r}_t = (r_{1t}, r_{2t}, \dots, r_{kt})'$ be the log returns of k assets at time t , where $\mathbf{a}'$ denotes the transpose of $\mathbf{a}$. For example, an investor holding stocks of IBM, Microsoft, Exxon Mobil, General Motors, and Wal-Mart may consider the five-dimensional daily log returns of these companies. Here r_{1t} denotes the daily log return of IBM stock, r_{2t} is that of Microsoft, and so on. As a second example, an investor who is interested in global investment may consider the return series of the S&P 500 index of the United States, the FTSE 100 index of the United

Kingdom, and the Nikkei 225 index of Japan. Here the series is three-dimensional, with r_{1t} denoting the return of the S&P 500 index, r_{2t} the return of the Financial Times Stock Exchange (FTSE) 100 index, and r_{3t} the return of the Nikkei 225. The goals of this chapter are (a) to explore the basic properties of $\mathbf{r}_t$ and (b) to study econometric models for analyzing the multivariate data $\{\mathbf{r}_t | t = 1, \dots, T\}$.

Many of the models and methods discussed in previous chapters can be generalized directly to the multivariate case. But there are situations in which the generalization requires some attention. In some situations, one needs new models and methods to handle the complicated relationships between multiple series. In this chapter, we discuss these issues with emphasis on intuition and applications. For statistical theory of multivariate time series analysis, readers are referred to Lütkepohl (2005) and Reinsel (1993).

8.1 WEAK STATIONARITY AND CROSS-CORRELATION MATRICES

Consider a k -dimensional time series $\mathbf{r}_t = (r_{1t}, \dots, r_{kt})'$. The series $\mathbf{r}_t$ is *weakly stationary* if its first and second moments are time invariant. In particular, the mean vector and covariance matrix of a weakly stationary series are constant over time. Unless stated explicitly to the contrary, we assume that the return series of financial assets are weakly stationary.

For a weakly stationary time series $\mathbf{r}_t$, we define its mean vector and covariance matrix as

$$\boldsymbol{\mu} = E(\mathbf{r}_t), \quad \boldsymbol{\Gamma}_0 = E[(\mathbf{r}_t - \boldsymbol{\mu})(\mathbf{r}_t - \boldsymbol{\mu})'], \quad (8.1)$$

where the expectation is taken element by element over the joint distribution of $\mathbf{r}_t$. The mean $\boldsymbol{\mu}$ is a k -dimensional vector consisting of the unconditional expectations of the components of $\mathbf{r}_t$. The covariance matrix $\boldsymbol{\Gamma}_0$ is a $k \times k$ matrix. The i th diagonal element of $\boldsymbol{\Gamma}_0$ is the variance of r_{it} , whereas the (i, j) th element of $\boldsymbol{\Gamma}_0$ is the covariance between r_{it} and r_{jt} . We write $\boldsymbol{\mu} = (\mu_1, \dots, \mu_k)'$ and $\boldsymbol{\Gamma}_0 = [\Gamma_{ij}(0)]$ when the elements are needed.

8.1.1 Cross-Correlation Matrices

Let $\mathbf{D}$ be a $k \times k$ diagonal matrix consisting of the standard deviations of r_{it} for $i = 1, \dots, k$. In other words, $\mathbf{D} = \text{diag}\{\sqrt{\Gamma_{11}(0)}, \dots, \sqrt{\Gamma_{kk}(0)}\}$. The concurrent, or lag-zero, cross-correlation matrix of $\mathbf{r}_t$ is defined as

$$\boldsymbol{\rho}_0 \equiv [\rho_{ij}(0)] = \mathbf{D}^{-1} \boldsymbol{\Gamma}_0 \mathbf{D}^{-1}.$$

More specifically, the (i, j) th element of $\boldsymbol{\rho}_0$ is

$$\rho_{ij}(0) = \frac{\Gamma_{ij}(0)}{\sqrt{\Gamma_{ii}(0)\Gamma_{jj}(0)}} = \frac{\text{Cov}(r_{it}, r_{jt})}{\text{std}(r_{it})\text{std}(r_{jt})},$$

which is the correlation coefficient between r_{it} and r_{jt} . In time series analysis, such a correlation coefficient is referred to as a concurrent, or contemporaneous, correlation coefficient because it is the correlation of the two series at time t . It is easy to see that $\rho_{ij}(0) = \rho_{ji}(0)$, $-1 \leq \rho_{ij}(0) \leq 1$, and $\rho_{ii}(0) = 1$ for $1 \leq i, j \leq k$. Thus, $\boldsymbol{\rho}(0)$ is a symmetric matrix with unit diagonal elements.

An important topic in multivariate time series analysis is the lead-lag relationships between component series. To this end, the cross-correlation matrices are used to measure the strength of linear dependence between time series. The lag- ℓ cross-covariance matrix of $\mathbf{r}_t$ is defined as

$$\boldsymbol{\Gamma}_\ell \equiv [\Gamma_{ij}(\ell)] = E[(\mathbf{r}_t - \boldsymbol{\mu})(\mathbf{r}_{t-\ell} - \boldsymbol{\mu})'], \quad (8.2)$$

where $\boldsymbol{\mu}$ is the mean vector of $\mathbf{r}_t$. Therefore, the (i, j) th element of $\boldsymbol{\Gamma}_\ell$ is the covariance between r_{it} and $r_{j,t-\ell}$. For a weakly stationary series, the cross-covariance matrix $\boldsymbol{\Gamma}_\ell$ is a function of ℓ , not the time index t .

The lag- ℓ cross-correlation matrix (CCM) of $\mathbf{r}_t$ is defined as

$$\boldsymbol{\rho}_\ell \equiv [\rho_{ij}(\ell)] = \mathbf{D}^{-1} \boldsymbol{\Gamma}_\ell \mathbf{D}^{-1}, \quad (8.3)$$

where, as before, $\mathbf{D}$ is the diagonal matrix of standard deviations of the individual series r_{it} . From the definition,

$$\rho_{ij}(\ell) = \frac{\Gamma_{ij}(\ell)}{\sqrt{\Gamma_{ii}(0)\Gamma_{jj}(0)}} = \frac{\text{Cov}(r_{it}, r_{j,t-\ell})}{\text{std}(r_{it})\text{std}(r_{jt})}, \quad (8.4)$$

which is the correlation coefficient between r_{it} and $r_{j,t-\ell}$. When $\ell > 0$, this correlation coefficient measures the linear dependence of r_{it} on $r_{j,t-\ell}$, which occurred prior to time t . Consequently, if $\rho_{ij}(\ell) \neq 0$ and $\ell > 0$, we say that the series r_{jt} *leads* the series r_{it} at lag ℓ . Similarly, $\rho_{ji}(\ell)$ measures the linear dependence of r_{jt} and $r_{i,t-\ell}$, and we say that the series r_{it} *leads* the series r_{jt} at lag ℓ if $\rho_{ji}(\ell) \neq 0$ and $\ell > 0$. Equation (8.4) also shows that the diagonal element $\rho_{ii}(\ell)$ is simply the lag- ℓ autocorrelation coefficient of r_{it} .

Based on this discussion, we obtain some important properties of the cross correlations when $\ell > 0$. First, in general, $\rho_{ij}(\ell) \neq \rho_{ji}(\ell)$ for $i \neq j$ because the two correlation coefficients measure different linear relationships between $\{r_{it}\}$ and $\{r_{jt}\}$. Therefore, $\boldsymbol{\Gamma}_\ell$ and $\boldsymbol{\rho}_\ell$ are in general not symmetric. Second, using $\text{Cov}(x, y) = \text{Cov}(y, x)$ and the weak stationarity assumption, we have

$$\text{Cov}(r_{it}, r_{j,t-\ell}) = \text{Cov}(r_{j,t-\ell}, r_{it}) = \text{Cov}(r_{jt}, r_{i,t+\ell}) = \text{Cov}(r_{jt}, r_{i,t-(-\ell)}),$$

so that $\Gamma_{ij}(\ell) = \Gamma_{ji}(-\ell)$. Because $\Gamma_{ji}(-\ell)$ is the (j, i) th element of the matrix $\boldsymbol{\Gamma}_{-\ell}$ and the equality holds for $1 \leq i, j \leq k$, we have $\boldsymbol{\Gamma}_\ell = \boldsymbol{\Gamma}'_{-\ell}$ and $\boldsymbol{\rho}_\ell = \boldsymbol{\rho}'_{-\ell}$. Consequently, unlike the univariate case, $\boldsymbol{\rho}_\ell \neq \boldsymbol{\rho}_{-\ell}$ for a general vector time series when $\ell > 0$. Because $\boldsymbol{\rho}_\ell = \boldsymbol{\rho}'_{-\ell}$, it suffices in practice to consider the cross-correlation matrices $\boldsymbol{\rho}_\ell$ for $\ell \geq 0$.

8.1.2 Linear Dependence

Considered jointly, the cross-correlation matrices $\{\rho_\ell | \ell = 0, 1, \dots\}$ of a weakly stationary vector time series contain the following information:

1. The diagonal elements $\{\rho_{ii}(\ell) | \ell = 0, 1, \dots\}$ are the autocorrelation function of r_{it} .
2. The off-diagonal element $\rho_{ij}(0)$ measures the concurrent linear relationship between r_{it} and r_{jt} .
3. For $\ell > 0$, the off-diagonal element $\rho_{ij}(\ell)$ measures the linear dependence of r_{it} on the past value $r_{j,t-\ell}$.

Therefore, if $\rho_{ij}(\ell) = 0$ for all $\ell > 0$, then r_{it} does not depend linearly on any past value $r_{j,t-\ell}$ of the r_{jt} series.

In general, the linear relationship between two time series $\{r_{it}\}$ and $\{r_{jt}\}$ can be summarized as follows:

1. r_{it} and r_{jt} have no linear relationship if $\rho_{ij}(\ell) = \rho_{ji}(\ell) = 0$ for all $\ell \geq 0$.
2. r_{it} and r_{jt} are concurrently correlated if $\rho_{ij}(0) \neq 0$.
3. r_{it} and r_{jt} have no lead-lag relationship if $\rho_{ij}(\ell) = 0$ and $\rho_{ji}(\ell) = 0$ for all $\ell > 0$. In this case, we say the two series are uncoupled.
4. There is a *unidirectional relationship* from r_{it} to r_{jt} if $\rho_{ij}(\ell) = 0$ for all $\ell > 0$, but $\rho_{ji}(v) \neq 0$ for some $v > 0$. In this case, r_{it} does not depend on any past value of r_{jt} , but r_{jt} depends on some past values of r_{it} .
5. There is a *feedback relationship* between r_{it} and r_{jt} if $\rho_{ij}(\ell) \neq 0$ for some $\ell > 0$ and $\rho_{ji}(v) \neq 0$ for some $v > 0$.

The conditions stated earlier are sufficient conditions. A more informative approach to study the relationship between time series is to build a multivariate model for the series because a properly specified model considers simultaneously the serial and cross correlations among the series.

8.1.3 Sample Cross-Correlation Matrices

Given the data $\{\mathbf{r}_t | t = 1, \dots, T\}$, the cross-covariance matrix $\mathbf{\Gamma}_\ell$ can be estimated by

$$\hat{\mathbf{\Gamma}}_\ell = \frac{1}{T} \sum_{t=\ell+1}^T (\mathbf{r}_t - \bar{\mathbf{r}})(\mathbf{r}_{t-\ell} - \bar{\mathbf{r}})', \quad \ell \geq 0, \quad (8.5)$$

where $\bar{\mathbf{r}} = (\sum_{t=1}^T \mathbf{r}_t)/T$ is the vector of sample means. The cross-correlation matrix ρ_ℓ is estimated by

$$\hat{\rho}_\ell = \hat{\mathbf{D}}^{-1} \hat{\mathbf{\Gamma}}_\ell \hat{\mathbf{D}}^{-1}, \quad \ell \geq 0, \quad (8.6)$$

where $\widehat{\mathbf{D}}$ is the $k \times k$ diagonal matrix of the sample standard deviations of the component series.

Similar to the univariate case, asymptotic properties of the sample cross-correlation matrix $\widehat{\boldsymbol{\rho}}_\ell$ have been investigated under various assumptions; see, for instance, Fuller (1976, Chapter 6). The estimate is consistent but is biased in a finite sample. For asset return series, the finite sample distribution of $\widehat{\boldsymbol{\rho}}_\ell$ is rather complicated partly because of the presence of conditional heteroscedasticity and high kurtosis. If the finite-sample distribution of cross correlations is needed, we recommend that proper bootstrap resampling methods be used to obtain an approximate estimate of the distribution. For many applications, a crude approximation of the variance of $\hat{\rho}_{ij}(\ell)$ is sufficient.

Example 8.1. Consider the monthly log returns of IBM stock and the S&P 500 index from January 1926 to December 2008 with 996 observations. The returns include dividend payments and are in percentages. Denote the returns of IBM stock and the S&P 500 index by r_{1t} and r_{2t} , respectively. These two returns form a bivariate time series $\mathbf{r}_t = (r_{1t}, r_{2t})'$. Figure 8.1 shows the time plots of $\mathbf{r}_t$. Figure 8.2 shows some scatterplots of the two series. The plots show that the two return series are concurrently correlated. Indeed, the sample concurrent correlation coefficient between the two returns is 0.65, which is statistically significant at the 5% level. However, the cross correlations at lag 1 are weak if any.

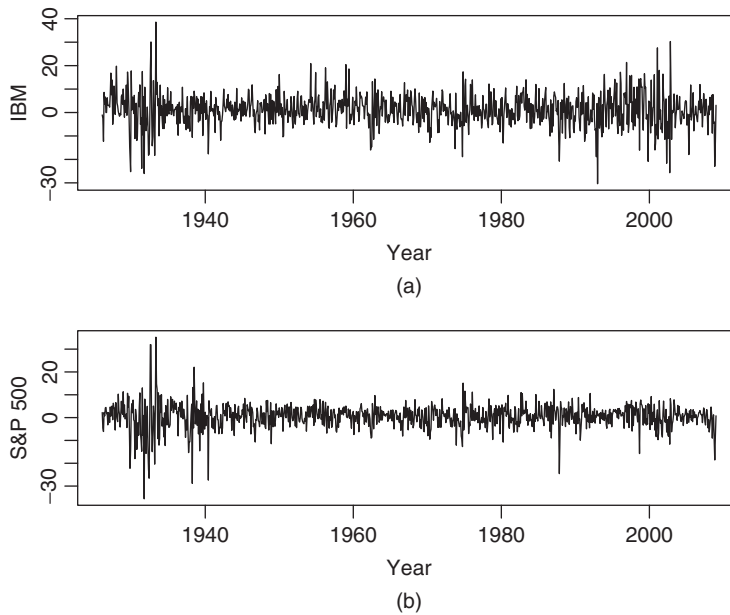


Figure 8.1 Time plots of monthly log returns, in percentages, for (a) IBM stock and (b) the S&P 500 index from January 1926 to December 2008.

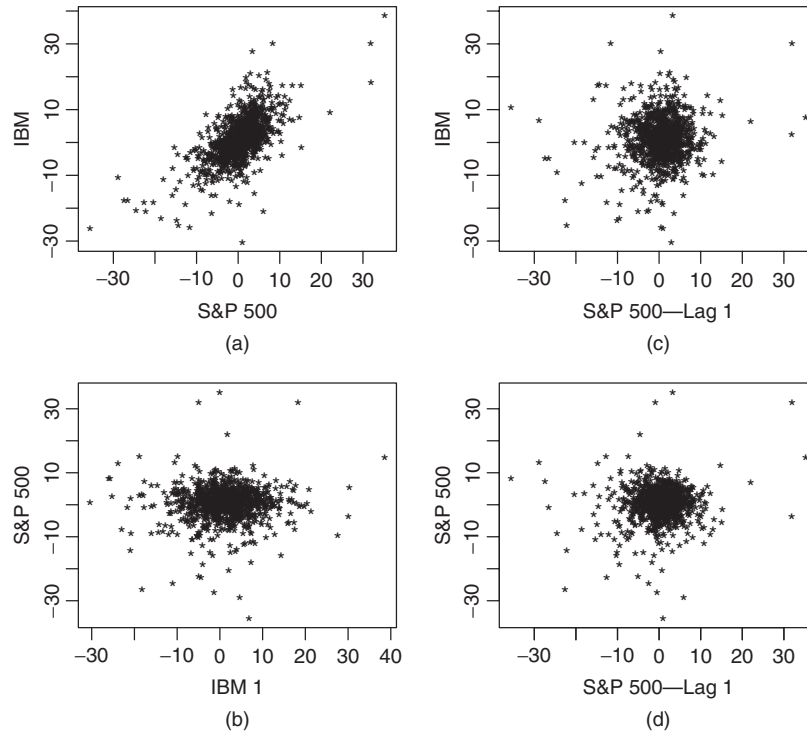


Figure 8.2 Some scatterplots for monthly log returns of IBM stock and S&P 500 index: (a) concurrent plot of IBM vs. S&P 500, (b) S&P 500 vs. lag-1 IBM, (c) IBM vs. lag-1 S&P 500, and (d) S&P 500 vs. lag-1 S&P 500.

Table 8.1 provides some summary statistics and cross-correlation matrices of the two series. For a bivariate series, each CCM is a 2×2 matrix with four correlations. Empirical experience indicates that it is rather hard to absorb simultaneously many cross-correlation matrices, especially when the dimension k is greater than 3. To overcome this difficulty, we use the simplifying notation of Tiao and Box (1981) and define a simplified cross-correlation matrix consisting of three symbols “+,” “−,” and “.” where they have the following meaning:

1. Plus sign (+) means that the corresponding correlation coefficient is greater than or equal to $2/\sqrt{T}$.
2. Minus sign (−) means that the corresponding correlation coefficient is less than or equal to $-2/\sqrt{T}$.
3. Period (.) means that the corresponding correlation coefficient is between $-2/\sqrt{T}$ and $2/\sqrt{T}$.

And $1/\sqrt{T}$ is the asymptotic 5% critical value of the sample correlation under the assumption that r_t is a white noise series.

TABLE 8.1 Summary Statistics and Cross-Correlation Matrices of Monthly Log Returns of IBM Stock and S&P 500 Index: January 1926 to December 2008

(a) Summary Statistics																			
Ticker		Mean		Standard Error		Skewness		Excess Kurtosis		Minimum		Maximum							
IBM		1.089		7.033		−0.068		2.622		−30.37		38.57							
SP5		0.430		5.537		−0.521		7.927		−35.59		35.22							
(b) Cross-Correlation Matrices																			
Lag 1		Lag 2		Lag 3		Lag 4		Lag 5											
0.04	0.10	0.00	−0.08	−0.01	−0.06	−0.03	−0.03	0.02	0.08										
0.04	0.08	0.02	−0.02	−0.06	−0.10	0.04	0.03	0.00	0.09										
(c) Simplified notation																			
$\begin{bmatrix} \cdot & + \\ \cdot & + \end{bmatrix}$				$\begin{bmatrix} \cdot & - \\ \cdot & \cdot \end{bmatrix}$				$\begin{bmatrix} \cdot & \cdot \\ \cdot & - \end{bmatrix}$				$\begin{bmatrix} \cdot & \cdot \\ \cdot & \cdot \end{bmatrix}$				$\begin{bmatrix} \cdot & + \\ \cdot & + \end{bmatrix}$			

Table 8.1(c) shows the simplified CCM for the monthly log returns of IBM stock and the S&P 500 index. It is easily seen that significant cross correlations at the approximate 5% level appear mainly at lags 1 and 3. An examination of the sample CCMs at these two lags indicates that (a) S&P 500 index returns have some marginal autocorrelations at lags 1, 2, 3, and 5 and (b) IBM stock returns depend weakly on the previous returns of the S&P 500 index. The latter observation is based on the significance of cross correlations at the (1, 2)th element of lag-1, lag-2 and lag-5 CCMs.

Figure 8.3 shows the sample autocorrelations and cross correlations of the two series. The upper-left plot is the sample ACF of IBM stock returns and the upper-right plot shows the dependence of IBM stock returns on the lagged S&P 500 index returns. The dashed lines in the plots are the asymptotic two standard error limits of the sample auto- and cross-correlation coefficients. From the plots, the dynamic relationship is weak between the two return series, but their contemporaneous correlation is statistically significant.

Example 8.2. Consider the simple returns of monthly indexes of U.S. government bonds with maturities in 30 years, 20 years, 10 years, 5 years, and 1 year. The data obtained from the CRSP database have 696 observations starting from January 1942 to December 1999. Let $\mathbf{r}_t = (r_{1t}, \dots, r_{5t})'$ be the return series with decreasing time to maturity. Figure 8.4 shows the time plots of $\mathbf{r}_t$ on the same scale. The variability of the 1-year bond returns is much smaller than that of returns with longer maturities. The sample means and standard deviations of the data are $\hat{\boldsymbol{\mu}} = 10^{-2}(0.43, 0.45, 0.45, 0.46, 0.44)'$ and $\hat{\boldsymbol{\sigma}} = 10^{-2}(2.53, 2.43, 1.97, 1.39, 0.53)'$.

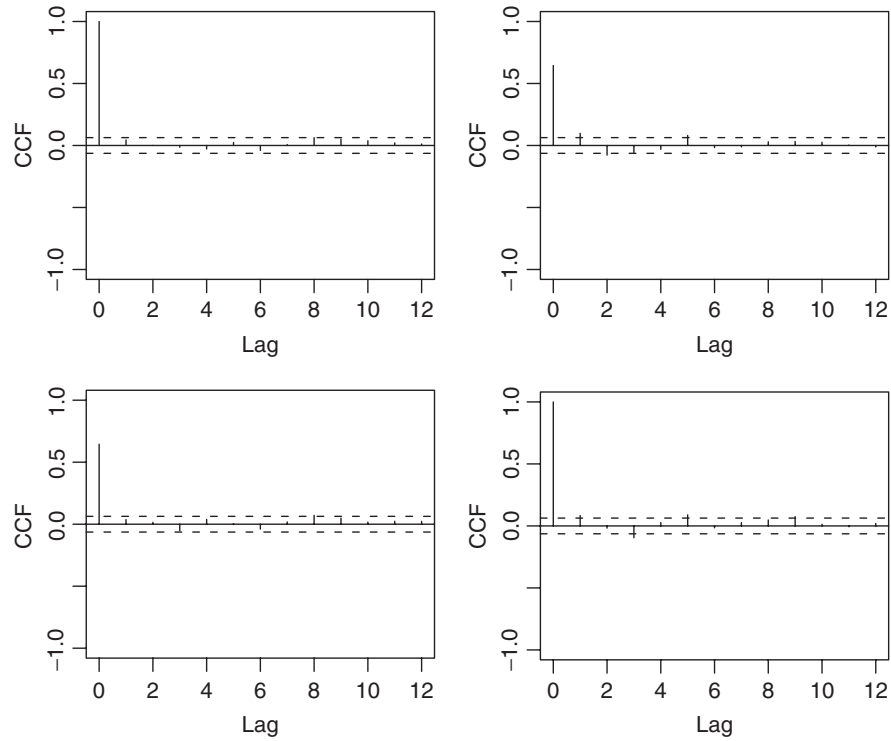


Figure 8.3 Sample auto- and cross-correlation functions (CCF) of two monthly log return series: (a) sample ACF of IBM stock returns, (b) cross-correlations between S&P 500 index and lagged IBM stock returns (lower left), (c) cross correlations between IBM stock and lagged S&P 500 index returns, and (d) sample ACF of S&P 500 index returns. Dashed lines denote 95% limits.

The concurrent correlation matrix of the series is

$$\hat{\rho}_0 = \begin{bmatrix} 1.00 & 0.98 & 0.92 & 0.85 & 0.63 \\ 0.98 & 1.00 & 0.91 & 0.86 & 0.64 \\ 0.92 & 0.91 & 1.00 & 0.90 & 0.68 \\ 0.85 & 0.86 & 0.90 & 1.00 & 0.82 \\ 0.63 & 0.64 & 0.68 & 0.82 & 1.00 \end{bmatrix}.$$

It is not surprising that (a) the series have high concurrent correlations, and (b) the correlations between long-term bonds are higher than those between short-term bonds.

Table 8.2 gives the lag-1 and lag-2 cross-correlation matrices of $\mathbf{r}_t$ and the corresponding simplified matrices. Most of the significant cross correlations are at lag 1, and the five return series appear to be intercorrelated. In addition, lag-1 and lag-2 sample ACFs of the 1-year bond returns are substantially higher than those of other series with longer maturities.

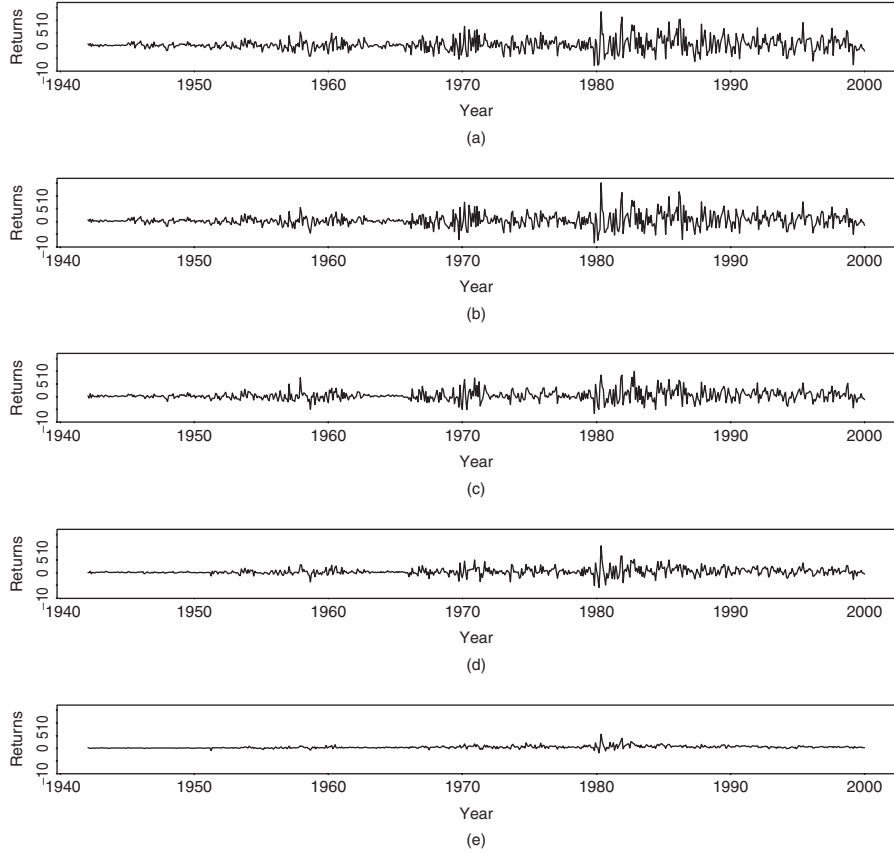


Figure 8.4 Time plots of monthly simple returns of five indexes of U.S. government bonds with maturities in (a) 30 years, (b) 20 years, (c) 10 years, (d) 5 years, and (e) 1 year. Sample period is from January 1942 to December 1999.

8.1.4 Multivariate Portmanteau Tests

The univariate Ljung–Box statistic $Q(m)$ has been generalized to the multivariate case by Hosking (1980, 1981) and Li and McLeod (1981). For a multivariate series, the null hypothesis of the test statistic is $H_0 : \rho_1 = \cdots = \rho_m = \mathbf{0}$, and the alternative hypothesis $H_a : \rho_i \neq \mathbf{0}$ for some $i \in \{1, \dots, m\}$. Thus, the statistic is used to test that there are no auto- and cross correlations in the vector series $\mathbf{r}_t$. The test statistic assumes the form

$$Q_k(m) = T^2 \sum_{\ell=1}^m \frac{1}{T-\ell} \text{tr}(\hat{\Gamma}_\ell' \hat{\Gamma}_0^{-1} \hat{\Gamma}_\ell \hat{\Gamma}_0^{-1}), \quad (8.7)$$

where T is the sample size, k is the dimension of $\mathbf{r}_t$, and $\text{tr}(\mathbf{A})$ is the trace of the matrix $\mathbf{A}$, which is the sum of the diagonal elements of $\mathbf{A}$. Under the null

TABLE 8.2 Sample Cross-Correlation Matrices of Monthly Simple Returns of Five Indexes of U.S. Government Bonds: January 1942 to December 1999

Lag 1					Lag 2				
Cross-Correlations									
0.10	0.08	0.11	0.12	0.16	−0.01	0.00	0.00	−0.03	0.03
0.10	0.08	0.12	0.14	0.17	−0.01	0.00	0.00	−0.04	0.02
0.09	0.08	0.09	0.13	0.18	0.01	0.01	0.01	−0.02	0.07
0.14	0.12	0.15	0.14	0.22	−0.02	−0.01	0.00	−0.04	0.07
0.17	0.15	0.21	0.22	0.40	−0.02	0.00	0.02	0.02	0.22
Simplified Cross-Correlation Matrices									
$\begin{bmatrix} + & + & + & + & + \\ + & + & + & + & + \\ + & + & + & + & + \\ + & + & + & + & + \\ \cdot & \cdot & \cdot & \cdot & \cdot \end{bmatrix}$					$\begin{bmatrix} \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & + \end{bmatrix}$				

hypothesis and some regularity conditions, $Q_k(m)$ follows asymptotically a chi-squared distribution with k^2m degrees of freedom.

Remark. The $Q_k(m)$ statistics can be rewritten in terms of the sample cross-correlation matrices $\widehat{\rho}_\ell$. Using the Kronecker product $\otimes$ and vectorization of matrices discussed in Appendix A of this chapter, we have

$$Q_k(m) = T^2 \sum_{\ell=1}^m \frac{1}{T-\ell} \mathbf{b}'_\ell (\widehat{\rho}_0^{-1} \otimes \widehat{\rho}_0^{-1}) \mathbf{b}_\ell,$$

where $\mathbf{b}_\ell = \text{vec}(\widehat{\rho}'_\ell)$. The test statistic proposed by Li and McLeod (1981) is

$$Q_k^*(m) = T \sum_{\ell=1}^m \mathbf{b}'_\ell (\widehat{\rho}_0^{-1} \otimes \widehat{\rho}_0^{-1}) \mathbf{b}_\ell + \frac{k^2 m(m+1)}{2T},$$

which is asymptotically equivalent to $Q_k(m)$. $\square$

Applying the $Q_k(m)$ statistics to the bivariate monthly log returns of IBM stock and the S&P 500 index of Example 8.1, we have $Q_2(1) = 9.81$, $Q_2(5) = 47.06$, and $Q_2(10) = 71.65$. Based on asymptotic chi-squared distributions with degrees of freedom 4, 20, and 40, the p values of these $Q_2(m)$ statistics are 0.044, 0.001, and 0.002, respectively. The portmanteau tests thus confirm the existence of serial dependence in the bivariate return series at the 5% significance level. For the five-dimensional monthly simple returns of bond indexes in Example 8.2, we have $Q_5(5) = 1065.63$, which is highly significant compared with a chi-squared distribution with 125 degrees of freedom.

The $Q_k(m)$ statistic is a joint test for checking the first m cross-correlation matrices of $\mathbf{r}_t$ being zero. If it rejects the null hypothesis, then we build a multivariate model for the series to study the lead–lag relationships between the component series. In what follows, we discuss some simple vector models useful for modeling the linear dynamic structure of a multivariate financial time series.

8.2 VECTOR AUTOREGRESSIVE MODELS

A simple vector model useful in modeling asset returns is the vector autoregressive (VAR) model. A multivariate time series $\mathbf{r}_t$ is a VAR process of order 1, or VAR(1) for short, if it follows the model

$$\mathbf{r}_t = \boldsymbol{\phi}_0 + \boldsymbol{\Phi} \mathbf{r}_{t-1} + \mathbf{a}_t, \quad (8.8)$$

where $\boldsymbol{\phi}_0$ is a k -dimensional vector, $\boldsymbol{\Phi}$ is a $k \times k$ matrix, and $\{\mathbf{a}_t\}$ is a sequence of serially uncorrelated random vectors with mean zero and covariance matrix $\boldsymbol{\Sigma}$. In application, the covariance matrix $\boldsymbol{\Sigma}$ is required to be positive definite; otherwise, the dimension of $\mathbf{r}_t$ can be reduced. In the literature, it is often assumed that $\mathbf{a}_t$ is multivariate normal.

Consider the bivariate case [i.e., $k = 2$, $\mathbf{r}_t = (r_{1t}, r_{2t})'$, and $\mathbf{a}_t = (a_{1t}, a_{2t})'$]. The VAR(1) model consists of the following two equations:

$$\begin{aligned} r_{1t} &= \phi_{10} + \Phi_{11}r_{1,t-1} + \Phi_{12}r_{2,t-1} + a_{1t}, \\ r_{2t} &= \phi_{20} + \Phi_{21}r_{1,t-1} + \Phi_{22}r_{2,t-1} + a_{2t}, \end{aligned}$$

where Φ_{ij} is the (i, j) th element of $\boldsymbol{\Phi}$ and ϕ_{i0} is the i th element of $\boldsymbol{\phi}_0$. Based on the first equation, Φ_{12} denotes the linear dependence of r_{1t} on $r_{2,t-1}$ in the presence of $r_{1,t-1}$. Therefore, Φ_{12} is the conditional effect of $r_{2,t-1}$ on r_{1t} given $r_{1,t-1}$. If $\Phi_{12} = 0$, then r_{1t} does not depend on $r_{2,t-1}$, and the model shows that r_{1t} only depends on its own past. Similarly, if $\Phi_{21} = 0$, then the second equation shows that r_{2t} does not depend on $r_{1,t-1}$ when $r_{2,t-1}$ is given.

Consider the two equations jointly. If $\Phi_{12} = 0$ and $\Phi_{21} \neq 0$, then there is a unidirectional relationship from r_{1t} to r_{2t} . If $\Phi_{12} = \Phi_{21} = 0$, then r_{1t} and r_{2t} are uncoupled. If $\Phi_{12} \neq 0$ and $\Phi_{21} \neq 0$, then there is a feedback relationship between the two series.

8.2.1 Reduced and Structural Forms

In general, the coefficient matrix $\boldsymbol{\Phi}$ of Eq. (8.8) measures the dynamic dependence of $\mathbf{r}_t$. The concurrent relationship between r_{1t} and r_{2t} is shown by the off-diagonal element σ_{12} of the covariance matrix $\boldsymbol{\Sigma}$ of $\mathbf{a}_t$. If $\sigma_{12} = 0$, then there is no concurrent linear relationship between the two component series. In the econometric literature, the VAR(1) model in Eq. (8.8) is called a *reduced-form* model because it

does not show explicitly the concurrent dependence between the component series. If necessary, an explicit expression involving the concurrent relationship can be deduced from the reduced-form model by a simple linear transformation. Because Σ is positive definite, there exists a lower triangular matrix L with unit diagonal elements and a diagonal matrix G such that $\Sigma = LGL'$; see Appendix A on Cholesky decomposition. Therefore, $L^{-1}\Sigma(L')^{-1} = G$.

Define $b_t = (b_{1t}, \dots, b_{kt})' = L^{-1}a_t$. Then

$$E(b_t) = L^{-1}E(a_t) = 0, \quad \text{Cov}(b_t) = L^{-1}\Sigma(L')^{-1} = L^{-1}\Sigma(L')^{-1} = G.$$

Since G is a diagonal matrix, the components of b_t are uncorrelated. Multiplying L^{-1} from the left to model (8.8), we obtain

$$L^{-1}r_t = L^{-1}\phi_0 + L^{-1}\Phi r_{t-1} + L^{-1}a_t = \phi_0^* + \Phi^* r_{t-1} + b_t, \quad (8.9)$$

where $\phi_0^* = L^{-1}\phi_0$ is a k -dimensional vector and $\Phi^* = L^{-1}\Phi$ is a $k \times k$ matrix. Because of the special matrix structure, the k th row of L^{-1} is in the form $(w_{k1}, w_{k2}, \dots, w_{k,k-1}, 1)$. Consequently, the k th equation of model (8.9) is

$$r_{kt} + \sum_{i=1}^{k-1} w_{ki} r_{it} = \phi_{k,0}^* + \sum_{i=1}^k \Phi_{ki}^* r_{i,t-1} + b_{kt}, \quad (8.10)$$

where $\phi_{k,0}^*$ is the k th element of ϕ_0^* and Φ_{ki}^* is the (k, i) th element of Φ^* . Because b_{kt} is uncorrelated with b_{it} for $1 \leq i < k$, Eq. (8.10) shows explicitly the concurrent linear dependence of r_{kt} on r_{it} , where $1 \leq i \leq k-1$. This equation is referred to as a *structural equation* for r_{kt} in the econometric literature.

For any other component r_{it} of r_t , we can rearrange the VAR(1) model so that r_{it} becomes the last component of r_t . The prior transformation method can then be applied to obtain a structural equation for r_{it} . Therefore, the reduced-form model (8.8) is equivalent to the structural form used in the econometric literature. In time series analysis, the reduced-form model is commonly used for two reasons. The first reason is ease in estimation. The second and main reason is that the concurrent correlations cannot be used in forecasting.

Example 8.3. To illustrate the transformation from a reduced-form model to structural equations, consider the bivariate AR(1) model

$$\begin{bmatrix} r_{1t} \\ r_{2t} \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix} + \begin{bmatrix} 0.2 & 0.3 \\ -0.6 & 1.1 \end{bmatrix} \begin{bmatrix} r_{1,t-1} \\ r_{2,t-1} \end{bmatrix} + \begin{bmatrix} a_{1t} \\ a_{2t} \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}.$$

For this particular covariance matrix Σ , the lower triangular matrix

$$L^{-1} = \begin{bmatrix} 1.0 & 0.0 \\ -0.5 & 1.0 \end{bmatrix}$$

provides a Cholesky decomposition [i.e., $\mathbf{L}^{-1}\mathbf{\Sigma}(\mathbf{L}')^{-1}$ is a diagonal matrix]. Premultiplying $\mathbf{L}^{-1}$ to the previous bivariate AR(1) model, we obtain

$$\begin{bmatrix} 1.0 & 0.0 \\ -0.5 & 1.0 \end{bmatrix} \begin{bmatrix} r_{1t} \\ r_{2t} \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.3 \end{bmatrix} + \begin{bmatrix} 0.2 & 0.3 \\ -0.7 & 0.95 \end{bmatrix} \begin{bmatrix} r_{1,t-1} \\ r_{2,t-1} \end{bmatrix} + \begin{bmatrix} b_{1t} \\ b_{2t} \end{bmatrix},$$

$$\mathbf{G} = \begin{bmatrix} 2 & 0 \\ 0 & 0.5 \end{bmatrix},$$

where $\mathbf{G} = \text{Cov}(\mathbf{b}_t)$. The second equation of this transformed model gives

$$r_{2t} = 0.3 + 0.5r_{1t} - 0.7r_{1,t-1} + 0.95r_{2,t-1} + b_{2t},$$

which shows explicitly the linear dependence of r_{2t} on r_{1t} .

Rearranging the order of elements in $\mathbf{r}_t$, the bivariate AR(1) model becomes

$$\begin{bmatrix} r_{2t} \\ r_{1t} \end{bmatrix} = \begin{bmatrix} 0.4 \\ 0.2 \end{bmatrix} + \begin{bmatrix} 1.1 & -0.6 \\ 0.3 & 0.2 \end{bmatrix} \begin{bmatrix} r_{2,t-1} \\ r_{1,t-1} \end{bmatrix} + \begin{bmatrix} a_{2t} \\ a_{1t} \end{bmatrix}, \quad \mathbf{\Sigma} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

The lower triangular matrix needed in the Cholesky decomposition of $\mathbf{\Sigma}$ becomes

$$\mathbf{L}^{-1} = \begin{bmatrix} 1.0 & 0.0 \\ -1.0 & 1.0 \end{bmatrix}.$$

Premultiplying $\mathbf{L}^{-1}$ to the earlier rearranged VAR(1) model, we obtain

$$\begin{bmatrix} 1.0 & 0.0 \\ -1.0 & 1.0 \end{bmatrix} \begin{bmatrix} r_{2t} \\ r_{1t} \end{bmatrix} = \begin{bmatrix} 0.4 \\ -0.2 \end{bmatrix} + \begin{bmatrix} 1.1 & -0.6 \\ -0.8 & 0.8 \end{bmatrix} \begin{bmatrix} r_{2,t-1} \\ r_{1,t-1} \end{bmatrix} + \begin{bmatrix} c_{1t} \\ c_{2t} \end{bmatrix},$$

$$\mathbf{G} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

where $\mathbf{G} = \text{Cov}(\mathbf{c}_t)$. The second equation now gives

$$r_{1t} = -0.2 + 1.0r_{2t} - 0.8r_{2,t-1} + 0.8r_{1,t-1} + c_{2t}.$$

Again this equation shows explicitly the concurrent linear dependence of r_{1t} on r_{2t} .

8.2.2 Stationarity Condition and Moments of a VAR(1) Model

Assume that the VAR(1) model in Eq. (8.8) is weakly stationary. Taking expectation of the model and using $E(\mathbf{a}_t) = \mathbf{0}$, we obtain

$$E(\mathbf{r}_t) = \boldsymbol{\phi}_0 + \boldsymbol{\Phi}E(\mathbf{r}_{t-1}).$$

Since $E(\mathbf{r}_t)$ is time invariant, we have

$$\boldsymbol{\mu} \equiv E(\mathbf{r}_t) = (\mathbf{I} - \boldsymbol{\Phi})^{-1}\boldsymbol{\phi}_0$$

provided that the matrix $\mathbf{I} - \Phi$ is nonsingular, where $\mathbf{I}$ is the $k \times k$ identity matrix.

Using $\phi_0 = (\mathbf{I} - \Phi)\mu$, the VAR(1) model in Eq. (8.8) can be written as

$$(\mathbf{r}_t - \mu) = \Phi(\mathbf{r}_{t-1} - \mu) + \mathbf{a}_t.$$

Let $\tilde{\mathbf{r}}_t = \mathbf{r}_t - \mu$ be the mean-corrected time series. Then the VAR(1) model becomes

$$\tilde{\mathbf{r}}_t = \Phi \tilde{\mathbf{r}}_{t-1} + \mathbf{a}_t. \quad (8.11)$$

This model can be used to derive properties of a VAR(1) model. By repeated substitutions, we can rewrite Eq. (8.11) as

$$\tilde{\mathbf{r}}_t = \mathbf{a}_t + \Phi \mathbf{a}_{t-1} + \Phi^2 \mathbf{a}_{t-2} + \Phi^3 \mathbf{a}_{t-3} + \cdots.$$

This expression shows several characteristics of a VAR(1) process. First, since $\mathbf{a}_t$ is serially uncorrelated, it follows that $\text{Cov}(\mathbf{a}_t, \mathbf{r}_{t-1}) = \mathbf{0}$. In fact, $\mathbf{a}_t$ is not correlated with $\mathbf{r}_{t-\ell}$ for all $\ell > 0$. For this reason, $\mathbf{a}_t$ is referred to as the *shock* or *innovation* of the series at time t . It turns out that, similar to the univariate case, $\mathbf{a}_t$ is uncorrelated with the past value $\mathbf{r}_{t-j}$ ($j > 0$) for all time series models. Second, postmultiplying the expression by $\mathbf{a}_t'$, taking expectation, and using the fact of no serial correlations in the $\mathbf{a}_t$ process, we obtain $\text{Cov}(\mathbf{r}_t, \mathbf{a}_t) = \Sigma$. Third, for a VAR(1) model, $\mathbf{r}_t$ depends on the past innovation $\mathbf{a}_{t-j}$ with coefficient matrix Φ^j . For such dependence to be meaningful, Φ^j must converge to zero as $j \rightarrow \infty$. This means that the k eigenvalues of Φ must be less than 1 in modulus; otherwise, Φ^j will either explode or converge to a nonzero matrix as $j \rightarrow \infty$. As a matter of fact, the requirement that all eigenvalues of Φ are less than 1 in modulus is the necessary and sufficient condition for weak stationarity of $\mathbf{r}_t$ provided that the covariance matrix of $\mathbf{a}_t$ exists. Notice that this stationarity condition reduces to that of the univariate AR(1) case in which the condition is $|\phi| < 1$. Furthermore, because

$$|\lambda \mathbf{I} - \Phi| = \lambda^k \left| \mathbf{I} - \Phi \frac{1}{\lambda} \right|,$$

the eigenvalues of Φ are the inverses of the zeros of the determinant $|\mathbf{I} - \Phi B|$. Thus, an equivalent sufficient and necessary condition for stationarity of $\mathbf{r}_t$ is that all zeros of the determinant $|\Phi(B)|$ are greater than one in modulus; that is, all zeros are outside the unit circle in the complex plane. Fourth, using the expression, we have

$$\text{Cov}(\mathbf{r}_t) = \Gamma_0 = \Sigma + \Phi \Sigma \Phi' + \Phi^2 \Sigma (\Phi^2)' + \cdots = \sum_{i=0}^{\infty} \Phi^i \Sigma (\Phi^i)',$$

where it is understood that $\Phi^0 = \mathbf{I}$, the $k \times k$ identity matrix.

Postmultiplying $\tilde{\mathbf{r}}'_{t-\ell}$ to Eq. (8.11), taking expectation, and using the result $\text{Cov}(\mathbf{a}_t, \mathbf{r}_{t-j}) = E(\mathbf{a}_t \tilde{\mathbf{r}}'_{t-j}) = \mathbf{0}$ for $j > 0$, we obtain

$$E(\tilde{\mathbf{r}}_t \tilde{\mathbf{r}}'_{t-\ell}) = \Phi E(\tilde{\mathbf{r}}_{t-1} \tilde{\mathbf{r}}'_{t-\ell}), \quad \ell > 0.$$

Therefore,

$$\Gamma_\ell = \Phi \Gamma_{\ell-1}, \quad \ell > 0, \quad (8.12)$$

where Γ_j is the lag- j cross-covariance matrix of $\mathbf{r}_t$. Again this result is a generalization of that of a univariate AR(1) process. By repeated substitutions, Eq. (8.12) shows that

$$\Gamma_\ell = \Phi^\ell \Gamma_0, \quad \text{for } \ell > 0.$$

Pre- and postmultiplying Eq. (8.12) by $\mathbf{D}^{-1/2}$, we obtain

$$\rho_\ell = \mathbf{D}^{-1/2} \Phi \Gamma_{\ell-1} \mathbf{D}^{-1/2} = \mathbf{D}^{-1/2} \Phi \mathbf{D}^{1/2} \mathbf{D}^{-1/2} \Gamma_{\ell-1} \mathbf{D}^{-1/2} = \Upsilon \rho_{\ell-1},$$

where $\Upsilon = \mathbf{D}^{-1/2} \Phi \mathbf{D}^{1/2}$. Consequently, the CCM of a VAR(1) model satisfies

$$\rho_\ell = \Upsilon^\ell \rho_0, \quad \text{for } \ell > 0.$$

8.2.3 Vector AR(p) Models

The generalization of VAR(1) to VAR(p) models is straightforward. The time series $\mathbf{r}_t$ follows a VAR(p) model if it satisfies

$$\mathbf{r}_t = \phi_0 + \Phi_1 \mathbf{r}_{t-1} + \cdots + \Phi_p \mathbf{r}_{t-p} + \mathbf{a}_t, \quad p > 0, \quad (8.13)$$

where ϕ_0 and $\mathbf{a}_t$ are defined as before, and Φ_j are $k \times k$ matrices. Using the back-shift operator B , the VAR(p) model can be written as

$$(\mathbf{I} - \Phi_1 B - \cdots - \Phi_p B^p) \mathbf{r}_t = \phi_0 + \mathbf{a}_t,$$

where $\mathbf{I}$ is the $k \times k$ identity matrix. This representation can be written in a compact form as

$$\Phi(B) \mathbf{r}_t = \phi_0 + \mathbf{a}_t,$$

where $\Phi(B) = \mathbf{I} - \Phi_1 B - \cdots - \Phi_p B^p$ is a matrix polynomial. If $\mathbf{r}_t$ is weakly stationary, then we have

$$\mu = E(\mathbf{r}_t) = (\mathbf{I} - \Phi_1 - \cdots - \Phi_p)^{-1} \phi_0 = [\Phi(1)]^{-1} \phi_0$$

provided that the inverse exists. Let $\tilde{\mathbf{r}}_t = \mathbf{r}_t - \boldsymbol{\mu}$. The VAR(p) model becomes

$$\tilde{\mathbf{r}}_t = \Phi_1 \tilde{\mathbf{r}}_{t-1} + \cdots + \Phi_p \tilde{\mathbf{r}}_{t-p} + \mathbf{a}_t. \quad (8.14)$$

Using this equation and the same techniques as those for VAR(1) models, we obtain that

- $\text{Cov}(\mathbf{r}_t, \mathbf{a}_t) = \boldsymbol{\Sigma}$, the covariance matrix of $\mathbf{a}_t$.
- $\text{Cov}(\mathbf{r}_{t-\ell}, \mathbf{a}_t) = \mathbf{0}$ for $\ell > 0$.
- $\boldsymbol{\Gamma}_\ell = \Phi_1 \boldsymbol{\Gamma}_{\ell-1} + \cdots + \Phi_p \boldsymbol{\Gamma}_{\ell-p}$ for $\ell > 0$.

The last property is called the moment equations of a VAR(p) model. It is a multivariate version of the Yule–Walker equation of a univariate AR(p) model. In terms of CCM, the moment equations become

$$\boldsymbol{\rho}_\ell = \Upsilon_1 \boldsymbol{\rho}_{\ell-1} + \cdots + \Upsilon_p \boldsymbol{\rho}_{\ell-p} \quad \text{for } \ell > 0,$$

where $\Upsilon_i = \mathbf{D}^{-1/2} \Phi_i \mathbf{D}^{1/2}$.

A simple approach to understanding properties of the VAR(p) model in Eq. (8.13) is to make use of the results of the VAR(1) model in Eq. (8.8). This can be achieved by transforming the VAR(p) model of $\mathbf{r}_t$ into a kp -dimensional VAR(1) model. Specifically, let $\mathbf{x}_t = (\tilde{\mathbf{r}}'_{t-p+1}, \tilde{\mathbf{r}}'_{t-p+2}, \dots, \tilde{\mathbf{r}}'_t)'$ and $\mathbf{b}_t = (0, \dots, 0, \mathbf{a}'_t)'$ be two kp -dimensional processes. The mean of $\mathbf{b}_t$ is zero and the covariance matrix of $\mathbf{b}_t$ is a $kp \times kp$ matrix with zero everywhere except for the lower right corner, which is $\boldsymbol{\Sigma}$. The VAR(p) model for $\mathbf{r}_t$ can then be written in the form

$$\mathbf{x}_t = \Phi^* \mathbf{x}_{t-1} + \mathbf{b}_t, \quad (8.15)$$

where Φ^* is a $kp \times kp$ matrix given by

$$\Phi^* = \begin{bmatrix} \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} & \mathbf{0} & \cdots & \mathbf{0} \\ \vdots & \vdots & \vdots & & & \vdots \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{I} \\ \Phi_p & \Phi_{p-1} & \Phi_{p-2} & \Phi_{p-3} & \cdots & \Phi_1 \end{bmatrix},$$

where $\mathbf{0}$ and $\mathbf{I}$ are the $k \times k$ zero matrix and identity matrix, respectively. In the literature, Φ^* is called the *companion* matrix of the matrix polynomial $\Phi(B)$.

Equation (8.15) is a VAR(1) model for $\mathbf{x}_t$, which contains $\mathbf{r}_t$ as its last k components. The results of a VAR(1) model shown in the previous section can now be used to derive properties of the VAR(p) model via Eq. (8.15). For example, from the definition, $\mathbf{x}_t$ is weakly stationary if and only if $\mathbf{r}_t$ is weakly stationary. Therefore, the necessary and sufficient condition of weak stationarity for the VAR(p)

model in Eq. (8.13) is that all eigenvalues of Φ^* in Eq. (8.15) are less than 1 in modulus. It is easy to show that $|\mathbf{I} - \Phi^*B| = |\Phi(B)|$. Therefore, similar to the VAR(1) case, the necessary and sufficient condition is equivalent to all zeros of the determinant $|\Phi(B)|$ being outside the unit circle.

Of particular relevance to financial time series analysis is the structure of the coefficient matrices Φ_ℓ of a VAR(p) model. For instance, if the (i, j) th element $\Phi_{ij}(\ell)$ of Φ_ℓ is zero for all ℓ , then r_{it} does not depend on the past values of r_{jt} . The structure of the coefficient matrices Φ_ℓ thus provides information on the lead-lag relationship between the components of $\mathbf{r}_t$.

8.2.4 Building a VAR(p) Model

We continue to use the iterative procedure of order specification, estimation, and model checking to build a vector AR model for a given time series. The concept of partial autocorrelation function of a univariate series can be generalized to specify the order p of a vector series. Consider the following consecutive VAR models:

$$\begin{aligned} \mathbf{r}_t &= \phi_0 + \Phi_1 \mathbf{r}_{t-1} + \mathbf{a}_t \\ \mathbf{r}_t &= \phi_0 + \Phi_1 \mathbf{r}_{t-1} + \Phi_2 \mathbf{r}_{t-2} + \mathbf{a}_t \\ &\vdots \\ \mathbf{r}_t &= \phi_0 + \Phi_1 \mathbf{r}_{t-1} + \cdots + \Phi_i \mathbf{r}_{t-i} + \mathbf{a}_t \\ &\vdots \end{aligned} \tag{8.16}$$

Parameters of these models can be estimated by the ordinary least-squares (OLS) method. This is called the multivariate linear regression estimation in multivariate statistical analysis; see Johnson and Wichern (1998).

For the i th equation in Eq. (8.16), let $\hat{\Phi}_j^{(i)}$ be the OLS estimate of Φ_j and $\hat{\phi}_0^{(i)}$ be the estimate of ϕ_0 , where the superscript (i) is used to denote that the estimates are for a VAR(i) model. Then the residual is

$$\hat{\mathbf{a}}_t^{(i)} = \mathbf{r}_t - \hat{\phi}_0^{(i)} - \hat{\Phi}_1^{(i)} \mathbf{r}_{t-1} - \cdots - \hat{\Phi}_i^{(i)} \mathbf{r}_{t-i}.$$

For $i = 0$, the residual is defined as $\hat{\mathbf{r}}_t^{(0)} = \mathbf{r}_t - \bar{\mathbf{r}}$, where $\bar{\mathbf{r}}$ is the sample mean of $\mathbf{r}_t$. The residual covariance matrix is defined as

$$\hat{\Sigma}_i = \frac{1}{T - 2i - 1} \sum_{t=i+1}^T \hat{\mathbf{a}}_t^{(i)} (\hat{\mathbf{a}}_t^{(i)})', \quad i \geq 0. \tag{8.17}$$

To specify the order p , one can test the hypothesis $H_0 : \Phi_\ell = \mathbf{0}$ versus the alternative hypothesis $H_a : \Phi_\ell \neq \mathbf{0}$ sequentially for $\ell = 1, 2, \dots$. For example, using

the first equation in Eq. (8.16), we can test the hypothesis $H_0 : \Phi_1 = \mathbf{0}$ versus the alternative hypothesis $H_a : \Phi_1 \neq \mathbf{0}$. The test statistic is

$$M(1) = - \left(T - k - \frac{5}{2} \right) \ln \left(\frac{|\widehat{\Sigma}_1|}{|\widehat{\Sigma}_0|} \right),$$

where $\widehat{\Sigma}_i$ is defined in Eq. (8.17) and $|A|$ denotes the determinant of the matrix A . Under some regularity conditions, the test statistic $M(1)$ is asymptotically a chi-squared distribution with k^2 degrees of freedom; see Tiao and Box (1981).

In general, we use the i th and $(i - 1)$ th equations in Eq. (8.16) to test $H_0 : \Phi_i = \mathbf{0}$ versus $H_a : \Phi_i \neq \mathbf{0}$; that is, testing a VAR(i) model versus a VAR($i - 1$) model. The test statistic is

$$M(i) = - \left(T - k - i - \frac{3}{2} \right) \ln \left(\frac{|\widehat{\Sigma}_i|}{|\widehat{\Sigma}_{i-1}|} \right). \quad (8.18)$$

Asymptotically, $M(i)$ is distributed as a chi-squared distribution with k^2 degrees of freedom.

Alternatively, one can use the Akaike information criterion (AIC) or its variants to select the order p . Assume that $\mathbf{a}_t$ is multivariate normal and consider the i th equation in Eq. (8.16). One can estimate the model by the maximum-likelihood (ML) method. For AR models, the OLS estimates $\widehat{\phi}_0$ and $\widehat{\phi}_j$ are equivalent to the (conditional) ML estimates. However, there are differences between the estimates of Σ . The ML estimate of Σ is

$$\tilde{\Sigma}_i = \frac{1}{T} \sum_{t=i+1}^T \widehat{\mathbf{a}}_t^{(i)} [\widehat{\mathbf{a}}_t^{(i)}]'. \quad (8.19)$$

The AIC of a VAR(i) model under the normality assumption is defined as

$$\text{AIC}(i) = \ln(|\tilde{\Sigma}_i|) + \frac{2k^2 i}{T}.$$

For a given vector time series, one selects the AR order p such that $\text{AIC}(p) = \min_{0 \leq i \leq p_0} \text{AIC}(i)$, where p_0 is a prespecified positive integer.

Other information criteria available for VAR(i) models are

$$\begin{aligned} \text{BIC}(i) &= \ln(|\tilde{\Sigma}_i|) + \frac{k^2 i \ln(T)}{T}, \\ \text{HQ}(i) &= \ln(|\tilde{\Sigma}_i|) + \frac{2k^2 i \ln[\ln(T)]}{T}. \end{aligned}$$

The HQ criterion is proposed by Hannan and Quinn (1979).

Example 8.4. Assuming that the bivariate series of monthly log returns of IBM stock and the S&P 500 index discussed in Example 8.1 follows a VAR model, we apply the $M(i)$ statistics and AIC to the data. Table 8.3 shows the results of these statistics. Both statistics indicate that a VAR(5) model might be adequate for the data. The $M(i)$ statistics are marginally significant at lags 1, 3, and 5 at the 5% level. The minimum of AIC occurs at order 5. For this particular instance, the $M(i)$ statistic is only marginally significant at the 1% level when $i = 2$, confirming the previous observation that the dynamic linear dependence between the two return series is weak.

Estimation and Model Checking

For a specified VAR model, one can estimate the parameters using either the OLS method or the ML method. The two methods are asymptotically equivalent. Under some regularity conditions, the estimates are asymptotically normal; see Reinsel (1993). A fitted model should then be checked carefully for any possible inadequacy. The $Q_k(m)$ statistic can be applied to the residual series to check the assumption that there are no serial or cross correlations in the residuals. For a fitted VAR(p) model, the $Q_k(m)$ statistic of the residuals is asymptotically a chi-squared distribution with $k^2m - g$ degrees of freedom, where g is the number of estimated parameters in the AR coefficient matrices; see Lütkepohl (2005).

Example 8.4 (Continued). Table 8.4(a) shows the estimation results of a VAR(5) model for the bivariate series of monthly log returns of IBM stock and the S&P 500 index. The specified model is in the form

$$\mathbf{r}_t = \phi_0 + \Phi_1 \mathbf{r}_{t-1} + \Phi_2 \mathbf{r}_{t-2} + \Phi_3 \mathbf{r}_{t-3} + \Phi_5 \mathbf{r}_{t-5} + \mathbf{a}_t, \quad (8.20)$$

where the first component of $\mathbf{r}_t$ denotes IBM stock returns. For this particular instance, we do not use AR coefficient matrix at lag 4 because of the weak serial dependence of the data. In general, when the $M(i)$ statistics and the AIC criterion specify a VAR(5) model, all five AR lags should be used. Table 8.4(b) shows the estimation results after some statistically insignificant parameters are set to zero. The $Q_k(m)$ statistics of the residual series for the fitted model in Table 8.4(b) give $Q_2(4) = 16.64$ and $Q_2(8) = 31.55$. Since the fitted VAR(5) model has six parameters in the AR coefficient matrices, these two $Q_k(m)$ statistics are distributed asymptotically as a chi-squared distribution with degrees of freedom 10 and 26,

TABLE 8.3 Order Specification Statistics for Monthly Log Returns of IBM Stock and S&P 500 Index from January 1926 to December 2008^a

Order	1	2	3	4	5	6
$M(i)$	10.76	13.41	10.34	7.78	12.07	1.93
AIC	6.795	6.789	6.786	6.786	6.782	6.788

^aThe 5% and 1% critical values of a chi-squared distribution with 4 degrees of freedom are 9.5 and 13.3.

respectively. The p -values of the test statistics are 0.083 and 0.208, and hence the fitted model is adequate at the 5% significance level. As shown by the univariate analysis, the return series are likely to have conditional heteroscedasticity. We discuss multivariate volatility in Chapter 10.

From the fitted model in Table 8.4(b), we make the following observations: (a) The concurrent correlation coefficient between the two innovational series is $24/\sqrt{48 \times 30} = 0.63$, which, as expected, is close to the sample correlation coefficient between r_{1t} and r_{2t} . (b) The two log return series have positive and significant means, implying that the log prices of the two series had an upward trend over the data span. (c) The model shows that

$$\begin{aligned} \text{IBM}_t &= 1.0 + 0.13\text{SP5}_{t-1} - 0.09\text{SP5}_{t-2} + 0.09\text{SP5}_{t-5} + a_{1t}, \\ \text{SP5}_t &= 0.4 + 0.08\text{SP5}_{t-1} - 0.06\text{SP5}_{t-3} + 0.09\text{SP5}_{t-5} + a_{2t}. \end{aligned}$$

Consequently, at the 5% significance level, there is a unidirectional dynamic relationship from the monthly S&P 500 index return to the IBM return. If the S&P 500 index represents the U.S. stock market, then IBM return is affected by the past movements of the market. However, past movements of IBM stock returns do not significantly affect the U.S. market, even though the two returns have substantial concurrent correlation. Finally, the fitted model can be written as

$$\begin{aligned} \begin{bmatrix} \text{IBM}_t \\ \text{SP5}_t \end{bmatrix} &= \begin{bmatrix} 1.0 \\ 0.4 \end{bmatrix} + \begin{bmatrix} 0.13 \\ 0.08 \end{bmatrix} \text{SP5}_{t-1} - \begin{bmatrix} 0.09 \\ 0 \end{bmatrix} \text{SP5}_{t-2} - \begin{bmatrix} 0 \\ 0.06 \end{bmatrix} \text{SP5}_{t-3} \\ &\quad + \begin{bmatrix} 0.09 \\ 0.09 \end{bmatrix} \text{SP5}_{t-5} + \begin{bmatrix} a_{1t} \\ a_{2t} \end{bmatrix}, \end{aligned}$$

indicating that SP5_t is the driving factor of the bivariate series.

TABLE 8.4 Estimation Results of a VAR(5) Model for the Monthly Log Returns, in Percentages, of IBM Stock and S&P 500 Index from January 1926 to December 2008

Parameter	ϕ_0	Φ_1			Φ_2		Φ_3		Φ_5		Σ	
<i>(a) Full Model</i>												
Estimate	1.0	−0.03	0.15	0.10	−0.17	0.05	−0.11	−0.06	0.14	48	24	
	0.4	−0.03	0.11	0.04	−0.04	0.02	−0.11	−0.07	0.15	24	30	
Standard error	0.23	0.04	0.05	0.04	0.05	0.04	0.05	0.04	0.05			
	0.18	0.03	0.04	0.03	0.04	0.03	0.04	0.03	0.04			
<i>(b) Simplified Model</i>												
Estimate	1.0	0	0.13	0	−0.09	0	0	0	0.09	48	24	
	0.4	0	0.08	0	0	0	−0.06	0	0.09	24	30	
Standard error	0.22	—	0.04	—	0.03	—	—	—	0.04			
	0.18	—	0.03	—	—	—	−0.06	—	0.03			

Forecasting

Treating a properly built model as the true model, one can apply the same techniques as those in the univariate analysis to produce forecasts and standard deviations of the associated forecast errors. For a VAR(p) model, the 1-step-ahead forecast at the time origin h is $\mathbf{r}_h(1) = \boldsymbol{\phi}_0 + \sum_{i=1}^p \boldsymbol{\Phi}_i \mathbf{r}_{h+1-i}$, and the associated forecast error is $\mathbf{e}_h(1) = \mathbf{a}_{h+1}$. The covariance matrix of the forecast error is $\boldsymbol{\Sigma}$. For 2-step-ahead forecasts, we substitute $\mathbf{r}_{h+1}$ by its forecast to obtain

$$\mathbf{r}_h(2) = \boldsymbol{\phi}_0 + \boldsymbol{\Phi}_1 \mathbf{r}_h(1) + \sum_{i=2}^p \boldsymbol{\Phi}_i \mathbf{r}_{h+2-i},$$

and the associated forecast error is

$$\mathbf{e}_h(2) = \mathbf{a}_{h+2} + \boldsymbol{\Phi}_1 [\mathbf{r}_t - \mathbf{r}_h(1)] = \mathbf{a}_{h+2} + \boldsymbol{\Phi}_1 \mathbf{a}_{h+1}.$$

The covariance matrix of the forecast error is $\boldsymbol{\Sigma} + \boldsymbol{\Phi}_1 \boldsymbol{\Sigma} \boldsymbol{\Phi}_1'$. If $\mathbf{r}_t$ is weakly stationary, then the ℓ -step-ahead forecast $\mathbf{r}_h(\ell)$ converges to its mean vector $\boldsymbol{\mu}$ as the forecast horizon ℓ increases and the covariance matrix of its forecast error converges to the covariance matrix of $\mathbf{r}_t$.

Table 8.5 provides 1-step- to 6-step-ahead forecasts of the monthly log returns, in percentages, of IBM stock and the S&P 500 index at the forecast origin $h = 996$. These forecasts are obtained by the refined VAR(5) model in Table 8.4(b). As expected, the standard errors of the forecasts converge to the sample standard errors 7.03 and 5.53, respectively, for the two log return series.

In summary, building a VAR model involves three steps: (a) Use the test statistic $M(i)$ or some information criterion to identify the order, (b) estimate the specified model by using the least-squares method and, if necessary, reestimate the model by removing statistically insignificant parameters, and (c) use the $Q_k(m)$ statistic of the residuals to check the adequacy of a fitted model. Other characteristics of the residual series, such as conditional heteroscedasticity and outliers, can also be checked. If the fitted model is adequate, then it can be used to obtain forecasts and make inference concerning the dynamic relationship between the variables.

We used SCA to perform the analysis in this section. The commands used include `miden`, `mtsm`, `mest`, and `mfore`, where the prefix `m` stands for multivariate. Details of the commands and output are shown below.

TABLE 8.5 Forecasts of a VAR(5) Model for Monthly Log Returns, in Percentages, of IBM Stock and S&P 500 Index: Forecast Origin Is December 2008

Step	1	2	3	4	5	6
IBM forecast	1.95	0.30	-0.82	0.14	1.16	1.29
Standard error	6.95	6.99	7.00	7.00	7.00	7.00
SP forecast	1.70	0.17	-1.26	-0.49	0.41	0.65
Standard error	5.48	5.50	5.50	5.51	5.51	5.53

SCA Demonstration

Output has been edited and % denotes explanation in the following:

```
input date, ibm, sp5. file 'm-ibmsp2608.txt'.
```

```
--% compute percentage log returns.
```

```
ibm=ln(ibm+1)*100
```

```
--
```

```
sp5=ln(sp5+1)*100
```

```
--% model identification
```

```
miden ibm,sp5. arfits 1 to 12.
```

```
TIME PERIOD ANALYZED . . . . . 1 TO 996
```

```
EFFECTIVE NUMBER OF OBSERVATIONS (NOBE) . . . 996
```

SERIES	NAME	MEAN	STD. ERROR
1	IBM	1.0891	7.0298
2	SP5	0.4301	5.5346

NOTE: THE APPROX. STD. ERROR FOR THE ESTIMATED CORRELATIONS BELOW

IS $(1/\text{NOBE}^{*.5}) = 0.03169$

SAMPLE CORRELATION MATRIX OF THE SERIES

1.00

0.65 1.00

SUMMARIES OF CROSS CORRELATION MATRICES USING +, -, ., WHERE

+ DENOTES A VALUE GREATER THAN $2/\text{SQRT}(\text{NOBE})$

- DENOTES A VALUE LESS THAN $-2/\text{SQRT}(\text{NOBE})$

. DENOTES A NON-SIGNIFICANT VALUE BASED ON THE ABOVE CRITERION

CROSS CORRELATION MATRICES IN TERMS OF +, -, .

LAGS 1 THROUGH 6

.	+	.	-	.	.	.	.	.	+	.	.
.	+	.	.	.	-	.	.	.	+	.	.

LAGS 7 THROUGH 12

.	.	+	.	.	.	.	.	.	.	.	.
.	.	+	.	.	+	.	.	.	.	.	.

===== STEPWISE AUTOREGRESSION SUMMARY =====

LAG	I	RESIDUAL VARIANCES	I	EIGENVAL. OF SIGMA	I	CHI-SQ TEST	I	AIC	I	SIGN. PAR. AR
1	I	.492E+02	I	.133E+02	I	10.76	I	6.795	I	. +
	I	.306E+02	I	.665E+02	I		I		I	. +
2	I	.486E+02	I	.133E+02	I	13.41	I	6.789	I	+ -

```

      I .306E+02 I .659E+02 I      I      I . .
-----+-----+-----+-----+-----+-----
    3 I .484E+02 I .132E+02 I    10.34 I    6.786 I . .
      I .303E+02 I .655E+02 I      I      I . -
-----+-----+-----+-----+-----+-----
    4 I .484E+02 I .131E+02 I    7.78 I    6.786 I . .
      I .302E+02 I .655E+02 I      I      I . .
-----+-----+-----+-----+-----+-----
    5 I .480E+02 I .131E+02 I    12.07 I    6.782 I . +
      I .299E+02 I .648E+02 I      I      I - +
-----+-----+-----+-----+-----+-----
    6 I .479E+02 I .131E+02 I    1.93 I    6.788 I . .
      I .298E+02 I .647E+02 I      I      I . .
-----+-----+-----+-----+-----+-----
    7 I .479E+02 I .130E+02 I    2.68 I    6.793 I . .
      I .298E+02 I .647E+02 I      I      I . .
-----+-----+-----+-----+-----+-----
    8 I .477E+02 I .130E+02 I    7.09 I    6.794 I . .
      I .296E+02 I .643E+02 I      I      I . .
-----+-----+-----+-----+-----+-----
    9 I .476E+02 I .130E+02 I    5.23 I    6.797 I . .
      I .295E+02 I .642E+02 I      I      I . .
-----+-----+-----+-----+-----+-----
   10 I .476E+02 I .130E+02 I    1.43 I    6.803 I . .
      I .295E+02 I .641E+02 I      I      I . .
-----+-----+-----+-----+-----+-----
   11 I .475E+02 I .130E+02 I    1.81 I    6.809 I . .
      I .294E+02 I .640E+02 I      I      I . .
-----+-----+-----+-----+-----+-----
   12 I .475E+02 I .129E+02 I    1.88 I    6.815 I . .
      I .294E+02 I .640E+02 I      I      I . .
-----+-----+-----+-----+-----+-----

```

NOTE:CHI-SQUARED CRITICAL VALUES WITH 4 DEGREES OF FREEDOM ARE
5 PERCENT: 9.5 1 PERCENT: 13.3

-- % model specification of a VAR(5) model without lag 4.

mtsm m1. series ibm, sp5. model @

(i-p2*b-p2*b**2-p3*b**3-p5*b**5)series=c+noise.

-- % estimation

mestim m1. hold resi(r1,r2).

-- % demonstration of setting zero constraint

p2(2,2)=0

--

cp2(2,2)=1

--

p3(1,2)=0

--

cp3(1,2)=1

--

mestim m1. hold resi(r1,r2)

FINAL MODEL SUMMARY WITH CONDITIONAL LIKELIHOOD PAR. EST.

----- CONSTANT VECTOR (STD ERROR) -----

1.039 (0.223)
0.390 (0.176)

----- PHI MATRICES -----

ESTIMATES OF PHI(1) MATRIX AND SIGNIFICANCE

.000 .129 . +
.000 .080 . +

STANDARD ERRORS

-- .040
-- .031

ESTIMATES OF PHI(2) MATRIX AND SIGNIFICANCE

.000 -.090 . -
.000 .000 . .

STANDARD ERRORS

-- .031
-- --

ESTIMATES OF PHI(3) MATRIX AND SIGNIFICANCE

.000 .000 . .
.000 -.061 . -

STANDARD ERRORS

-- --
-- .024

ESTIMATES OF PHI(5) MATRIX AND SIGNIFICANCE

.000 .093 . +
.000 .087 . +

STANDARD ERRORS

-- .040
-- .032

ERROR COVARIANCE MATRIX

	1	2
1	48.328570	
2	24.361464	30.027406

-- % compute residual cross-correlation matrices

miden r1,r2. maxl 12.

-- % prediction

mfore m1. nofs 6.

6 FORECASTS, BEGINNING AT ORIGIN = 996

SERIES:	IBM	SP5
TIME	FORECAST	STD ERR
997	1.954	6.952
	FORECAST	STD ERR
	1.698	5.480

998	0.304	6.988	0.173	5.497
999	-0.815	7.001	-1.263	5.497
1000	0.138	7.001	-0.494	5.507
1001	1.162	7.002	0.408	5.508
1002	1.294	7.022	0.649	5.528

8.2.5 Impulse Response Function

Similar to the univariate case, a VAR(p) model can be written as a linear function of the past innovations, that is,

$$\mathbf{r}_t = \boldsymbol{\mu} + \mathbf{a}_t + \boldsymbol{\Psi}_1 \mathbf{a}_{t-1} + \boldsymbol{\Psi}_2 \mathbf{a}_{t-2} + \cdots, \quad (8.21)$$

where $\boldsymbol{\mu} = [\boldsymbol{\Phi}(1)]^{-1} \boldsymbol{\phi}_0$ provided that the inverse exists, and the coefficient matrices $\boldsymbol{\Psi}_i$ can be obtained by equating the coefficients of B^i in the equation

$$(\mathbf{I} - \boldsymbol{\Phi}_1 B - \cdots - \boldsymbol{\Phi}_p B^p)(\mathbf{I} + \boldsymbol{\Psi}_1 B + \boldsymbol{\Psi}_2 B^2 + \cdots) = \mathbf{I},$$

where $\mathbf{I}$ is the identity matrix. This is a moving-average representation of $\mathbf{r}_t$ with the coefficient matrix $\boldsymbol{\Psi}_i$ being the impact of the past innovation $\mathbf{a}_{t-i}$ on $\mathbf{r}_t$. Equivalently, $\boldsymbol{\Psi}_i$ is the effect of $\mathbf{a}_t$ on the future observation $\mathbf{r}_{t+i}$. Therefore, $\boldsymbol{\Psi}_i$ is often referred to as the *impulse response function* of $\mathbf{r}_t$. However, since the components of $\mathbf{a}_t$ are often correlated, the interpretation of elements in $\boldsymbol{\Psi}_i$ of Eq. (8.21) is not straightforward. To aid interpretation, one can use the Cholesky decomposition mentioned earlier to transform the innovations so that the resulting components are uncorrelated. Specifically, there exists a lower triangular matrix $\mathbf{L}$ such that $\boldsymbol{\Sigma} = \mathbf{LGL}'$, where $\mathbf{G}$ is a diagonal matrix and the diagonal elements of $\mathbf{L}$ are unity. See Eq. (8.9). Let $\mathbf{b}_t = \mathbf{L}^{-1} \mathbf{a}_t$. Then, $\text{Cov}(\mathbf{b}_t) = \mathbf{G}$ so that the elements b_{jt} are uncorrelated. Rewrite Eq. (8.21) as

$$\begin{aligned} \mathbf{r}_t &= \boldsymbol{\mu} + \mathbf{a}_t + \boldsymbol{\Psi}_1 \mathbf{a}_{t-1} + \boldsymbol{\Psi}_2 \mathbf{a}_{t-2} + \cdots \\ &= \boldsymbol{\mu} + \mathbf{L} \mathbf{L}^{-1} \mathbf{a}_t + \boldsymbol{\Psi}_1 \mathbf{L} \mathbf{L}^{-1} \mathbf{a}_{t-1} + \boldsymbol{\Psi}_2 \mathbf{L} \mathbf{L}^{-1} \mathbf{a}_{t-2} + \cdots \\ &= \boldsymbol{\mu} + \boldsymbol{\Psi}_0^* \mathbf{b}_t + \boldsymbol{\Psi}_1^* \mathbf{b}_{t-1} + \boldsymbol{\Psi}_2^* \mathbf{b}_{t-2} + \cdots, \end{aligned} \quad (8.22)$$

where $\boldsymbol{\Psi}_0^* = \mathbf{L}$ and $\boldsymbol{\Psi}_i^* = \boldsymbol{\Psi}_i \mathbf{L}$. The coefficient matrices $\boldsymbol{\Psi}_i^*$ are called the *impulse response function* of $\mathbf{r}_t$ with respect to the orthogonal innovations $\mathbf{b}_t$. Specifically, the (i, j) th element of $\boldsymbol{\Psi}_\ell^*$; that is, $\Psi_{ij}^*(\ell)$, is the impact of $b_{j,t}$ on the future observation $r_{i,t+\ell}$. In practice, one can further normalize the orthogonal innovation $\mathbf{b}_t$ such that the variance of b_{it} is one. A weakness of the above orthogonalization is that the result depends on the ordering of the components of $\mathbf{r}_t$. In particular, $b_{1t} = a_{1t}$ so that a_{1t} is not transformed. Different orderings of the components of $\mathbf{r}_t$ may lead to different impulse response functions. Interpretation of the impulse response function is, therefore, associated with the innovation series $\mathbf{b}_t$.

Both SCA and S-Plus enable one to obtain the impulse response function of a fitted VAR model. To demonstrate analysis of VAR models in S-Plus, we again use

the monthly log return series of IBM stock and the S&P 500 index of Example 8.1. For details of S-Plus commands, see Zivot and Wang (2003).

S-Plus Demonstration

The following output has been edited and % denotes explanation:

```
> module(finmetrics)
> da=read.table("m-ibmsp2608.txt",header=T) % Load data
> ibm=log(da[,2]+1)*100 % Compute percentage log returns
> sp5=log(da[,3]+1)*100
> y=cbind(ibm,sp5) % Create a vector series
> y1=data.frame(y) % Create a data frame
> ord.choice=VAR(y1,max.ar=10) % Order selection using BIC
> names(ord.choice)
[1] "R"      "coef"   "fitted"  "residuals" "Sigma" "df.resid"
[7] "rank"   "call"   "ar.order" "n.na"      "terms"  "Y0"
[13] "info"
> ord.choice$ar.order % selected order
[1] 1
> ord.choice$info
      ar(1)      ar(2)      ar(3)      ar(4)      ar(5)      ar(6)
BIC 12325.41 12339.42 12356.58 12376.28 12391.57 12417.2
      ar(7)      ar(8)      ar(9)      ar(10)
BIC 12442.03 12462.52 12484.78 12510.91
> ord=VAR(y1,max.ar=10,criterion='AIC') % Using AIC
> ord$ar.order
[1] 5
> ord$info
      ar(1)      ar(2)      ar(3)      ar(4)      ar(5)      ar(6)
AIC 12296.04 12290.48 12288.07 12288.2 12283.91 12289.96
      ar(7)      ar(8)      ar(9)      ar(10)
AIC 12295.22 12296.13 12298.82 12305.37
```

The AIC selects a VAR(5) model as before, but BIC selects a VAR(1) model. For simplicity, we shall use VAR(1) specification in the demonstration. Note that different normalizations are used between the two packages so that the values of information criteria appear to be different; see the AIC in Table 8.3. This is not important because normalization does not affect order selection. Turn to estimation.

```
> var1.fit=VAR(y~ar(1)) % Estimate a VAR(1) model
> summary(var1.fit)
Call:
VAR(formula = y ~ ar(1))
Coefficients:
              ibm      sp5
(Intercept)  1.0614  0.4087
      (std.err)  0.2249  0.1773
```

```

(t.stat)    4.7198  2.3053

ibm.lag1    -0.0320 -0.0223
(std.err)   0.0413  0.0326
(t.stat)    -0.7728 -0.6855

sp5.lag1     0.1503  0.1020
(std.err)    0.0525  0.0414
(t.stat)     2.8612  2.4637

Regression Diagnostics:
               ibm      sp5
R-squared    0.0101  0.0075
Adj. R-squared 0.0081  0.0055
Resid. Scale 7.0078  5.5247

Information Criteria:
      logL      AIC      BIC      HQ
-6193.988 12399.977 12429.393 12411.159

               total residual
Degree of freedom:  995      992

> plot(var1.fit)
Make a plot selection (or 0 to exit):
1: plot: All
2: plot: Response and Fitted Values
3: plot: Residuals
....
8: plot: PACF of Squared Residuals
Selection: 3

```

The fitted model is

$$IBM_t = 1.06 - 0.03IBM_{t-1} + 0.15SP5_{t-1} + a_{1t},$$

$$SP5_t = 0.41 - 0.02IBM_{t-1} + 0.10SP5_{t-1} + a_{2t}.$$

Based on t statistics of the coefficient estimates, only the lagged variable $SP5_{t-1}$ is informative in both equations. Figure 8.5 shows the time plots of the two residual series, where the two horizontal lines indicate the two standard error limits. As expected, there exist clusters of outlying observations.

Next, we compute 1-step- to 6-step-ahead forecasts and the impulse response function of the fitted VAR(1) model when the IBM stock return is the first component of $\mathbf{r}_t$. Compared with those of a VAR(5) model in Table 8.5, the forecasts of the VAR(1) model converge faster to the sample mean of the series.

```

> var1.pred=predict(var1.fit,n.predict=6) % Compute forecasts
> summary(var1.pred)

```

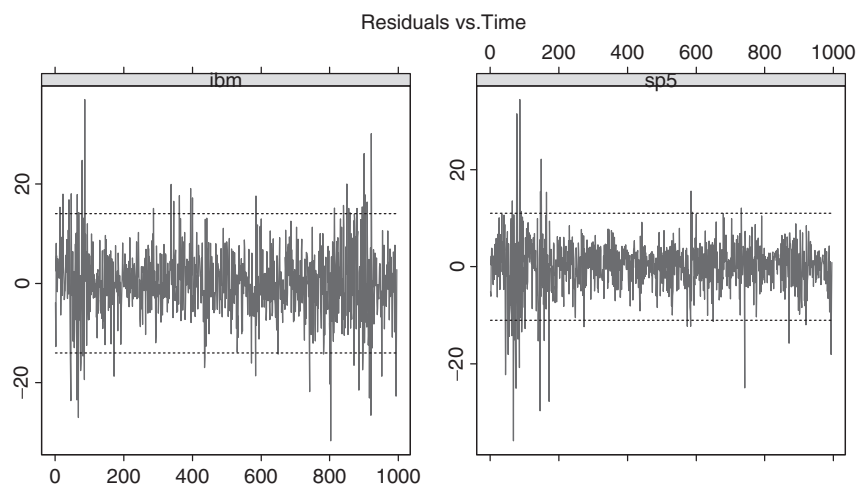


Figure 8.5 Residual plots of fitting a VAR(1) model to the monthly log returns, in percentages, of IBM stock and S&P 500 index. Sample period is from January 1926 to December 2008.

Predicted Values with Standard Errors:

```

          ibm    sp5
1-step-ahead 1.0798 0.4192
  (std.err)  7.0078 5.5247
2-step-ahead 1.0899 0.4274
  (std.err)  7.0434 5.5453
3-step-ahead 1.0908 0.4280
  (std.err)  7.0436 5.5454

```

...

```

6-step-ahead 1.0909 0.4280
  (std.err)  7.0436 5.5454

```

```
> plot(var1.pred,y,n.old=12) % Obtain forecast plot
```

```
% Below is to compute the impulse response function
```

```
> var1.irf=impRes(var1.fit,period=6,std.err='asymptotic')
```

```
> summary(var1.irf)
```

Impulse Response Function:

(with responses in rows, and innovations in columns)

```
, , lag.0
```

```

          ibm    sp5
      ibm 6.9973 0.0000
(std.err) 0.1569 0.0000
      sp5 3.5432 4.2280
(std.err) 0.1558 0.0948

```

```
, , lag.1
```

```

          ibm    sp5
      ibm 0.3088 0.6353
(std.err) 0.2217 0.2221

```

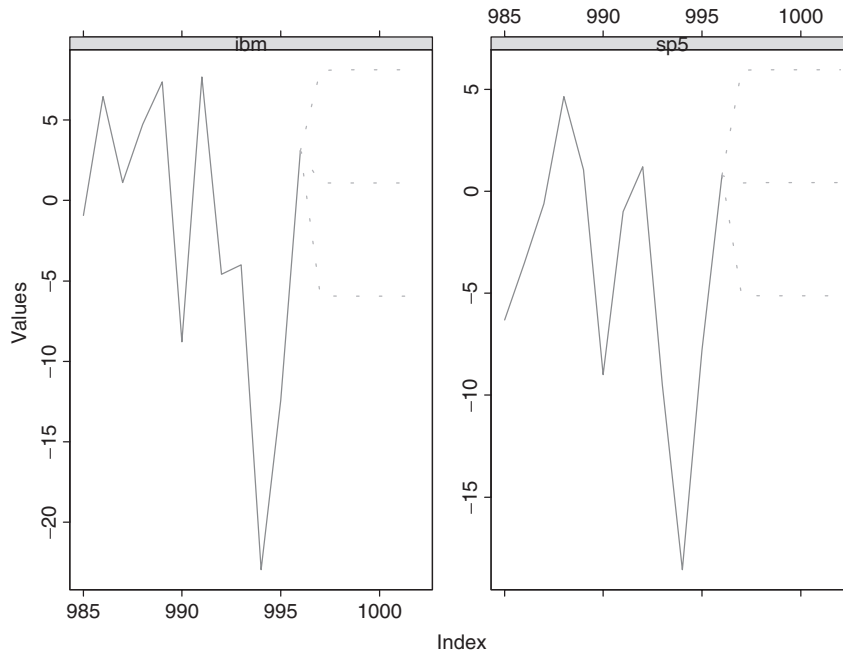


Figure 8.6 Forecasting plots of fitted VAR(1) model to monthly log returns, in percentages, of IBM stock and S&P 500 index. Sample period is from January 1926 to December 2008.

```

      sp5 0.2050 0.4312
(std.err) 0.1746 0.1750
.....
> plot(var1.irf)

```

Figure 8.6 shows the forecasts and their pointwise 95% confidence intervals along with the last 12 data points of the series. Figure 8.7 shows the impulse response functions of the fitted VAR(1) model where the IBM stock return is the first component of $\mathbf{r}_t$. Since the dynamic dependence of the returns is weak, the impulse response functions exhibit simple patterns and decay quickly.

8.3 VECTOR MOVING-AVERAGE MODELS

A vector moving-average model of order q , or VMA(q), is in the form

$$\mathbf{r}_t = \boldsymbol{\theta}_0 + \mathbf{a}_t - \boldsymbol{\Theta}_1 \mathbf{a}_{t-1} - \cdots - \boldsymbol{\Theta}_q \mathbf{a}_{t-q} \quad \text{or} \quad \mathbf{r}_t = \boldsymbol{\theta}_0 + \boldsymbol{\Theta}(B) \mathbf{a}_t, \quad (8.23)$$

where $\boldsymbol{\theta}_0$ is a k -dimensional vector, $\boldsymbol{\Theta}_i$ are $k \times k$ matrices, and $\boldsymbol{\Theta}(B) = \mathbf{I} - \boldsymbol{\Theta}_1 B - \cdots - \boldsymbol{\Theta}_q B^q$ is the MA matrix polynomial in the back-shift operator B . Similar to the univariate case, VMA(q) processes are weakly stationary provided

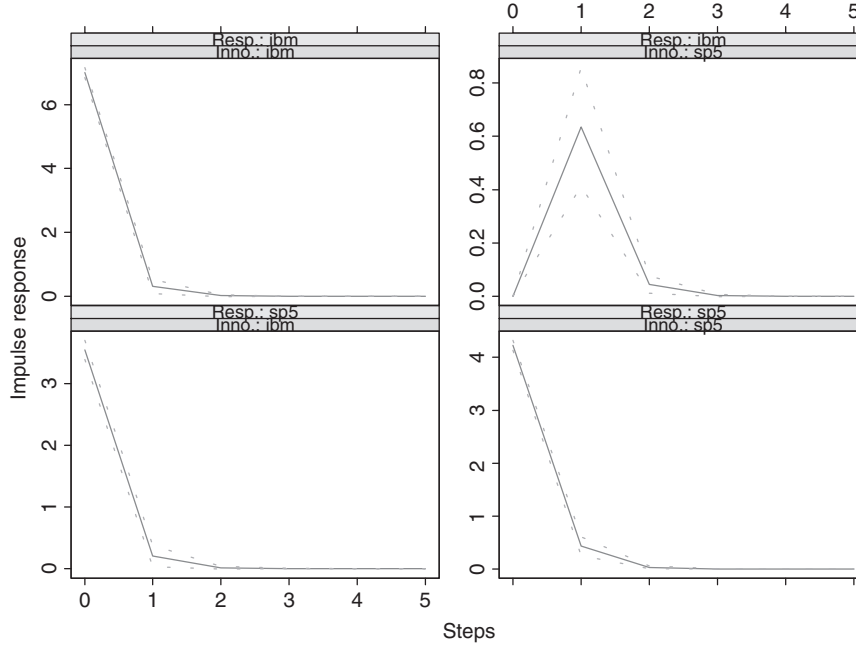


Figure 8.7 Plots of impulse response functions of orthogonal innovations for fitted VAR(1) model to monthly log returns, in percentages, of IBM stock and S&P 500 index. Sample period is from January 1926 to December 2008.

that the covariance matrix Σ of $\mathbf{a}_t$ exists. Taking expectation of Eq. (8.23), we obtain that $\boldsymbol{\mu} = E(\mathbf{r}_t) = \boldsymbol{\theta}_0$. Thus, the constant vector $\boldsymbol{\theta}_0$ is the mean vector of $\mathbf{r}_t$ for a VMA model.

Let $\tilde{\mathbf{r}}_t = \mathbf{r}_t - \boldsymbol{\theta}_0$ be the mean-corrected VAR(q) process. Then using Eq. (8.23) and the fact that $\{\mathbf{a}_t\}$ has no serial correlations, we have

1. $\text{Cov}(\mathbf{r}_t, \mathbf{a}_t) = \Sigma$.
2. $\Gamma_0 = \Sigma + \Theta_1 \Sigma \Theta_1' + \cdots + \Theta_q \Sigma \Theta_q'$.
3. $\Gamma_\ell = \mathbf{0}$ if $\ell > q$.
4. $\Gamma_\ell = \sum_{j=\ell}^q \Theta_j \Sigma \Theta_{j-\ell}'$ if $1 \leq \ell \leq q$, where $\Theta_0 = -I$.

Since $\Gamma_\ell = \mathbf{0}$ for $\ell > q$, the cross-correlation matrices (CCMs) of a VMA(q) process $\mathbf{r}_t$ satisfy

$$\rho_\ell = \mathbf{0}, \quad \ell > q. \quad (8.24)$$

Therefore, similar to the univariate case, the sample CCMs can be used to identify the order of a VMA process.

To better understand the VMA processes, let us consider the bivariate MA(1) model

$$\mathbf{r}_t = \boldsymbol{\theta}_0 + \mathbf{a}_t - \boldsymbol{\Theta} \mathbf{a}_{t-1} = \boldsymbol{\mu} + \mathbf{a}_t - \boldsymbol{\Theta} \mathbf{a}_{t-1}, \quad (8.25)$$

where, for simplicity, the subscript of $\boldsymbol{\Theta}_1$ is removed. This model can be written explicitly as

$$\begin{bmatrix} r_{1t} \\ r_{2t} \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} + \begin{bmatrix} a_{1t} \\ a_{2t} \end{bmatrix} - \begin{bmatrix} \Theta_{11} & \Theta_{12} \\ \Theta_{21} & \Theta_{22} \end{bmatrix} \begin{bmatrix} a_{1,t-1} \\ a_{2,t-1} \end{bmatrix}. \quad (8.26)$$

It says that the current return series $\mathbf{r}_t$ only depends on the current and past shocks. Therefore, the model is a finite-memory model.

Consider the equation for r_{1t} in Eq. (8.26). The parameter Θ_{12} denotes the linear dependence of r_{1t} on $a_{2,t-1}$ in the presence of $a_{1,t-1}$. If $\Theta_{12} = 0$, then r_{1t} does not depend on the lagged values of a_{2t} and, hence, the lagged values of r_{2t} . Similarly, if $\Theta_{21} = 0$, then r_{2t} does not depend on the past values of r_{1t} . The off-diagonal elements of $\boldsymbol{\Theta}$ thus show the dynamic dependence between the component series. For this simple VMA(1) model, we can classify the relationships between r_{1t} and r_{2t} as follows:

1. They are uncoupled series if $\Theta_{12} = \Theta_{21} = 0$.
2. There is a unidirectional dynamic relationship from r_{1t} to r_{2t} if $\Theta'_{12} = 0$, but $\Theta_{21} \neq 0$. The opposite unidirectional relationship holds if $\Theta_{21} = 0$, but $\Theta_{12} \neq 0$.
3. There is a feedback relationship between r_{1t} and r_{2t} if $\Theta_{12} \neq 0$ and $\Theta_{21} \neq 0$.

Finally, the concurrent correlation between r_{it} is the same as that between a_{it} . The previous classification can be generalized to a VMA(q) model.

Estimation

Unlike the VAR models, estimation of VMA models is much more involved; see Hillmer and Tiao (1979), Lütkepohl (2005), and the references therein. For the likelihood approach, there are two methods available. The first is the conditional-likelihood method that assumes that $\mathbf{a}_t = \mathbf{0}$ for $t \leq 0$. The second is the exact-likelihood method that treats $\mathbf{a}_t$ with $t \leq 0$ as additional parameters of the model. To gain some insight into the problem of estimation, we consider the VMA(1) model in Eq. (8.25). Suppose that the data are $\{\mathbf{r}_t | t = 1, \dots, T\}$ and $\mathbf{a}_t$ is multivariate normal. For a VMA(1) model, the data depend on $\mathbf{a}_0$.

Conditional MLE

The conditional-likelihood method assumes that $\mathbf{a}_0 = \mathbf{0}$. Under such an assumption and rewriting the model as $\mathbf{a}_t = \mathbf{r}_t - \boldsymbol{\theta}_0 + \boldsymbol{\Theta} \mathbf{a}_{t-1}$, we can compute the shock $\mathbf{a}_t$ recursively as

$$\mathbf{a}_1 = \mathbf{r}_1 - \boldsymbol{\theta}_0, \quad \mathbf{a}_2 = \mathbf{r}_2 - \boldsymbol{\theta}_0 + \boldsymbol{\Theta}_1 \mathbf{a}_1, \quad \dots$$

Consequently, the likelihood function of the data becomes

$$f(\mathbf{r}_1, \dots, \mathbf{r}_T | \boldsymbol{\theta}_0, \boldsymbol{\Theta}, \boldsymbol{\Sigma}) = \prod_{t=1}^T \frac{1}{(2\pi)^{k/2} |\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2} \mathbf{a}_t' \boldsymbol{\Sigma}^{-1} \mathbf{a}_t\right),$$

which can be evaluated to obtain the parameter estimates.

Exact MLE

For the exact-likelihood method, $\mathbf{a}_0$ is an unknown vector that must be estimated from the data to evaluate the likelihood function. For simplicity, let $\tilde{\mathbf{r}}_t = \mathbf{r}_t - \boldsymbol{\theta}_0$ be the mean-corrected series. Using $\tilde{\mathbf{r}}_t$ and Eq. (8.25), we have

$$\mathbf{a}_t = \tilde{\mathbf{r}}_t + \boldsymbol{\Theta} \mathbf{a}_{t-1}. \quad (8.27)$$

By repeated substitutions, $\mathbf{a}_0$ is related to all $\tilde{\mathbf{r}}_t$ as

$$\begin{aligned} \mathbf{a}_1 &= \tilde{\mathbf{r}}_1 + \boldsymbol{\Theta} \mathbf{a}_0, \\ \mathbf{a}_2 &= \tilde{\mathbf{r}}_2 + \boldsymbol{\Theta} \mathbf{a}_1 = \tilde{\mathbf{r}}_2 + \boldsymbol{\Theta} \tilde{\mathbf{r}}_1 + \boldsymbol{\Theta}^2 \mathbf{a}_0, \\ &\vdots \\ \mathbf{a}_T &= \tilde{\mathbf{r}}_T + \boldsymbol{\Theta} \tilde{\mathbf{r}}_{T-1} + \dots + \boldsymbol{\Theta}^{T-1} \tilde{\mathbf{r}}_1 + \boldsymbol{\Theta}^T \mathbf{a}_0. \end{aligned} \quad (8.28)$$

Thus, $\mathbf{a}_0$ is a linear function of the data if $\boldsymbol{\theta}_0$ and $\boldsymbol{\Theta}$ are given. This result enables us to estimate $\mathbf{a}_0$ using the data and initial estimates of $\boldsymbol{\theta}_0$ and $\boldsymbol{\Theta}$. More specifically, given $\boldsymbol{\theta}_0$, $\boldsymbol{\Theta}$, and the data, we can define

$$\mathbf{r}_t^* = \tilde{\mathbf{r}}_t + \boldsymbol{\Theta} \tilde{\mathbf{r}}_{t-1} + \dots + \boldsymbol{\Theta}^{t-1} \tilde{\mathbf{r}}_1, \quad \text{for } t = 1, 2, \dots, T.$$

Equation (8.28) can then be rewritten as

$$\begin{aligned} \mathbf{r}_1^* &= -\boldsymbol{\Theta} \mathbf{a}_0 + \mathbf{a}_1, \\ \mathbf{r}_2^* &= -\boldsymbol{\Theta}^2 \mathbf{a}_0 + \mathbf{a}_2, \\ &\vdots \\ \mathbf{r}_T^* &= -\boldsymbol{\Theta}^T \mathbf{a}_0 + \mathbf{a}_T. \end{aligned}$$

This is in the form of a multiple linear regression with parameter vector $\mathbf{a}_0$, even though the covariance matrix $\boldsymbol{\Sigma}$ of $\mathbf{a}_t$ may not be a diagonal matrix. If initial estimate of $\boldsymbol{\Sigma}$ is also available, one can premultiply each equation of the prior system by $\boldsymbol{\Sigma}^{-1/2}$, which is the square root matrix of $\boldsymbol{\Sigma}$. The resulting system is indeed a multiple linear regression, and the ordinary least-squares method can be used to obtain an estimate of $\mathbf{a}_0$. Denote the estimate by $\hat{\mathbf{a}}_0$.

Using the estimate $\hat{\mathbf{a}}_0$, we can compute the shocks $\mathbf{a}_t$ recursively as

$$\mathbf{a}_1 = \mathbf{r}_1 - \boldsymbol{\theta}_0 + \boldsymbol{\Theta}\hat{\mathbf{a}}_0, \quad \mathbf{a}_2 = \mathbf{r}_2 - \boldsymbol{\theta}_0 + \boldsymbol{\Theta}\mathbf{a}_1, \quad \dots$$

This recursion is a linear transformation from $(\mathbf{a}_0, \mathbf{r}_1, \dots, \mathbf{r}_T)$ to $(\mathbf{a}_0, \mathbf{a}_1, \dots, \mathbf{a}_T)$, from which we can (a) obtain the joint distribution of $\mathbf{a}_0$ and the data, and (2) integrate out $\mathbf{a}_0$ to derive the exact-likelihood function of the data. The resulting likelihood function can then be evaluated to obtain the exact ML estimates. For details, see Hillmer and Tiao (1979).

In summary, the exact-likelihood method works as follows. Given initial estimates of $\boldsymbol{\theta}_0$, $\boldsymbol{\Theta}$, and $\boldsymbol{\Sigma}$, one uses Eq. (8.28) to derive an estimate of $\mathbf{a}_0$. This estimate is in turn used to compute $\mathbf{a}_t$ recursively using Eq. (8.27) and starting with $\mathbf{a}_1 = \hat{\mathbf{r}}_1 + \boldsymbol{\Theta}\hat{\mathbf{a}}_0$. The resulting $\{\mathbf{a}_t\}_{t=1}^T$ are then used to evaluate the exact-likelihood function of the data to update the estimates of $\boldsymbol{\theta}_0$, $\boldsymbol{\Theta}$, and $\boldsymbol{\Sigma}$. The whole process is then repeated until the estimates converge. This iterative method to evaluate the exact-likelihood function applies to the general VMA(q) models.

From the previous discussion, the exact-likelihood method requires more intensive computation than the conditional-likelihood approach does. But it provides more accurate parameter estimates, especially when some eigenvalues of $\boldsymbol{\Theta}$ are close to 1 in modulus. Hillmer and Tiao (1979) provide some comparison between the conditional- and exact-likelihood estimations of VMA models. In multivariate time series analysis, the exact maximum-likelihood method becomes important if one suspects that the data might have been overdifferenced. Overdifferencing may occur in many situations (e.g., differencing individual components of a cointegrated system; see discussion later on cointegration).

In summary, building a VMA model involves three steps: (a) Use the sample cross-correlation matrices to specify the order q —for a VMA(q) model, $\rho_\ell = \mathbf{0}$ for $\ell > q$; (b) estimate the specified model by using either the conditional- or exact-likelihood method—the exact method is preferred when the sample size is not large; and (c) the fitted model should be checked for adequacy [e.g., applying the $Q_k(m)$ statistics to the residual series]. Finally, forecasts of a VMA model can be obtained by using the same procedure as a univariate MA model.

Example 8.5. Consider again the bivariate series of monthly log returns in percentages of IBM stock and the S&P 500 index from January 1926 to December 2008. Since significant cross correlations occur mainly at lags 1, 2, 3 and 5, we employ the VMA(5) model

$$\mathbf{r}_t = \boldsymbol{\theta}_0 + \mathbf{a}_t - \boldsymbol{\Theta}_1\mathbf{a}_{t-1} - \boldsymbol{\Theta}_2\mathbf{a}_{t-2} - \boldsymbol{\Theta}_3\mathbf{a}_{t-3} - \boldsymbol{\Theta}_5\mathbf{a}_{t-5} \quad (8.29)$$

for the data. Table 8.6 shows the estimation results of the model. The $Q_k(m)$ statistics for the residuals of the simplified model give $Q_2(4) = 16.00$ and $Q_2(8) = 29.46$. Compared with chi-squared distributions with 10 and 26 degrees of freedom, the p values of these statistics are 0.10 and 0.291, respectively. Thus, the model is adequate at the 5% significance level.

TABLE 8.6 Estimation Results for Monthly Log Returns of IBM Stock and S&P 500 Index Using the Vector Moving-Average Model in Eq. (8.29)^a

Parameter	θ_0	Θ_1		Θ_2		Θ_3		Θ_5	
<i>(a) Full Model with Conditional-Likelihood Method</i>									
Estimate	1.1	0.02	−0.15	−0.09	0.15	−0.05	0.11	0.05	−0.15
	0.4	0.02	−0.10	−0.04	0.04	−0.01	0.11	0.07	−0.15
Standard error	0.24	0.04	0.05	0.04	0.05	0.04	0.05	0.04	0.05
	0.19	0.03	0.04	0.03	0.04	0.03	0.04	0.03	0.04
<i>(b) Full Model with Exact-Likelihood Method</i>									
Estimate	1.1	0.02	−0.05	−0.09	0.15	−0.05	0.11	0.05	−0.15
	0.4	0.02	−0.10	−0.04	0.04	−0.01	0.11	0.07	−0.15
Standard error	0.24	0.04	0.05	0.04	0.05	0.04	0.05	0.04	0.05
	0.19	0.03	0.04	0.03	0.04	0.03	0.04	0.03	0.04
<i>(c) Simplified Model with Exact-Likelihood Method</i>									
Estimate	1.1	0.0	−0.13	0.0	0.08	0.0	0.0	0.0	−0.10
	0.4	0.0	−0.09	0.0	0.0	0.0	0.07	0.0	−0.09
Standard error	0.24	—	0.04	—	0.03	—	—	—	0.04
	0.19	—	0.03	—	—	—	0.02	—	0.03

^aThe sample period is from January 1926 to December 2008. The residual covariance matrix is not shown as it is similar to that in Table 8.4

From Table 8.6, we make the following observations:

1. The difference between conditional- and exact-likelihood estimates is small for this particular example. This is not surprising because the sample size is not small and, more important, the dynamic structure of the data is weak.
2. The VMA(5) model provides essentially the same dynamic relationship for the series as that of the VAR(5) model in Example 8.4. The monthly log return of IBM stock depends on the previous returns of the S&P 500 index. The market return, in contrast, does not depend on lagged returns of IBM stock. In other words, the dynamic structure of the data is driven by the market return, not by the IBM return. The concurrent correlation between the two returns remains strong, however.

8.4 VECTOR ARMA MODELS

Univariate ARMA models can also be generalized to handle vector time series. The resulting models are called VARMA models. The generalization, however, encounters some new issues that do not occur in developing VAR and VMA models. One of the issues is the *identifiability* problem. Unlike the univariate ARMA models, VARMA models may not be uniquely defined. For example, the VMA(1) model

$$\begin{bmatrix} r_{1t} \\ r_{2t} \end{bmatrix} = \begin{bmatrix} a_{1t} \\ a_{2t} \end{bmatrix} - \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a_{1,t-1} \\ a_{2,t-1} \end{bmatrix}$$

is *identical* to the VAR(1) model

$$\begin{bmatrix} r_{1t} \\ r_{2t} \end{bmatrix} - \begin{bmatrix} 0 & -2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} r_{1,t-1} \\ r_{2,t-1} \end{bmatrix} = \begin{bmatrix} a_{1t} \\ a_{2t} \end{bmatrix}.$$

The equivalence of the two models can easily be seen by examining their component models. For the VMA(1) model, we have

$$r_{1t} = a_{1t} - 2a_{2,t-1}, \quad r_{2t} = a_{2t}.$$

For the VAR(1) model, the equations are

$$r_{1t} + 2r_{2,t-1} = a_{1t}, \quad r_{2t} = a_{2t}.$$

From the model for r_{2t} , we have $r_{2,t-1} = a_{2,t-1}$. Therefore, the models for r_{1t} are identical. This type of identifiability problem is harmless because either model can be used in a real application.

Another type of identifiability problem is more troublesome. Consider the VARMA(1,1) model

$$\begin{bmatrix} r_{1t} \\ r_{2t} \end{bmatrix} - \begin{bmatrix} 0.8 & -2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} r_{1,t-1} \\ r_{2,t-1} \end{bmatrix} = \begin{bmatrix} a_{1t} \\ a_{2t} \end{bmatrix} - \begin{bmatrix} -0.5 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a_{1,t-1} \\ a_{2,t-1} \end{bmatrix}.$$

This model is identical to the VARMA(1,1) model

$$\begin{bmatrix} r_{1t} \\ r_{2t} \end{bmatrix} - \begin{bmatrix} 0.8 & -2 + \eta \\ 0 & \omega \end{bmatrix} \begin{bmatrix} r_{1,t-1} \\ r_{2,t-1} \end{bmatrix} = \begin{bmatrix} a_{1t} \\ a_{2t} \end{bmatrix} - \begin{bmatrix} -0.5 & \eta \\ 0 & \omega \end{bmatrix} \begin{bmatrix} a_{1,t-1} \\ a_{2,t-1} \end{bmatrix},$$

for any nonzero ω and η . In this particular instance, the equivalence occurs because we have $r_{2t} = a_{2t}$ in both models. The effects of the parameters ω and η on the system cancel out between AR and MA parts of the second model. Such an identifiability problem is serious because, without proper constraints, the likelihood function of a vector ARMA(1,1) model for the data is not uniquely defined, resulting in a situation similar to the exact multicollinearity in a regression analysis. This type of identifiability problem can occur in a vector model even if none of the components is a white noise series.

These two simple examples highlight the new issues involved in the generalization to VARMA models. Building a VARMA model for a given data set thus requires some attention. In the time series literature, methods of *structural specification* have been proposed to overcome the identifiability problem; see Tiao and Tsay (1989), Tsay (1991), and the references therein. We do not discuss the detail of structural specification here because VAR and VMA models are sufficient in most financial applications. When VARMA models are used, only lower order models are entertained [e.g., a VARMA(1,1) or VARMA(2,1) model] especially when the time series involved are not seasonal.

A VARMA(p, q) model can be written as

$$\Phi(B)r_t = \phi_0 + \Theta(B)a_t,$$

where $\Phi(B) = I - \Phi_1 B - \dots - \Phi_p B^p$ and $\Theta(B) = I - \Theta_1 B - \dots - \Theta_q B^q$ are two $k \times k$ matrix polynomials. We assume that the two matrix polynomials have no left common factors; otherwise, the model can be simplified. The necessary and sufficient condition of weak stationarity for r_t is the same as that for the VAR(p) model with matrix polynomial $\Phi(B)$. For $v > 0$, the (i, j) th elements of the coefficient matrices Φ_v and Θ_v measure the linear dependence of r_{1t} on $r_{j,t-v}$ and $a_{j,t-v}$, respectively. If the (i, j) th element is zero for all AR and MA coefficient matrices, then r_{it} does not depend on the lagged values of r_{jt} . However, the converse proposition does not hold in a VARMA model. In other words, nonzero coefficients at the (i, j) th position of AR and MA matrices may exist even when r_{it} does not depend on any lagged value of r_{jt} .

To illustrate, consider the following bivariate model

$$\begin{bmatrix} \Phi_{11}(B) & \Phi_{12}(B) \\ \Phi_{21}(B) & \Phi_{22}(B) \end{bmatrix} \begin{bmatrix} r_{1t} \\ r_{2t} \end{bmatrix} = \begin{bmatrix} \Theta_{11}(B) & \Theta_{12}(B) \\ \Theta_{21}(B) & \Theta_{22}(B) \end{bmatrix} \begin{bmatrix} a_{1t} \\ a_{2t} \end{bmatrix}.$$

Here the necessary and sufficient conditions for the existence of a unidirectional dynamic relationship from r_{1t} to r_{2t} are

$$\Phi_{22}(B)\Theta_{12}(B) - \Phi_{12}(B)\Theta_{22}(B) = 0,$$

but

$$\Phi_{11}(B)\Theta_{21}(B) - \Phi_{21}(B)\Theta_{11}(B) \neq 0. \quad (8.30)$$

These conditions can be obtained as follows. Letting

$$\Omega(B) = |\Phi(B)| = \Phi_{11}(B)\Phi_{22}(B) - \Phi_{12}(B)\Phi_{21}(B)$$

be the determinant of the AR matrix polynomial and premultiplying the model by the matrix

$$\begin{bmatrix} \Phi_{22}(B) & -\Phi_{12}(B) \\ -\Phi_{21}(B) & \Phi_{11}(B) \end{bmatrix},$$

we can rewrite the bivariate model as

$$\begin{aligned} \Omega(B) \begin{bmatrix} r_{1t} \\ r_{2t} \end{bmatrix} &= \begin{bmatrix} \Phi_{22}(B)\Theta_{11}(B) - \Phi_{12}(B)\Theta_{21}(B) & \Phi_{22}(B)\Theta_{12}(B) - \Phi_{12}(B)\Theta_{22}(B) \\ \Phi_{11}(B)\Theta_{21}(B) - \Phi_{21}(B)\Theta_{11}(B) & \Phi_{11}(B)\Theta_{22}(B) - \Phi_{21}(B)\Theta_{12}(B) \end{bmatrix} \\ &\quad \times \begin{bmatrix} a_{1t} \\ a_{2t} \end{bmatrix}. \end{aligned}$$

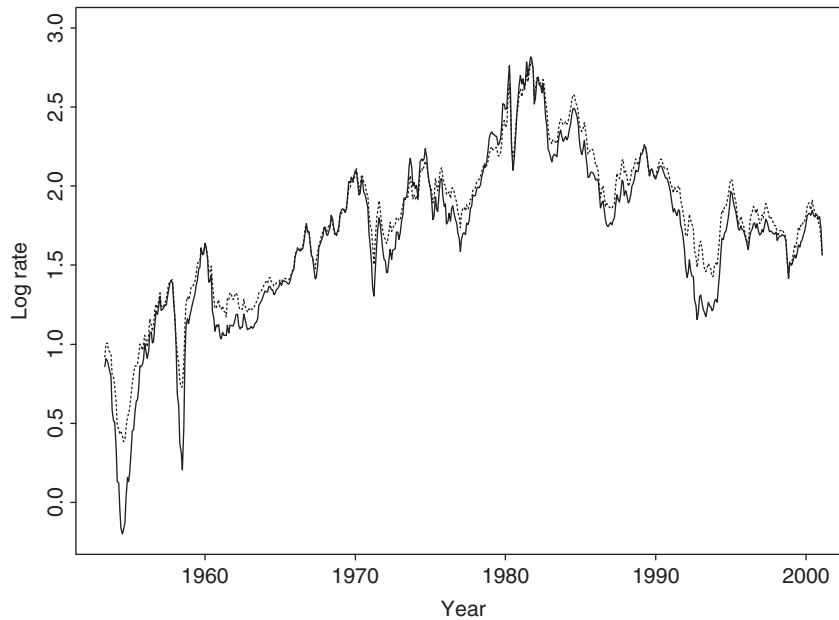


Figure 8.8 Time plots of log U.S. monthly interest rates from April 1953 to January 2001. Solid line denotes 1-year Treasury constant maturity rate and dashed line denotes 3-year rate.

Consider the equation for r_{1t} . The first condition in Eq. (8.30) shows that r_{1t} does not depend on any past value of a_{2t} or r_{2t} . From the equation for r_{2t} , the second condition in Eq. (8.30) implies that r_{2t} indeed depends on some past values of a_{1t} . Based on Eq. (8.30), $\Theta_{12}(B) = \Phi_{12}(B) = 0$ is a sufficient, but not necessary, condition for the unidirectional relationship from r_{1t} to r_{2t} .

Estimation of a VARMA model can be carried out by either the conditional or exact maximum-likelihood method. The $Q_k(m)$ statistic continues to apply to the residual series of a fitted model, but the degrees of freedom of its asymptotic chi-squared distribution are $k^2m - g$, where g is the number of estimated parameters in both the AR and MA coefficient matrices.

Example 8.6. To demonstrate VARMA modeling, we consider two U.S. monthly interest rate series. The first series is the 1-year Treasury constant maturity rate, and the second series is the 3-year Treasury constant maturity rate. The data are obtained from the Federal Reserve Bank of St. Louis, and the sampling period is from April 1953 to January 2001. There are 574 observations. To ensure the positiveness of U.S. interest rates, we analyze the log series. Figure 8.8 shows the time plots of the two log interest rate series. The solid line denotes the 1-year maturity rate. The two series moved closely in the sampling period.

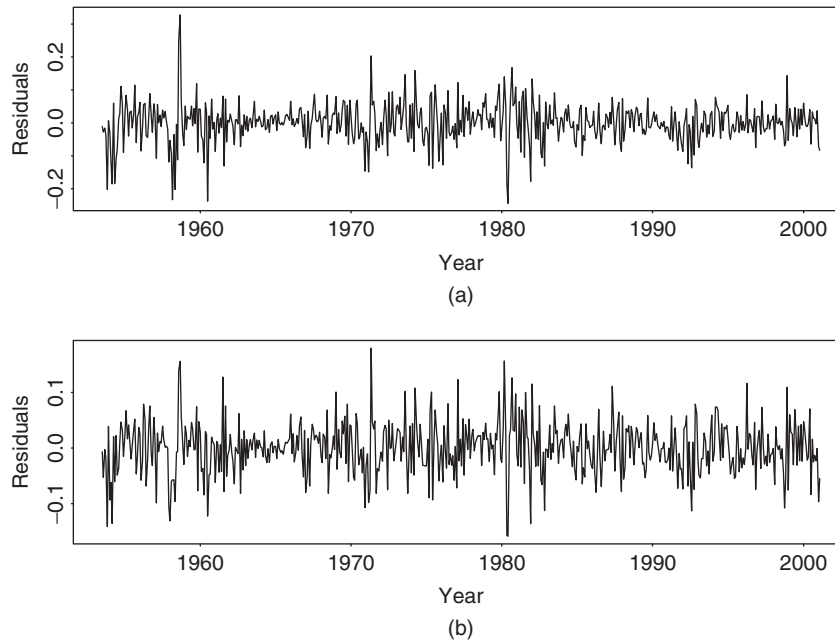
The $M(i)$ statistics and AIC criterion specify a VAR(4) model for the data. However, we employ a VARMA(2,1) model because the two models provide similar

TABLE 8.7 Parameter Estimates of VARMA(2,1) Model for Two Monthly U.S. Interest Rate Series Based on Exact-Likelihood Method

Parameter	Φ_1		Φ_2		ϕ_0	Θ_1		$\Sigma \times 10^3$	
Estimate	1.82	-0.97	-0.84	0.98	0.028	0.90	-1.66	3.58	2.50
	—	0.99	—	—	0.025	—	-0.47	2.50	2.19
Standard error	0.03	0.08	0.03	0.08	0.014	0.03	0.10		
	—	0.01	—	—	0.011	—	0.04		

fits. Table 8.7 shows the parameter estimates of the VARMA(2,1) model obtained by the exact-likelihood method. We removed the insignificant parameters and reestimated the simplified model. The residual series of the fitted model has some minor serial and cross correlations at lags 7 and 11. Figure 8.9 shows the residual plots and indicates the existence of some outlying data points. The model can be further improved, but it seems to capture the dynamic structure of the data reasonably well.

The final VARMA(2,1) model shows some interesting characteristics of the data. First, the interest rate series are highly contemporaneously correlated. The concurrent correlation coefficient is $2.5/\sqrt{3.58 \times 2.19} = 0.893$. Second, there is a

**Figure 8.9** Residual plots for log U.S. monthly interest rate series of Example 8.6. Fitted model is VARMA(2,1): (a) 1-year rate and (b) 3-year rate.

unidirectional linear relationship from the 3-year rate to the 1-year rate because the (2, 1)th elements of all AR and MA matrices are zero, but some (1, 2)th element is not zero. As a matter of fact, the model in Table 8.7 shows that

$$\begin{aligned} r_{3t} &= 0.025 + 0.99r_{3,t-1} + a_{3t} + 0.47a_{3,t-1}, \\ r_{1t} &= 0.028 + 1.82r_{1,t-1} - 0.84r_{1,t-2} - 0.97r_{3,t-1} + 0.98r_{3,t-2} \\ &\quad + a_{1t} - 0.90a_{1,t-1} + 1.66a_{3,t-1}, \end{aligned}$$

where r_{it} is the log series of i -year interest rate and a_{it} is the corresponding shock series. Therefore, the 3-year interest rate does not depend on the past values of the 1-year rate, but the 1-year rate depends on the past values of the 3-year rate. Third, the two interest rate series appear to be unit-root nonstationary. Using the back-shift operator B , the model can be rewritten approximately as

$$\begin{aligned} (1 - B)r_{3t} &= 0.03 + (1 + 0.47B)a_{3t}, \\ (1 - B)(1 - 0.82B)r_{1t} &= 0.03 - 0.97B(1 - B)r_{3,t} + (1 - 0.9B)a_{1t} + 1.66Ba_{3,t}. \end{aligned}$$

Finally, the SCA commands used in the analysis are given in Appendix C.

8.4.1 Marginal Models of Components

Given a vector model for $\mathbf{r}_t$, the implied univariate models for the components r_{it} are the *marginal* models. For a k -dimensional ARMA(p, q) model, the marginal models are ARMA($kp, (k-1)p + q$). This result can be obtained in two steps. First, the marginal model of a VMA(q) model is univariate MA(q). Assume that $\mathbf{r}_t$ is a VMA(q) process. Because the cross-correlation matrix of $\mathbf{r}_t$ vanishes after lag q (i.e., $\rho_\ell = \mathbf{0}$ for $\ell > q$), the ACF of r_{it} is zero beyond lag q . Therefore, r_{it} is an MA process and its univariate model is in the form $r_{it} = \theta_{i,0} + \sum_{j=1}^q \theta_{i,j}b_{i,t-j}$, where $\{b_{it}\}$ is a sequence of uncorrelated random variables with mean zero and variance σ_{ib}^2 . The parameters $\theta_{i,j}$ and σ_{ib} are functions of the parameters of the VMA model for $\mathbf{r}_t$.

The second step to obtain the result is to diagonalize the AR matrix polynomial of a VARMA(p, q) model. For illustration, consider the bivariate AR(1) model

$$\begin{bmatrix} 1 - \Phi_{11}B & -\Phi_{12}B \\ -\Phi_{21}B & 1 - \Phi_{22}B \end{bmatrix} \begin{bmatrix} r_{1t} \\ r_{2t} \end{bmatrix} = \begin{bmatrix} a_{1t} \\ a_{2t} \end{bmatrix}.$$

Premultiplying the model by the matrix polynomial

$$\begin{bmatrix} 1 - \Phi_{22}B & \Phi_{12}B \\ \Phi_{21}B & 1 - \Phi_{11}B \end{bmatrix},$$

we obtain

$$[(1 - \Phi_{11}B)(1 - \Phi_{22}B) - \Phi_{12}\Phi_{22}B^2] \begin{bmatrix} r_{1t} \\ r_{2t} \end{bmatrix} = \begin{bmatrix} 1 - \Phi_{22}B & -\Phi_{12}B \\ -\Phi_{21}B & 1 - \Phi_{11}B \end{bmatrix} \begin{bmatrix} a_{1t} \\ a_{2t} \end{bmatrix}.$$

The left-hand side of the prior equation shows that the univariate AR polynomials for r_{it} are of order 2. In contrast, the right-hand side of the equation is in a VMA(1) form. Using the result of VMA models in step 1, we show that the univariate model for r_{it} is ARMA(2,1). The technique generalizes easily to the k -dimensional VAR(1) model, and the marginal models are ARMA($k, k - 1$). More generally, for a k -dimensional VAR(p) model, the marginal models are ARMA[$kp, (k - 1)p$]. The result for VARMA models follows directly from those of VMA and VAR models.

The order [$kp, (k - 1)p + q$] is the maximum order (i.e., the upper bound) for the marginal models. The actual marginal order of r_{it} can be much lower.

8.5 UNIT-ROOT NONSTATIONARITY AND COINTEGRATION

When modeling several unit-root nonstationary time series jointly, one may encounter the case of *cointegration*. Consider the bivariate ARMA(1,1) model

$$\begin{bmatrix} x_{1t} \\ x_{2t} \end{bmatrix} - \begin{bmatrix} 0.5 & -1.0 \\ -0.25 & 0.5 \end{bmatrix} \begin{bmatrix} x_{1,t-1} \\ x_{2,t-1} \end{bmatrix} = \begin{bmatrix} a_{1t} \\ a_{2t} \end{bmatrix} - \begin{bmatrix} 0.2 & -0.4 \\ -0.1 & 0.2 \end{bmatrix} \begin{bmatrix} a_{1,t-1} \\ a_{2,t-1} \end{bmatrix}, \quad (8.31)$$

where the covariance matrix Σ of the shock $\mathbf{a}_t$ is positive definite. This is not a weakly stationary model because the two eigenvalues of the AR coefficient matrix are 0 and 1. Figure 8.10 shows the time plots of a simulated series of the model with 200 data points and $\Sigma = \mathbf{I}$, whereas Figure 8.11 shows the sample autocorrelations of the two component series x_{it} . It is easy to see that the two series have high autocorrelations and exhibit features of unit-root nonstationarity. The two marginal models of $\mathbf{x}_t$ are indeed unit-root nonstationary. Rewrite the model as

$$\begin{bmatrix} 1 - 0.5B & B \\ 0.25B & 1 - 0.5B \end{bmatrix} \begin{bmatrix} x_{1t} \\ x_{2t} \end{bmatrix} = \begin{bmatrix} 1 - 0.2B & 0.4B \\ 0.1B & 1 - 0.2B \end{bmatrix} \begin{bmatrix} a_{1t} \\ a_{2t} \end{bmatrix}.$$

Premultiplying the above equation by

$$\begin{bmatrix} 1 - 0.5B & -B \\ -0.25B & 1 - 0.5B \end{bmatrix},$$

we obtain the result

$$\begin{bmatrix} 1 - B & 0 \\ 0 & 1 - B \end{bmatrix} \begin{bmatrix} x_{1t} \\ x_{2t} \end{bmatrix} = \begin{bmatrix} 1 - 0.7B & -0.6B \\ -0.15B & 1 - 0.7B \end{bmatrix} \begin{bmatrix} a_{1t} \\ a_{2t} \end{bmatrix}.$$

Therefore, each component x_{it} of the model is unit-root nonstationary and follows an ARIMA(0,1,1) model.

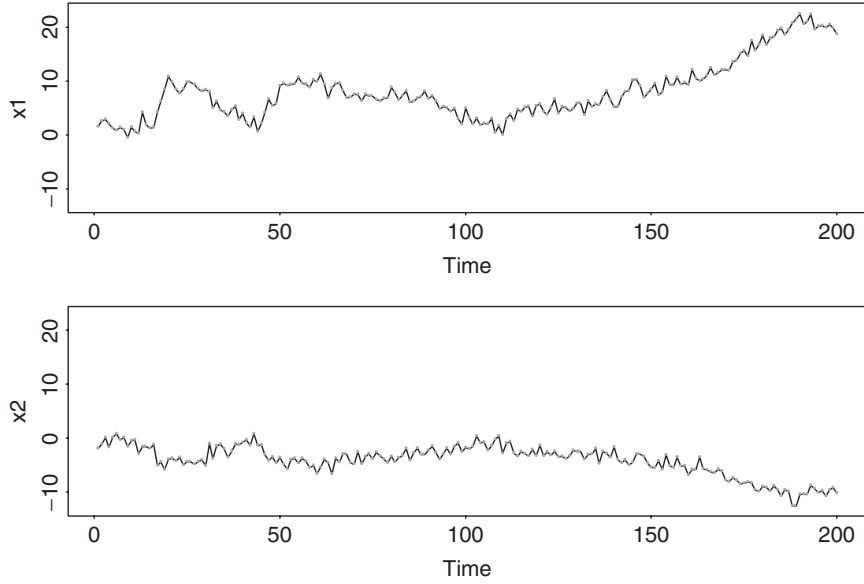


Figure 8.10 Time plots of simulated series based on model (8.31) with identity covariance matrix for shocks.

However, we can consider a linear transformation by defining

$$\begin{bmatrix} y_{1t} \\ y_{2t} \end{bmatrix} = \begin{bmatrix} 1.0 & -2.0 \\ 0.5 & 1.0 \end{bmatrix} \begin{bmatrix} x_{1t} \\ x_{2t} \end{bmatrix} \equiv \mathbf{L} \mathbf{x}_t,$$

$$\begin{bmatrix} b_{1t} \\ b_{2t} \end{bmatrix} = \begin{bmatrix} 1.0 & -2.0 \\ 0.5 & 1.0 \end{bmatrix} \begin{bmatrix} a_{1t} \\ a_{2t} \end{bmatrix} \equiv \mathbf{L} \mathbf{a}_t.$$

The VARMA model of the transformed series $\mathbf{y}_t$ can be obtained as follows:

$$\begin{aligned} \mathbf{L} \mathbf{x}_t &= \mathbf{L} \Phi \mathbf{x}_{t-1} + \mathbf{L} \mathbf{a}_t - \mathbf{L} \Theta \mathbf{a}_{t-1} \\ &= \mathbf{L} \Phi \mathbf{L}^{-1} \mathbf{L} \mathbf{x}_{t-1} + \mathbf{L} \mathbf{a}_t - \mathbf{L} \Theta \mathbf{L}^{-1} \mathbf{L} \mathbf{a}_{t-1} \\ &= \mathbf{L} \Phi \mathbf{L}^{-1} (\mathbf{L} \mathbf{x}_{t-1}) + \mathbf{b}_t - \mathbf{L} \Theta \mathbf{L}^{-1} \mathbf{b}_{t-1}. \end{aligned}$$

Thus, the model for $\mathbf{y}_t$ is

$$\begin{bmatrix} y_{1t} \\ y_{2t} \end{bmatrix} - \begin{bmatrix} 1.0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix} = \begin{bmatrix} b_{1t} \\ b_{2t} \end{bmatrix} - \begin{bmatrix} 0.4 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} b_{1,t-1} \\ b_{2,t-1} \end{bmatrix}. \quad (8.32)$$

From the prior model, we see that (a) y_{1t} and y_{2t} are uncoupled series with concurrent correlation equal to that between the shocks b_{1t} and b_{2t} , (b) y_{1t} follows a univariate ARIMA(0,1,1) model, and (c) y_{2t} is a white noise series (i.e., $y_{2t} = b_{2t}$).

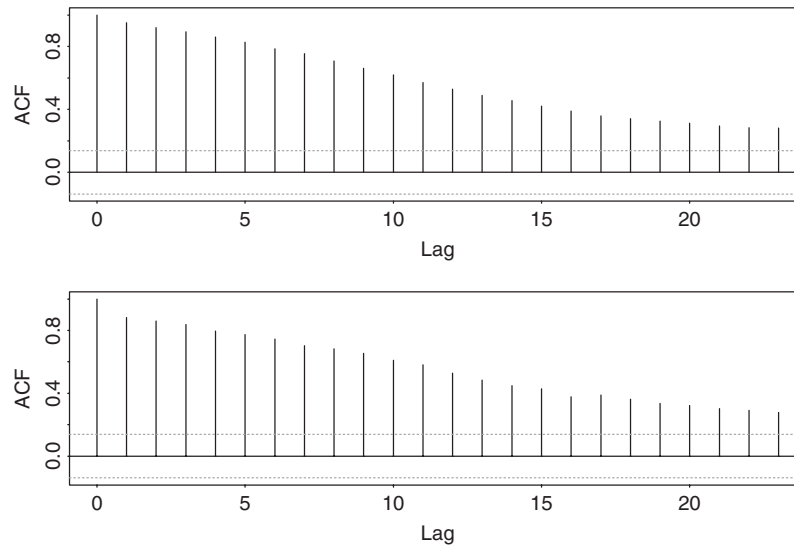


Figure 8.11 Sample autocorrelation functions of two simulated component series. There are 200 observations, and model is given by Eq. (8.31) with identity covariance matrix for shocks.

In particular, the model in Eq. (8.32) shows that there is *only* a single unit root in the system. Consequently, the unit roots of x_{1t} and x_{2t} are introduced by the unit root of y_{1t} . In the literature, y_{1t} is referred to as the *common trend* of x_{1t} and x_{2t} .

The phenomenon that both x_{1t} and x_{2t} are unit-root nonstationary, but there is only a single unit root in the vector series, is referred to as *cointegration* in the econometric and time series literature. Another way to define cointegration is to focus on linear transformations of unit-root nonstationary series. For the simulated example of model (8.31), the transformation shows that the linear combination $y_{2t} = 0.5x_{1t} + x_{2t}$ does not have a unit root. Consequently, x_{1t} and x_{2t} are cointegrated if (a) both of them are unit-root nonstationary, and (b) they have a linear combination that is unit-root stationary.

Generally speaking, for a k -dimensional unit-root nonstationary time series, cointegration exists if there are less than k unit roots in the system. Let h be the number of unit roots in the k -dimensional series $\mathbf{x}_t$. Cointegration exists if $0 < h < k$, and the quantity $k - h$ is called the number of cointegrating factors. Alternatively, the number of cointegrating factors is the number of different linear combinations that are unit-root stationary. The linear combinations are called the cointegrating vectors. For the prior simulated example, $y_{2t} = (0.5, 1)\mathbf{x}_t$ so that $(0.5, 1)'$ is a cointegrating vector for the system. For more discussions on cointegration and cointegration tests, see Box and Tiao (1977), Engle and Granger (1987), Stock and Watson (1988), and Johansen (1988). We discuss cointegrated VAR models in Section 8.6.

The concept of cointegration is interesting and has attracted a lot of attention in the literature. However, there are difficulties in testing for cointegration in a real

application. The main source of difficulties is that cointegration tests overlook the scaling effects of the component series. Interested readers are referred to Cochrane (1988) and Tiao, Tsay, and Wang (1993) for further discussion.

While I have some misgivings on the practical value of cointegration tests, the idea of cointegration is highly relevant in financial study. For example, consider the stock of Finnish Nokia Corporation. Its price on the Helsinki Stock Market must move in unison with the price of its American Depositary Receipts on the New York Stock Exchange; otherwise there exists some arbitrage opportunity for investors. If the stock price has a unit root, then the two price series must be cointegrated. In practice, such a cointegration can exist after adjusting for transaction costs and exchange rate risk. We discuss issues like this later in Section 8.7.

8.5.1 An Error Correction Form

Because there are more unit-root nonstationary components than the number of unit roots in a cointegrated system, differencing individual components to achieve stationarity results in overdifferencing. Overdifferencing leads to the problem of unit roots in the MA matrix polynomial, which in turn may encounter difficulties in parameter estimation. If the MA matrix polynomial contains unit roots, the vector time series is said to be noninvertible.

Engle and Granger (1987) discuss an error correction representation for a cointegrated system that overcomes the difficulty of estimating noninvertible VARMA models. Consider the cointegrated system in Eq. (8.31). Let $\Delta \mathbf{x}_t = \mathbf{x}_t - \mathbf{x}_{t-1}$ be the differenced series. Subtracting $\mathbf{x}_{t-1}$ from both sides of the equation, we obtain a model for $\Delta \mathbf{x}_t$ as

$$\begin{aligned} & \begin{bmatrix} \Delta x_{1t} \\ \Delta x_{2t} \end{bmatrix} \\ &= \begin{bmatrix} -0.5 & -1.0 \\ -0.25 & -0.5 \end{bmatrix} \begin{bmatrix} x_{1,t-1} \\ x_{2,t-1} \end{bmatrix} + \begin{bmatrix} a_{1t} \\ a_{2t} \end{bmatrix} - \begin{bmatrix} 0.2 & -0.4 \\ -0.1 & 0.2 \end{bmatrix} \begin{bmatrix} a_{1,t-1} \\ a_{2,t-1} \end{bmatrix} \\ &= \begin{bmatrix} -1 \\ -0.5 \end{bmatrix} [0.5, 1.0] \begin{bmatrix} x_{1,t-1} \\ x_{2,t-1} \end{bmatrix} + \begin{bmatrix} a_{1t} \\ a_{2t} \end{bmatrix} - \begin{bmatrix} 0.2 & -0.4 \\ -0.1 & 0.2 \end{bmatrix} \begin{bmatrix} a_{1,t-1} \\ a_{2,t-1} \end{bmatrix}. \end{aligned}$$

This is a stationary model because both $\Delta \mathbf{x}_t$ and $[0.5, 1.0]\mathbf{x}_t = y_{2t}$ are unit-root stationary. Because $\mathbf{x}_{t-1}$ is used on the right-hand side of the previous equation, the MA matrix polynomial is the same as before and, hence, the model does not encounter the problem of noninvertibility. Such a formulation is referred to as an error correction model for $\Delta \mathbf{x}_t$, and it can be extended to the general cointegrated VARMA model. For a cointegrated VARMA(p, q) model with m cointegrating factors ($m < k$), an error correction representation is

$$\Delta \mathbf{x}_t = \alpha \beta' \mathbf{x}_{t-1} + \sum_{i=1}^{p-1} \Phi_i^* \Delta \mathbf{x}_{t-i} + \mathbf{a}_t - \sum_{j=1}^q \Theta_j \mathbf{a}_{t-j}, \quad (8.33)$$

where α and β are $k \times m$ full-rank matrices. The AR coefficient matrices Φ_i^* are functions of the original coefficient matrices Φ_j . Specifically, we have

$$\begin{aligned}\Phi_j^* &= - \sum_{i=j+1}^p \Phi_i, \quad j = 1, \dots, p-1, \\ \alpha\beta' &= \Phi_p + \Phi_{p-1} + \dots + \Phi_1 - I = -\Phi(1).\end{aligned}\tag{8.34}$$

These results can be obtained by equating coefficient matrices of the AR matrix polynomials. The time series $\beta'x_t$ is unit-root stationary, and the columns of β are the cointegrating vectors of x_t .

Existence of the stationary series $\beta'x_{t-1}$ in the error correction representation (8.33) is natural. It can be regarded as a “compensation” term for the overdifferentiated system Δx_t . The stationarity of $\beta'x_{t-1}$ can be justified as follows. The theory of unit-root time series shows that the sample correlation coefficient between a unit-root nonstationary series and a stationary series converges to zero as the sample size goes to infinity; see Tsay and Tiao (1990) and the references therein. In an error correction representation, x_{t-1} is unit-root nonstationary, but Δx_t is stationary. Therefore, the only way that Δx_t can relate meaningfully to x_{t-1} is through a stationary series $\beta'x_{t-1}$.

Remark. Our discussion of cointegration assumes that all unit roots are of multiplicity 1, but the concept can be extended to cases in which the unit roots have different multiplicities. Also, if the number of cointegrating factors m is given, then the error correction model in Eq. (8.33) can be estimated by likelihood methods. We discuss the simple case of cointegrated VAR models in the next section. Finally, there are many ways to construct an error correction representation. In fact, one can use any $\alpha\beta'x_{t-v}$ for $1 \leq v \leq p$ in Eq. (8.33) with some modifications to the AR coefficient matrices Φ_i^* . $\square$

8.6 COINTEGRATED VAR MODELS

To better understand cointegration, we focus on VAR models for their simplicity in estimation. Consider a k -dimensional VAR(p) time series x_t with possible time trend so that the model is

$$x_t = \mu_t + \Phi_1 x_{t-1} + \dots + \Phi_p x_{t-p} + a_t, \tag{8.35}$$

where the innovation a_t is assumed to be Gaussian and $\mu_t = \mu_0 + \mu_1 t$, where μ_0 and μ_1 are k -dimensional constant vectors. Write $\Phi(B) = I - \Phi_1 B - \dots - \Phi_p B^p$. Recall that if all zeros of the determinant $|\Phi(B)|$ are outside the unit circle, then x_t is unit-root stationary. In the literature, a unit-root stationary series is said to be an $I(0)$ process; that is, it is not integrated. If $|\Phi(1)| = 0$, then x_t is unit-root nonstationary. For simplicity, we assume that x_t is at most an integrated process of

order 1; that is, an $I(1)$ process. This means that $(1 - B)x_{it}$ is unit-root stationary if x_{it} itself is not.

An error correction model (ECM) for the $\text{VAR}(p)$ process $\mathbf{x}_t$ is

$$\Delta \mathbf{x}_t = \boldsymbol{\mu}_t + \boldsymbol{\Pi} \mathbf{x}_{t-1} + \boldsymbol{\Phi}_1^* \Delta \mathbf{x}_{t-1} + \cdots + \boldsymbol{\Phi}_{p-1}^* \Delta \mathbf{x}_{t-p+1} + \mathbf{a}_t, \quad (8.36)$$

where $\boldsymbol{\Phi}_j^*$ are defined in Eq. (8.34) and $\boldsymbol{\Pi} = \boldsymbol{\alpha} \boldsymbol{\beta}' = -\boldsymbol{\Phi}(1)$. We refer to the term $\boldsymbol{\Pi} \mathbf{x}_{t-1}$ of Eq. (8.36) as the *error correction term*, which plays a key role in cointegration study. Notice that $\boldsymbol{\Phi}_i$ can be recovered from the ECM representation via

$$\begin{aligned} \boldsymbol{\Phi}_1 &= \mathbf{I} + \boldsymbol{\Pi} + \boldsymbol{\Phi}_1^*, \\ \boldsymbol{\Phi}_i &= \boldsymbol{\Phi}_i^* - \boldsymbol{\Phi}_{i-1}^*, \quad i = 2, \dots, p, \end{aligned}$$

where $\boldsymbol{\Phi}_p^* = \mathbf{0}$, the zero matrix. Based on the assumption that $\mathbf{x}_t$ is at most $I(1)$, $\Delta \mathbf{x}_t$ of Eq. (8.36) is an $I(0)$ process.

If $\mathbf{x}_t$ contains unit roots, then $|\boldsymbol{\Phi}(1)| = 0$ so that $\boldsymbol{\Pi} = -\boldsymbol{\Phi}(1)$ is singular. Therefore, three cases are of interest in considering the ECM in Eq. (8.36):

1. $\text{Rank}(\boldsymbol{\Pi}) = 0$. This implies $\boldsymbol{\Pi} = \mathbf{0}$ and $\mathbf{x}_t$ is not cointegrated. The ECM of Eq. (8.36) reduces to

$$\Delta \mathbf{x}_t = \boldsymbol{\mu}_t + \boldsymbol{\Phi}_1^* \Delta \mathbf{x}_{t-1} + \cdots + \boldsymbol{\Phi}_{p-1}^* \Delta \mathbf{x}_{t-p+1} + \mathbf{a}_t,$$

so that $\Delta \mathbf{x}_t$ follows a $\text{VAR}(p-1)$ model with deterministic trend $\boldsymbol{\mu}_t$.

2. $\text{Rank}(\boldsymbol{\Pi}) = k$. This implies that $|\boldsymbol{\Phi}(1)| \neq 0$ and $\mathbf{x}_t$ contains no unit roots; that is, $\mathbf{x}_t$ is $I(0)$. The ECM model is not informative and one studies $\mathbf{x}_t$ directly.
3. $0 < \text{Rank}(\boldsymbol{\Pi}) = m < k$. In this case, one can write $\boldsymbol{\Pi}$ as

$$\boldsymbol{\Pi} = \boldsymbol{\alpha} \boldsymbol{\beta}', \quad (8.37)$$

where $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are $k \times m$ matrices with $\text{Rank}(\boldsymbol{\alpha}) = \text{Rank}(\boldsymbol{\beta}) = m$. The ECM of Eq. (8.36) becomes

$$\Delta \mathbf{x}_t = \boldsymbol{\mu}_t + \boldsymbol{\alpha} \boldsymbol{\beta}' \mathbf{x}_{t-1} + \boldsymbol{\Phi}_1^* \Delta \mathbf{x}_{t-1} + \cdots + \boldsymbol{\Phi}_{p-1}^* \Delta \mathbf{x}_{t-p+1} + \mathbf{a}_t. \quad (8.38)$$

This means that $\mathbf{x}_t$ is cointegrated with m linearly independent cointegrating vectors, $\mathbf{w}_t = \boldsymbol{\beta}' \mathbf{x}_t$, and has $k - m$ unit roots that give $k - m$ common stochastic trends of $\mathbf{x}_t$.

If $\mathbf{x}_t$ is cointegrated with $\text{Rank}(\boldsymbol{\Pi}) = m$, then a simple way to obtain a presentation of the $k - m$ common trends is to obtain an orthogonal complement matrix $\boldsymbol{\alpha}_\perp$ of $\boldsymbol{\alpha}$; that is, $\boldsymbol{\alpha}_\perp$ is a $k \times (k - m)$ matrix such that $\boldsymbol{\alpha}_\perp' \boldsymbol{\alpha} = \mathbf{0}$, a $(k - m) \times m$ zero matrix, and use $\mathbf{y}_t = \boldsymbol{\alpha}_\perp' \mathbf{x}_t$. To see this, one can premultiply the ECM by $\boldsymbol{\alpha}_\perp'$

and use $\Pi = \alpha\beta'$ to see that there would be no error correction term in the resulting equation. Consequently, the $(k - m)$ -dimensional series y_t should have $k - m$ unit roots. For illustration, consider the bivariate example of Section 8.5.1. For this special series, $\alpha = (-1, -0.5)'$ and $\alpha_\perp = (1, -2)'$. Therefore, $y_t = (1, -2)x_t = x_{1t} - 2x_{2t}$, which is precisely the unit-root nonstationary series y_{1t} in Eq. (8.32).

Note that the factorization in Eq. (8.37) is not unique because for any $m \times m$ orthogonal matrix Ω satisfying $\Omega\Omega' = I$, we have a

$$\alpha\beta' = \alpha\Omega\Omega'\beta' = (\alpha\Omega)(\beta\Omega)' \equiv \alpha_*\beta'_*,$$

where both α_* and β_* are also of rank m . Additional constraints are needed to uniquely identify α and β . It is common to require that $\beta' = [I_m, \beta'_1]$, where I_m is the $m \times m$ identity matrix and β_1 is a $(k - m) \times m$ matrix. In practice, this may require reordering of the elements of x_t such that the first m components all have a unit root. The elements of α and β must also satisfy other constraints for the process $w_t = \beta'x_t$ to be unit-root stationary. For example, consider the case of a bivariate VAR(1) model with one cointegrating vector. Here $k = 2$, $m = 1$, and the ECM is

$$\Delta x_t = \mu_t + \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} [1, \beta_1]x_{t-1} + a_t.$$

Premultiplying the prior equation by β' , using $w_{t-i} = \beta'x_{t-i}$, and moving w_{t-1} to the right-hand side of the equation, we obtain

$$w_t = \beta'\mu_t + (1 + \alpha_1 + \alpha_2\beta_1)w_{t-1} + b_t,$$

where $b_t = \beta'a_t$. This implies that w_t is a stationary AR(1) process. Consequently, α_1 and β_1 must satisfy the stationarity constraint $|1 + \alpha_1 + \alpha_2\beta_1| < 1$.

The prior discussion shows that the rank of Π in the ECM of Eq. (8.36) is the number of cointegrating vectors. Thus, to test for cointegration, one can examine the rank of Π . This is the approach taken by Johansen (1988, 1995) and Reinsel and Ahn (1992).

8.6.1 Specification of the Deterministic Function

Similar to the univariate case, the limiting distributions of cointegration tests depend on the deterministic function μ_t . In this section, we discuss some specifications of μ_t that have been proposed in the literature. To understand some of the statements made below, keep in mind that $\alpha'_\perp x_t$ provides a presentation for the common stochastic trends of x_t if it is cointegrated.

1. $\mu_t = 0$: In this case, all the component series of x_t are $I(1)$ without drift and the stationary series $w_t = \beta'x_t$ has mean zero.

2. $\mu_t = \mu_0 = \alpha c_0$, where c_0 is an m -dimensional nonzero constant vector. The ECM becomes

$$\Delta x_t = \alpha(\beta' x_{t-1} + c_0) + \Phi_1^* \Delta x_{t-1} + \cdots + \Phi_{p-1}^* \Delta x_{t-p+1} + a_t,$$

so that the components of x_t are $I(1)$ without drift, but w_t have a nonzero mean $-c_0$. This is referred to as the case of restricted constant.

3. $\mu_t = \mu_0$, which is nonzero. Here the component series of x_t are $I(1)$ with drift μ_0 and w_t may have a nonzero mean.
4. $\mu_t = \mu_0 + \alpha c_1 t$, where c_1 is a nonzero vector. The ECM becomes

$$\Delta x_t = \mu_0 + \alpha(\beta' x_{t-1} + c_1 t) + \Phi_1^* \Delta x_{t-1} + \cdots + \Phi_{p-1}^* \Delta x_{t-p+1} + a_t,$$

so that the components of x_t are $I(1)$ with drift μ_0 and w_t has a linear time trend related to $c_1 t$. This is the case of restricted trend.

5. $\mu_t = \mu_0 + \mu_1 t$, where μ_i are nonzero. Here both the constant and trend are unrestricted. The components of x_t are $I(1)$ and have a quadratic time trend and w_t have a linear trend.

Obviously, the last case is not common in empirical work. The first case is not common for economic time series but may represent the log price series of some assets. The third case is also useful in modeling asset prices.

8.6.2 Maximum-Likelihood Estimation

In this section, we briefly outline the maximum-likelihood estimation (MLE) of a cointegrated VAR(p) model. Suppose that the data are $\{x_t | t = 1, \dots, T\}$. Without loss of generality, we write $\mu_t = \mu d_t$, where $d_t = [1, t]'$, and it is understood that μ_t depends on the specification of the previous section. For a given m , which is the rank of Π , the ECM model becomes

$$\Delta x_t = \mu d_t + \alpha \beta' x_{t-1} + \Phi_1^* \Delta x_{t-1} + \cdots + \Phi_{p-1}^* \Delta x_{t-p+1} + a_t, \quad (8.39)$$

where $t = p+1, \dots, T$. A key step in the estimation is to concentrate the likelihood function with respect to the deterministic term and the stationary effects. This is done by considering the following two multivariate linear regressions:

$$\Delta x_t = \gamma_0 d_t + \Omega_1 \Delta x_{t-1} + \cdots + \Omega_{p-1} \Delta x_{t-p+1} + u_t, \quad (8.40)$$

$$x_{t-1} = \gamma_1 d_t + \Xi_1 \Delta x_{t-1} + \cdots + \Xi_{p-1} \Delta x_{t-p+1} + v_t. \quad (8.41)$$

Let $\hat{u}_t$ and $\hat{v}_t$ be the residuals of Eqs. (8.40) and (8.41), respectively. Define the sample covariance matrices

$$S_{00} = \frac{1}{T-p} \sum_{t=p+1}^T \hat{u}_t \hat{u}_t', \quad S_{01} = \frac{1}{T-p} \sum_{t=p+1}^T \hat{u}_t \hat{v}_t', \quad S_{11} = \frac{1}{T-p} \sum_{t=p+1}^T \hat{v}_t \hat{v}_t'.$$

Next, compute the eigenvalues and eigenvectors of $S_{10}S_{00}^{-1}S_{01}$ with respect to S_{11} . This amounts to solving the eigenvalue problem

$$[\lambda S_{11} - S_{10}S_{00}^{-1}S_{01}] = 0.$$

Denote the eigenvalue and eigenvector pairs by $(\hat{\lambda}_i, \mathbf{e}_i)$, where $\hat{\lambda}_1 > \hat{\lambda}_2 > \dots > \hat{\lambda}_k$. Here the eigenvectors are normalized so that $\mathbf{e}'S_{11}\mathbf{e} = \mathbf{I}$, where $\mathbf{e} = [\mathbf{e}_1, \dots, \mathbf{e}_k]$ is the matrix of eigenvectors.

The unnormalized MLE of the cointegrating vector $\boldsymbol{\beta}$ is $\hat{\boldsymbol{\beta}} = [\mathbf{e}_1, \dots, \mathbf{e}_m]$ and from which we can obtain an MLE for $\boldsymbol{\beta}$ that satisfies the identifying constraint and normalization condition. Denote the resulting estimate by $\hat{\boldsymbol{\beta}}_c$ with the subscript c signifying constraints. The MLE of other parameters can then be obtained by the multivariate linear regression

$$\Delta \mathbf{x}_t = \boldsymbol{\mu} \mathbf{d}_t + \boldsymbol{\alpha} \hat{\boldsymbol{\beta}}_c' \mathbf{x}_{t-1} + \boldsymbol{\Phi}_1^* \Delta \mathbf{x}_{t-1} + \dots + \boldsymbol{\Phi}_{p-1}^* \Delta \mathbf{x}_{t-p+1} + \mathbf{a}_t.$$

The maximized value of the likelihood function based on m cointegrating vectors is

$$L_{\max}^{-2/T} \propto |S_{00}| \prod_{i=1}^m (1 - \hat{\lambda}_i).$$

This value is used in the maximum-likelihood ratio test for testing $\text{Rank}(\boldsymbol{\Pi}) = m$. Finally, estimates of the orthogonal complements of $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ can be obtained using

$$\hat{\boldsymbol{\alpha}}_{\perp} = S_{00}^{-1} S_{11} [\mathbf{e}_{m+1}, \dots, \mathbf{e}_k], \quad \hat{\boldsymbol{\beta}}_{\perp} = S_{11} [\mathbf{e}_{m+1}, \dots, \mathbf{e}_k].$$

8.6.3 Cointegration Test

For a specified deterministic term $\boldsymbol{\mu}_t$, we now discuss the maximum-likelihood test for testing the rank of the $\boldsymbol{\Pi}$ matrix in Eq. (8.36). Let $H(m)$ be the null hypothesis that the rank of $\boldsymbol{\Pi}$ is m . For example, under $H(0)$, $\text{Rank}(\boldsymbol{\Pi}) = 0$ so that $\boldsymbol{\Pi} = \mathbf{0}$ and there is no cointegration. The hypotheses of interest are

$$H(0) \subset \dots \subset H(m) \subset \dots \subset H(k).$$

For testing purpose, the ECM in Eq. (8.39) becomes

$$\Delta \mathbf{x}_t = \boldsymbol{\mu} \mathbf{d}_t + \boldsymbol{\Pi} \mathbf{x}_{t-1} + \boldsymbol{\Phi}_1^* \Delta \mathbf{x}_{t-1} + \dots + \boldsymbol{\Phi}_{p-1}^* \Delta \mathbf{x}_{t-p+1} + \mathbf{a}_t,$$

where $t = p + 1, \dots, T$. Our goal is to test the rank of $\boldsymbol{\Pi}$. Mathematically, the rank of $\boldsymbol{\Pi}$ is the number of nonzero eigenvalues of $\boldsymbol{\Pi}$, which can be obtained if a consistent estimate of $\boldsymbol{\Pi}$ is available. Based on the prior equation, which is in the form of a multivariate linear regression, we see that $\boldsymbol{\Pi}$ is related to the covariance matrix between $\mathbf{x}_{t-1}$ and $\Delta \mathbf{x}_t$ after adjusting for the effects of $\mathbf{d}_t$ and $\Delta \mathbf{x}_{t-i}$ for i

$= 1, \dots, p-1$. The necessary adjustments can be achieved by the techniques of multivariate linear regression shown in the previous section. Indeed, the adjusted series of $\mathbf{x}_{t-1}$ and $\Delta \mathbf{x}_t$ are $\hat{\mathbf{v}}_t$ and $\hat{\mathbf{u}}_t$, respectively. The equation of interest for the cointegration test then becomes

$$\hat{\mathbf{u}}_t = \mathbf{\Pi} \hat{\mathbf{v}}_t + \mathbf{a}_t.$$

Under the normality assumption, the likelihood ratio test for testing the rank of $\mathbf{\Pi}$ in the prior equation can be done by using the canonical correlation analysis between $\hat{\mathbf{u}}_t$ and $\hat{\mathbf{v}}_t$. See Johnson and Wichern (1998) for information on canonical correlation analysis. The associated canonical correlations are the *partial canonical correlations* between $\Delta \mathbf{x}_{t-1}$ and $\mathbf{x}_{t-1}$ because the effects of $\mathbf{d}_t$ and $\Delta \mathbf{x}_{t-i}$ have been adjusted. The quantities $\{\hat{\lambda}_i\}$ are the squared canonical correlations between $\hat{\mathbf{u}}_t$ and $\hat{\mathbf{v}}_t$.

Consider the hypotheses

$$H_0 : \text{Rank}(\mathbf{\Pi}) = m \quad \text{versus} \quad H_a : \text{Rank}(\mathbf{\Pi}) > m.$$

Johansen (1988) proposes the likelihood ratio (LR) statistic

$$\text{LR}_{\text{tr}}(m) = -(T-p) \sum_{i=m+1}^k \ln(1 - \hat{\lambda}_i) \quad (8.42)$$

to perform the test. If $\text{Rank}(\mathbf{\Pi}) = m$, then $\hat{\lambda}_i$ should be small for $i > m$ and hence $\text{LR}_{\text{tr}}(m)$ should be small. This test is referred to as the *trace* cointegration test. Due the presence of unit roots, the asymptotic distribution of $\text{LR}_{\text{tr}}(m)$ is not chi squared but a function of standard Brownian motions. Thus, critical values of $\text{LR}_{\text{tr}}(m)$ must be obtained via simulation.

Johansen (1988) also considers a sequential procedure to determine the number of cointegrating vectors. Specifically, the hypotheses of interest are

$$H_0 : \text{Rank}(\mathbf{\Pi}) = m \quad \text{versus} \quad H_a : \text{Rank}(\mathbf{\Pi}) = m + 1.$$

The LR ratio test statistic, called the maximum eigenvalue statistic, is

$$\text{LR}_{\text{max}}(m) = -(T-p) \ln(1 - \hat{\lambda}_{m+1}).$$

Again, critical values of the test statistics are nonstandard and must be evaluated via simulation.

8.6.4 Forecasting of Cointegrated VAR Models

The fitted ECM can be used to produce forecasts. First, conditioned on the estimated parameters, the ECM equation can be used to produce forecasts of the differenced series $\Delta \mathbf{x}_t$. Such forecasts can in turn be used to obtain forecasts of $\mathbf{x}_t$. A difference between ECM forecasts and the traditional VAR forecasts is that the ECM approach imposes the cointegration relationships in producing the forecasts.

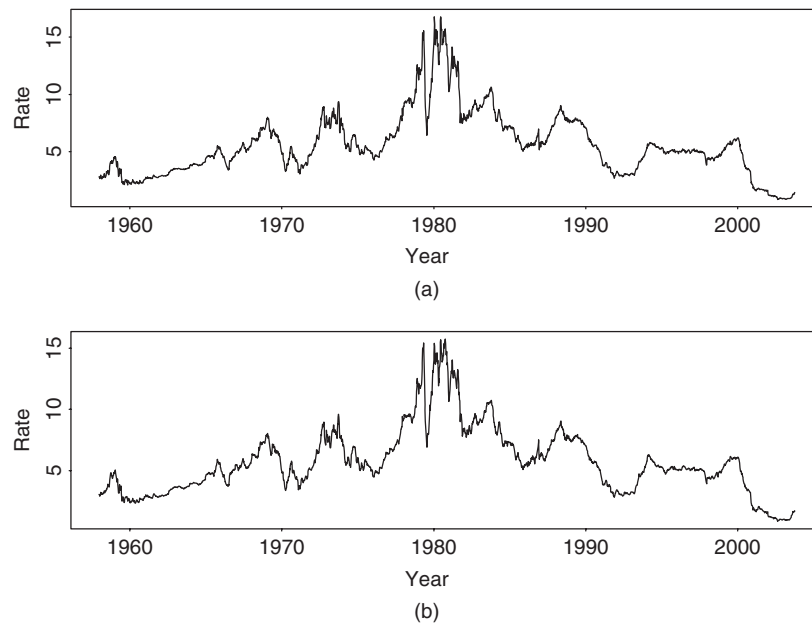


Figure 8.12 Time plots of weekly U.S. interest rate from December 12, 1958, to August 6, 2004. (a) The 3-month Treasury bill rate and (b) 6-month Treasury bill rate. Rates are from secondary market.

8.6.5 An Example

To demonstrate the analysis of cointegrated VAR models, we consider two weekly U.S. short-term interest rates. The series are the 3-month Treasury bill (TB) rate and 6-month Treasury bill rate from December 12, 1958, to August 6, 2004, for 2383 observations. The TB rates are from the secondary market and obtained from the Federal Reserve Bank of St. Louis. Figure 8.12 shows the time plots of the interest rates. As expected, the two series move closely together.

Our analysis uses the S-Plus software with commands VAR for VAR analysis, coint for cointegration test, and VECM for vector error correction estimation. Denote the two series by `tb3m` and `tb6m` and define the vector series $\mathbf{x}_t = (\text{tb3m}_t, \text{tb6m}_t)'$. The augmented Dickey–Fuller unit-root tests fail to reject the hypothesis of a unit root in the individual series; see Chapter 2. Indeed, the test statistics are -2.34 and -2.33 with p value about 0.16 for the 3-month and 6-month interest rate when an AR(3) model is used. Thus, we proceed to VAR modeling.

For the bivariate series $\mathbf{x}_t$, the BIC criterion selects a VAR(3) model:

```
> x=cbind(tb3m,tb6m)
> y=data.frame(x)
> ord.choice$ar.order
[1] 3
```

To perform a cointegration test, we choose a restricted constant for μ_t because there is no reason a priori to believe the existence of a drift in the U.S. interest rate. Both Johansen's tests confirm that the two series are cointegrated with one cointegrating vector when a VAR(3) model is entertained.

```
> cointst.rc=coint(x,trend='rc', lags=2) % lags = p-1.
> cointst.rc
Call:
coint(Y = x, lags = 2, trend = "rc")
```

```
Trend Specification:
H1*(r): Restricted constant
```

```
Trace tests sign. at the 5% level are flagged by ' + '.
Trace tests sign. at the 1% level are flagged by ' ++ '.
Max Eig. tests sign. at the 5% level are flagged by ' * '.
Max Eig. tests sign. at the 1% level are flagged by ' *** '.
```

```
Tests for Cointegration Rank:
      Eigenvalue Trace Stat  95% CV  99% CV
H(0)+++ 0.0322      83.2712    19.96   24.60
H(1)     0.0023      5.4936     9.24   12.97
```

```
      Max Stat  95% CV  99% CV
H(0)+++ 77.7776   15.67   20.20
H(1)     5.4936    9.24   12.97
```

Next, we perform the maximum-likelihood estimation of the specified cointegrated VAR(3) model using an ECM presentation. The results are as follows:

```
> vecm.fit=VECM(cointst.rc)
> summary(vecm.fit)
Call:
VECM(test = cointst.rc)
```

```
Cointegrating Vectors:
```

```
      coint.1
      1.0000

      tb6m   -1.0124
(std.err)    0.0086
(t.stat)  -118.2799
```

```
Intercept*    0.2254
(std.err)     0.0545
(t.stat)      4.1382
```

```
VECM Coefficients:
```

```

          tb3m    tb6m
coint.1 -0.0949 -0.0211
(std.err) 0.0199 0.0179
(t.stat) -4.7590 -1.1775

tb3m.lag1 0.0466 -0.0419
(std.err) 0.0480 0.0432
(t.stat) 0.9696 -0.9699

tb6m.lag1 0.2650 0.3164
(std.err) 0.0538 0.0484
(t.stat) 4.9263 6.5385

tb3m.lag2 -0.2067 -0.0346
(std.err) 0.0481 0.0433
(t.stat) -4.2984 -0.8005

tb6m.lag2 0.2547 0.0994
(std.err) 0.0543 0.0488
(t.stat) 4.6936 2.0356

Regression Diagnostics:
          tb3m    tb6m
R-squared 0.1081 0.0913
Adj. R-squared 0.1066 0.0898
Resid. Scale 0.2009 0.1807

> plot(vecm.fit)
Make a plot selection (or 0 to exit):

1: plot: All
2: plot: Response and Fitted Values
3: plot: Residuals
...
13: plot: PACF of Squared Cointegrating Residuals
Selection:

```

As expected, the output shows that the stationary series is $w_t \approx \text{tb3m}_t - \text{tb6m}_t$ and the mean of w_t is about -0.225 . The fitted ECM is

$$\begin{aligned} \Delta \mathbf{x}_t = & \begin{bmatrix} -0.09 \\ -0.02 \end{bmatrix} (w_{t-1} + 0.23) + \begin{bmatrix} 0.05 & 0.27 \\ -0.04 & 0.32 \end{bmatrix} \Delta \mathbf{x}_{t-1} \\ & + \begin{bmatrix} -0.21 & 0.25 \\ -0.03 & 0.10 \end{bmatrix} \Delta \mathbf{x}_{t-2} + \mathbf{a}_t, \end{aligned}$$

and the estimated standard errors of a_{it} are 0.20 and 0.18, respectively. Adequacy of the fitted ECM can be examined via various plots. For illustration, Figure 8.13 shows the cointegrating residuals. Some large residuals are shown

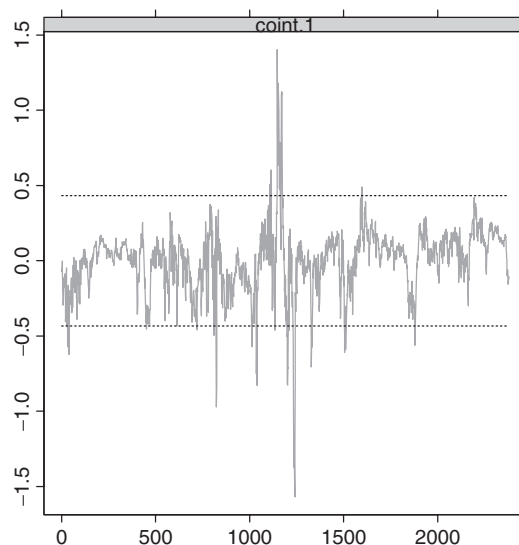


Figure 8.13 Time plot of cointegrating residuals for an ECM fit to weekly U.S. interest rate series. Data span is from December 12, 1958, to August 6, 2004.

in the plot, which occurred in the early 1980s when the interest rates were high and volatile.

Finally, we use the fitted ECM to produce 1-step- to 10-step-ahead forecasts for both Δx_t and x_t . The forecast origin is August 6, 2004.

```
> vecm.fst=predict(vecm.fit, n.predict=10)
> summary(vecm.fst)
Predicted Values with Standard Errors:
```

	tb3m	tb6m
1-step-ahead	-0.0378	-0.0642
(std.err)	0.2009	0.1807
2-step-ahead	-0.0870	-0.0864
(std.err)	0.3222	0.2927
...		
10-step-ahead	-0.2276	-0.1314
(std.err)	0.8460	0.8157

```
> plot(vecm.fst,xold=diff(x),n.old=12)

> vecm.fit.level=VECM(cointst.rc,levels=T)
> vecm.fst.level=predict(vecm.fit.level, n.predict=10)
> summary(vecm.fst.level)

Predicted Values with Standard Errors:
```

	tb3m	tb6m
1-step-ahead	1.4501	1.7057

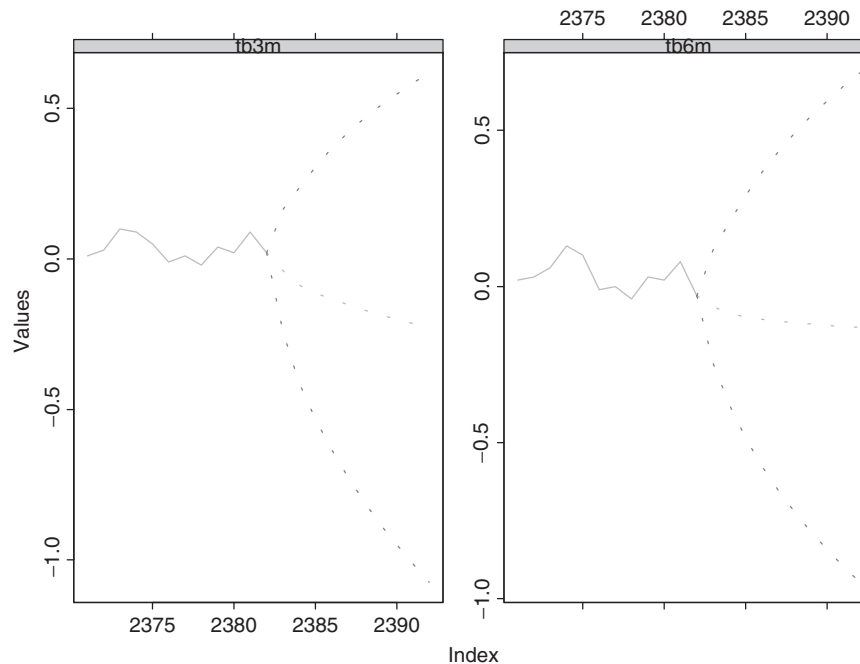


Figure 8.14 Forecasting plots of fitted ECM model for weekly U.S. interest rate series. Forecasts are for differenced series and forecast origin is August 6, 2004.

```
(std.err) 0.2009 0.1807
2-step-ahead 1.4420 1.7017
(std.err) 0.3222 0.2927
...
10-step-ahead 1.4722 1.7078
(std.err) 0.8460 0.8157
> plot(vecm.fst.level, xold=x, n.old=50)
```

The forecasts are shown in Figures 8.14 and 8.15 for the differenced data and the original series, respectively, along with some observed data points. The dashed lines in the plots are pointwise 95% confidence intervals. Because of unit-root nonstationarity, the intervals are wide and not informative.

Remark. The package `urca` of R can be used to perform Johansen's cointegration test. The command is `ca.jo`. It requires specification of some sub-commands. See the section of pairs trading for demonstration. $\square$

8.7 THRESHOLD COINTEGRATION AND ARBITRAGE

In this section, we focus on detecting arbitrage opportunities in index trading by using multivariate time series methods. We also demonstrate that simple univariate

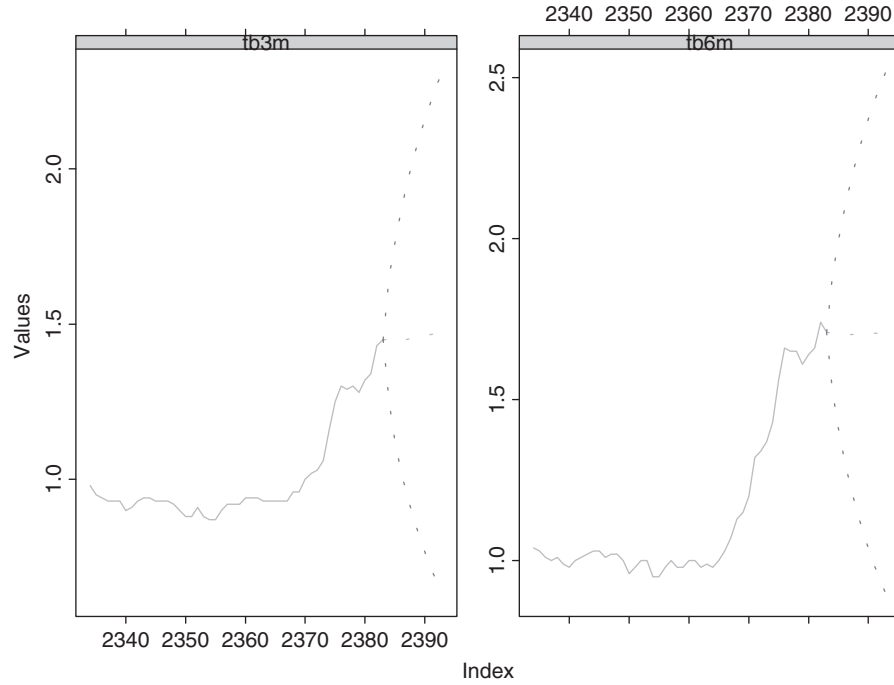


Figure 8.15 Forecasting plots of fitted ECM model for weekly U.S. interest rate series. Forecasts are for interest rates and forecast origin is August 6, 2004.

nonlinear models of Chapter 4. can be extended naturally to the multivariate case in conjunction with the idea of cointegration.

Our study considers the relationship between the price of the S&P 500 index futures and the price of the shares underlying the index on the cash market. Let $f_{t,\ell}$ be the log price of the index futures at time t with maturity ℓ , and let s_t be the log price of the shares underlying the index on the cash market at time t . A version of the *cost-of-carry model* in the finance literature states

$$f_{t,\ell} - s_t = (r_{t,\ell} - q_{t,\ell})(\ell - t) + z_t^*, \quad (8.43)$$

where $r_{t,\ell}$ is the risk-free interest rate, $q_{t,\ell}$ is the dividend yield with respect to the cash price at time t , and $(\ell - t)$ is the time to maturity of the futures contract; see Brenner and Kroner (1995), Dwyer, Locke, and Yu (1996), and the references therein.

The z_t^* process of model (8.43) must be unit-root stationary; otherwise there exist *persistent* arbitrage opportunities. Here an arbitrage trading consists of simultaneously buying (short-selling) the security index and selling (buying) the index futures whenever the log prices diverge by more than the cost of carrying the index over time until maturity of the futures contract. Under the weak stationarity of z_t^* ,

for arbitrage to be profitable, z_t^* must exceed a certain value in modulus determined by transaction costs and other economic and risk factors.

It is commonly believed that the $f_{t,\ell}$ and s_t series of the S&P 500 index contain a unit root, but Eq. (8.43) indicates that they are cointegrated after adjusting for the effect of interest rate and dividend yield. The cointegrating vector is $(1, -1)$ after the adjustment, and the cointegrated series is z_t^* . Therefore, one should use an error correction form to model the return series $\mathbf{r}_t = (\Delta f_t, \Delta s_t)'$, where $\Delta f_t = f_{t,\ell} - f_{t-1,\ell}$ and $\Delta s_t = s_t - s_{t-1}$, where for ease in notation we drop the maturity time ℓ from the subscript of Δf_t .

8.7.1 Multivariate Threshold Model

In practice, arbitrage tradings affect the dynamic of the market, and hence the model for $\mathbf{r}_t$ may vary over time depending on the presence or absence of arbitrage tradings. Consequently, the prior discussions lead naturally to the following model:

$$\mathbf{r}_t = \begin{cases} \mathbf{c}_1 + \sum_{i=1}^p \Phi_i^{(1)} \mathbf{r}_{t-i} + \beta_1 z_{t-1} + \mathbf{a}_t^{(1)} & \text{if } z_{t-1} \leq \gamma_1, \\ \mathbf{c}_2 + \sum_{i=1}^p \Phi_i^{(2)} \mathbf{r}_{t-i} + \beta_2 z_{t-1} + \mathbf{a}_t^{(2)} & \text{if } \gamma_1 < z_{t-1} \leq \gamma_2, \\ \mathbf{c}_3 + \sum_{i=1}^p \Phi_i^{(3)} \mathbf{r}_{t-i} + \beta_3 z_{t-1} + \mathbf{a}_t^{(3)} & \text{if } \gamma_2 < z_{t-1}, \end{cases} \quad (8.44)$$

where $z_t = 100z_t^*$, $\gamma_1 < 0 < \gamma_2$ are two real numbers, and $\{\mathbf{a}_t^{(i)}\}$ are sequences of two-dimensional white noises and are independent of each other. Here we use $z_t = 100z_t^*$ because the actual value of z_t^* is relatively small.

The model in Eq. (8.44) is referred to as a multivariate threshold model with three regimes. The two real numbers γ_1 and γ_2 are the thresholds and z_{t-1} is the threshold variable. The threshold variable z_{t-1} is supported by the data; see Tsay (1998). In general, one can select z_{t-d} as a threshold variable by considering $d \in \{1, \dots, d_0\}$, where d_0 is a prespecified positive integer.

Model (8.44) is a generalization of the threshold autoregressive model of Chapter 4. It is also a generalization of the error correlation model of Eq. (8.33). As mentioned earlier, an arbitrage trading is profitable only when z_t^* or, equivalently, z_t is large in modulus. Therefore, arbitrage tradings only occurred in regimes 1 and 3 of model (8.44). As such, the dynamic relationship between $f_{t,\ell}$ and s_t in regime 2 is determined mainly by the normal market force, and hence the two series behave more or less like a random walk. In other words, the two log prices in the middle regime should be free from arbitrage effects and, hence, free from the cointegration constraint. From an econometric viewpoint, this means that the estimate of β_2 in the middle regime should be insignificant.

In summary, we expect that the cointegration effects between the log price of the futures and the log price of security index on the cash market are significant in regimes 1 and 3, but insignificant in regime 2. This phenomenon is referred to as a *threshold cointegration*; see Balke and Fomby (1997).

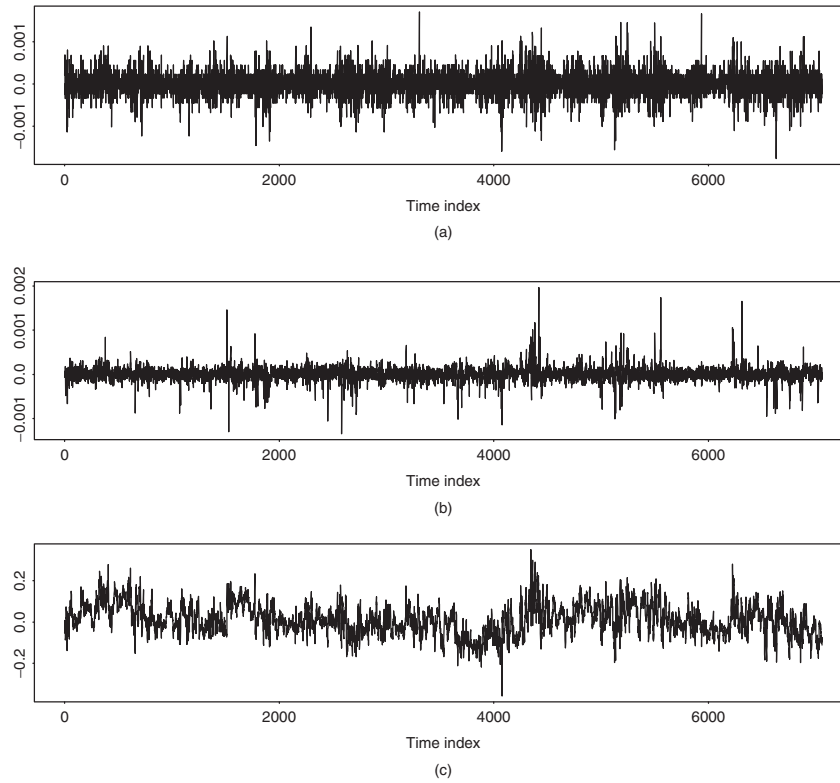


Figure 8.16 Time plots of 1-minute log returns of S&P 500 index futures and cash prices and associated threshold variable in May 1993: (a) log returns of index futures, (b) log returns of index cash prices, and (c) z_t series.

8.7.2 The Data

The data used in this case study are the intraday transaction data of the S&P 500 index in May 1993 and its June futures contract traded at the Chicago Mercantile Exchange; see Forbes, Kalb, and Kofman (1999), who used the data to construct a minute-by-minute bivariate price series with 7060 observations. To avoid the undue influence of unusual returns, I replaced 10 extreme values (5 on each side) by the simple average of their two nearest neighbors. This step does not affect the qualitative conclusion of the analysis but may affect the conditional heteroscedasticity in the data. For simplicity, we do not consider conditional heteroscedasticity in the study. Figure 8.16 shows the time plots of the log returns of the index futures and cash prices and the associated threshold variable $z_t = 100z_t^*$ of model (8.43).

8.7.3 Estimation

A formal specification of the multivariate threshold model in Eq. (8.44) includes selecting the threshold variable, determining the number of regimes, and choosing

the order p for each regime. Interested readers are referred to Tsay (1998) and Forbes, Kalb, and Kofman (1999). The thresholds γ_1 and γ_2 can be estimated by using some information criteria [e.g., the Akaike information criterion (AIC) or the sum of squares of residuals]. Assuming $p = 8$, $d \in \{1, 2, 3, 4\}$, $\gamma_1 \in [-0.15, -0.02]$, and $\gamma_2 \in [0.025, 0.145]$, and using a grid search method with 300 points on each of the two intervals, the AIC selects z_{t-1} as the threshold variable with thresholds $\hat{\gamma}_1 = -0.0226$ and $\hat{\gamma}_2 = 0.0377$. Details of the parameter estimates are given in Table 8.8.

From Table 8.8, we make the following observations. First, the t ratios of $\hat{\beta}_2$ in the middle regime show that, as expected, the estimates are insignificant at the 5% level, confirming that there is no cointegration between the two log prices in the absence of arbitrage opportunities. Second, Δf_t depends negatively on Δf_{t-1} in all three regimes. This is in agreement with the bid–ask bounce discussed in Chapter 5. Third, past log returns of the index futures seem to be more informative than the past log returns of the cash prices because there are more significant t ratios in Δf_{t-i} than in Δs_{t-i} . This is reasonable because futures series are in general more liquid. For more information on index arbitrage, see Dwyer, Locke, and Yu (1996).

8.8 PAIRS TRADING

Pairs trading is a market-neutral trading strategy. There are several versions of pairs trading in the equity markets. In this section, we focus on the statistical arbitrage pairs trading, which makes use of the ideas of cointegration and error correction model discussed in the chapter. Our discussion will be brief. For more information concerning pairs trading and statistical arbitrage, see Vidyamurthy (2004) and Pole (2007).

The general theme for trading in the equity markets is to buy undervalued stocks and sell overvalued ones. However, the true price of a stock is hard to assess. Pairs trading attempts to resolve this difficulty using the idea of relative pricing. Based on the arbitrage pricing theory (APT) in finance, if two stocks have similar characteristics, then the prices of both stocks must be more or less the same. If the prices differ, then it is likely that one of the stocks is overpriced and the other underpriced. Pairs trading involves selling the higher priced stock and buying the lower priced stock with the hope that the mispricing will correct itself in the future. Note that the true prices of the two stocks are not important. The observed prices may be wrong. What is important is that the observed prices be the same. The gap (properly scaled) between the two observed prices is called the spread. For pairs trading, the greater the spread, the larger the magnitude of mispricing and the greater the profit potential. Before discussing a trading strategy, we first introduce the theoretical framework.

8.8.1 Theoretical Framework

Consider two stocks. Let P_{it} be the observed price of stock i at time t and $p_{it} = \ln(P_{it})$ be the corresponding log price. As mentioned in earlier chapters, it is

TABLE 8.8 Least-Squares Estimates and Their t Ratios of Multivariate Threshold Model in Eq. (8.43) for S&P 500 Index Data in May 1993^a

	Regime 1		Regime 2		Regime 3	
	Δf_t	Δs_t	Δf_t	Δs_t	Δf_t	Δs_t
ϕ_0	0.00002	0.00005	0.00000	0.00000	-0.00001	-0.00005
t	(1.47)	(7.64)	(-0.07)	(0.53)	(-0.74)	(-6.37)
Δf_{t-1}	-0.08468	0.07098	-0.03861	0.04037	-0.04102	0.02305
t	(-3.83)	(6.15)	(-1.53)	(3.98)	(-1.72)	(1.96)
Δf_{t-2}	-0.00450	0.15899	0.04478	0.08621	-0.02069	0.09898
t	(-0.20)	(13.36)	(1.85)	(8.88)	(-0.87)	(8.45)
Δf_{t-3}	0.02274	0.11911	0.07251	0.09752	0.00365	0.08455
t	(0.95)	(9.53)	(3.08)	(10.32)	(0.15)	(7.02)
Δf_{t-4}	0.02429	0.08141	0.01418	0.06827	-0.02759	0.07699
t	(0.99)	(6.35)	(0.60)	(7.24)	(-1.13)	(6.37)
Δf_{t-5}	0.00340	0.08936	0.01185	0.04831	-0.00638	0.05004
t	(0.14)	(7.10)	(0.51)	(5.13)	(-0.26)	(4.07)
Δf_{t-6}	0.00098	0.07291	0.01251	0.03580	-0.03941	0.02615
t	(0.04)	(5.64)	(0.54)	(3.84)	(-1.62)	(2.18)
Δf_{t-7}	-0.00372	0.05201	0.02989	0.04837	-0.02031	0.02293
t	(-0.15)	(4.01)	(1.34)	(5.42)	(-0.85)	(1.95)
Δf_{t-8}	0.00043	0.00954	0.01812	0.02196	-0.04422	0.00462
t	(0.02)	(0.76)	(0.85)	(2.57)	(-1.90)	(0.40)
Δs_{t-1}	-0.08419	0.00264	-0.07618	-0.05633	0.06664	0.11143
t	(-2.01)	(0.12)	(-1.70)	(-3.14)	(1.49)	(5.05)
Δs_{t-2}	-0.05103	0.00256	-0.10920	-0.01521	0.04099	-0.01179
t	(-1.18)	(0.11)	(-2.59)	(-0.90)	(0.92)	(-0.53)
Δs_{t-3}	0.07275	-0.03631	-0.00504	0.01174	-0.01948	-0.01829
t	(1.65)	(-1.58)	(-0.12)	(0.71)	(-0.44)	(-0.84)
Δs_{t-4}	0.04706	0.01438	0.02751	0.01490	0.01646	0.00367
t	(1.03)	(0.60)	(0.71)	(0.96)	(0.37)	(0.17)
Δs_{t-5}	0.08118	0.02111	0.03943	0.02330	-0.03430	-0.00462
t	(1.77)	(0.88)	(0.97)	(1.43)	(-0.83)	(-0.23)
Δs_{t-6}	0.04390	0.04569	0.01690	0.01919	0.06084	-0.00392
t	(0.96)	(1.92)	(0.44)	(1.25)	(1.45)	(-0.19)
Δs_{t-7}	-0.03033	0.02051	-0.08647	0.00270	-0.00491	0.03597
t	(-0.70)	(0.91)	(-2.09)	(0.16)	(-0.13)	(1.90)
Δs_{t-8}	-0.02920	0.03018	0.01887	-0.00213	0.00030	0.02171
t	(-0.68)	(1.34)	(0.49)	(-0.14)	(0.01)	(1.14)
z_{t-1}	0.00024	0.00097	-0.00010	0.00012	0.00025	0.00086
t	(1.34)	(10.47)	(-0.30)	(0.86)	(1.41)	(9.75)

^aThe numbers of data points for the three regimes are 2234, 2410, and 2408, respectively.

reasonable to assume that p_{it} is unit-root nonstationary and follows a random-walk model; that is, $p_{it} = p_{i,t-1} + r_{it}$, where $\{r_{it}\}$ is the return and forms a sequence of uncorrelated innovations. If the two stocks have similar risk factors, then they should have similar returns based on APT. Therefore, p_{1t} and p_{2t} are likely to be driven by a common component and are cointegrated. In other words, there exists a linear combination $w_t = p_{1t} - \gamma p_{2t}$, which is unit-root stationary and, hence, mean reverting. The two price series $\{p_{1t}\}$ and $\{p_{2t}\}$ thus assume an error correction form

$$\begin{bmatrix} p_{1t} - p_{1,t-1} \\ p_{2t} - p_{2,t-1} \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} (w_{t-1} - \mu_w) + \begin{bmatrix} \epsilon_{1t} \\ \epsilon_{2t} \end{bmatrix}, \quad (8.45)$$

where $\mu_w = E(w_t)$ denotes the mean of w_t . The four parameters γ , μ_w , α_1 , and α_2 can be estimated, for instance, by the maximum-likelihood or least-squares methods; see Section 8.6.2. We refer to the stationary series w_t as the *spread* between the two log stock prices.

The left-hand side of Eq. (8.45) consists the log returns of the two stocks. The equation says that the returns depend on w_{t-1} , which is the stationary. Specifically, $w_{t-1} - \mu_w$ denotes the deviation from the log-run equilibrium between the two stocks. Equation (8.45) shows that, for cointegrated stocks, the returns depend on the past deviation from equilibrium. The coefficients α_1 and α_2 show the effect of past deviation on the returns r_{1t} and r_{2t} , respectively. In practice, α_1 and α_2 should have opposite signs, indicating reversion to the equilibrium.

Next, consider a portfolio with long one share of stock 1 and short γ shares of stock 2. The return of the portfolio for a given time period i is

$$\begin{aligned} r_{p,t+i} &= (p_{1,t+i} - p_{1t}) - \gamma(p_{2,t+i} - p_{2t}) \\ &= (p_{1,t+i} - \gamma p_{2,t+i}) - (p_{1t} - \gamma p_{2t}) \\ &= w_{t+i} - w_t. \end{aligned}$$

Therefore, the return $r_{p,t+i}$ of the portfolio is the increment of the spread in the time period. As expected, the return of the portfolio does not depend on the mean of w_t .

8.8.2 Trading Strategy

The idea behind a pairs-trading strategy is to trade on the oscillations about the equilibrium value of the spread. The oscillations in spread occur because the spread is mean reverting. Since the equilibrium value is the mean of w_t , that is, μ_w , we can put on a trade when w_t deviates substantially from its mean and unwind the trade when the equilibrium is restored. In practice, how big the deviation needs to be in order for the trading to be profitable depends on several factors. Trading costs, marginal interest rates, and bid-ask spreads of the two stocks are three obvious factors. Mathematically, let η be the cost involved in carrying out a pairs trading. Let Δ be a target deviation of w_t from its mean μ_w for pairs trading. Then, conditioned on $2\Delta > \eta$, a simple trading strategy is as follows:

- Buy a share of stock 1 and short γ shares of stock 2 at time t if $w_t = p_{1t} - \gamma p_{2t} = \mu_w - \Delta$.
- Unwind the position at time $t + i$ ($i > 0$) if $w_{t+i} = p_{1,t+i} - \gamma p_{2,t+i} = \mu_w + \Delta$.

One can identify the time point t so long as Δ is not too large compared with the standard deviation of w_t . The time point $t + i$ will occur because of the mean reverting of the spread series. In this particular instance, the return of the portfolio $w_{t+i} - w_t = 2\Delta$ and the net profit of the trade is $2\Delta - \eta > 0$.

Discussion. The aforementioned trading strategy is just one of many possibilities. For instance, if $\Delta > \eta$, one can unwind the position when $w_{t+i} = \mu_w$. The net profit of the pairs trading then is $\Delta - \eta > 0$. This may result in more transactions and trading costs, but it shortens the holding period of the portfolio. If Δ is negative, then one can short one share of stock 1 and buy γ shares of stock 2 to make a net profit $-2\Delta - \eta$. The quantity η is the threshold for trading and is likely to depend on several factors such as transaction fees and bid–ask spreads of the two stocks. $\square$

8.8.3 Simple Illustration

To demonstrate pairs trading, we consider two stocks traded on the New York Stock Exchange. The two companies are the Billiton Ltd. of Australia and the Vale S.A. of Brazil with stock symbols BHP and VALE, respectively. BHP of Australia is a natural resources company with business in Australia, the Americas, and Southern Africa. Vale of Brazil is a worldwide metals and mining company. Thus, both multinational companies belong to the natural resources industry and encounter similar risk factors. The daily prices of the two stocks were downloaded from Yahoo Finance, and we employ adjusted closing prices from July 1, 2002, to March 31, 2006, in our study.

Figure 8.17 shows the time plots of the daily log prices of the two stocks (adjusted closing prices). The upper plot is for the BHP stock. From the plots, the prices of the two stocks exhibit certain characteristics of comovement. Let p_{1t} and p_{2t} be the daily log closing prices of BHP and VALE, respectively. We analyze the series using both the least-squares and maximum-likelihood methods.

Least-Squares Estimation

A simple way to verify that the two stocks are suitable for pairs trading is to check the cointegration of their log stock prices. To this end, we consider the simple linear regression $p_{1t} = \beta_0 + \beta_1 p_{2t} + w_t$, where w_t denotes the residual series. For the BHP and VALE stocks, we have

$$p_{1t} = 1.823 + 0.717 p_{2t} + \hat{w}_t, \quad \sigma_w = 0.044.$$

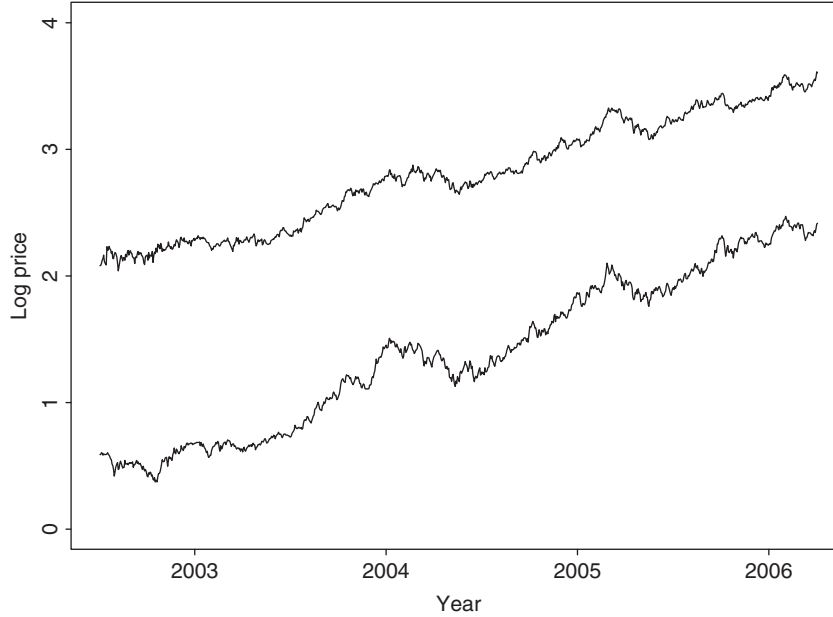


Figure 8.17 Daily log (adjusted) closing prices of BHP and VALE stocks from July 1, 2002, to March 31, 2006. Upper plot is for BHP stock.

Figure 8.18(a) shows the time plot of the residual series $\hat{w}_t$. The plot shows that the residual series has certain characteristics of a stationary time series. In particular, it has mean zero and fluctuates around its mean within a fixed range. Figure 8.18(b) gives the sample ACF of $\hat{w}_t$. The ACFs decay exponentially, supporting that $\hat{w}_t$ is indeed stationary. To further confirm the stationarity assertion, we fit an AR(2) model to $\hat{w}_t$ and obtain

$$(1 - 0.805B - 0.122B^2)\hat{w}_t = a_t, \quad \sigma_a = 0.018.$$

Following the discussion of Chapter 2, we can obtain the two characteristic roots of the fitted AR(2) model. Indeed, the model can be rewritten as $(1 - 0.935B)(1 - 0.130B)\hat{w}_t = a_t$. Hence, $\hat{w}_t$ is stationary. Finally, we conduct an augmented Dickey–Fuller unit-root test on $\hat{w}_t$ using an AR(2) model and find that the test statistic is -6.04 with a p value of 0.01 . The unit-root hypothesis is clearly rejected.

Maximum-Likelihood Estimation

A formal approach to verify the cointegration of the two log stock prices is to perform a cointegration test. Let $\mathbf{x}_t = (p_{1t}, p_{2t})'$. Using information criteria, a VAR(1) model is specified for $\mathbf{x}_t$. We then conduct cointegration tests with restricted and unrestricted constant. Both tests give similar results so that we only report the results for the case of restricted constant.

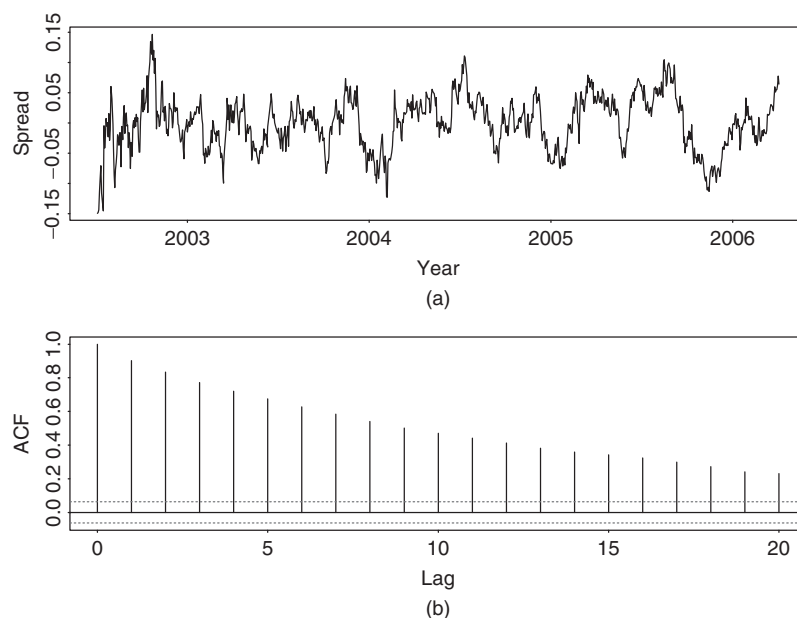


Figure 8.18 Results of least-squares estimation: (a) Time plot of the estimated spread between BHP and VALE daily log stock prices. (b) Sample autocorrelation functions of estimated spread.

```
> coint2=coint(xx,trend="rc")
> coint2
coint(Y = xt, trend = "rc")
```

Trend Specification:
H1*(r): Restricted constant

Trace tests signif. at the 5% level are flagged by ' + '.
Trace tests signif. at the 1% level are flagged by '++'.
Max Eigenvalue tests signif. at the 5% level are
flagged by ' * '.
Max Eigenvalue tests signif. at the 1% level are
flagged by '***'.

Tests for Cointegration Rank:

	Eigenvalue	TraceSt	95%-CV	99%-CV	Max-St	95%-CV	99%-CV
CV							
H(0)+++	0.0415	47.7400	19.960	24.600	39.965	15.670	20.200
H(1)	0.0082	7.7748	9.240	12.970	7.774	9.240	12.970

The test confirms that x_t is cointegrated. Next, we perform the maximum-likelihood estimation of the error correction model. The results are given below:

```
> n3=VECM(coint2)
> summary(n3)
```



```

VECM(test = coint2)

Cointegrating Vectors:
      coint.1
      1.0000

      vale   -0.7177
(std.err)    0.0112
(t.stat)    -64.0913

Intercept*   -1.8144
(std.err)    0.0169
(t.stat)    -107.0430

VECM Coefficients:
      bhp      vale
coint.1 -0.0671  0.0263
(std.err) 0.0145  0.0168
(t.stat) -4.6462  1.5659

bhp.lag1 -0.1119  0.0659
(std.err) 0.0366  0.0425
(t.stat) -3.0596  1.5516

vale.lag1 0.0732  0.0445
(std.err) 0.0320  0.0371
(t.stat) 2.2920  1.1986

Regression Diagnostics:
      bhp      vale
R-squared 0.0370  0.0104
Adj. R-squared 0.0350  0.0083
Resid. Scale 0.0193  0.0224

```

Based on the estimation result, we have the model

$$\Delta \mathbf{x}_t = \begin{bmatrix} -0.067 \\ 0.026 \end{bmatrix} (w_{t-1} - 1.81) + \begin{bmatrix} -0.11 & 0.07 \\ 0.07 & 0.04 \end{bmatrix} \Delta \mathbf{x}_{t-1} + \mathbf{a}_t,$$

where the estimated standard errors of a_{it} are 0.019 and 0.022, respectively. In addition, the spread series is $w_t = p_{1t} - 0.718p_{2t}$, which is stationary with mean 1.81. Clearly, the result is very close to that of the least-squares estimation. In particular, the γ parameter for the pairs trading is $\hat{\gamma} = 0.718$. Also, as expected, α_1 is negative whereas α_2 is positive.

Trading Strategy

Since the standard error of the spread series w_t is 0.044, we can select $\Delta = 0.045$, which is slightly greater than one standard error of w_t , for pairs trading. This choice

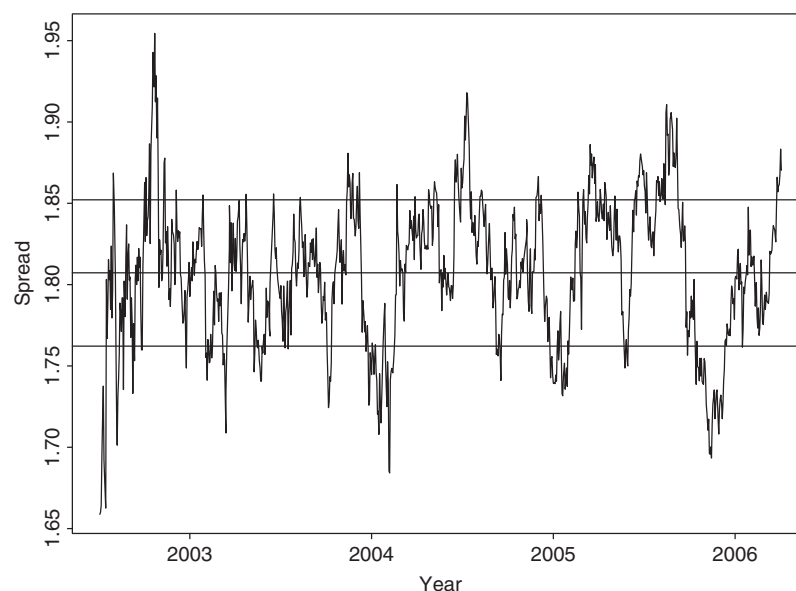


Figure 8.19 Time plot of fitted spread series between daily log prices of BHP and VALE stocks. Three horizontal lines denotes μ_w , $\mu_w + 0.045$, and $\mu_w - 0.045$ with $\mu_w = E(w_t)$.

of Δ ensures that the probability for the spread w_t to deviate Δ away from its mean is not small. In fact, under the normality assumption, the probability is about 30%. Figure 8.19 shows the time plot of the spread series w_t of the fitted error correction model. Three horizontal lines are imposed on the plot. They are μ_w , $\mu_w + 0.045$, and $\mu_w - 0.045$ with the latter two serving as boundaries for pairs trading. Since w_t varies from the lower boundary to the upper one (or from the upper boundary to the lower one) several times, there are many pairs-trading opportunities. From the discussion of Section 8.8.2, the log return of each pairs trading is $2\Delta = 0.09$, which is not small. A more realistic demonstration is to implement the trading in a out-of-sample period. However, the example shows that pairs trading is feasible.

Finally, an important question in pairs trading is to identify the cointegrated pairs of stocks. There are some procedures available in the literature. It seems reasonable to consider pairs of stocks that have similar risk factors. In other words, one should make use of finance theory to guide the selection.

R Demonstration

The following output has been edited:

```
> library(urca)
> help(ca.jo)
> da=read.table("d-bhp0206.txt",header=T)
> dal=read.table("d-vale0206.txt",header=T)
> bhp=log(da[,9])
```

```

> vale=log(da1[,9])
> m1=lm(bhp~vale)
> summary(m1)
Call:
lm(formula = bhp ~ vale)

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.822648    0.003662   497.7  >2e-16 ***
vale          0.716664    0.002354   304.4  >2e-16 ***
---
Residual standard error: 0.04421 on 944 degrees of freedom
Multiple R-squared:  0.9899,    Adjusted R-squared:  0.9899
F-statistic: 9.266e+04 on 1 and 944 DF,  p-value: < 2.2e-16

> wt=m1$residuals
> m3=arima(wt,order=c(2,0,0),include.mean=F)
> m3
Call:
arima(x = wt, order = c(2, 0, 0), include.mean = F)

Coefficients:
          ar1      ar2
          0.8051  0.1219
s.e.      0.0322  0.0325

sigma^2 estimated as 0.0003326: log likelihood=2444.76
> p1=c(1,-m3$coef)
> x=polyroot(p1)
> x
[1] 1.069100+0i -7.675365-0i
> 1/Mod(x)
[1] 0.9353661 0.1302870

> xt=cbind(bhp,vale)
> mm=ar(xt)
> mm$order
[1] 2
> cot=ca.jo(xt,ecdet="const",type='trace',K=2,
spec='transitory')
> summary(cot)
#####
# Johansen-Procedure #
#####
Test type: trace statistic, without linear trend and
            constant in cointegration

Eigenvalues (lambda):

```

```
[1] 4.148282e-02 8.206470e-03 -4.610389e-18
```

Values of teststatistic and critical values of test:

	test	10pct	5pct	1pct
r <= 1		7.78	7.52	9.24 12.97
r = 0		47.77	17.85	19.96 24.60

Eigenvectors, normalised to first column:
(These are the cointegration relations)

	bhp.l1	vale.l1	constant
bhp.l1	1.000000	1.000000	1.000000
vale.l1	-0.717704	-0.7327542	2.047274
constant	-1.828460	-1.5411890	-5.712629

Weights W:
(This is the loading matrix)

	bhp.l1	vale.l1	constant
bhp.d	-0.06731196	0.004568985	9.341093e-18
vale.d	0.02545606	0.007541565	1.015639e-18

```
> col=ca.jo(xt,ecdet="const",type='eigen',K=2,
spec='transitory')
> summary(col)
#####
# Johansen-Procedure #
#####
Test type: maximal eigenvalue statistic (lambda max), without
linear trend and constant in cointegration
```

Eigenvalues (lambda):
[1] 4.148282e-02 8.206470e-03 -4.610389e-18

Values of teststatistic and critical values of test:

	test	10pct	5pct	1pct
r <= 1		7.78	7.52	9.24 12.97
r = 0		40.00	13.75	15.67 20.20

Eigenvectors, normalised to first column:
(These are the cointegration relations)

	bhp.l1	vale.l1	constant
bhp.l1	1.000000	1.000000	1.000000
vale.l1	-0.717704	-0.7327542	2.047274
constant	-1.828460	-1.5411890	-5.712629

Weights W:
(This is the loading matrix)

```

          bhp.l1      vale.l1      constant
bhp.d   -0.06731196  0.004568985  9.341093e-18
vale.d   0.02545606  0.007541565  1.015639e-18

```

APPENDIX A: REVIEW OF VECTORS AND MATRICES

In this appendix, we briefly review some algebra and properties of vectors and matrices. No proofs are given as they can be found in standard textbooks on matrices (e.g., Graybill, 1969).

An $m \times n$ real-valued matrix is an $m \times n$ array of real numbers. For example,

$$A = \begin{bmatrix} 2 & 5 & 8 \\ -1 & 3 & 4 \end{bmatrix}$$

is a 2×3 matrix. This matrix has two rows and three columns. In general, an $m \times n$ matrix is written as

$$A \equiv [a_{ij}] = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1,n-1} & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2,n-1} & a_{2n} \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{m,n-1} & a_{mn} \end{bmatrix}. \quad (8.46)$$

The positive integers m and n are the *row dimension* and *column dimension* of A . The real number a_{ij} is referred to as the (i, j) th element of A . In particular, the elements a_{ii} are the *diagonal elements* of the matrix.

An $m \times 1$ matrix forms an m -dimensional column vector, and a $1 \times n$ matrix is an n -dimensional row vector. In the literature, a vector is often meant to be a column vector. If $m = n$, then the matrix is a square matrix. If $a_{ij} = 0$ for $i \neq j$ and $m = n$, then the matrix A is a *diagonal matrix*. If $a_{ij} = 0$ for $i \neq j$ and $a_{ii} = 1$ for all i , then A is the $m \times m$ *identity matrix*, which is commonly denoted by I_m or simply I if the dimension is clear.

The $n \times m$ matrix

$$A' = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m-1,1} & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m-1,2} & a_{m2} \\ \vdots & \vdots & & \vdots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{m-1,n} & a_{mn} \end{bmatrix}$$

is the *transpose* of the matrix A . For example,

$$\begin{bmatrix} 2 & -1 \\ 5 & 3 \\ 8 & 4 \end{bmatrix} \text{ is the transpose of } \begin{bmatrix} 2 & 5 & 8 \\ -1 & 3 & 4 \end{bmatrix}.$$

We use the notation $A' = [a'_{ij}]$ to denote the transpose of A . From the definition, $a'_{ij} = a_{ji}$ and $(A')' = A$. If $A' = A$, then A is a *symmetric matrix*.

Basic Operations

Suppose that $A = [a_{ij}]_{m \times n}$ and $C = [c_{ij}]_{p \times q}$ are two matrices with dimensions given in the subscript. Let b be a real number. Some basic matrix operations are defined next:

- Addition: $A + C = [a_{ij} + c_{ij}]_{m \times n}$ if $m = p$ and $n = q$.
- Subtraction: $A - C = [a_{ij} - c_{ij}]_{m \times n}$ if $m = p$ and $n = q$.
- Scalar multiplication: $bA = [ba_{ij}]_{m \times n}$.
- Multiplication: $AC = [\sum_{v=1}^n a_{iv}c_{vj}]_{m \times q}$ provided that $n = p$.

When the dimensions of matrices satisfy the condition for multiplication to take place, the two matrices are said to be *conformable*. An example of matrix multiplication is

$$\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ -1 & 2 & -4 \end{bmatrix} = \begin{bmatrix} 2 \cdot 1 - 1 \cdot 1 & 2 \cdot 2 + 1 \cdot 2 & 2 \cdot 3 - 1 \cdot 4 \\ 1 \cdot 1 - 1 \cdot 1 & 1 \cdot 2 + 1 \cdot 2 & 1 \cdot 3 - 1 \cdot 4 \end{bmatrix} \\ = \begin{bmatrix} 1 & 6 & 2 \\ 0 & 4 & -1 \end{bmatrix}.$$

Important rules of matrix operations include (a) $(AC)' = C'A'$ and (b) $AC \neq CA$ in general.

Inverse, Trace, Eigenvalue, and Eigenvector

A square matrix $A_{m \times m}$ is *nonsingular* or *invertible* if there exists a unique matrix $C_{m \times m}$ such that $AC = CA = I_m$, the $m \times m$ identity matrix. In this case, C is called the *inverse* matrix of A and is denoted by $C = A^{-1}$.

The trace of $A_{m \times m}$ is the sum of its diagonal elements [i.e., $\text{tr}(A) = \sum_{i=1}^m a_{ii}$]. It is easy to see that (a) $\text{tr}(A + C) = \text{tr}(A) + \text{tr}(C)$, (b) $\text{tr}(A) = \text{tr}(A')$, and (c) $\text{tr}(AC) = \text{tr}(CA)$ provided that the two matrices are conformable.

A number λ and an $m \times 1$ vector b , possibly complex valued, are a right *eigenvalue* and *eigenvector* pair of the matrix A if $Ab = \lambda b$. There are m possible eigenvalues for the matrix A . For a real-valued matrix A , complex eigenvalues occur in conjugated pairs. The matrix A is nonsingular if and only if all of its eigenvalues are nonzero. Denote the eigenvalues by $\{\lambda_i | i = 1, \dots, m\}$: We have $\text{tr}(A) = \sum_{i=1}^m \lambda_i$. In addition, the *determinant* of the matrix A can be defined as $|A| = \prod_{i=1}^m \lambda_i$. For a general definition of determinant of a matrix, see a standard textbook on matrices (e.g., Graybill, 1969).

Finally, the rank of the matrix $A_{m \times n}$ is the number of nonzero eigenvalues of the symmetric matrix AA' . Also, for a nonsingular matrix A , $(A^{-1})' = (A')^{-1}$.

Positive-Definite Matrix

A square matrix A ($m \times m$) is a *positive-definite* matrix if (a) A is symmetric and (b) all eigenvalues of A are positive. Alternatively, A is a positive-definite matrix if for any nonzero m -dimensional vector b , we have $b'Ab > 0$.

Useful properties of a positive-definite matrix A include (a) all eigenvalues of A are real and positive, and (b) the matrix can be decomposed as

$$A = P\Lambda P',$$

where Λ is a diagonal matrix consisting of all eigenvalues of A and P is an $m \times m$ matrix consisting of the m right eigenvectors of A . It is common to write the eigenvalues as $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_m$ and the eigenvectors as $e_1, \dots, e_m$ such that $Ae_i = \lambda_i e_i$ and $e_i'e_i = 1$. In addition, these eigenvectors are orthogonal to each other—namely, $e_i'e_j = 0$ if $i \neq j$ —if the eigenvalues are distinct. The matrix P is an *orthogonal* matrix and the decomposition is referred to as the *spectral decomposition* of the matrix A . Consider, for example, the simple 2×2 matrix

$$\Sigma = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix},$$

which is positive definite. Simple calculations show that

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Therefore, 3 and 1 are eigenvalues of Σ with normalized eigenvectors $(1/\sqrt{2}, 1/\sqrt{2})'$ and $(1/\sqrt{2}, -1/\sqrt{2})'$, respectively. It is easy to verify that the spectral decomposition holds—that is,

$$\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}.$$

For a symmetric matrix A , there exists a lower triangular matrix L with diagonal elements being 1 and a diagonal matrix G such that $A = LGL'$; see Chapter 1 of Strang (1980). If A is positive definite, then the diagonal elements of G are positive. In this case, we have

$$A = L\sqrt{G}\sqrt{G}L' = (L\sqrt{G})(L\sqrt{G})',$$

where $L\sqrt{G}$ is again a lower triangular matrix and the square root is taken element by element. Such a decomposition is called the *Cholesky decomposition* of A . This decomposition shows that a positive-definite matrix A can be diagonalized as

$$L^{-1}A(L')^{-1} = L^{-1}A(L^{-1})' = G.$$

Since L is a lower triangular matrix with unit diagonal elements, L^{-1} is also lower triangular matrix with unit diagonal elements. Consider again the prior 2×2 matrix Σ . It is easy to verify that

$$L = \begin{bmatrix} 1.0 & 0.0 \\ 0.5 & 1.0 \end{bmatrix} \quad \text{and} \quad G = \begin{bmatrix} 2.0 & 0.0 \\ 0.0 & 1.5 \end{bmatrix}$$

satisfy $\Sigma = LGL'$. In addition,

$$L^{-1} = \begin{bmatrix} 1.0 & 0.0 \\ -0.5 & 1.0 \end{bmatrix} \quad \text{and} \quad L^{-1}\Sigma(L^{-1})' = G.$$

Vectorization and Kronecker Product

Writing an $m \times n$ matrix A in its columns as $A = [a_1, \dots, a_n]$, we define the stacking operation as $\text{vec}(A) = (a_1', a_2', \dots, a_n')'$, which is an $mn \times 1$ vector. For two matrices $A_{m \times n}$ and $C_{p \times q}$, the Kronecker product between A and C is

$$A \otimes C = \begin{bmatrix} a_{11}C & a_{12}C & \cdots & a_{1n}C \\ a_{21}C & a_{22}C & \cdots & a_{2n}C \\ \vdots & \vdots & & \vdots \\ a_{m1}C & a_{m2}C & \cdots & a_{mn}C \end{bmatrix}_{mp \times nq}.$$

For example, assume that

$$A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}, \quad C = \begin{bmatrix} 4 & -1 & 3 \\ -2 & 5 & 2 \end{bmatrix}.$$

Then $\text{vec}(A) = (2, -1, 1, 3)'$, $\text{vec}(C) = (4, -2, -1, 5, 3, 2)'$, and

$$A \otimes C = \begin{bmatrix} 8 & -2 & 6 & 4 & -1 & 3 \\ -4 & 10 & 4 & -2 & 5 & 2 \\ -4 & 1 & -3 & 12 & -3 & 9 \\ 2 & -5 & -2 & -6 & 15 & 6 \end{bmatrix}.$$

Assuming that the dimensions are appropriate, we have the following useful properties for the two operators:

1. $A \otimes C \neq C \otimes A$ in general.
2. $(A \otimes C)' = A' \otimes C'$.
3. $A \otimes (C + D) = A \otimes C + A \otimes D$.
4. $(A \otimes C)(F \otimes G) = (AF) \otimes (CG)$.
5. If A and C are invertible, then $(A \otimes C)^{-1} = A^{-1} \otimes C^{-1}$.
6. For square matrices A and C , $\text{tr}(A \otimes C) = \text{tr}(A)\text{tr}(C)$.

7. $\text{vec}(\mathbf{A} + \mathbf{C}) = \text{vec}(\mathbf{A}) + \text{vec}(\mathbf{C})$.
8. $\text{vec}(\mathbf{ABC}) = (\mathbf{C}' \otimes \mathbf{A}) \text{vec}(\mathbf{B})$.
9. $\text{tr}(\mathbf{AC}) = \text{vec}(\mathbf{C}')' \text{vec}(\mathbf{A}) = \text{vec}(\mathbf{A}')' \text{vec}(\mathbf{C})$.
10. $\text{tr}(\mathbf{ABC}) = \text{vec}(\mathbf{A}')' (\mathbf{C}' \otimes \mathbf{I}) \text{vec}(\mathbf{B}) = \text{vec}(\mathbf{A}')' (\mathbf{I} \otimes \mathbf{B}) \text{vec}(\mathbf{C})$
 $= \text{vec}(\mathbf{B}')' (\mathbf{A}' \otimes \mathbf{I}) \text{vec}(\mathbf{C}) = \text{vec}(\mathbf{B}')' (\mathbf{I} \otimes \mathbf{C}) \text{vec}(\mathbf{A})$
 $= \text{vec}(\mathbf{C}')' (\mathbf{B}' \otimes \mathbf{I}) \text{vec}(\mathbf{A}) = \text{vec}(\mathbf{C}')' (\mathbf{I} \otimes \mathbf{A}) \text{vec}(\mathbf{B})$.

In multivariate statistical analysis, we often deal with symmetric matrices. It is therefore convenient to generalize the stacking operation to the *half-stacking* operation, which consists of elements on or below the main diagonal. Specifically, for a symmetric square matrix $\mathbf{A} = [a_{ij}]_{k \times k}$, define

$$\text{vech}(\mathbf{A}) = (\mathbf{a}'_1, \mathbf{a}'_{2*}, \dots, \mathbf{a}'_{k*})',$$

where $\mathbf{a}_1$ is the first column of $\mathbf{A}$, and $\mathbf{a}_{i*} = (a_{ii}, a_{i+1,i}, \dots, a_{ki})'$ is a $(k - i + 1)$ -dimensional vector. The dimension of $\text{vech}(\mathbf{A})$ is $k(k + 1)/2$. For example, suppose that $k = 3$. Then we have $\text{vech}(\mathbf{A}) = (a_{11}, a_{21}, a_{31}, a_{22}, a_{32}, a_{33})'$, which is a six-dimensional vector.

APPENDIX B: MULTIVARIATE NORMAL DISTRIBUTIONS

A k -dimensional random vector $\mathbf{x} = (x_1, \dots, x_k)'$ follows a multivariate normal distribution with mean $\boldsymbol{\mu} = (\mu_1, \dots, \mu_k)'$ and positive-definite covariance matrix $\boldsymbol{\Sigma} = [\sigma_{ij}]$ if its probability density function (pdf) is

$$f(\mathbf{x} | \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{k/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right]. \quad (8.47)$$

We use the notation $\mathbf{x} \sim N_k(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ to denote that $\mathbf{x}$ follows such a distribution. This normal distribution plays an important role in multivariate statistical analysis and it has several nice properties. Here we consider only those properties that are relevant to our study. Interested readers are referred to Johnson and Wichern (1998) for details.

To gain insight into multivariate normal distributions, consider the bivariate case (i.e., $k = 2$). In this case, we have

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{bmatrix}, \quad \boldsymbol{\Sigma}^{-1} = \frac{1}{\sigma_{11}\sigma_{22} - \sigma_{12}^2} \begin{bmatrix} \sigma_{22} & -\sigma_{12} \\ -\sigma_{12} & \sigma_{11} \end{bmatrix}.$$

Using the correlation coefficient $\rho = \sigma_{12}/(\sigma_1\sigma_2)$, where $\sigma_i = \sqrt{\sigma_{ii}}$ is the standard deviation of x_i , we have $\sigma_{12} = \rho\sqrt{\sigma_{11}\sigma_{22}}$ and $|\boldsymbol{\Sigma}| = \sigma_{11}\sigma_{22}(1 - \rho^2)$. The pdf of $\mathbf{x}$ then becomes

$$f(x_1, x_2 | \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1 - \rho^2}} \exp \left\{ -\frac{1}{2(1 - \rho^2)} [Q(\mathbf{x}, \boldsymbol{\mu}, \boldsymbol{\Sigma})] \right\},$$

where

$$Q(\mathbf{x}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \left(\frac{x_1 - \mu_1}{\sigma_1} \right)^2 + \left(\frac{x_2 - \mu_2}{\sigma_2} \right)^2 - 2\rho \left(\frac{x_1 - \mu_1}{\sigma_1} \right) \left(\frac{x_2 - \mu_2}{\sigma_2} \right).$$

Chapter 4 of Johnson and Wichern (1998) contains some plots of this pdf function.

Let $\mathbf{c} = (c_1, \dots, c_k)'$ be a nonzero k -dimensional vector. Partition the random vector as $\mathbf{x} = (\mathbf{x}'_1, \mathbf{x}'_2)'$, where $\mathbf{x}_1 = (x_1, \dots, x_p)'$ and $\mathbf{x}_2 = (x_{p+1}, \dots, x_k)'$ with $1 \leq p < k$. Also partition $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ accordingly as

$$\begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix} \sim N \left(\begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{bmatrix} \right).$$

Some properties of $\mathbf{x}$ are as follows:

1. $\mathbf{c}'\mathbf{x} \sim N(\mathbf{c}'\boldsymbol{\mu}, \mathbf{c}'\boldsymbol{\Sigma}\mathbf{c})$. That is, any nonzero linear combination of $\mathbf{x}$ is univariate normal. The inverse of this property also holds. Specifically, if $\mathbf{c}'\mathbf{x}$ is univariate normal for any nonzero vector $\mathbf{c}$, then $\mathbf{x}$ is multivariate normal.
2. The marginal distribution of $\mathbf{x}_i$ is normal. In fact, $\mathbf{x}_i \sim N_{k_i}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_{ii})$ for $i = 1$ and 2 , where $k_1 = p$ and $k_2 = k - p$.
3. $\boldsymbol{\Sigma}_{12} = \mathbf{0}$ if and only if $\mathbf{x}_1$ and $\mathbf{x}_2$ are independent.
4. The random variable $y = (\mathbf{x} - \boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})$ follows a chi-squared distribution with m degrees of freedom.
5. The conditional distribution of $\mathbf{x}_1$ given $\mathbf{x}_2 = \mathbf{b}$ is also normally distributed as

$$(\mathbf{x}_1|\mathbf{x}_2 = \mathbf{b}) \sim N_p[\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{b} - \boldsymbol{\mu}_2), \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}].$$

The last property is useful in many scientific areas. For instance, it forms the basis for time series forecasting under the normality assumption and for recursive least-squares estimation.

APPENDIX C: SOME SCA COMMANDS

The following SCA commands are used in the analysis of Example 8.6:

```
input x1,x2. file 'm-gsln3-5301.txt' % Load data
--
r1=ln(x1) % Take log transformation
--
r2=ln(x2)
--
miden r1,r2. no ccm. arfits 1 to 8.
-- % Denote the model by v21.
mtsm v21. series r1,r2. model (i-p1*b-p2*b**2)series= @
```

```

c+(i-t1*b)noise.
--
mestim v21.    % Initial estimation
--
p1(2,1)=0      % Set zero constraints
--
cp1(2,1)=1
--
p2(2,1)=0
--
cp2(2,1)=1
--
p2(2,2)=0
--
cp2(2,2)=1
--
t1(2,1)=0
--
ct1(2,1)=1
-- % Refine estimation and store residuals
mestim v21. method exact. hold resi(res1,res2)
--
miden res1,res2.

```

EXERCISES

- 8.1. Consider the monthly log stock returns, in percentages and including dividends, of Merck & Company, Johnson & Johnson, General Electric, General Motors, Ford Motor Company, and value-weighted index from January 1960 to December 2008; see the file `m-mrk2vw.txt`.
 - (a) Compute the sample mean, covariance matrix, and correlation matrix of the data.
 - (b) Test the hypothesis $H_0 : \rho_1 = \cdots = \rho_6 = \mathbf{0}$, where ρ_i is the lag- i cross-correlation matrix of the data. Draw conclusions based on the 5% significance level.
 - (c) Is there any lead-lag relationship among the six return series?
- 8.2. The Federal Reserve Bank of St. Louis publishes selected interest rates and U.S. financial data on its website: <http://research.stlouisfed.org/fred2/>. Consider the monthly 1-year and 10-year Treasury constant maturity rates from April 1953 to October 2009 for 679 observations; see the file `m-gs1n10.txt`. The rates are in percentages.
 - (a) Let $c_t = r_t - r_{t-1}$ be the change series of the monthly interest rate r_t . Build a bivariate autoregressive model for the two change series. Discuss the implications of the model. Transform the model into a structural form.

- (b) Build a bivariate moving-average model for the two change series. Discuss the implications of the model and compare it with the bivariate AR model built earlier.
- 8.3. Again consider the monthly 1-year and 10-year Treasury constant maturity rates from April 1953 to October 2009. Consider the log series of the data and build a VARMA model for the series. Discuss the implications of the model obtained.
- 8.4. Again consider the monthly 1-year and 10-year Treasury constant maturity rates from April 1953 to October 2009. Are the two interest rate series threshold cointegrated? Use the interest spread $s_t = r_{10,t} - r_{1,t}$ as the threshold variable, where r_{it} is the i -year Treasury constant maturity rate. If they are threshold cointegrated, build a multivariate threshold model for the two series.
- 8.5. The bivariate AR(4) model $\mathbf{x}_t - \Phi_4 \mathbf{x}_{t-4} = \phi_0 + \mathbf{a}_t$ is a special seasonal model with periodicity 4, where $\{\mathbf{a}_t\}$ is a sequence of independent and identically distributed normal random vectors with mean zero and covariance matrix Σ . Such a seasonal model may be useful in studying quarterly earnings of a company. (a) Assume that $\mathbf{x}_t$ is weakly stationary. Derive the mean vector and covariance matrix of $\mathbf{x}_t$. (b) Derive the necessary and sufficient condition of weak stationarity for $\mathbf{x}_t$. (c) Show that $\Gamma_\ell = \Phi_4 \Gamma_{\ell-4}$ for $\ell > 0$, where Γ_ℓ is the lag- ℓ autocovariance matrix of $\mathbf{x}_t$.
- 8.6. The bivariate MA(4) model $\mathbf{x}_t = \mathbf{a}_t - \Theta_4 \mathbf{a}_{t-4}$ is another seasonal model with periodicity 4, where $\{\mathbf{a}_t\}$ is a sequence of independent and identically distributed normal random vectors with mean zero and covariance matrix Σ . Derive the covariance matrices Γ_ℓ of $\mathbf{x}_t$ for $\ell = 0, \dots, 5$.
- 8.7. Consider the monthly U.S. 1-year and 3-year Treasury constant maturity rates from April 1953 to March 2004. The data can be obtained from the Federal Reserve Bank of St. Louis or from the file `m-gs1n3-5304.txt` (1-year, 3-year, dates). See also Example 8.6, which uses a shorter data span. Here we use the interest rates directly without the log transformation and define $\mathbf{x}_t = (x_{1t}, x_{2t})'$, where x_{1t} is the 1-year maturity rate and x_{2t} is the 3-year maturity rate.
- (a) Identify a VAR model for the bivariate interest rate series. Write down the fitted model.
- (b) Compute the impulse response functions of the fitted VAR model. It suffices to use the first 6 lags.
- (c) Use the fitted VAR model to produce 1-step- to 12-step-ahead forecasts of the interest rates, assuming that the forecast origin is March 2004.
- (d) Are the two interest rate series cointegrated, when a restricted constant term is used? Use 5% significance level to perform the test.
- (e) If the series are cointegrated, build an ECM for the series. Write down the fitted model.

- (f) Use the fitted ECM to produce 1-step- to 12-step-ahead forecasts of the interest rates, assuming that the forecast origin is March 2004.
- (g) Compare the forecasts produced by the VAR model and the ECM.

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State-Space Models and Kalman Filter

The state-space model provides a flexible approach to time series analysis, especially for simplifying maximum-likelihood estimation and handling missing values. In this chapter, we discuss the relationship between the state-space model and the ARIMA model, the Kalman filter algorithm, various smoothing methods, and some applications. We begin with a simple model that shows the basic ideas of the state-space approach to time series analysis before introducing the general state-space model. For demonstrations, we use the model to analyze realized volatility series of asset returns, the time-varying coefficient market models, and the quarterly earnings per share of a company.

There are many books on statistical analysis using the state-space model. Durbin and Koopman (2001) provide a recent treatment of the approach, Kim and Nelson (1999) focus on economic applications and regime switching, and Anderson and Moore (1979) give a nice summary of theory and applications of the approach for engineering and optimal control. Many time series textbooks include the Kalman filter and state-space model. For example, Chan (2002), Shumway and Stoffer (2000), Hamilton (1994), and Harvey (1993) all have chapters on the topic. West and Harrison (1997) provide a Bayesian treatment with emphasis on forecasting, and Kitagawa and Gersch (1996) use a smoothing prior approach.

The derivation of Kalman filter and smoothing algorithms necessarily involves heavy notation. Therefore, Section 11.4 could be dry for readers who are interested mainly in the concept and applications of state-space models and can be skipped on the first read.

11.1 LOCAL TREND MODEL

Consider the univariate time series y_t satisfying

$$y_t = \mu_t + e_t, \quad e_t \sim N(0, \sigma_e^2), \quad (11.1)$$

$$\mu_{t+1} = \mu_t + \eta_t, \quad \eta_t \sim N(0, \sigma_\eta^2), \quad (11.2)$$

where $\{e_t\}$ and $\{\eta_t\}$ are two independent Gaussian white noise series and $t = 1, \dots, T$. The initial value μ_1 is either given or follows a known distribution, and it is independent of $\{e_t\}$ and $\{\eta_t\}$ for $t > 0$. Here μ_t is a pure *random walk* of Chapter 2 with initial value μ_1 , and y_t is an observed version of μ_t with added noise e_t . In the literature, μ_t is referred to as the *trend* of the series, which is not directly observable, and y_t is the observed data with observational noise e_t . The dynamic dependence of y_t is governed by that of μ_t because $\{e_t\}$ is not serially correlated.

The model in Eqs. (11.1) and (11.2) can readily be used to analyze realized volatility of an asset price; see Example 11.1. Here μ_t represents the underlying log volatility of the asset price and y_t is the logarithm of realized volatility. The true log volatility is not directly observed but evolves over time according to a random-walk model. On the other hand, y_t is constructed from high-frequency transactions data and subjected to the influence of market microstructure noises. The standard deviation of e_t denotes the scale used to measure the impact of market microstructure noises.

The model in Eqs. (11.1) and (11.2) is a special *linear Gaussian state-space model*. The variable μ_t is called the *state* of the system at time t and is not directly observed. Equation (11.1) provides the link between the data y_t and the state μ_t and is called the *observation equation* with *measurement error* e_t . Equation (11.2) governs the time evolution of the state variable and is the *state equation* (or *state transition equation*) with innovation η_t . The model is also called a *local-level model* in Durbin and Koopman (2001, Chapter 2), which is a simple case of the *structural time series model* of Harvey (1993).

Relationship to ARIMA Model

If there is no measurement error in Eq. (11.1), that is, $\sigma_e = 0$, then $y_t = \mu_t$, which is an ARIMA(0,1,0) model. If $\sigma_e > 0$, that is, there exist measurement errors, then y_t is an ARIMA(0,1,1) model satisfying

$$(1 - B)y_t = (1 - \theta B)a_t, \quad (11.3)$$

where $\{a_t\}$ is a Gaussian white noise with mean zero and variance σ_a^2 . The values of θ and σ_a are determined by σ_e and σ_η . This result can be derived as follows.

From Eq. (11.2), we have

$$(1 - B)\mu_{t+1} = \eta_t \quad \text{or} \quad \mu_{t+1} = \frac{1}{1 - B}\eta_t.$$

Using this result, Eq. (11.1) can be written as

$$y_t = \frac{1}{1-B} \eta_{t-1} + e_t.$$

Multiplying by $(1-B)$, we have

$$(1-B)y_t = \eta_{t-1} + e_t - e_{t-1}.$$

Let $(1-B)y_t = w_t$. We have $w_t = \eta_{t-1} + e_t - e_{t-1}$. Under the model assumptions, it is easy to see that (a) w_t is Gaussian, (b) $\text{Var}(w_t) = 2\sigma_e^2 + \sigma_\eta^2$, (c) $\text{Cov}(w_t, w_{t-1}) = -\sigma_e^2$, and (d) $\text{Cov}(w_t, w_{t-j}) = 0$ for $j > 1$. Consequently, w_t follows an MA(1) model and can be written as $w_t = (1-\theta B)a_t$. By equating the variance and lag-1 autocovariance of $w_t = (1-\theta B)a_t = \eta_{t-1} + e_t - e_{t-1}$, we have

$$(1+\theta^2)\sigma_a^2 = 2\sigma_e^2 + \sigma_\eta^2, \quad (11.4)$$

$$\theta\sigma_a^2 = \sigma_e^2. \quad (11.5)$$

For given σ_e^2 and σ_η^2 , one considers the ratio of the prior two equations to form a quadratic function of θ . This quadratic form has two solutions so one should select the one that satisfies $|\theta| < 1$. The value of σ_a^2 can then be easily obtained. Thus, the state-space model in Eqs. (11.1) and (11.2) is also an ARIMA(0,1,1) model, which is the simple exponential smoothing model of Chapter 2.

On the other hand, for an ARIMA(0,1,1) model with positive θ , one can use the prior two identities to solve for σ_e^2 and σ_η^2 , and obtain a local trend model. If θ is negative, then the model can still be put in a state-space form without the observational error, that is, $\sigma_e = 0$. In fact, as will be seen later, an ARIMA model can be transformed into state-space models in many ways. Thus, the linear state-space model is closely related to the ARIMA model.

In practice, what one observes is the y_t series. Thus, based on the data alone, the decision of using ARIMA models or linear state-space models is not critical. Both model representations have pros and cons. The objective of data analysis, substantive issues, and experience all play a role in choosing a statistical model.

Example 11.1. To illustrate the ideas of the state-space model and Kalman filter, we consider the intradaily realized volatility of Alcoa stock from January 2, 2003, to May 7, 2004, for 340 observations. The daily realized volatility used is the sum of squares of intraday 10-minute log returns measured in percentage. No overnight returns or the first 10-minute intraday returns are used. See Chapter 3 for more information about realized volatility. The series used in the demonstration is the logarithm of the daily realized volatility.

Figure 11.1 shows the time plot of the logarithms of the realized volatility of Alcoa stock from January 2, 2003, to May 7, 2004. The transactions data are obtained from the TAQ database of the NYSE. If ARIMA models are entertained,

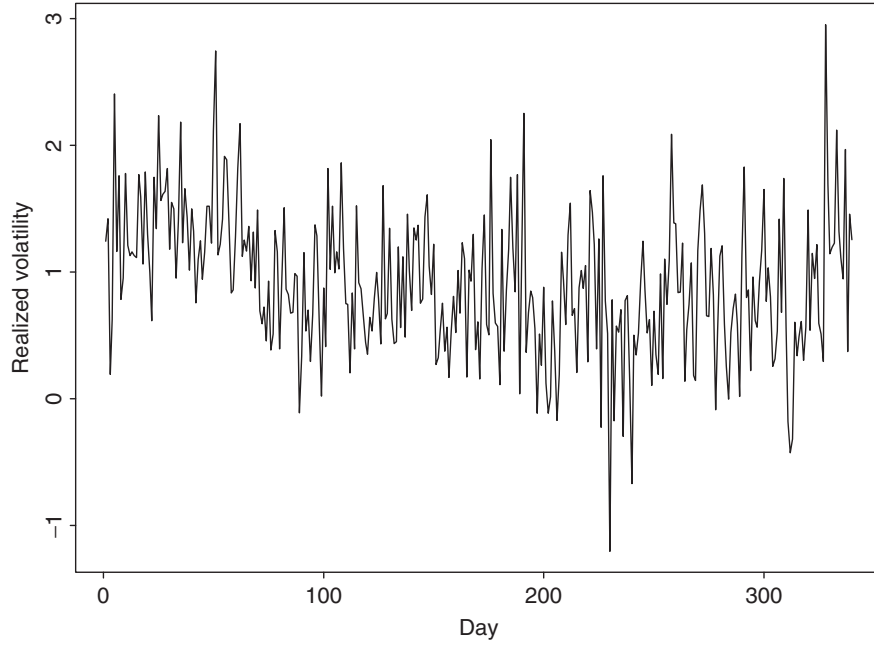


Figure 11.1 Time plot of logarithms of intradaily realized volatility of Alcoa stock from January 2, 2003, to May 7, 2004. Realized volatility is computed from intraday 10-minute log returns measured in percentage.

we obtain an ARIMA(0,1,1) model

$$(1 - B)y_t = (1 - 0.858B)a_t, \quad \hat{\sigma}_a = 0.5184, \quad (11.6)$$

where y_t is the log realized volatility, and the standard error of $\hat{\theta}$ is 0.028. The residuals show $Q(12) = 12.4$ with a p value of 0.33, indicating that there is no significant serial correlation in the residuals. Similarly, the squared residuals give $Q(12) = 8.2$ with a p value of 0.77, suggesting no ARCH effects in the series.

Since $\hat{\theta}$ is positive, we can transform the ARIMA(0,1,1) model into a local trend model in Eqs. (11.1) and (11.2). The maximum-likelihood estimates (MLE) of the two parameters are $\hat{\sigma}_\eta = 0.0735$ and $\hat{\sigma}_e = 0.4803$. The measurement errors have a larger variance than the state innovations, confirming that intraday high-frequency returns are subject to measurement errors. Details of estimation will be discussed in Section 11.1.7. Here we treat the two estimates as given and use the model to demonstrate application of the Kalman filter. Note that using the model in Eq. (11.6) and the relation in Eqs. (11.4) and (11.5), we obtain $\sigma_e = 0.480$ and $\sigma_\eta = 0.0736$. These values are close to the MLE shown above.

11.1.1 Statistical Inference

Return to the state-space model in Eqs. (11.1) and (11.2). The aim of the analysis is to infer properties of the state μ_t from the data $\{y_t|t = 1, \dots, T\}$ and the model. Three types of inference are commonly discussed in the literature. They are *filtering*, *prediction*, and *smoothing*. Let $F_t = \{y_1, \dots, y_t\}$ be the information available at time t (inclusive) and assume that the model is known, including all parameters. The three types of inference can briefly be described as follows:

- *Filtering*. Filtering means to recover the state variable μ_t given F_t , that is, to remove the measurement errors from the data.
- *Prediction*. Prediction means to forecast μ_{t+h} or y_{t+h} for $h > 0$ given F_t , where t is the forecast origin.
- *Smoothing*. Smoothing is to estimate μ_t given F_T , where $T > t$.

A simple analogy of the three types of inference is reading a handwritten note. Filtering is figuring out the word you are reading based on knowledge accumulated from the beginning of the note, predicting is to guess the next word, and smoothing is deciphering a particular word once you have read through the note.

To describe the inference more precisely, we introduce some notation. Let $\mu_{t|j} = E(\mu_t|F_j)$ and $\Sigma_{t|j} = \text{Var}(\mu_t|F_j)$ be, respectively, the conditional mean and variance of μ_t given F_j . Similarly, $y_{t|j}$ denotes the conditional mean of y_t given F_j . Furthermore, let $v_t = y_t - y_{t|t-1}$ and $V_t = \text{Var}(v_t|F_{t-1})$ be the 1-step-ahead forecast error and its variance of y_t given F_{t-1} . Note that the forecast error v_t is independent of F_{t-1} so that the conditional variance is the same as the unconditional variance; that is, $\text{Var}(v_t|F_{t-1}) = \text{Var}(v_t)$. From Eq. (11.1),

$$y_{t|t-1} = E(y_t|F_{t-1}) = E(\mu_t + e_t|F_{t-1}) = E(\mu_t|F_{t-1}) = \mu_{t|t-1}.$$

Consequently,

$$v_t = y_t - y_{t|t-1} = y_t - \mu_{t|t-1} \quad (11.7)$$

and

$$\begin{aligned} V_t &= \text{Var}(y_t - \mu_{t|t-1}|F_{t-1}) = \text{Var}(\mu_t + e_t - \mu_{t|t-1}|F_{t-1}) \\ &= \text{Var}(\mu_t - \mu_{t|t-1}|F_{t-1}) + \text{Var}(e_t|F_{t-1}) = \Sigma_{t|t-1} + \sigma_e^2. \end{aligned} \quad (11.8)$$

It is also easy to see that

$$\begin{aligned} E(v_t) &= E[E(v_t|F_{t-1})] = E[E(y_t - y_{t|t-1}|F_{t-1})] = E[y_{t|t-1} - y_{t|t-1}] = 0, \\ \text{Cov}(v_t, y_j) &= E(v_t y_j) = E[E(v_t y_j|F_{t-1})] = E[y_j E(v_t|F_{t-1})] = 0, \quad j < t. \end{aligned}$$

Thus, as expected, the 1-step-ahead forecast error is uncorrelated (hence, independent) with y_j for $j < t$. Furthermore, for the linear model in Eqs. (11.1) and (11.2), $\mu_{t|t} = E(\mu_t|F_t) = E(\mu_t|F_{t-1}, v_t)$ and $\Sigma_{t|t} = \text{Var}(\mu_t|F_t) = \text{Var}(\mu_t|F_{t-1}, v_t)$. In other words, the information set F_t can be written as $F_t = \{F_{t-1}, y_t\} = \{F_{t-1}, v_t\}$.

The following properties of multivariate normal distribution are useful in studying the Kalman filter under normality. They can be shown via the multivariate linear regression method or factorization of the joint density. See, also, Appendix B of Chapter 8. For random vectors $\mathbf{w}$ and $\mathbf{m}$, denote the mean vectors and covariance matrix as $E(\mathbf{w}) = \boldsymbol{\mu}_w$, $E(\mathbf{m}) = \boldsymbol{\mu}_m$, and $\text{Cov}(\mathbf{m}, \mathbf{w}) = \boldsymbol{\Sigma}_{mw}$, respectively.

Theorem 11.1. Suppose that $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{z}$ are three random vectors such that their joint distribution is multivariate normal. In addition, assume that the diagonal block covariance matrix $\boldsymbol{\Sigma}_{ww}$ is nonsingular for $w = x, y, z$, and $\boldsymbol{\Sigma}_{yz} = \mathbf{0}$. Then,

1. $E(\mathbf{x}|\mathbf{y}) = \boldsymbol{\mu}_x + \boldsymbol{\Sigma}_{xy}\boldsymbol{\Sigma}_{yy}^{-1}(\mathbf{y} - \boldsymbol{\mu}_y)$.
2. $\text{Var}(\mathbf{x}|\mathbf{y}) = \boldsymbol{\Sigma}_{xx} - \boldsymbol{\Sigma}_{xx}\boldsymbol{\Sigma}_{yy}^{-1}\boldsymbol{\Sigma}_{yx}$.
3. $E(\mathbf{x}|\mathbf{y}, \mathbf{z}) = E(\mathbf{x}|\mathbf{y}) + \boldsymbol{\Sigma}_{xz}\boldsymbol{\Sigma}_{zz}^{-1}(\mathbf{z} - \boldsymbol{\mu}_z)$.
4. $\text{Var}(\mathbf{x}|\mathbf{y}, \mathbf{z}) = \text{Var}(\mathbf{x}|\mathbf{y}) - \boldsymbol{\Sigma}_{xz}\boldsymbol{\Sigma}_{zz}^{-1}\boldsymbol{\Sigma}_{zx}$.

11.1.2 Kalman Filter

The goal of the *Kalman filter* is to update knowledge of the state variable recursively when a new data point becomes available. That is, knowing the conditional distribution of μ_t given F_{t-1} and the new data y_t , we would like to obtain the conditional distribution of μ_t given F_t , where, as before, $F_j = \{y_1, \dots, y_j\}$. Since $F_t = \{F_{t-1}, v_t\}$, giving y_t and F_{t-1} is equivalent to giving v_t and F_{t-1} . Consequently, to derive the Kalman filter, it suffices to consider the joint conditional distribution of $(\mu_t, v_t)'$ given F_{t-1} before applying Theorem 11.1.

The conditional distribution of v_t given F_{t-1} is normal with mean zero and variance given in Eq. (11.8), and that of μ_t given F_{t-1} is also normal with mean $\mu_{t|t-1}$ and variance $\Sigma_{t|t-1}$. Furthermore, the joint distribution of $(\mu_t, v_t)'$ given F_{t-1} is also normal. Thus, what remains to be solved is the conditional covariance between μ_t and v_t given F_{t-1} . From the definition,

$$\begin{aligned}
 \text{Cov}(\mu_t, v_t|F_{t-1}) &= E(\mu_t v_t|F_{t-1}) = E[\mu_t(y_t - \mu_{t|t-1})|F_{t-1}] \quad [\text{by Eq. (11.7)}] \\
 &= E[\mu_t(\mu_t + e_t - \mu_{t|t-1})|F_{t-1}] \\
 &= E[\mu_t(\mu_t - \mu_{t|t-1})|F_{t-1}] + E(\mu_t e_t|F_{t-1}) \\
 &= E[(\mu_t - \mu_{t|t-1})^2|F_{t-1}] = \text{Var}(\mu_t|F_{t-1}) = \Sigma_{t|t-1}, \quad (11.9)
 \end{aligned}$$

where we have used the fact that $E[\mu_{t|t-1}(\mu_t - \mu_{t|t-1})|F_{t-1}] = 0$. Putting the results together, we have

$$\begin{bmatrix} \mu_t \\ v_t \end{bmatrix}_{F_{t-1}} \sim N \left(\begin{bmatrix} \mu_{t|t-1} \\ 0 \end{bmatrix}, \begin{bmatrix} \Sigma_{t|t-1} & \Sigma_{t|t-1} \\ \Sigma_{t|t-1} & V_t \end{bmatrix} \right).$$

By Theorem 11.1, the conditional distribution of μ_t given F_t is normal with mean and variance

$$\mu_{t|t} = \mu_{t|t-1} + \frac{\Sigma_{t|t-1} v_t}{V_t} = \mu_{t|t-1} + K_t v_t, \quad (11.10)$$

$$\Sigma_{t|t} = \Sigma_{t|t-1} - \frac{\Sigma_{t|t-1}^2}{V_t} = \Sigma_{t|t-1} (1 - K_t), \quad (11.11)$$

where $K_t = \Sigma_{t|t-1} / V_t$ is commonly referred to as the *Kalman gain*, which is the regression coefficient of μ_t on v_t . From Eq. (11.10), Kalman gain is the factor that governs the contribution of the new shock v_t to the state variable μ_t .

Next, one can make use of the knowledge of μ_t given F_t to predict μ_{t+1} via Eq. (11.2). Specifically, we have

$$\mu_{t+1|t} = E(\mu_{t+1} | F_t) = E(\mu_t | F_t) = \mu_{t|t}, \quad (11.12)$$

$$\Sigma_{t+1|t} = \text{Var}(\mu_{t+1} | F_t) = \text{Var}(\mu_t | F_t) + \text{Var}(\eta_t) = \Sigma_{t|t} + \sigma_\eta^2. \quad (11.13)$$

Once the new data y_{t+1} is observed, one can repeat the above procedure to update knowledge of μ_{t+1} . This is the famous *Kalman filter* algorithm proposed by Kalman (1960).

In summary, putting Eqs. (11.7) and (11.13) together and conditioning on the initial assumption that μ_1 is distributed as $N(\mu_{1|0}, \Sigma_{1|0})$, the Kalman filter for the local trend model is as follows:

$$\begin{aligned} v_t &= y_t - \mu_{t|t-1}, \\ V_t &= \Sigma_{t|t-1} + \sigma_\epsilon^2, \\ K_t &= \Sigma_{t|t-1} / V_t, \\ \mu_{t+1|t} &= \mu_{t|t-1} + K_t v_t, \\ \Sigma_{t+1|t} &= \Sigma_{t|t-1} (1 - K_t) + \sigma_\eta^2, \quad t = 1, \dots, T. \end{aligned} \quad (11.14)$$

There are many ways to derive the Kalman filter. We use Theorem 11.1, which describes some properties of multivariate normal distribution, for its simplicity. In practice, the choice of initial values $\Sigma_{1|0}$ and $\mu_{1|0}$ requires some attention and we shall discuss it later in Section 11.1.6. For the local trend model in Eqs. (11.1) and (11.2), the two parameters σ_ϵ and σ_η can be estimated via the maximum-likelihood method. Again, the Kalman filter is useful in evaluating the likelihood function of the data in estimation. We shall discuss estimation in Section 11.1.7.

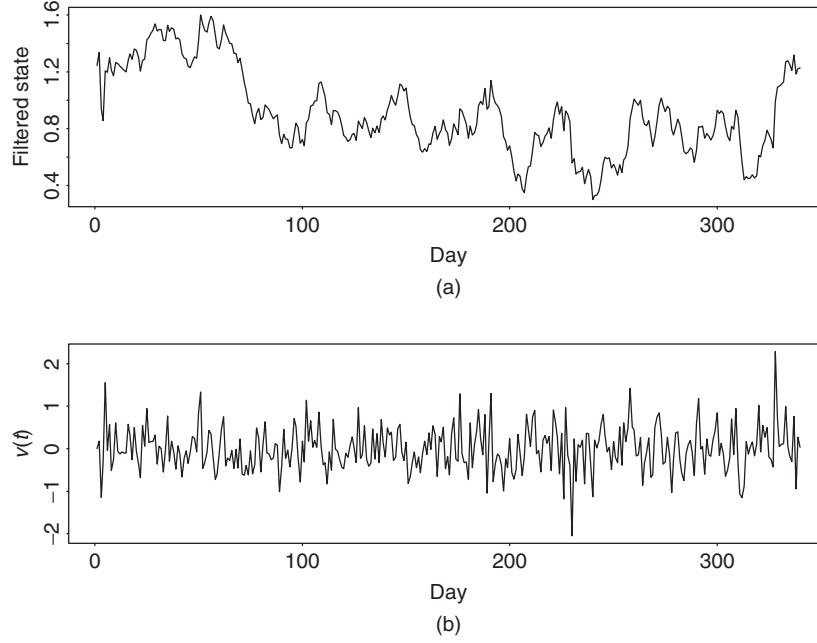


Figure 11.2 Time plots of output of Kalman filter applied to daily realized log volatility of Alcoa stock based on local trend state-space model: (a) filtered state $\mu_{t|t}$ and (b) 1-step-ahead forecast error v_t .

Example 11.1 (Continued). To illustrate application of the Kalman filter, we use the fitted state-space model for daily realized volatility of Alcoa stock returns and apply the Kalman filter algorithm to the data with $\Sigma_{1|0} = \infty$ and $\mu_{1|0} = 0$. The choice of these initial values will be discussed in Section 11.1.6. Figure 11.2(a) shows the time plot of the filtered state variable $\mu_{t|t}$, and Figure 11.2(b) is the time plot of the 1-step-ahead forecast error v_t . Compared with Figure 11.1, the filtered states are smoother. The forecast errors appear to be stable and center around zero. These forecast errors are out-of-sample 1-step-ahead prediction errors.

11.1.3 Properties of Forecast Error

The 1-step-ahead forecast errors $\{v_t\}$ are useful in many applications, hence it pays to study carefully their properties. Given the initial values $\Sigma_{1|0}$ and $\mu_{1|0}$, which are independent of y_t , the Kalman filter enables us to compute v_t recursively as a linear function of $\{y_1, \dots, y_t\}$. Specifically, by repeated substitutions,

$$v_1 = y_1 - \mu_{1|0},$$

$$v_2 = y_2 - \mu_{2|1} = y_2 - \mu_{1|0} - K_1(y_1 - \mu_{1|0}),$$

$$v_3 = y_3 - \mu_{3|2} = y_3 - \mu_{1|0} - K_2(y_2 - \mu_{1|0}) - K_1(1 - K_2)(y_1 - \mu_{1|0}),$$

and so on. This transformation can be written in matrix form as

$$\mathbf{v} = \mathbf{K}(\mathbf{y} - \mu_{1|0}\mathbf{1}_T), \quad (11.15)$$

where $\mathbf{v} = (v_1, \dots, v_T)'$, $\mathbf{y} = (y_1, \dots, y_T)'$, $\mathbf{1}_T$ is the T -dimensional vector of ones, and $\mathbf{K}$ is a lower triangular matrix defined as

$$\mathbf{K} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ k_{21} & 1 & 0 & \cdots & 0 \\ k_{31} & k_{32} & 1 & & 0 \\ \vdots & \vdots & & \ddots & \vdots \\ k_{T1} & k_{T2} & k_{T3} & \cdots & 1 \end{bmatrix},$$

where $k_{i,i-1} = -K_{i-1}$ and $k_{ij} = -(1 - K_{i-1})(1 - K_{i-2}) \cdots (1 - K_{j+1})K_j$ for $i = 2, \dots, T$ and $j = 1, \dots, i-2$. It should be noted that, from the definition, the Kalman gain K_i does not depend on $\mu_{1|0}$ or the data $\{y_1, \dots, y_i\}$; it depends on $\Sigma_{1|0}$ and σ_e^2 and σ_η^2 .

The transformation in Eq. (11.5) has several important implications. First, $\{v_t\}$ are mutually independent under the normality assumption. To show this, consider the joint probability density function of the data

$$p(y_1, \dots, y_T) = p(y_1) \prod_{j=2}^T p(y_j | F_{j-1}).$$

Equation (11.15) indicates that the transformation from y_t to v_t has a unit Jacobian so that $p(\mathbf{v}) = p(\mathbf{y})$. Furthermore, since $\mu_{1|0}$ is given, $p(v_1) = p(y_1)$. Consequently, the joint probability density function of $\mathbf{v}$ is

$$p(\mathbf{v}) = p(\mathbf{y}) = p(y_1) \prod_{j=2}^T p(y_j | F_{j-1}) = p(v_1) \prod_j^T p(v_j) = \prod_{j=1}^T p(v_j).$$

This shows that $\{v_t\}$ are mutually independent.

Second, the Kalman filter provides a Cholesky decomposition of the covariance matrix of $\mathbf{y}$. To see this, let $\mathbf{\Omega} = \text{Cov}(\mathbf{y})$. Equation (11.15) shows that $\text{Cov}(\mathbf{v}) = \mathbf{K}\mathbf{\Omega}\mathbf{K}'$. On the other hand, $\{v_t\}$ are mutually independent with $\text{Var}(v_t) = V_t$. Therefore, $\mathbf{K}\mathbf{\Omega}\mathbf{K}' = \text{diag}\{V_1, \dots, V_T\}$, which is precisely a Cholesky decomposition of $\mathbf{\Omega}$. The elements k_{ij} of the matrix $\mathbf{K}$ thus have some nice interpretations; see Chapter 10.

State Error Recursion

Turn to the estimation error of the state variable μ_t . Define

$$x_t = \mu_t - \mu_{t|t-1}$$

as the forecast error of the state variable μ_t given data F_{t-1} . From Section 11.1.1, $\text{Var}(x_t|F_{t-1}) = \Sigma_{t|t-1}$. From the Kalman filter in Eq. (11.14),

$$v_t = y_t - \mu_{t|t-1} = \mu_t + e_t - \mu_{t|t-1} = x_t + e_t,$$

and

$$\begin{aligned} x_{t+1} &= \mu_{t+1} - \mu_{t+1|t} = \mu_t + \eta_t - (\mu_{t|t-1} + K_t v_t) \\ &= x_t + \eta_t - K_t v_t = x_t + \eta_t - K_t(x_t + e_t) = L_t x_t + \eta_t - K_t e_t, \end{aligned}$$

where $L_t = 1 - K_t = 1 - \Sigma_{t|t-1}/V_t = (V_t - \Sigma_{t|t-1})/V_t = \sigma_e^2/V_t$. Consequently, for the state errors, we have

$$v_t = x_t + e_t, \quad x_{t+1} = L_t x_t + \eta_t - K_t e_t, \quad t = 1, \dots, T, \quad (11.16)$$

where $x_1 = \mu_1 - \mu_{1|0}$. Equation (11.16) is in the form of a time-varying state-space model with x_t being the state variable and v_t the observation.

11.1.4 State Smoothing

Next we consider the estimation of the state variables $\{\mu_1, \dots, \mu_T\}$ given the data F_T and the model. That is, given the state-space model in Eqs. (11.1) and (11.2), we wish to obtain the conditional distribution $\mu_t|F_T$ for all t . To this end, we first recall some facts available about the model:

- All distributions involved are normal so that we can write the conditional distribution of μ_t given F_T as $N(\mu_{t|T}, \Sigma_{t|T})$, where $t \leq T$. We refer to $\mu_{t|T}$ as the *smoothed state* at time t and $\Sigma_{t|T}$ as the *smoothed state variance*.
- Based on the properties of $\{v_t\}$ shown in Section 11.1.3, $\{v_1, \dots, v_T\}$ are mutually independent and are linear functions of $\{y_1, \dots, y_T\}$.
- If $y_1, \dots, y_T$ are fixed, then F_{t-1} and $\{v_t, \dots, v_T\}$ are fixed, and vice versa.
- $\{v_t, \dots, v_T\}$ are independent of F_{t-1} with mean zero and variance $\text{Var}(v_j) = V_j$ for $j \geq t$.

Applying Theorem 11.1(3) to the conditional joint distribution of $(\mu_t, v_t, \dots, v_T)$ given F_{t-1} , we have

$$\begin{aligned} \mu_{t|T} &= E(\mu_t|F_T) = E(\mu_t|F_{t-1}, v_t, \dots, v_T) \\ &= E(\mu_t|F_{t-1}) + \text{Cov}[\mu_t, (v_t, \dots, v_T)'] \text{Cov}[(v_t, \dots, v_T)']^{-1} (v_t, \dots, v_T)' \\ &= \mu_{t|t-1} + \begin{bmatrix} \text{Cov}(\mu_t, v_t) \\ \text{Cov}(\mu_t, v_{t+1}) \\ \vdots \\ \text{Cov}(\mu_t, v_T) \end{bmatrix}' \begin{bmatrix} V_t & 0 & \cdots & 0 \\ 0 & V_{t+1} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & V_T \end{bmatrix}^{-1} \begin{bmatrix} v_t \\ v_{t+1} \\ \vdots \\ v_T \end{bmatrix} \end{aligned}$$

$$= \mu_{t|t-1} + \sum_{j=t}^T \text{Cov}(\mu_t, v_j) V_j^{-1} v_j. \quad (11.17)$$

From the definition and independence of $\{v_t\}$, $\text{Cov}(\mu_t, v_j) = \text{Cov}(x_t, v_j)$ for $j = t, \dots, T$, and

$$\text{Cov}(x_t, v_t) = E[x_t(x_t + e_t)] = \text{Var}(x_t) = \Sigma_{t|t-1},$$

$$\text{Cov}(x_t, v_{t+1}) = E[x_t(x_{t+1} + e_{t+1})] = E[x_t(L_t x_t + \eta_t - K_t e_t)] = \Sigma_{t|t-1} L_t.$$

Similarly, we have

$$\text{Cov}(x_t, v_{t+2}) = E[x_t(x_{t+2} + e_{t+2})] = \dots = \Sigma_{t|t-1} L_t L_{t+1},$$

$$\vdots$$

$$\text{Cov}(x_t, v_T) = E[x_t(x_T + e_T)] = \dots = \Sigma_{t|t-1} \prod_{j=t}^{T-1} L_j.$$

Consequently, Eq. (11.17) becomes

$$\begin{aligned} \mu_{t|T} &= \mu_{t|t-1} + \Sigma_{t|t-1} \frac{v_t}{V_t} + \Sigma_{t|t-1} L_t \frac{v_{t+1}}{V_{t+1}} + \Sigma_{t|t-1} L_t L_{t+1} \frac{v_{t+2}}{V_{t+2}} + \dots \\ &\equiv \mu_{t|t-1} + \Sigma_{t|t-1} q_{t-1}, \end{aligned}$$

where

$$q_{t-1} = \frac{v_t}{V_t} + L_t \frac{v_{t+1}}{V_{t+1}} + L_t L_{t+1} \frac{v_{t+2}}{V_{t+2}} + \dots + \left(\prod_{j=t}^{T-1} L_j \right) \frac{v_T}{V_T} \quad (11.18)$$

is a weighted linear combination of the innovations $\{v_t, \dots, v_T\}$. This weighted sum satisfies

$$\begin{aligned} q_{t-1} &= \frac{v_t}{V_t} + L_t \left[\frac{v_{t+1}}{V_{t+1}} + L_{t+1} \frac{v_{t+2}}{V_{t+2}} + \dots + \left(\prod_{j=t+1}^{T-1} L_j \right) \frac{v_T}{V_T} \right] \\ &= \frac{v_t}{V_t} + L_t q_t. \end{aligned}$$

Therefore, using the initial value $q_T = 0$, we have the backward recursion

$$q_{t-1} = \frac{v_t}{V_t} + L_t q_t, \quad t = T, T-1, \dots, 1. \quad (11.19)$$

Putting Eqs. (11.17) and (11.19) together, we have a backward recursive algorithm to compute the smoothed state variables:

$$q_{t-1} = V_t^{-1}v_t + L_t q_t, \quad \mu_{t|T} = \mu_{t|t-1} + \Sigma_{t|t-1} q_{t-1}, \quad t = T, \dots, 1, \quad (11.20)$$

where $q_T = 0$, and $\mu_{t|t-1}$, $\Sigma_{t|t-1}$ and L_t are available from the Kalman filter in Eq. (11.14).

Smoothed State Variance

The variance of the smoothed state variable $\mu_{t|T}$ can be derived in a similar manner via Theorem 11.1(4). Specifically, letting $\mathbf{v}_t^T = (v_t, \dots, v_T)'$, we have

$$\begin{aligned} \Sigma_{t|T} &= \text{Var}(\mu_t | F_T) = \text{Var}(\mu_t | F_{t-1}, v_t, \dots, v_T) \\ &= \text{Var}(\mu_t | F_{t-1}) - \text{Cov}[\mu_t, (\mathbf{v}_t^T)'] \text{Cov}[(\mathbf{v}_t^T)]^{-1} \text{Cov}[\mu_t, (\mathbf{v}_t^T)] \\ &= \Sigma_{t|t-1} - \sum_{j=t}^T [\text{Cov}(\mu_t, v_j)]^2 V_j^{-1}, \end{aligned} \quad (11.21)$$

where $\text{Cov}(\mu_t, v_j) = \text{Cov}(x_t, v_j)$ are given earlier after Eq. (11.17). Thus,

$$\begin{aligned} \Sigma_{t|T} &= \Sigma_{t|t-1} - \Sigma_{t|t-1}^2 \frac{1}{V_t} - \Sigma_{t|t-1}^2 L_t^2 \frac{1}{V_{t+1}} - \dots - \Sigma_{t|t-1}^2 \left(\prod_{j=t}^{T-1} L_j^2 \right) \frac{1}{V_T} \\ &\equiv \Sigma_{t|t-1} - \Sigma_{t|t-1}^2 M_{t-1}, \end{aligned} \quad (11.22)$$

where

$$M_{t-1} = \frac{1}{V_t} + L_t^2 \frac{1}{V_{t+1}} + L_t^2 L_{t+1}^2 \frac{1}{V_{t+2}} + \dots + \left(\prod_{j=t}^{T-1} L_j^2 \right) \frac{1}{V_T},$$

is a weighted linear combination of the inverses of variances of the 1-step-ahead forecast errors after time $t-1$. Let $M_T = 0$ because no 1-step-ahead forecast error is available after time index T . The statistic M_{t-1} can be written as

$$\begin{aligned} M_{t-1} &= \frac{1}{V_t} + L_t^2 \left[\frac{1}{V_{t+1}} + L_{t+1}^2 \frac{1}{V_{t+2}} + \dots + \left(\prod_{j=t+1}^{T-1} L_j^2 \right) \frac{1}{V_T} \right] \\ &= \frac{1}{V_t} + L_t^2 M_t, \quad t = T, T-1, \dots, 1. \end{aligned}$$

Note that from the independence of $\{v_t\}$ and Eq. (11.18), we have

$$\text{Var}(q_{t-1}) = \frac{1}{V_t} + L_t^2 \frac{1}{V_{t+1}} + \dots + \left(\prod_{j=t}^{T-1} L_j^2 \right) \frac{1}{V_T} = M_{t-1}.$$

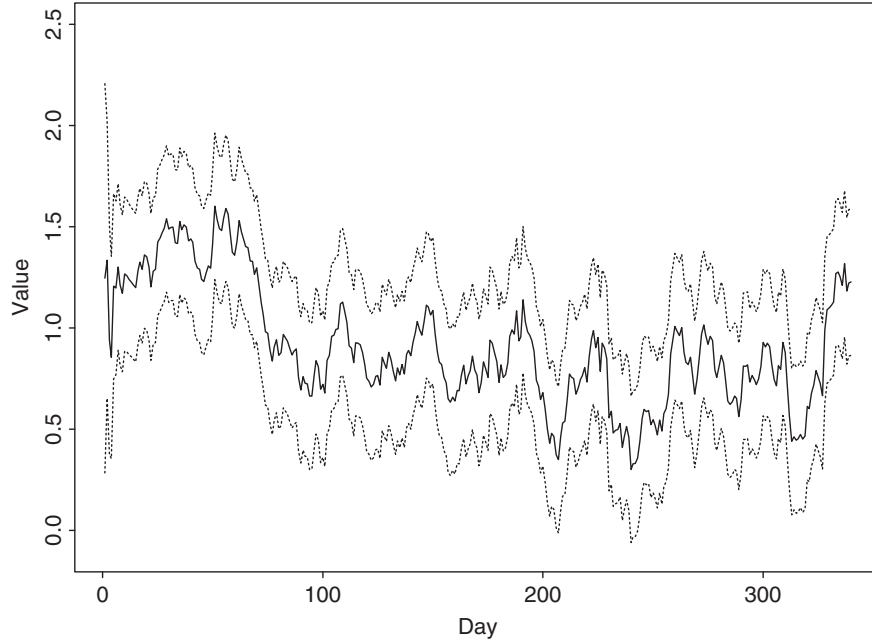


Figure 11.3 Filtered state variable $\mu_{t|t}$ and its 95% pointwise confidence interval for daily log realized volatility of Alcoa stock returns based on fitted local-trend state-space model.

Combining the results, variances of the smoothed state variables can be computed efficiently via the backward recursion

$$M_{t-1} = V_t^{-1} + L_t^2 M_t, \quad \Sigma_{t|T} = \Sigma_{t|t-1} - \Sigma_{t|t-1}^2 M_{t-1}, \quad t = T, \dots, 1, \quad (11.23)$$

where $M_T = 0$.

Example 11.1 (Continued). Applying the Kalman filter and state-smoothing algorithms in Eqs. (11.20) and (11.23) to the daily realized volatility of Alcoa stock using the fitted state-space model, we can easily compute the filtered state $\mu_{t|t}$ and the smoothed state $\mu_{t|T}$ and their variances. Figure 11.3 shows the filtered state variable and its 95% pointwise confidence interval, whereas Figure 11.4 provides the time plot of smoothed state variable and its 95% pointwise confidence interval. As expected, the smoothed state variables are smoother than the filtered state variables. The confidence intervals for the smoothed state variables are also narrower than those of the filtered state variables. Note that the width of the 95% confidence interval of $\mu_{1|1}$ depends on the initial value $\Sigma_{1|0}$.

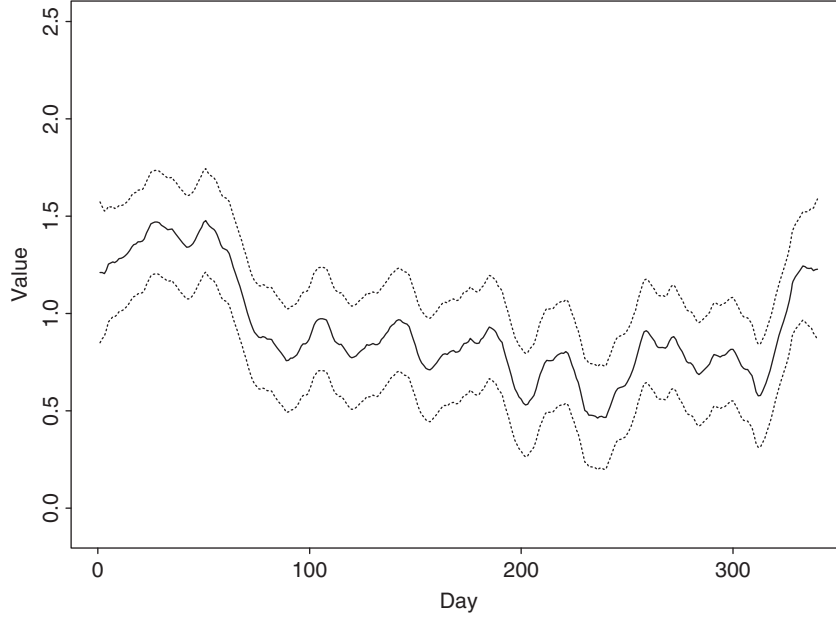


Figure 11.4 Smoothed state variable $\mu_{t|T}$ and its 95% pointwise confidence interval for daily log realized volatility of Alcoa stock returns based on fitted local-trend state-space model.

11.1.5 Missing Values

An advantage of the state-space model is in handling missing values. Suppose that the observations $\{y_t\}_{t=\ell+1}^{\ell+h}$ are missing, where $h \geq 1$ and $1 \leq \ell < T$. There are several ways to handle missing values in state-space formulation. Here we discuss a method that keeps the original time scale and model form. For $t \in \{\ell + 1, \dots, \ell + h\}$, we can use Eq. (11.2) to express μ_t as a linear combination of $\mu_{\ell+1}$ and $\{\eta_j\}_{j=\ell+1}^{t-1}$. Specifically,

$$\mu_t = \mu_{t-1} + \eta_{t-1} = \dots = \mu_{\ell+1} + \sum_{j=\ell+1}^{t-1} \eta_j,$$

where it is understood that the summation term is zero if its lower limit is greater than its upper limit. Therefore, for $t \in \{\ell + 1, \dots, \ell + h\}$,

$$\begin{aligned} E(\mu_t | F_{t-1}) &= E(\mu_t | F_\ell) = \mu_{\ell+1 | \ell}, \\ \text{Var}(\mu_t | F_{t-1}) &= \text{Var}(\mu_t | F_\ell) = \Sigma_{\ell+1 | \ell} + (t - \ell - 1)\sigma_\eta^2. \end{aligned}$$

Consequently, we have

$$\mu_{t|t-1} = \mu_{t-1|t-2}, \quad \Sigma_{t|t-1} = \Sigma_{t-1|t-2} + \sigma_\eta^2, \quad (11.24)$$

for $t = \ell + 2, \dots, \ell + h$. These results show that we can continue to apply the Kalman filter algorithm in Eq. (11.14) by taking $v_t = 0$ and $K_t = 0$ for $t = \ell + 1, \dots, \ell + h$. This is rather natural because when y_t is missing, there is no new innovation or new Kalman gain so that $v_t = 0$ and $K_t = 0$.

11.1.6 Effect of Initialization

In this section, we consider the effects of initial condition $\mu_1 \sim N(\mu_{1|0}, \Sigma_{1|0})$ on the Kalman filter and state smoothing. From the Kalman filter in Eq. (11.14),

$$v_1 = y_1 - \mu_{1|0}, \quad V_1 = \Sigma_{1|0} + \sigma_e^2,$$

and, by Eqs. (11.10)–(11.13),

$$\begin{aligned} \mu_{2|1} &= \mu_{1|0} + \frac{\Sigma_{1|0}}{V_1} v_1 = \mu_{1|0} + \frac{\Sigma_{1|0}}{\Sigma_{1|0} + \sigma_e^2} (y_1 - \mu_{1|0}), \\ \Sigma_{2|1} &= \Sigma_{1|0} \left(1 - \frac{\Sigma_{1|0}}{\Sigma_{1|0} + \sigma_e^2} \right) + \sigma_\eta^2 = \frac{\Sigma_{1|0}}{\Sigma_{1|0} + \sigma_e^2} \sigma_e^2 + \sigma_\eta^2. \end{aligned}$$

Therefore, letting $\Sigma_{1|0}$ increase to infinity, we have $\mu_{2|1} = y_1$ and $\Sigma_{2|1} = \sigma_e^2 + \sigma_\eta^2$. This is equivalent to treating y_1 as fixed and assuming $\mu_1 \sim N(y_1, \sigma_e^2)$. In the literature, this approach to initializing the Kalman filter is called *diffuse initialization* because a very large $\Sigma_{1|0}$ means one is uncertain about the initial condition.

Next, turn to the effect of diffuse initialization on state smoothing. It is obvious that based on the results of Kalman filtering, state smoothing is not affected by the diffuse initialization for $t = T, \dots, 2$. Thus, we focus on μ_1 given F_T . From Eq. (11.20) and the definition of $L_1 = 1 - K_1 = V_1^{-1} \sigma_e^2$,

$$\begin{aligned} \mu_{1|T} &= \mu_{1|0} + \Sigma_{1|0} q_0 \\ &= \mu_{1|0} + \Sigma_{1|0} \left[\frac{1}{\Sigma_{1|0} + \sigma_e^2} v_1 + \left(1 - \frac{\Sigma_{1|0}}{\Sigma_{1|0} + \sigma_e^2} \right) q_1 \right] \\ &= \mu_{1|0} + \frac{\Sigma_{1|0}}{\Sigma_{1|0} + \sigma_e^2} (v_1 + \sigma_e^2 q_1). \end{aligned}$$

Letting $\Sigma_{1|0} \rightarrow \infty$, we have $\mu_{1|T} = \mu_{1|0} + v_1 + \sigma_e^2 q_1 = y_1 + \sigma_e^2 q_1$. Furthermore, from Eq. (11.23) and using $V_1 = \Sigma_{1|0} + \sigma_e^2$, we have

$$\begin{aligned} \Sigma_{1|T} &= \Sigma_{1|0} - \Sigma_{1|0}^2 \left[\frac{1}{\Sigma_{1|0} + \sigma_e^2} + \left(1 - \frac{\Sigma_{1|0}}{\Sigma_{1|0} + \sigma_e^2} \right)^2 M_1 \right] \\ &= \Sigma_{1|0} \left(1 - \frac{\Sigma_{1|0}}{\Sigma_{1|0} + \sigma_e^2} \right) - \left(1 - \frac{\Sigma_{1|0}}{\Sigma_{1|0} + \sigma_e^2} \right)^2 \Sigma_{1|0}^2 M_1 \\ &= \left(\frac{\Sigma_{1|0}}{\Sigma_{1|0} + \sigma_e^2} \right) \sigma_e^2 - \left(\frac{\Sigma_{1|0}}{\Sigma_{1|0} + \sigma_e^2} \right)^2 \sigma_e^4 M_1. \end{aligned}$$

Thus, letting $\Sigma_{1|0} \rightarrow \infty$, we obtain $\Sigma_{1|T} = \sigma_e^2 - \sigma_e^4 M_1$.

Based on the prior discussion, we suggest using diffuse initialization when little is known about the initial value μ_1 . However, it might be hard to justify the use of a random variable with infinite variance in real applications. If necessary, one can treat μ_1 as an additional parameter of the state-space model and estimate it jointly with other parameters. This latter approach is closely related to the exact maximum-likelihood estimation of Chapters 2 and 8.

11.1.7 Estimation

In this section, we consider the estimation of σ_e and σ_η of the local trend model in Eqs. (11.1) and (11.2). Based on properties of forecast errors discussed in Section 11.1.3, the Kalman filter provides an efficient way to evaluate the likelihood function of the data for estimation. Specifically, the likelihood function under normality is

$$\begin{aligned} p(y_1, \dots, y_T | \sigma_e, \sigma_\eta) &= p(y_1 | \sigma_e, \sigma_\eta) \prod_{t=2}^T (y_t | F_{t-1}, \sigma_e, \sigma_\eta) \\ &= p(y_1 | \sigma_e, \sigma_\eta) \prod_{t=2}^T (v_t | F_{t-1}, \sigma_e, \sigma_\eta), \end{aligned}$$

where $y_1 \sim N(\mu_{1|0}, V_1)$ and $v_t = (y_t - \mu_{t|t-1}) \sim N(0, V_t)$. Consequently, assuming $\mu_{1|0}$ and $\Sigma_{1|0}$ are known, and taking the logarithms, we have

$$\ln[L(\sigma_e, \sigma_\eta)] = -\frac{T}{2} \ln(2\pi) - \frac{1}{2} \sum_{t=1}^T \left[\ln(V_t) + \frac{v_t^2}{V_t} \right], \quad (11.25)$$

which involves v_t and V_t . Therefore, the log-likelihood function, including cases with missing values, can be evaluated recursively via the Kalman filter. Many software packages perform state-space model estimation via a Kalman filter algorithm such as Matlab, RATS, and S-Plus. In this chapter, we use the SsfPack program developed by Koopman, Shephard, and Doornik (1999) and available in S-Plus and OX. Both Ssfpack and OX are free and can be downloaded from their websites.

11.1.8 S-Plus Commands Used

We provide here the SsfPack commands used to perform analysis of the daily realized volatility of Aloc stock returns. Only brief explanations are given. For further details of the commands used, see Durbin and Koopman (2001, Section 6.6). S-Plus uses specific notation to specify a state-space model; see Table 11.1. The notation must be followed closely. In Table 11.2, we give some commands and their functions.

TABLE 11.1 State-Space Form and Notation in S-Plus

State-Space Parameter	S-Plus Name
δ	mDelta
Φ	mPhi
Ω	mOmega
Σ	mSigma

TABLE 11.2 Some Commands of SsfPack Package

Command	Function
SsfFit	Maximum-likelihood estimation
CheckSsf	Create “Ssf” object in S-Plus
KalmanFil	Perform Kalman filtering
KalmanSmo	Perform state smoothing
SsfMomentEst with task “STFIL”	Compute filtered state and variance
SsfMomentEst with task “STSMO”	Compute smoothed state and variance
SsfCondDens with task “STSMO”	Compute smoothed state without variance

In our analysis, we first perform maximum-likelihood estimation of the state-space model in Eqs. (11.1) and (11.2) to obtain estimates of σ_e and σ_η . The initial values used are $\Sigma_{1|0} = -1$ and $\mu_{1|0} = 0$, where -1 signifies diffuse initialization, that is, $\Sigma_{1|0}$ is very large. We then treat the fitted model as given to perform Kalman filtering and state smoothing.

SsfPack and S-Plus Commands for State-Space Model

```
> da = read.table(file='aa-rv-0304.txt',header=F) % load data
> y = log(da[,1]) % log(RV)
> ltm.start=c(3,1) % Initial parameter values
> P1 = -1 % Initialization of Kalman filter
> a1 = 0
> ltm.m=function(parm){ % Specify a function for the
+ sigma.eta=parm[1] % local trend model.
+ sigma.e=parm[2]
+ ssf.m=list(mPhi=as.matrix(c(1,1)),
+ mOmega=diag(c(sigma.eta^2,sigma.e^2)),
+ mSigma=as.matrix(c(P1,a1)))
+ CheckSsf(ssf.m)
+ }
% perform estimation
> ltm.mle=SsfFit(ltm.start,y,"ltm.m",lower=c(0,0),
+ upper=c(100,100))
> ltm.mle$parameters
```

```

[1] 0.07350827 0.48026284
> sigma.eta=ltm.mle$parameters[1]
> sigma.eta
[1] 0.07350827
> sigma.e=ltm.mle$parameters[2]
> sigma.e
[1] 0.4802628
% Specify a state-space model in S-Plus.
> ssf.ltm.list=list(mPhi=as.matrix(c(1,1)),
+ mOmega=diag(c(sigma.eta^2,sigma.e^2)),
+ mSigma=as.matrix(c(P1,a1)))
% check validity of the specified model.
> ssf.ltm=CheckSsf(ssf.ltm.list)
> ssf.ltm
$mPhi:
      [,1]
[1,]      1
[2,]      1
$mOmega:
      [,1] [,2]
[1,] 0.0054035 0.0000000
[2,] 0.0000000 0.2306524
$mSigma:
      [,1]
[1,]     -1
[2,]      0
$mDelta:
      [,1]
[1,]      0
[2,]      0
$mJPhi:
[1] 0
$mJOmega:
[1] 0
$mJDelta:
[1] 0
$mX:
[1] 0
$cT:
[1] 0
$cX:
[1] 0
$cY:
[1] 1
$cSt:
[1] 1
attr(, "class"):
[1] "ssf"
% Apply Kalman filter

```



```

> KalmanFil.ltm=KalmanFil(y,ssf.ltm,task="STFIL")
> names(KalmanFil.ltm)
[1] "mOut"      "innov"     "std.innov"  "mGain"     "loglike"
[6] "loglike.conc" "dVar"      "mEst"       "mOffP"     "task"
[11] "err"       "call"
> par(mfcol=c(2,1)) % Obtain plot
> plot(KalmanFil.ltm$mEst[,1],xlab='day',
+ ylab='filtered state',type='l')
> title(main='(a) Filtered state variable')
> plot(KalmanFil.ltm$mOut[,1],xlab='day',
+ ylab='v(t)',type='l')
> title(main='(b) Prediction error')
% Obtain residuals and their variances
> KalmanSmo.ltm=KalmanSmo(KalmanFil.ltm,ssf.ltm)
> names(KalmanSmo.ltm)
[1] "state.residuals" "response.residuals" "state.variance"
[4] "response.variance" "aux.residuals"      "scores"
[7] "call"
% Filtered states
> FiledEst.ltm=SsfMomentEst(y,ssf.ltm,task="STFIL")
> names(FiledEst.ltm)
[1] "state.moment" "state.variance" "response.moment"
[4] "response.variance" "task"
% Smoothed states
> SmoedEst.ltm=SsfMomentEst(y,ssf.ltm,task="STSMO")
> names(SmoedEst.ltm)
[1] "state.moment" "state.variance" "response.moment"
[4] "response.variance" "task"
% Obtain plots of filtered and smoothed states with 95% C.I.
> up=FiledEst.ltm$ state.moment+
+ 2*sqrt(FiledEst.ltm$ state.variance)
> lw=FiledEst.ltm$ state.moment-
+ 2*sqrt(FiledEst.ltm$ state.variance)
> par(mfcol=c(1,1))
> plot(FiledEst.ltm$ state.moment,type='l',xlab='day',
+ ylab='value',ylim=c(-0.1,2.5))
> lines(1:340,up,lty=2)
> lines(1:340,lw,lty=2)
> title(main='Filed state variable')
> up=SmoedEst.ltm$ state.moment+
+ 2*sqrt(SmoedEst.ltm$ state.variance)
> lw=SmoedEst.ltm$ state.moment-
+ 2*sqrt(SmoedEst.ltm$ state.variance)
> plot(SmoedEst.ltm$ state.moment,type='l',xlab='day',
+ ylab='value',ylim=c(-0.1,2.5))
> lines(1:340,up,lty=2)
> lines(1:340,lw,lty=2)
> title(main='Smoothed state variable')
% Model checking via standardized residuals

```

```
> resi=KalmanFil.ltm$ mOut[,1]*sqrt(KalmanFil.ltm$ mOut[,3])
> archTest(resi)
> autocorTest(resi)
```

For the daily realized volatility of Alcoa stock returns, the fitted local trend model is adequate based on residual analysis. Specifically, given the parameter estimates, we use the Kalman filter to obtain the 1-step-ahead forecast error v_t and its variance V_t . We then compute the standardized forecast error $\tilde{v}_t = v_t/\sqrt{V_t}$ and check the serial correlations and ARCH effects of $\{\tilde{v}_t\}$. We found that $Q(25) = 23.37(0.56)$ for the standardized forecast errors, and the LM test statistic for ARCH effect is 18.48(0.82) for 25 lags, where the number in parentheses denotes p value.

11.2 LINEAR STATE-SPACE MODELS

We now consider the general state-space model. Many dynamic time series models in economics and finance can be represented in state-space form. Examples include the ARIMA models, dynamic linear models with unobserved components, time-varying regression models, and stochastic volatility models. A general Gaussian linear state-space model assumes the form

$$s_{t+1} = d_t + T_t s_t + R_t \eta_t, \quad (11.26)$$

$$y_t = c_t + Z_t s_t + e_t, \quad (11.27)$$

where $s_t = (s_{1t}, \dots, s_{mt})'$ is an m -dimensional state vector, $y_t = (y_{1t}, \dots, y_{kt})'$ is a k -dimensional observation vector, d_t and c_t are m - and k -dimensional deterministic vectors, T_t and Z_t are $m \times m$ and $k \times m$ coefficient matrices, R_t is an $m \times n$ matrix often consisting of a subset of columns of the $m \times m$ identity matrix, and $\{\eta_t\}$ and $\{e_t\}$ are n - and k -dimensional Gaussian white noise series such that

$$\eta_t \sim N(\mathbf{0}, Q_t), \quad e_t \sim N(\mathbf{0}, H_t),$$

where Q_t and H_t are positive-definite matrices. We assume that $\{e_t\}$ and $\{\eta_t\}$ are independent, but this condition can be relaxed if necessary. The initial state s_1 is $N(\mu_{1|0}, \Sigma_{1|0})$, where $\mu_{1|0}$ and $\Sigma_{1|0}$ are given, and is independent of e_t and η_t for $t > 0$.

Equation (11.27) is the *measurement* or *observation* equation that relates the vector of observations y_t to the state vector s_t , the explanatory variable c_t , and the measurement error e_t . Equation (11.26) is the *state* or *transition* equation that describes a first-order Markov Chain to govern the state transition with innovation η_t . The matrices T_t , R_t , Q_t , Z_t , and H_t are known and referred to as *system matrices*. These matrices are often sparse, and they can be functions of some parameters θ , which can be estimated by the maximum-likelihood method.

The state-space model in Eqs. (11.26) and (11.27) can be rewritten in a compact form as

$$\begin{bmatrix} s_{t+1} \\ y_t \end{bmatrix} = \delta_t + \Phi_t s_t + u_t, \quad (11.28)$$

where

$$\delta_t = \begin{bmatrix} d_t \\ c_t \end{bmatrix}, \quad \Phi_t = \begin{bmatrix} T_t \\ Z_t \end{bmatrix}, \quad u_t = \begin{bmatrix} R_t \eta_t \\ e_t \end{bmatrix},$$

and $\{u_t\}$ is a sequence of Gaussian white noises with mean zero and covariance matrix

$$\Omega_t = \text{Cov}(u_t) = \begin{bmatrix} R_t Q_t R_t' & \mathbf{0} \\ \mathbf{0} & H_t \end{bmatrix}$$

The case of diffuse initialization is achieved by using

$$\Sigma_{1|0} = \Sigma_* + \lambda \Sigma_\infty,$$

where Σ_* and Σ_∞ are $m \times m$ symmetric positive-definite matrices and λ is a large real number, which can approach infinity. In S-Plus and SsfPack, the notation

$$\Sigma = \begin{bmatrix} \Sigma_{1|0} \\ \mu'_{1|0} \end{bmatrix}_{(m+1) \times m}$$

is used; see the notation in Table 11.1.

In many applications, the system matrices are time invariant. However, these matrices can be time varying, making the state-space model flexible.

11.3 MODEL TRANSFORMATION

To appreciate the flexibility of the state-space model, we rewrite some well-known econometric and financial models in state-space form.

11.3.1 CAPM with Time-Varying Coefficients

First, consider the capital asset pricing model (CAPM) with time-varying intercept and slope. The model is

$$\begin{aligned} r_t &= \alpha_t + \beta_t r_{M,t} + e_t, & e_t &\sim N(0, \sigma_e^2), \\ \alpha_{t+1} &= \alpha_t + \eta_t, & \eta_t &\sim N(0, \sigma_\eta^2), \\ \beta_{t+1} &= \beta_t + \epsilon_t, & \epsilon_t &\sim N(0, \sigma_\epsilon^2), \end{aligned} \quad (11.29)$$

where r_t is the excess return of an asset, $r_{M,t}$ is the excess return of the market, and the innovations $\{e_t, \eta_t, \epsilon_t\}$ are mutually independent. This CAPM allows for

time-varying α and β that evolve as a random walk over time. We can easily rewrite the model as

$$\begin{bmatrix} \alpha_{t+1} \\ \beta_{t+1} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \alpha_t \\ \beta_t \end{bmatrix} + \begin{bmatrix} \eta_t \\ \epsilon_t \end{bmatrix},$$

$$r_t = [1, r_{M,t}] \begin{bmatrix} \alpha_t \\ \beta_t \end{bmatrix} + e_t.$$

Thus, the time-varying CAPM is a special case of the state-space model with $s_t = (\alpha_t, \beta_t)'$, $T_t = R_t = I_2$, the 2×2 identity matrix, $d_t = \mathbf{0}$, $c_t = 0$, $Z_t = (1, r_{M,t})$, $H_t = \sigma_e^2$, and $Q_t = \text{diag}\{\sigma_\eta^2, \sigma_\epsilon^2\}$. Furthermore, in the form of Eq. (11.28), we have $\delta_t = \mathbf{0}$, $u_t = (\eta_t, \epsilon_t, e_t)'$,

$$\Phi_t = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & r_{M,t} \end{bmatrix}, \quad \Omega_t = \begin{bmatrix} \sigma_\eta^2 & 0 & 0 \\ 0 & \sigma_\epsilon^2 & 0 \\ 0 & 0 & \sigma_e^2 \end{bmatrix}.$$

If diffuse initialization is used, then

$$\Sigma = \begin{bmatrix} -1 & 0 \\ 0 & -1 \\ 0 & 0 \end{bmatrix}.$$

SsfPack/S-Plus Specification of Time-Varying Models

For the CAPM in Eq. (11.29), Φ_t contains $r_{M,t}$, which is time varying. Some special input is required to specify such a model in SsfPack. Basically, it requires two additional variables: (a) a data matrix X that stores Z_t and (b) an index matrix for Φ_t that identifies Z_t from the data matrix. The notation for index matrices of the state-space model in Eq. (11.28) is given in Table 11.3. Note that the matrix J_Φ must have the same dimension as Φ_t . The elements of J_Φ are all set to -1 except the elements for which the corresponding elements of Φ_t are time varying. The nonnegative index value of J_Φ indicates the column of the data matrix X , which contains the time-varying values.

To illustrate, consider the monthly simple excess returns of General Motors stock from January 1990 to December 2003 used in Chapter 9. The monthly simple

TABLE 11.3 Notation and Name Used in SsfPack/S-Plus for Time-Varying State-Space Model

Index Matrix	Name Used in SsfPack/S-Plus
J_δ	mJDelta
J_Φ	mJPhi
J_Ω	mJOmega
Time-Varying Data Matrix	Name Used in SsfPack/S-Plus
X	mX

excess return of the S&P 500 composite index is used as the market return. The specification of a time-varying CAPM requires values of the variances σ_η^2 , σ_ϵ^2 , and σ_e^2 . Suppose that $(\sigma_\eta, \sigma_\epsilon, \sigma_e) = (0.02, 0.04, 0.1)$. The state-space specification for the CAPM under SsfPack/S-Plus is given below:

```
> X.mtx=cbind(1,sp) % Here ``sp`` is market excess returns.
> Phi.t = rbind(diag(2),rep(0,2))
> Sigma=-Phi.t
> sigma.eta=.02
> sigma.ep=.04
> sigma.e=.1
> Omega=diag(c(sigma.eta^2,sigma.ep^2,sigma.e^2))
> JPhi = matrix(-1,3,2) % Create a 3-by-2 matrix of -1.
> JPhi[3,1]=1
> JPhi[3,2]=2
> ssf.tv.capm=list(mPhi=Phi.t,
+ mOmega=Omega,
+ mJPhi=JPhi,
+ mSigma=Sigma,
+ mX=X.mtx)
> ssf.tv.capm
$mPhi:
      [,1] [,2]
[1,]     1     0
[2,]     0     1
[3,]     0     0
$mOmega:
      [,1] [,2] [,3]
[1,] 4e-04 0.0000 0.00
[2,] 0e+00 0.0016 0.00
[3,] 0e+00 0.0000 0.01
$mJPhi:
      [,1] [,2]
[1,]    -1    -1
[2,]    -1    -1
[3,]     1     2
$mSigma:
      [,1] [,2]
[1,]    -1     0
[2,]     0    -1
[3,]     0     0
$mX:
numeric matrix: 168 rows, 2 columns.
      sp
[1,] 1 -0.075187
...
[168,] 1 0.05002
```

11.3.2 ARMA Models

Consider a zero-mean ARMA(p, q) process y_t of Chapter 2:

$$\phi(B)y_t = \theta(B)a_t, \quad a_t \sim N(0, \sigma_a^2), \quad (11.30)$$

where $\phi(B) = 1 - \sum_{i=1}^p \phi_i B^i$ and $\theta(B) = 1 - \sum_{j=1}^q \theta_j B^j$, and p and q are non-negative integers. There are many ways to transform such an ARMA model into a state-space form. We discuss three methods available in the literature. Let $m = \max(p, q + 1)$ and rewrite the ARMA model in Eq. (11.30) as

$$y_t = \sum_{i=1}^m \phi_i y_{t-i} + a_t - \sum_{j=1}^{m-1} \theta_j a_{t-j}, \quad (11.31)$$

where $\phi_i = 0$ for $i > p$ and $\theta_j = 0$ for $j > q$. In particular, $\theta_m = 0$ because $m > q$.

Akaike's Approach

Akaike (1975) defines the state vector s_t as the minimum collection of variables that contains all the information needed to produce forecasts at the forecast origin t . It turns out that, for the ARMA process in Eq. (11.30) with $m = \max(p, q + 1)$, $s_t = (y_{t|t}, y_{t+1|t}, \dots, y_{t+m-1|t})'$, where $y_{t+j|t} = E(y_{t+j}|F_t)$ is the conditional expectation of y_{t+j} given $F_t = \{y_1, \dots, y_t\}$. Since $y_{t|t} = y_t$, the first element of s_t is y_t . Thus, the observation equation is

$$y_t = Zs_t, \quad (11.32)$$

where $Z = (1, 0, \dots, 0)_{1 \times m}$. We derive the transition equation in several steps. First, from the definition,

$$s_{1,t+1} = y_{t+1} = y_{t+1|t} + (y_{t+1} - y_{t+1|t}) = s_{2t} + a_{t+1}, \quad (11.33)$$

where s_{it} is the i th element of s_t . Next, consider the MA representation of ARMA models given in Chapter 2. That is,

$$y_t = a_t + \psi_1 a_{t-1} + \psi_2 a_{t-2} + \dots = \sum_{i=0}^{\infty} \psi_i a_{t-i},$$

where $\psi_0 = 1$ and other ψ weights can be obtained by equating coefficients of B^i in $1 + \sum_{i=1}^{\infty} \psi_i B^i = \theta(B)/\phi(B)$. In particular, we have

$$\begin{aligned} \psi_1 &= \phi_1 - \theta_1, \\ \psi_2 &= \phi_1 \psi_1 + \phi_2 - \theta_2, \\ &\vdots \end{aligned}$$

$$\begin{aligned}
\psi_{m-1} &= \phi_1 \psi_{m-2} + \phi_2 \psi_{m-3} + \cdots + \phi_{m-2} \psi_1 + \phi_{m-1} - \theta_{m-1} \\
&= \sum_{i=1}^{m-1} \phi_i \psi_{m-1-i} - \theta_{m-1}.
\end{aligned} \tag{11.34}$$

Using the MA representation, we have, for $j > 0$,

$$\begin{aligned}
y_{t+j|t} &= E(y_{t+j}|F_t) = E\left(\sum_{i=0}^{\infty} \psi_i a_{t+j-i}|F_t\right) \\
&= \psi_j a_t + \psi_{j+1} a_{t-1} + \psi_{j+2} a_{t-2} + \cdots
\end{aligned}$$

and

$$\begin{aligned}
y_{t+j|t+1} &= E(y_{t+j}|F_{t+1}) = \psi_{j-1} a_{t+1} + \psi_j a_t + \psi_{j+1} a_{t-1} + \cdots \\
&= \psi_{j-1} a_{t+1} + y_{t+j|t}.
\end{aligned}$$

Thus, for $j > 0$, we have

$$y_{t+j|t+1} = y_{t+j|t} + \psi_{j-1} a_{t+1}. \tag{11.35}$$

This result is referred to as the forecast updating formula of ARMA models. It provides a simple way to update the forecast from origin t to origin $t+1$ when y_{t+1} becomes available. The new information of y_{t+1} is contained in the innovation a_{t+1} , and the time- t forecast is revised based on this new information with weight ψ_{j-1} to compute the time- $(t+1)$ forecast.

Finally, from Eq. (11.31) and using $E(a_{t+j}|F_{t+1}) = 0$ for $j > 1$, we have

$$y_{t+m|t+1} = \sum_{i=1}^m \phi_i y_{t+m-i|t+1} - \theta_{m-1} a_{t+1}.$$

Taking Eq. (11.35), the prior equation becomes

$$\begin{aligned}
y_{t+m|t+1} &= \sum_{i=1}^{m-1} \phi_i (y_{t+m-i|t} + \psi_{m-i-1} a_{t+1}) + \phi_m y_{t|t} - \theta_{m-1} a_{t+1} \\
&= \sum_{i=1}^m \phi_i y_{t+m-i|t} + \left(\sum_{i=1}^{m-1} \phi_i \psi_{m-1-i} - \theta_{m-1} \right) a_{t+1} \\
&= \sum_{i=1}^m \phi_i y_{t+m-i|t} + \psi_{m-1} a_{t+1},
\end{aligned} \tag{11.36}$$

where the last equality uses Eq. (11.34). Combining Eqs. (11.33) and (11.35) for $j = 2, \dots, m-1$, and (11.36) together, we have

$$\begin{bmatrix} y_{t+1} \\ y_{t+2|t+1} \\ \vdots \\ y_{t+m-1|t+1} \\ y_{t+m|t+1} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & & 0 \\ \vdots & & & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ \phi_m & \phi_{m-1} & \phi_{m-2} & \cdots & \phi_1 \end{bmatrix} \begin{bmatrix} y_t \\ y_{t+1|t} \\ \vdots \\ y_{t+m-2|t} \\ y_{t+m-1|t} \end{bmatrix} + \begin{bmatrix} 1 \\ \psi_1 \\ \vdots \\ \psi_{m-2} \\ \psi_{m-1} \end{bmatrix} a_{t+1}. \quad (11.37)$$

Thus, the transition equation of Akaike's approach is

$$s_{t+1} = \mathbf{T}s_t + \mathbf{R}\eta_t, \quad \eta_t \sim N(0, \sigma_a^2), \quad (11.38)$$

where $\eta_t = a_{t+1}$, and $\mathbf{T}$ and $\mathbf{R}$ are the coefficient matrices in Eq. (11.37).

Harvey's Approach

Harvey (1993, Section 4.4) provides a state-space form with an m -dimensional state vector s_t , the first element of which is y_t , that is, $s_{1t} = y_t$. The other elements of s_t are obtained recursively. From the ARMA($m, m-1$) model, we have

$$\begin{aligned} y_{t+1} &= \phi_1 y_t + \sum_{i=2}^m \phi_i y_{t+1-i} - \sum_{j=1}^{m-1} \theta_j a_{t+1-j} + a_{t+1} \\ &\equiv \phi_1 s_{1t} + s_{2t} + \eta_t, \end{aligned}$$

where $s_{2t} = \sum_{i=2}^m \phi_i y_{t+1-i} - \sum_{j=1}^{m-1} \theta_j a_{t+1-j}$, $\eta_t = a_{t+1}$, and as defined earlier $s_{1t} = y_t$. Focusing on $s_{2,t+1}$, we have

$$\begin{aligned} s_{2,t+1} &= \sum_{i=2}^m \phi_i y_{t+2-i} - \sum_{j=1}^{m-1} \theta_j a_{t+2-j} \\ &= \phi_2 y_t + \sum_{i=3}^m \phi_i y_{t+2-i} - \sum_{j=2}^{m-1} \theta_j a_{t+2-j} - \theta_1 a_{t+1} \\ &\equiv \phi_2 s_{1t} + s_{3t} + (-\theta_1)\eta_t, \end{aligned}$$

where $s_{3t} = \sum_{i=3}^m \phi_i y_{t+2-i} - \sum_{j=2}^{m-1} \theta_j a_{t+2-j}$. Next, considering $s_{3,t+1}$, we have

$$\begin{aligned} s_{3,t+1} &= \sum_{i=3}^m \phi_i y_{t+3-i} - \sum_{j=2}^{m-1} \theta_j a_{t+3-j} \\ &= \phi_3 y_t + \sum_{i=4}^m \phi_i y_{t+3-i} - \sum_{j=3}^{m-1} \theta_j a_{t+3-j} + (-\theta_2) a_{t+1} \\ &\equiv \phi_3 s_{1t} + s_{4t} + (-\theta_2) \eta_t, \end{aligned}$$

where $s_{4t} = \sum_{i=4}^m \phi_i y_{t+3-i} - \sum_{j=3}^{m-1} \theta_j a_{t+3-j}$. Repeating the procedure, we have $s_{mt} = \sum_{i=m}^m \phi_i y_{t+m-1-i} - \sum_{j=m-1}^{m-1} \theta_j a_{t+m-1-j} = \phi_m y_{t-1} - \theta_{m-1} a_t$. Finally,

$$\begin{aligned} s_{m,t+1} &= \phi_m y_t - \theta_{m-1} a_{t+1} \\ &= \phi_m s_{1t} + (-\theta_{m-1}) \eta_t. \end{aligned}$$

Putting the prior equations together, we have a state-space form

$$s_{t+1} = \mathbf{T} s_t + \mathbf{R} \eta_t, \quad \eta_t \sim N(0, \sigma_a^2), \quad (11.39)$$

$$y_t = \mathbf{Z} s_t, \quad (11.40)$$

where the system matrices are time invariant defined as $\mathbf{Z} = (1, 0, \dots, 0)_{1 \times m}$,

$$\mathbf{T} = \begin{bmatrix} \phi_1 & 1 & 0 & \cdots & 0 \\ \phi_2 & 0 & 1 & & 0 \\ \vdots & & & & \vdots \\ \phi_{m-1} & 0 & 0 & & 1 \\ \phi_m & 0 & 0 & \cdots & 0 \end{bmatrix}, \quad \mathbf{R} = \begin{bmatrix} 1 \\ -\theta_1 \\ \vdots \\ -\theta_{m-1} \end{bmatrix},$$

and $\mathbf{d}_t$, $\mathbf{c}_t$, and $\mathbf{H}_t$ are all zero. The model in Eqs. (11.39) and (11.40) has no measurement errors. It has an advantage that the AR and MA coefficients are directly used in the system matrices.

Aoki's Approach

Aoki (1987, Chapter 4) discusses several ways to convert an ARMA model into a state-space form. First, consider the MA model, that is, $y_t = \theta(B)a_t$. In this case, we can simply define $\mathbf{s}_t = (a_{t-q}, a_{t-q+2}, \dots, a_{t-1})'$ and obtain the state-space form

$$\begin{bmatrix} a_{t-q+1} \\ a_{t-q+2} \\ \vdots \\ a_{t-1} \\ a_t \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & & 1 \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} a_{t-q} \\ a_{t-q+1} \\ \vdots \\ a_{t-2} \\ a_{t-1} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} a_t, \quad (11.41)$$

$$y_t = (-\theta_q, -\theta_{q-1}, \dots, -\theta_1)s_t + a_t.$$

Note that, in this particular case, a_t appears in both state and measurement equations.

Next, consider the AR model, that is, $\phi(B)z_t = a_t$. Aoki (1987) introduces two methods. The first method is a straightforward one by defining $s_t = (z_{t-p+1}, \dots, z_t)'$ to obtain

$$\begin{bmatrix} z_{t-p+2} \\ z_{t-p+3} \\ \vdots \\ z_t \\ z_{t+1} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & & 1 \\ \phi_p & \phi_{p-1} & \phi_{p-2} & \cdots & \phi_1 \end{bmatrix} \begin{bmatrix} z_{t-p+1} \\ z_{t-p+2} \\ \vdots \\ z_{t+1} \\ z_t \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} a_{t+1}, \quad (11.42)$$

$$z_t = (0, 0, \dots, 0, 1)s_t.$$

The second method defines the state vector in the same way as the first method except that a_t is removed from the last element; that is, $s_t = z_t - a_t$ if $p = 1$ and $s_t = (z_{t-p+1}, \dots, z_{t-1}, z_t - a_t)'$ if $p > 1$. Simple algebra shows that

$$\begin{bmatrix} z_{t-p+2} \\ z_{t-p+3} \\ \vdots \\ z_t \\ z_{t+1} - a_{t+1} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & & 1 \\ \phi_p & \phi_{p-1} & \phi_{p-2} & \cdots & \phi_1 \end{bmatrix} \begin{bmatrix} z_{t-p+1} \\ z_{t-p+2} \\ \vdots \\ z_{t-1} \\ z_t - a_t \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ \phi_1 \end{bmatrix} a_t, \quad (11.43)$$

$$z_t = (0, 0, \dots, 0, 1)s_t + a_t.$$

Again, a_t appears in both transition and measurement equations.

Turn to the ARMA(p, q) model $\phi(B)y_t = \theta(B)a_t$. For simplicity, we assume $q < p$ and introduce an auxiliary variable $z_t = [1/\phi(B)]a_t$. Then, we have

$$\phi(B)z_t = a_t, \quad y_t = \theta(B)z_t.$$

Since z_t is an AR(p) model, we can use the transition equation in Eq. (11.42) or (11.43). If Eq. (11.42) is used, we can use $y_t = \theta(B)z_t$ to construct the measurement equation as

$$y_t = (-\theta_{p-1}, -\theta_{p-2}, \dots, -\theta_1, 1)s_t, \quad (11.44)$$

where it is understood that $p > q$ and $\theta_j = 0$ for $j > q$. On the other hand, if Eq. (11.43) is used as the transition equation, we construct the measurement equation as

$$y_t = (-\theta_{p-1}, -\theta_{p-2}, \dots, -\theta_1, 1)s_t + a_t. \quad (11.45)$$

In summary, there are many state-space representations for an ARMA model. Each representation has its pros and cons. For estimation and forecasting purposes, one can choose any one of those representations. On the other hand, for a time-invariant coefficient state-space model in Eqs. (11.26) and (11.27), one can use the Cayley–Hamilton theorem to show that the observation y_t follows an ARMA(m, m) model, where m is the dimension of the state vector.

SsfPack Command

In SsfPack/S-Plus, a command `GetSsfArma` can be used to transform an ARMA model into a state-space form. Harvey's approach is used. To illustrate, consider the AR(1) model

$$y_t = 0.6y_{t-1} + a_t, \quad a_t \sim N(0, 0.4^2).$$

The state-space form of the model is

```
> ssf.ar1 = GetSsfArma(ar=0.6, sigma=0.4)
> ssf.ar1
$mPhi:
      [,1]
[1,]  0.6
[2,]  1.0
$mOmega:
      [,1] [,2]
[1,] 0.16   0
[2,] 0.00   0
$mSigma:
      [,1]
[1,] 0.25
[2,] 0.00
```

Since the AR(1) model is stationary, the program uses $\Sigma_{1|0} = \text{Var}(y_t) = (0.4)^2 / (1 - 0.6^2) = 0.25$ and $\mu_{1|0} = 0$. These values appear in the matrix `mSigma`.

As a second example, consider the ARMA(2,1) model

$$y_t = 1.2y_{t-1} - 0.35y_{t-2} + a_t - 0.25a_{t-1}, \quad a_t \sim N(0, 1.1^2).$$

The state-space form of the model is

```
> arma21.m = list(ar=c(1.2, -0.35), ma=c(-0.25), sigma=1.1)
> ssf.arma21 = GetSsfArma(model=arma21.m)
> ssf.arma21
$mPhi:
      [,1] [,2]
[1,]  1.20   1
[2,] -0.35   0
[3,]  1.00   0
$mOmega:
      [,1] [,2] [,3]
[1,]  1.2100 -0.302500  0
[2,] -0.3025  0.075625  0
[3,]  0.0000  0.000000  0
$mSigma:
      [,1] [,2]
[1,]  4.060709 -1.4874057
[2,] -1.487406  0.5730618
[3,]  0.000000  0.0000000
```

As expected, the output shows that

$$\mathbf{T} = \begin{bmatrix} 1.2 & 1 \\ -0.35 & 0 \end{bmatrix}, \quad \mathbf{Z} = (1, 0),$$

and `mPhi` and `mOmega` follow the format of Eq. (11.28), and the covariance matrix of $(s_{1t}, s_{2t})'$ is used in `mSigma`, where $s_{1t} = y_t$ and $s_{2t} = -0.35y_{t-1} - 0.25y_{t-2}$. Note that in `SsfPack`, the MA polynomial of an ARMA model assumes the form $\theta(B) = 1 + \theta_1 B + \dots + \theta_q B^q$, not the form $\theta(B) = 1 - \theta_1 B - \dots - \theta_q B^q$ commonly used in the literature.

11.3.3 Linear Regression Model

Multiple linear regression models can also be represented in state-space form. Consider the model

$$y_t = \mathbf{x}_t' \boldsymbol{\beta} + e_t, \quad e_t \sim N(0, \sigma_e^2),$$

where $\mathbf{x}_t$ is a p -dimensional explanatory variable and $\boldsymbol{\beta}$ is a p -dimensional parameter vector. Let $s_t = \boldsymbol{\beta}$ for all t . Then the model can be written as

$$\begin{bmatrix} s_{t+1} \\ y_t \end{bmatrix} = \begin{bmatrix} \mathbf{I}_p \\ \mathbf{x}'_t \end{bmatrix} s_t + \begin{bmatrix} \mathbf{0}_p \\ e_t \end{bmatrix}. \quad (11.46)$$

Thus, the system matrices are $\mathbf{T}_t = \mathbf{I}_p$, $\mathbf{Z}_t = \mathbf{x}'_t$, $\mathbf{d}_t = \mathbf{0}$, $c_t = 0$, $\mathbf{Q}_t = \mathbf{0}$, and $\mathbf{H}_t = \sigma_e^2$. Since the state vector is fixed, a diffuse initialization should be used.

One can extend the regression model so that $\boldsymbol{\beta}_t$ is random, say,

$$\boldsymbol{\beta}_{t+1} = \boldsymbol{\beta}_t + \mathbf{R}_t \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim N(\mathbf{0}, 1),$$

and $\mathbf{R}_t = (\sigma_1, \dots, \sigma_p)'$ with $\sigma_i \geq 0$. If $\sigma_i = 0$, then β_i is time invariant.

SsfPack Command

In SsfPack, the command `GetSsfReg` creates a state-space form for the multiple linear regression model. The command has an input argument that contains the data matrix of explanatory variables. To illustrate, consider the simple market model

$$r_t = \beta_0 + \beta_1 r_{M,t} + e_t, \quad t = 1, \dots, 168,$$

where r_t is the return of an asset and $r_{M,t}$ is the market return, for example, the S&P 500 composite index return. The state-space form can be obtained as

```
> ssf.reg=GetSsfReg(cbind(1,sp)) % 'sp' is market return.
> ssf.reg
$mPhi:
      [,1] [,2]
[1,]      1      0
[2,]      0      1
[3,]      0      0
$mOmega:
      [,1] [,2] [,3]
[1,]      0      0      0
[2,]      0      0      0
[3,]      0      0      1
$mSigma:
      [,1] [,2]
[1,]     -1      0
[2,]      0     -1
[3,]      0      0
$mJPhi:
      [,1] [,2]
[1,]     -1     -1
[2,]     -1     -1
[3,]      1      2
$mX:
```

numeric matrix: 168 rows, 2 columns.

```

      sp
[1,] 1 -0.075187
...
[168,] 1 0.05002

```

11.3.4 Linear Regression Models with ARMA Errors

Consider the regression model with ARMA(p, q) errors:

$$y_t = \mathbf{x}_t' \boldsymbol{\beta} + z_t, \quad \phi(B)z_t = \theta(B)a_t, \quad (11.47)$$

where $a_t \sim N(0, \sigma_a^2)$ and $\mathbf{x}_t$ is a k -dimensional vector of explanatory variables. A special case of this model is the nonzero mean ARMA(p, q) model in which $\mathbf{x}_t = 1$ for all t and $\boldsymbol{\beta}$ becomes a scalar parameter. Let s_t be a state vector for the z_t series, for example, that defined in Eq. (11.39). We can define a state vector s_t^* for y_t as

$$s_t^* = \begin{bmatrix} s_t \\ \boldsymbol{\beta}_t \end{bmatrix}, \quad (11.48)$$

where $\boldsymbol{\beta}_t = \boldsymbol{\beta}$ for all t . Then, a state-space form for y_t is

$$s_{t+1}^* = \mathbf{T}^* s_t^* + \mathbf{R}^* \eta_t, \quad (11.49)$$

$$y_t = \mathbf{Z}_t^* s_t^*, \quad (11.50)$$

where $\mathbf{Z}_t^* = (1, 0, \dots, 0, \mathbf{x}_t')$, $m = \max(p, q + 1)$, and

$$\mathbf{T}^* = \begin{bmatrix} \mathbf{T} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_k \end{bmatrix}, \quad \mathbf{R}^* = \begin{bmatrix} \mathbf{R} \\ \mathbf{0} \end{bmatrix},$$

where $\mathbf{T}$ and $\mathbf{R}$ are defined in Eq. (11.39). In a compact form, we have the state-space model

$$\begin{bmatrix} s_{t+1}^* \\ y_t \end{bmatrix} = \begin{bmatrix} \mathbf{T}^* \\ \mathbf{Z}_t^* \end{bmatrix} s_t^* + \begin{bmatrix} \mathbf{R}^* \eta_t \\ 0 \end{bmatrix}.$$

SsfPack Command

SsdPack uses the command `GetSsfRegArma` to construct a state-space form for linear regression models with ARMA errors. The arguments of the command can be found using the command `args(GetSsfRegArma)`. They consist of a data matrix for the explanatory variables and ARMA model specification. To illustrate, consider the model

$$y_t = \beta_0 + \beta_1 x_t + z_t, \quad t = 1, \dots, 168,$$

$$z_t = 1.2z_{t-1} - 0.35z_{t-2} + a_t - 0.25a_{t-1}, \quad a_t \sim N(0, \sigma_a^2).$$

We use the notation X to denote the $T \times 2$ matrix of regressors $(1, x_t)$. A state-space form for the prior model can be obtained as

```
> ssf.reg.arma21=GetSsfRegArma(X,ar=c(1.2,-0.35),
+ ma=c(-0.25))
> ssf.reg.arma21
$mPhi:
      [,1] [,2] [,3] [,4]
[1,]  1.20   1    0    0
[2,] -0.35   0    0    0
[3,]  0.00   0    1    0
[4,]  0.00   0    0    1
[5,]  1.00   0    0    0
$mOmega:
      [,1] [,2] [,3] [,4] [,5]
[1,]  1.00 -0.2500   0    0    0
[2,] -0.25  0.0625   0    0    0
[3,]  0.00  0.0000   0    0    0
[4,]  0.00  0.0000   0    0    0
[5,]  0.00  0.0000   0    0    0
$mSigma:
      [,1] [,2] [,3] [,4]
[1,]  3.35595 -1.229260   0    0
[2,] -1.22926  0.473604   0    0
[3,]  0.00000  0.000000  -1    0
[4,]  0.00000  0.000000   0   -1
[5,]  0.00000  0.000000   0    0
$mJPhi:
      [,1] [,2] [,3] [,4]
[1,]   -1   -1   -1   -1
[2,]   -1   -1   -1   -1
[3,]   -1   -1   -1   -1
[4,]   -1   -1   -1   -1
[5,]   -1   -1    1    2
$mX:
numeric matrix: 168 rows, 2 columns.
      xt
[1,]  1 0.4993
...
[168,]  1 0.7561
```

11.3.5 Scalar Unobserved Component Model

The basic univariate unobserved component model, or the *structural time series model* (STSM), assumes the form

$$y_t = \mu_t + \gamma_t + \bar{\omega}_t + e_t, \quad (11.51)$$

where μ_t , γ_t , and $\bar{\omega}_t$ represent the unobserved *trend*, *seasonal*, and *cycle* components, respectively, and e_t is the unobserved *irregular* component. In the literature, a nonstationary (possibly double-unit-root) model is commonly used for the trend component:

$$\begin{aligned} \mu_{t+1} &= \mu_t + \beta_t + \eta_t, & \eta_t &\sim N(0, \sigma_\eta^2), \\ \beta_t &= \beta_{t-1} + \varsigma_t, & \varsigma_t &\sim N(0, \sigma_\varsigma^2), \end{aligned} \quad (11.52)$$

where $\mu_1 \sim N(0, \xi)$ and $\beta_1 \sim N(0, \xi)$ with ξ a large real number, for example, $\xi = 10^8$. See, for instance, Kitagawa and Gersch (1996). If $\sigma_\varsigma = 0$, then μ_t follows a random walk with drift β_1 . If $\sigma_\varsigma = \sigma_\eta = 0$, then μ_t represents a linear deterministic trend.

The seasonal component γ_t assumes the form

$$(1 + B + \cdots + B^{s-1})\gamma_t = \omega_t, \quad \omega_t \sim N(0, \sigma_\omega^2), \quad (11.53)$$

where s is the number of seasons in a year, that is, the period of the seasonality. If $\sigma_\omega = 0$, then the seasonal pattern is deterministic. The cycle component is postulated as

$$\begin{bmatrix} \bar{\omega}_{t+1} \\ \bar{\omega}_{t+1}^* \end{bmatrix} = \delta \begin{bmatrix} \cos(\lambda_c) & \sin(\lambda_c) \\ -\sin(\lambda_c) & \cos(\lambda_c) \end{bmatrix} \begin{bmatrix} \bar{\omega}_t \\ \bar{\omega}_t^* \end{bmatrix} + \begin{bmatrix} \varepsilon_t \\ \varepsilon_t^* \end{bmatrix}, \quad (11.54)$$

where

$$\begin{bmatrix} \varepsilon_t \\ \varepsilon_t^* \end{bmatrix} \sim N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \sigma_\varepsilon^2 (1 - \delta^2) \mathbf{I}_2 \right),$$

$\bar{\omega}_0 \sim N(0, \sigma_\varepsilon^2)$, $\bar{\omega}_0^* \sim N(0, \sigma_\varepsilon^2)$, and $\text{Cov}(\bar{\omega}_0, \bar{\omega}_0^*) = 0$, $\delta \in (0, 1]$ is called a *damping* factor, and the frequency of the cycle is $\lambda_c = 2\pi/q$ with q being the period. If $\delta = 1$, then the cycle becomes a deterministic sine–cosine wave.

SsfPack/S-Plus Command

The command `GetSsfStsm` constructs a state-space form for the structural time series model. It allows for 10 cycle components; see the output of the command `args(GetSsfStsm)`. Table 11.4 provides a summary of the arguments and their corresponding symbols of the model. To illustrate, consider the local trend model in Eqs. (11.1) and (11.2) with $\sigma_e = 0.4$ and $\sigma_\eta = 0.2$. This is a special case of the scalar unobserved component model. One can obtain a state-space form as

TABLE 11.4 Arguments of Command GetSsfStsm in SsfPack/S-Plus

Argument	STSM parameter
irregular	σ_e
level	σ_η
slope	σ_ζ
seasonalDummy	σ_ω, s
seasonalTrig	σ_ω, s
SeasonalHS	σ_ω, s
Cycle0	$\sigma_\varepsilon, \lambda_c, \delta$
$\vdots$	$\vdots$
Cycle9	$\sigma_\varepsilon, \lambda_c, \delta$

```

> ssf.stsm=GetSsfStsm(irregular=0.4,level=0.2)
> ssf.stsm
$mPhi:
      [,1]
[1,]     1
[2,]     1
$mOmega:
      [,1] [,2]
[1,] 0.04 0.00
[2,] 0.00 0.16
$mSigma:
      [,1]
[1,]    -1
[2,]     0

```

11.4 KALMAN FILTER AND SMOOTHING

In this section, we study the Kalman filter and various smoothing methods for the general state-space model in Eqs. (11.26) and (11.27). The derivation follows closely the steps taken in Section 11.1. For readers interested in applications, this section can be skipped at the first read. A good reference for this section is Durbin and Koopman (2001, Chapter 4).

11.4.1 Kalman Filter

Recall that the aim of the Kalman filter is to obtain recursively the conditional distribution of s_{t+1} given the data $F_t = \{y_1, \dots, y_t\}$ and the model. Since the conditional distribution involved is normal, it suffices to study the conditional mean and covariance matrix. Let $s_{j|i}$ and $\Sigma_{j|i}$ be the conditional mean and covariance matrix of s_j given F_i , that is, $s_j|F_i \sim N(s_{j|i}, \Sigma_{j|i})$. From Eq. (11.26),

$$s_{t+1|t} = E(d_t + T_t s_t + R_t \eta_t | F_t) = d_t + T_t s_{t|t}, \quad (11.55)$$

$$\Sigma_{t+1|t} = \text{Var}(T_t s_t + R_t \eta_t | F_t) = T_t \Sigma_{t|t} T_t' + R_t Q_t R_t'. \quad (11.56)$$

Similarly to that of Section 11.1, let $y_{t|t-1}$ be the conditional mean of y_t given F_{t-1} . From Eq. (11.27),

$$y_{t|t-1} = c_t + Z_t s_{t|t-1}.$$

Let

$$v_t = y_t - y_{t|t-1} = y_t - (c_t + Z_t s_{t|t-1}) = Z_t(s_t - s_{t|t-1}) + e_t, \quad (11.57)$$

be the 1-step-ahead forecast error of y_t given F_{t-1} . It is easy to see that (a) $E(v_t | F_{t-1}) = \mathbf{0}$; (b) v_t is independent of F_{t-1} , that is, $\text{Cov}(v_t, y_j) = \mathbf{0}$ for $1 \leq j < t$; and (c) $\{v_t\}$ is a sequence of independent normal random vectors. Also, let $V_t = \text{Var}(v_t | F_{t-1}) = \text{Var}(v_t)$ be the covariance matrix of the 1-step-ahead forecast error. From Eq. (11.57), we have

$$V_t = \text{Var}[Z_t(s_t - s_{t|t-1}) + e_t] = Z_t \Sigma_{t|t-1} Z_t' + H_t. \quad (11.58)$$

Since $F_t = \{F_{t-1}, y_t\} = \{F_{t-1}, v_t\}$, we can apply Theorem 11.1 to obtain

$$\begin{aligned} s_{t|t} &= E(s_t | F_t) = E(s_t | F_{t-1}, v_t) \\ &= E(s_t | F_{t-1}) + \text{Cov}(s_t, v_t) [\text{Var}(v_t)]^{-1} v_t \\ &= s_{t|t-1} + C_t V_t^{-1} v_t, \end{aligned} \quad (11.59)$$

where $C_t = \text{Cov}(s_t, v_t | F_{t-1})$ given by

$$\begin{aligned} C_t &= \text{Cov}(s_t, v_t | F_{t-1}) = \text{Cov}[s_t, Z_t(s_t - s_{t|t-1}) + e_t | F_{t-1}] \\ &= \text{Cov}[s_t, Z_t(s_t - s_{t|t-1}) | F_{t-1}] = \Sigma_{t|t-1} Z_t'. \end{aligned}$$

Here we assume that V_t is invertible because H_t is. Using Eqs. (11.55) and (11.59), we obtain

$$s_{t+1|t} = d_t + T_t s_{t|t-1} + T_t C_t V_t^{-1} v_t = d_t + T_t s_{t|t-1} + K_t v_t, \quad (11.60)$$

where

$$K_t = T_t C_t V_t^{-1} = T_t \Sigma_{t|t-1} Z_t' V_t^{-1}, \quad (11.61)$$

which is the *Kalman gain* at time t . Applying Theorem 11.1(2), we have

$$\begin{aligned}
 \Sigma_{t|t} &= \text{Var}(s_t | F_{t-1}, v_t) \\
 &= \text{Var}(s_t | F_{t-1}) - \text{Cov}(s_t, v_t) [\text{Var}(v_t)]^{-1} \text{Cov}(s_t, v_t)' \\
 &= \Sigma_{t|t-1} - C_t V_t^{-1} C_t' \\
 &= \Sigma_{t|t-1} - \Sigma_{t|t-1} Z_t' V_t^{-1} Z_t \Sigma_{t|t-1}.
 \end{aligned} \tag{11.62}$$

Plugging Eq. (11.62) into Eq. (11.56) and using Eq. (11.61), we obtain

$$\Sigma_{t+1|t} = T_t \Sigma_{t|t-1} L_t' + R_t Q_t R_t', \tag{11.63}$$

where

$$L_t = T_t - K_t Z_t.$$

Putting the prior equations together, we obtain the celebrated Kalman filter for the state-space model in Eqs. (11.26) and (11.27). Given the starting values $s_{1|0}$ and $\Sigma_{1|0}$, the Kalman filter algorithm is

$$\begin{aligned}
 v_t &= y_t - c_t - Z_t s_{t|t-1}, \\
 V_t &= Z_t \Sigma_{t|t-1} Z_t' + H_t, \\
 K_t &= T_t \Sigma_{t|t-1} Z_t' V_t^{-1}, \\
 L_t &= T_t - K_t Z_t, \\
 s_{t+1|t} &= d_t + T_t s_{t|t-1} + K_t v_t, \\
 \Sigma_{t+1|t} &= T_t \Sigma_{t|t-1} L_t' + R_t Q_t R_t', \quad t = 1, \dots, T.
 \end{aligned} \tag{11.64}$$

If the filtered quantities $s_{t|t}$ and $\Sigma_{t|t}$ are also of interest, then we modify the filter to include the contemporaneous filtering equations in Eqs. (11.59) and (11.62). The resulting algorithm is

$$\begin{aligned}
 v_t &= y_t - c_t - Z_t s_{t|t-1}, \\
 C_t &= \Sigma_{t|t-1} Z_t', \\
 V_t &= Z_t \Sigma_{t|t-1} Z_t' + H_t = Z_t C_t + H_t, \\
 s_{t|t} &= s_{t|t-1} + C_t V_t^{-1} v_t, \\
 \Sigma_{t|t} &= \Sigma_{t|t-1} - C_t V_t^{-1} C_t', \\
 s_{t+1|t} &= d_t + T_t s_{t|t}, \\
 \Sigma_{t+1|t} &= T_t \Sigma_{t|t} T_t' + R_t Q_t R_t'.
 \end{aligned}$$

Steady State

If the state-space model is time invariant, that is, all system matrices are time invariant, then the matrices $\Sigma_{t|t-1}$ converge to a constant matrix Σ_* , which is a solution of the matrix equation

$$\Sigma_* = T\Sigma_*T' - T\Sigma_*ZV^{-1}Z\Sigma_*T' + RQR',$$

where $V = Z\Sigma_*Z' + H$. The solution that is reached after convergence to Σ_* is referred to as the *steady-state solution* of the Kalman filter. Once the steady state is reached, V_t , K_t , and $\Sigma_{t+1|t}$ are all constant. This can lead to considerable saving in computing time.

11.4.2 State Estimation Error and Forecast Error

Define the state prediction error as

$$\mathbf{x}_t = \mathbf{s}_t - \mathbf{s}_{t|t-1}.$$

From the definition, the covariance matrix of $\mathbf{x}_t$ is $\text{Var}(\mathbf{x}_t|F_{t-1}) = \text{Var}(\mathbf{s}_t|F_{t-1}) = \Sigma_{t|t-1}$. Following Section 11.1, we investigate properties of $\mathbf{x}_t$. First, from Eq. (11.57),

$$\mathbf{v}_t = \mathbf{Z}_t(\mathbf{s}_t - \mathbf{s}_{t|t-1}) + \mathbf{e}_t = \mathbf{Z}_t\mathbf{x}_t + \mathbf{e}_t.$$

Second, from Eqs. (11.64) and (11.26), and the prior equation, we have

$$\begin{aligned} \mathbf{x}_{t+1} &= \mathbf{s}_{t+1} - \mathbf{s}_{t+1|t} \\ &= \mathbf{T}_t(\mathbf{s}_t - \mathbf{s}_{t|t-1}) + \mathbf{R}_t\boldsymbol{\eta}_t - \mathbf{K}_t\mathbf{v}_t \\ &= \mathbf{T}_t\mathbf{x}_t + \mathbf{R}_t\boldsymbol{\eta}_t - \mathbf{K}_t(\mathbf{Z}_t\mathbf{x}_t + \mathbf{e}_t) \\ &= \mathbf{L}_t\mathbf{x}_t + \mathbf{R}_t\boldsymbol{\eta}_t - \mathbf{K}_t\mathbf{e}_t, \end{aligned}$$

where, as before, $\mathbf{L}_t = \mathbf{T}_t - \mathbf{K}_t\mathbf{Z}_t$. Consequently, we obtain a state-space form for $\mathbf{v}_t$ as

$$\mathbf{v}_t = \mathbf{Z}_t\mathbf{x}_t + \mathbf{e}_t, \quad \mathbf{x}_{t+1} = \mathbf{L}_t\mathbf{x}_t + \mathbf{R}_t\boldsymbol{\eta}_t - \mathbf{K}_t\mathbf{e}_t, \quad (11.65)$$

with $\mathbf{x}_1 = \mathbf{s}_1 - \mathbf{s}_{1|0}$ for $t = 1, \dots, T$.

Finally, similar to the local-trend model in Section 11.1, we can show that the 1-step-ahead forecast errors $\{\mathbf{v}_t\}$ are independent of each other and $\{\mathbf{v}_t, \dots, \mathbf{v}_T\}$ is independent of F_{t-1} .

11.4.3 State Smoothing

State smoothing focuses on the conditional distribution of s_t given F_T . Notice that (a) F_{t-1} and $\{v_t, \dots, v_T\}$ are independent and (b) v_t are serially independent. We can apply Theorem 11.1 to the joint distribution of s_t and $\{v_t, \dots, v_T\}$ given F_{t-1} and obtain

$$\begin{aligned} s_{t|T} &= E(s_t|F_T) = E(s_t|F_{t-1}, v_t, \dots, v_T) \\ &= E(s_t|F_{t-1}) + \sum_{j=t}^T \text{Cov}(s_t, v_j)[\text{Var}(v_j)]^{-1} v_j \\ &= s_{t|t-1} + \sum_{j=t}^T \text{Cov}(s_t, v_j) V_j^{-1} v_j, \end{aligned} \quad (11.66)$$

where the covariance matrices are conditional on F_{t-1} . The covariance matrices $\text{Cov}(s_t, v_j)$ for $j = t, \dots, T$ can be derived as follows. By Eq. (11.65),

$$\begin{aligned} \text{Cov}(s_t, v_j) &= E(s_t v_j') \\ &= E[s_t (Z_j x_j + e_j)'] = E(s_t x_j') Z_j', \quad j = t, \dots, T. \end{aligned} \quad (11.67)$$

Furthermore,

$$\begin{aligned} E(s_t x_t') &= E[s_t (s_t - s_{t|t-1})'] = \text{Var}(s_t) = \Sigma_{t|t-1}, \\ E(s_t x_{t+1}') &= E[s_t (L_t x_t + R_t \eta_t - K_t e_t)'] = \Sigma_{t|t-1} L_t', \\ E(s_t x_{t+2}') &= \Sigma_{t|t-1} L_t' L_{t+1}', \\ &\vdots \\ E(s_t x_T') &= \Sigma_{t|t-1} L_t' \cdots L_{T-1}'. \end{aligned} \quad (11.68)$$

Plugging the prior two equations into Eq. (11.66), we have

$$\begin{aligned} s_{T|T} &= s_{T|T-1} + \Sigma_{T|T-1} Z_T' V_T^{-1} v_T, \\ s_{T-1|T} &= s_{T-1|T-2} + \Sigma_{T-1|T-2} Z_{T-1}' V_{T-1}^{-1} v_{T-1} + \Sigma_{T-1|T-2} L_{T-1}' Z_T' V_T^{-1} v_T, \\ s_{t|T} &= s_{t|t-1} + \Sigma_{t|t-1} Z_t' V_t^{-1} v_t + \Sigma_{t|t-1} L_t' Z_{t+1}' V_{t+1}^{-1} v_{t+1} \\ &\quad + \cdots + \Sigma_{t|t-1} L_t' L_{t+1}' \cdots L_{T-1}' Z_T' V_T^{-1} v_T, \end{aligned}$$

for $t = T-2, T-3, \dots, 1$, where it is understood that $L_t' \cdots L_{T-1}' = I_m$ when $t = T$. These smoothed state vectors can be expressed as

$$s_{t|T} = s_{t|t-1} + \Sigma_{t|t-1} q_{t-1}, \quad (11.69)$$

where $\mathbf{q}_{T-1} = \mathbf{Z}'_T \mathbf{V}_T^{-1} \mathbf{v}_T$, $\mathbf{q}_{T-2} = \mathbf{Z}'_{T-1} \mathbf{V}_{T-1}^{-1} \mathbf{v}_{T-1} + \mathbf{L}'_{T-1} \mathbf{Z}'_T \mathbf{V}_T^{-1} \mathbf{v}_T$, and

$$\mathbf{q}_{t-1} = \mathbf{Z}'_t \mathbf{V}_t^{-1} \mathbf{v}_t + \mathbf{L}'_t \mathbf{Z}'_{t+1} \mathbf{V}_{t+1}^{-1} \mathbf{v}_{t+1} + \cdots + \mathbf{L}'_t \mathbf{L}'_{t+1} \cdots \mathbf{L}'_{T-1} \mathbf{Z}'_T \mathbf{V}_T^{-1} \mathbf{v}_T,$$

for $t = T-2, T-3, \dots, 1$. The quantity $\mathbf{q}_{t-1}$ is a weighted sum of the 1-step-ahead forecast errors $\mathbf{v}_j$ occurring after time $t-1$. From the definition in the prior equation, $\mathbf{q}_t$ can be computed recursively backward as

$$\mathbf{q}_{t-1} = \mathbf{Z}'_t \mathbf{V}_t^{-1} \mathbf{v}_t + \mathbf{L}'_t \mathbf{q}_t, \quad t = T, \dots, 1, \quad (11.70)$$

with $\mathbf{q}_T = \mathbf{0}$. Putting the equations together, we have a backward recursion for the smoothed state vectors as

$$\mathbf{q}_{t-1} = \mathbf{Z}'_t \mathbf{V}_t^{-1} \mathbf{v}_t + \mathbf{L}'_t \mathbf{q}_t, \quad s_{t|T} = s_{t|t-1} + \boldsymbol{\Sigma}_{t|t-1} \mathbf{q}_{t-1}, \quad t = T, \dots, 1, \quad (11.71)$$

starting with $\mathbf{q}_T = \mathbf{0}$, where $s_{t|t-1}$, $\boldsymbol{\Sigma}_{t|t-1}$, $\mathbf{L}_t$, and $\mathbf{V}_t$ are available from the Kalman filter. This algorithm is referred to as the *fixed interval smoother* in the literature; see de Jong (1989) and the references therein.

Covariance Matrix of Smoothed State Vector

Next, we derive the covariance matrices of the smoothed state vectors. Applying Theorem 11.1(4) to the conditional joint distribution of s_t and $\{\mathbf{v}_t, \dots, \mathbf{v}_T\}$ given F_{t-1} , we have

$$\boldsymbol{\Sigma}_{t|T} = \boldsymbol{\Sigma}_{t|t-1} - \sum_{j=t}^T \text{Cov}(s_t, \mathbf{v}_j) [\text{Var}(\mathbf{v}_j)]^{-1} [\text{Cov}(s_t, \mathbf{v}_j)]'.$$

Using the covariance matrices in Eqs. (11.67) and (11.68), we further obtain

$$\begin{aligned} \boldsymbol{\Sigma}_{t|T} &= \boldsymbol{\Sigma}_{t|t-1} - \boldsymbol{\Sigma}_{t|t-1} \mathbf{Z}'_t \mathbf{V}_t^{-1} \mathbf{Z}_t \boldsymbol{\Sigma}_{t|t-1} - \boldsymbol{\Sigma}_{t|t-1} \mathbf{L}'_t \mathbf{Z}'_{t+1} \mathbf{V}_{t+1}^{-1} \mathbf{Z}_{t+1} \mathbf{L}_t \boldsymbol{\Sigma}_{t|t-1} \\ &\quad - \cdots - \boldsymbol{\Sigma}_{t|t-1} \mathbf{L}'_t \cdots \mathbf{L}'_{T-1} \mathbf{Z}'_T \mathbf{V}_T^{-1} \mathbf{Z}_T \mathbf{L}_{T-1} \cdots \mathbf{L}_t \boldsymbol{\Sigma}_{t|t-1} \\ &= \boldsymbol{\Sigma}_{t|t-1} - \boldsymbol{\Sigma}_{t|t-1} \mathbf{M}_{t-1} \boldsymbol{\Sigma}_{t|t-1}, \end{aligned}$$

where

$$\begin{aligned} \mathbf{M}_{t-1} &= \mathbf{Z}'_t \mathbf{V}_t^{-1} \mathbf{Z}_t + \mathbf{L}'_t \mathbf{Z}'_{t+1} \mathbf{V}_{t+1}^{-1} \mathbf{Z}_{t+1} \mathbf{L}_t \\ &\quad + \cdots + \mathbf{L}'_t \cdots \mathbf{L}'_{T-1} \mathbf{Z}'_T \mathbf{V}_T^{-1} \mathbf{Z}_T \mathbf{L}_{T-1} \cdots \mathbf{L}_t. \end{aligned}$$

Again, $\mathbf{L}'_t \cdots \mathbf{L}_{T-1} = \mathbf{I}_m$ when $t = T$. From its definition, the $\mathbf{M}_{t-1}$ matrix satisfies

$$\mathbf{M}_{t-1} = \mathbf{Z}'_t \mathbf{V}_t^{-1} \mathbf{Z}_t + \mathbf{L}'_t \mathbf{M}_t \mathbf{L}_t, \quad t = T, \dots, 1, \quad (11.72)$$

with the starting value $\mathbf{M}_T = \mathbf{0}$. Collecting the results, we obtain a backward recursion to compute $\Sigma_{t|T}$ as

$$\mathbf{M}_{t-1} = \mathbf{Z}_t' \mathbf{V}_t^{-1} \mathbf{Z}_t + \mathbf{L}_t' \mathbf{M}_t \mathbf{L}_t, \quad \Sigma_{t|T} = \Sigma_{t|t-1} - \Sigma_{t|t-1} \mathbf{M}_{t-1} \Sigma_{t|t-1}, \quad (11.73)$$

for $t = T, \dots, 1$ with $\mathbf{M}_T = \mathbf{0}$. Note that, like that of the local trend model in Section 11.1, $\mathbf{M}_t = \text{Var}(\mathbf{q}_t)$.

Combining the two backward recursions of smoothed state vectors, we have

$$\begin{aligned} \mathbf{q}_{t-1} &= \mathbf{Z}_t' \mathbf{V}_t^{-1} \mathbf{v}_t + \mathbf{L}_t' \mathbf{q}_t, \\ \mathbf{s}_{t|T} &= \mathbf{s}_{t|t-1} + \Sigma_{t|t-1} \mathbf{q}_{t-1}, \\ \mathbf{M}_{t-1} &= \mathbf{Z}_t' \mathbf{V}_t^{-1} \mathbf{Z}_t + \mathbf{L}_t' \mathbf{M}_t \mathbf{L}_t, \\ \Sigma_{t|T} &= \Sigma_{t|t-1} - \Sigma_{t|t-1} \mathbf{M}_{t-1} \Sigma_{t|t-1}, \quad t = T, \dots, 1, \end{aligned} \quad (11.74)$$

with $\mathbf{q}_T = \mathbf{0}$ and $\mathbf{M}_T = \mathbf{0}$.

Suppose that the state-space model in Eqs. (11.26) and (11.27) is known. Application of the Kalman filter and state smoothing can proceed in two steps. First, the Kalman filter in Eq. (11.64) is used for $t = 1, \dots, T$ and the quantities $\mathbf{v}_t, \mathbf{V}_t, \mathbf{K}_t, \mathbf{s}_{t|t-1}$, and $\Sigma_{t|t-1}$ are stored. Second, the state smoothing algorithm in Eq. (11.74) is applied for $t = T, T-1, \dots, 1$ to obtain $\mathbf{s}_{t|T}$ and $\Sigma_{t|T}$.

11.4.4 Disturbance Smoothing

Let $\mathbf{e}_{t|T} = E(\mathbf{e}_t | F_T)$ and $\boldsymbol{\eta}_{t|T} = E(\boldsymbol{\eta}_t | F_T)$ be the smoothed disturbances of the observation and transition equation, respectively. These *smoothed disturbances* are useful in many applications, for example, in model checking. In this section, we study recursive algorithms to compute smoothed disturbances and their covariance matrices. Again, applying Theorem 11.1 to the conditional joint distribution of $\mathbf{e}_t$ and $\{\mathbf{v}_t, \dots, \mathbf{v}_T\}$ given F_{t-1} , we obtain

$$\mathbf{e}_{t|T} = E(\mathbf{e}_t | F_{t-1}, \mathbf{v}_t, \dots, \mathbf{v}_T) = \sum_{j=t}^T E(\mathbf{e}_t \mathbf{v}_j') \mathbf{V}_j^{-1} \mathbf{v}_j, \quad (11.75)$$

where $E(\mathbf{e}_t | F_{t-1}) = \mathbf{0}$ is used. Using Eq. (11.65),

$$E(\mathbf{e}_t \mathbf{v}_j') = E(\mathbf{e}_t \mathbf{x}_j') \mathbf{Z}_j' + E(\mathbf{e}_t \mathbf{e}_j').$$

Since $E(\mathbf{e}_t \mathbf{x}_t') = \mathbf{0}$, we have

$$E(\mathbf{e}_t \mathbf{v}_j') = \begin{cases} \mathbf{H}_t, & \text{if } j = t, \\ E(\mathbf{e}_t \mathbf{x}_j') \mathbf{Z}_j', & \text{for } j = t+1, \dots, T. \end{cases} \quad (11.76)$$

Using Eq. (11.65) repeatedly and the independence between $\{e_t\}$ and $\{\eta_t\}$, we obtain

$$\begin{aligned} E(e_t x'_{t+1}) &= -H_t K'_t, \\ E(e_t x'_{t+2}) &= -H_t K'_t L'_{t+1}, \\ &\vdots \\ E(e_t x'_T) &= -H_t K'_t L'_{t+1} \cdots L'_{T-1}, \end{aligned} \quad (11.77)$$

where it is understood that $L'_{t+1} \cdots L'_{T-1} = I_m$ if $t = T - 1$. Based on Eqs. (11.76) and (11.77),

$$\begin{aligned} e_{t|T} &= H_t (V_t^{-1} v_t - K'_t Z'_{t+1} V_{t+1}^{-1} v_{t+1} - \cdots - K'_t L'_{t+1} \cdots L'_{T-1} Z'_T V_T^{-1} v_T) \\ &= H_t (V_t^{-1} v_t - K'_t q_t) \\ &= H_t o_t, \quad t = T, \dots, 1, \end{aligned} \quad (11.78)$$

where q_t is defined in Eq. (11.69) and $o_t = V_t^{-1} v_t - K'_t q_t$. We refer to o_t as the *smoothing measurement error*.

The smoothed disturbance $\eta_{t|T}$ can be derived analogously, and we have

$$\eta_{t|T} = \sum_{j=t}^T E(\eta_t v'_j) V_j^{-1} v_j. \quad (11.79)$$

The state-space form in Eq. (11.69) gives

$$E(\eta_t v'_j) = \begin{cases} Q_t R'_t Z'_{t+1}, & \text{if } j = t + 1, \\ E(\eta_t x'_j) Z'_j, & \text{if } j = t + 2, \dots, T, \end{cases}$$

where

$$\begin{aligned} E(\eta_t x'_{t+2}) &= Q_t R'_t L'_{t+1}, \\ E(\eta_t x'_{t+3}) &= Q_t R'_t L'_{t+1} L'_{t+2}, \\ &\vdots \\ E(\eta_t x'_T) &= Q_t R'_t L'_{t+1} \cdots L'_{T-1}, \end{aligned}$$

for $t = 1, \dots, T$. Consequently, Eq. (11.79) implies

$$\begin{aligned} \eta_{t|T} &= Q_t R'_t (Z'_{t+1} V_{t+1}^{-1} v_{t+1} + L'_{t+1} Z'_{t+2} V_{t+2}^{-1} v_{t+2} \\ &\quad + \cdots + L'_{t+1} \cdots L'_{T-1} Z'_T V_T^{-1} v_T) \\ &= Q_t R'_t q_t, \quad t = T, \dots, 1, \end{aligned} \quad (11.80)$$

where $\mathbf{q}_t$ is defined earlier in Eq. (11.70).

Koopman (1993) uses the smoothed disturbance $\eta_{t|T}$ to derive a new recursion for computing $\mathbf{s}_{t|T}$. From the transition equation in Eq. (11.26),

$$\mathbf{s}_{t+1|T} = \mathbf{d}_t + \mathbf{T}_t \mathbf{s}_{t|T} + \mathbf{R}_t \eta_{t|T}.$$

Using Eq. (11.80), we have

$$\mathbf{s}_{t+1|T} = \mathbf{d}_t + \mathbf{T}_t \mathbf{s}_{t|T} + \mathbf{R}_t \mathbf{Q}_t \mathbf{R}_t' \mathbf{q}_t, \quad t = 1, \dots, T, \quad (11.81)$$

where the initial value is $\mathbf{s}_{1|T} = \mathbf{s}_{1|0} + \mathbf{\Sigma}_{1|0} \mathbf{q}_0$ with $\mathbf{q}_0$ obtained from the recursion in Eq. (11.70).

Covariance Matrices of Smoothed Disturbances

The covariance matrix of the smoothed disturbance can also be obtained using Theorem 11.1. Specifically,

$$\begin{aligned} \text{Var}(\mathbf{e}_t | F_T) &= \text{Var}(\mathbf{e}_t | F_{t-1}, \mathbf{v}_t, \dots, \mathbf{v}_T) \\ &= \text{Var}(\mathbf{e}_t | F_{t-1}) - \sum_{j=t}^T \text{Cov}(\mathbf{e}_t, \mathbf{v}_j) \mathbf{V}_j^{-1} [\text{Cov}(\mathbf{e}_t, \mathbf{v}_j)]'. \end{aligned}$$

Note that $\text{Cov}(\mathbf{e}_t, \mathbf{v}_j) = E(\mathbf{e}_t \mathbf{v}_j')$, which is given in Eq. (11.76). Thus, we have

$$\begin{aligned} \text{Var}(\mathbf{e}_t | F_T) &= \mathbf{H}_t - \mathbf{H}_t (\mathbf{V}_t^{-1} + \mathbf{K}_t' \mathbf{Z}_{t+1}' \mathbf{V}_{t+1}^{-1} \mathbf{Z}_{t+1} \mathbf{K}_t \\ &\quad + \mathbf{K}_t' \mathbf{L}_{t+1}' \mathbf{Z}_{t+2}' \mathbf{V}_{t+2}^{-1} \mathbf{Z}_{t+2} \mathbf{L}_{t+1} \mathbf{K}_t \\ &\quad + \dots + \mathbf{K}_t' \mathbf{L}_{t+1}' \dots \mathbf{L}_{T-1}' \mathbf{Z}_T' \mathbf{V}_T^{-1} \mathbf{Z}_T \mathbf{L}_{T-1} \dots \mathbf{L}_{t+1} \mathbf{K}_t) \mathbf{H}_t \\ &= \mathbf{H}_t - \mathbf{H}_t (\mathbf{V}_t^{-1} + \mathbf{K}_t' \mathbf{M}_t \mathbf{K}_t) \mathbf{H}_t \\ &= \mathbf{H}_t - \mathbf{H}_t \mathbf{N}_t \mathbf{H}_t, \end{aligned}$$

where $\mathbf{N}_t = \mathbf{V}_t^{-1} + \mathbf{K}_t' \mathbf{M}_t \mathbf{K}_t$, where $\mathbf{M}_t$ is given in Eq. (11.72). Similarly,

$$\text{Var}(\eta_t | F_T) = \text{Var}(\eta_t) - \sum_{j=t}^T \text{Cov}(\eta_t, \mathbf{v}_j) \mathbf{V}_j^{-1} [\text{Cov}(\eta_t, \mathbf{v}_j)]^{-1},$$

where $\text{Cov}(\eta_t, \mathbf{v}_j) = E(\eta_t \mathbf{v}_j')$, which is given before when we derived the formula for $\eta_{t|T}$. Consequently,

$$\begin{aligned} \text{Var}(\eta_t | F_T) &= \mathbf{Q}_t - \mathbf{Q}_t \mathbf{R}_t' (\mathbf{Z}_{t+1}' \mathbf{V}_{t+1}^{-1} \mathbf{Z}_{t+1} + \mathbf{L}_{t+1}' \mathbf{Z}_{t+2}' \mathbf{V}_{t+2}^{-1} \mathbf{Z}_{t+2} \mathbf{L}_{t+1} \\ &\quad + \dots + \mathbf{L}_{t+1}' \dots \mathbf{L}_{T-1}' \mathbf{Z}_T' \mathbf{V}_T^{-1} \mathbf{Z}_T \mathbf{L}_{T-1} \dots \mathbf{L}_{t+1}) \mathbf{R}_t \mathbf{Q}_t \\ &= \mathbf{Q}_t - \mathbf{Q}_t \mathbf{R}_t' \mathbf{M}_t \mathbf{R}_t \mathbf{Q}_t. \end{aligned}$$

In summary, the disturbance smoothing algorithm is as follows:

$$\begin{aligned}
 \mathbf{e}_{t|T} &= \mathbf{H}_t(\mathbf{V}_t^{-1}\mathbf{v}_t - \mathbf{K}_t'\mathbf{q}_t), \\
 \boldsymbol{\eta}_{t|T} &= \mathbf{Q}_t\mathbf{R}_t'\mathbf{q}_t, \\
 \mathbf{q}_{t-1} &= \mathbf{Z}_t'\mathbf{V}_t^{-1}\mathbf{v}_t + \mathbf{L}_t'\mathbf{q}_t, \\
 \text{Var}(\mathbf{e}_t|F_T) &= \mathbf{H}_t - \mathbf{H}_t(\mathbf{V}_t^{-1} + \mathbf{K}_t'\mathbf{M}_t\mathbf{K}_t)\mathbf{H}_t, \\
 \text{Var}(\boldsymbol{\eta}_t|F_T) &= \mathbf{Q}_t - \mathbf{Q}_t\mathbf{R}_t'\mathbf{M}_t\mathbf{R}_t\mathbf{Q}_t, \\
 \mathbf{M}_{t-1} &= \mathbf{Z}_t'\mathbf{V}_t^{-1}\mathbf{Z}_t + \mathbf{L}_t'\mathbf{M}_t\mathbf{L}_t, \quad t = T, \dots, 1,
 \end{aligned} \tag{11.82}$$

where $\mathbf{q}_T = \mathbf{0}$ and $\mathbf{M}_T = \mathbf{0}$.

11.5 MISSING VALUES

For the general state-space model in Eqs. (11.26) and (11.27), we consider two cases of missing values. First, suppose that similar to the local trend model in Section 11.1 the observations $\mathbf{y}_t$ at $t = \ell + 1, \dots, \ell + h$ are missing. In this case, there is no new information available at these time points and we set

$$\mathbf{v}_t = \mathbf{0}, \quad \mathbf{K}_t = \mathbf{0}, \quad \text{for } t = \ell + 1, \dots, \ell + h.$$

The Kalman filter in Eq. (11.64) can then proceed as usual. That is,

$$\mathbf{s}_{t+1|t} = \mathbf{d}_t + \mathbf{T}_t\mathbf{s}_{t|t-1}, \quad \boldsymbol{\Sigma}_{t+1|t} = \mathbf{T}_t\boldsymbol{\Sigma}_{t|t-1}\mathbf{T}_t' + \mathbf{R}_t\mathbf{Q}_t\mathbf{R}_t',$$

for $t = \ell + 1, \dots, \ell + h$. Similarly, the smoothed state vectors can be computed as usual via Eq. (11.74) with

$$\mathbf{q}_{t-1} = \mathbf{T}_t'\mathbf{q}_t, \quad \mathbf{M}_{t-1} = \mathbf{T}_t'\mathbf{M}_t\mathbf{T}_t,$$

for $t = \ell + 1, \dots, \ell + h$.

In the second case, some components of $\mathbf{y}_t$ are missing. Let $\mathbf{y}_t^* = \mathbf{J}\mathbf{y}_t$ be the vector of observed data at time t , where $\mathbf{J}$ is an indicator matrix identifying the observed data. More specifically, rows of $\mathbf{J}$ are a subset of the rows of the $k \times k$ identity matrix. In this case, the observation equation (11.27) of the model can be transformed as

$$\mathbf{y}_t^* = \mathbf{c}_t^* + \mathbf{Z}_t^*\mathbf{s}_t + \mathbf{e}_t^*,$$

where $\mathbf{c}_t^* = \mathbf{J}\mathbf{c}_t$, $\mathbf{Z}_t^* = \mathbf{J}\mathbf{Z}_t$, and $\mathbf{e}_t^* = \mathbf{J}\mathbf{e}_t$ with covariance matrix $\text{Var}(\mathbf{e}_t^*) = \mathbf{H}_t^* = \mathbf{J}\mathbf{H}_t\mathbf{J}'$. The Kalman filter and state-smoothing recursion continue to apply except that the modified observation equation is used at time t . Consequently, the ease in handling missing values is a nice feature of the state-space model.

11.6 FORECASTING

Suppose that the forecast origin is t and we are interested in predicting $\mathbf{y}_{t+j}$ for $j = 1, \dots, h$, where $h > 0$. Also, we adopt the minimum mean-squared error forecasts. Similar to the ARMA models, the j -step-ahead forecast $\mathbf{y}_t(j)$ turns out to be the expected value of $\mathbf{y}_{t+j}$ given F_t and the model. That is, $\mathbf{y}_t(j) = E(\mathbf{y}_{t+j}|F_t)$. In what follows, we show that these forecasts and the covariance matrices of the associated forecast errors can be obtained via the Kalman filter in Eq. (11.64) by treating $\{\mathbf{y}_{t+1}, \dots, \mathbf{y}_{t+h}\}$ as missing values, that is, the first case in Section 11.5.

Consider the 1-step-ahead forecast. From Eq. (11.27),

$$\mathbf{y}_t(1) = E(\mathbf{y}_{t+1}|F_t) = \mathbf{c}_{t+1} + \mathbf{Z}_{t+1}\mathbf{s}_{t+1|t},$$

where $\mathbf{s}_{t+1|t}$ is available via the Kalman filter at the forecast origin t . The associated forecast error is

$$\mathbf{e}_t(1) = \mathbf{y}_{t+1} - \mathbf{y}_t(1) = \mathbf{Z}_{t+1}(\mathbf{s}_{t+1} - \mathbf{s}_{t+1|t}) + \mathbf{e}_{t+1}.$$

Therefore, the covariance matrix of the 1-step-ahead forecast error is

$$\text{Var}[\mathbf{e}_t(1)] = \mathbf{Z}_{t+1}\mathbf{\Sigma}_{t+1|t}\mathbf{Z}_{t+1}' + \mathbf{H}_{t+1}.$$

This is precisely the covariance matrix $\mathbf{V}_{t+1}$ of the Kalman filter in Eq. (11.64). Thus, we have showed the case for $h = 1$.

Now, for $h > 1$, we consider 1-step- to h -step-ahead forecasts sequentially. From Eq. (11.27), the j -step-ahead forecast is

$$\mathbf{y}_t(j) = \mathbf{c}_{t+j} + \mathbf{Z}_{t+j}\mathbf{s}_{t+j|t}, \quad (11.83)$$

and the associated forecast error is

$$\mathbf{e}_t(j) = \mathbf{Z}_{t+j}(\mathbf{s}_{t+j} - \mathbf{s}_{t+j|t}) + \mathbf{e}_{t+j}.$$

Recall that $\mathbf{s}_{t+j|t}$ and $\mathbf{\Sigma}_{t+j|t}$ are, respectively, the conditional mean and covariance matrix of $\mathbf{s}_{t+j}$ given F_t . The prior equation says that

$$\text{Var}[\mathbf{e}_t(j)] = \mathbf{Z}_{t+j}\mathbf{\Sigma}_{t+j|t}\mathbf{Z}_{t+j}' + \mathbf{H}_{t+j}. \quad (11.84)$$

Furthermore, from Eq. (11.26),

$$\mathbf{s}_{t+j+1|t} = \mathbf{d}_{t+j} + \mathbf{T}_{t+j}\mathbf{s}_{t+j|t},$$

which in turn implies that

$$\mathbf{s}_{t+j+1} - \mathbf{s}_{t+j+1|t} = \mathbf{T}_{t+j}(\mathbf{s}_{t+j} - \mathbf{s}_{t+j|t}) + \mathbf{R}_{t+j}\boldsymbol{\eta}_{t+j}.$$

Consequently,

$$\Sigma_{t+j+1|t} = T_{t+j} \Sigma_{t+j|t} T_{t+j}' + R_{t+j} Q_{t+j} R_{t+j}'. \quad (11.85)$$

Note that $\text{Var}[e_t(j)] = V_{t+j}$ and Eqs. (11.83) and (11.85) are the recursion of the Kalman filter in Eq. (11.64) for $t+j$ with $j = 1, \dots, h$ when $v_{t+j} = \mathbf{0}$ and $K_{t+j} = \mathbf{0}$. Thus, the forecast $y_t(j)$ and the covariance matrix of its forecast error $e_t(j)$ can be obtained via the Kalman filter with missing values.

Finally, the prediction error series $\{v_t\}$ can be used to evaluate the likelihood function for estimation and the standardized prediction errors $D_t^{-1/2} v_t$ can be used for model checking, where $D_t = \text{diag}\{V_t(1, 1), \dots, V_t(k, k)\}$ with $V_t(i, i)$ being the (i, i) th element of V_t .

11.7 APPLICATION

In this section, we consider some applications of the state-space model in finance and business. Our objectives are to highlight the applicability of the model and to demonstrate the practical implementation of the analysis in S-Plus with SsfPack.

Example 11.2. Consider the CAPM for the monthly simple excess returns of General Motors (GM) stock from January 1990 to December 2003; see Chapter 9. We use the simple excess returns of the S&P 500 composite index as the market returns. The returns are in percentages. Our illustration starts with a simple market model

$$r_t = \alpha + \beta r_{M,t} + e_t, \quad e_t \sim N(0, \sigma_e^2), \quad (11.86)$$

for $t = 1, \dots, 168$. This is a fixed-coefficient model and can easily be estimated by the ordinary least-squares (OLS) method. Denote the GM stock return and the market return by gm and sp, respectively. The result follows:

```
> da=read.table('`m-gmsp-excess-9003.txt`',header=F)
> gm=da[,1]
> sp=da[,2]
> fit=OLS(gm~sp)
> summary(fit)
Call:
OLS(formula = gm~sp)
Coefficients:
                Value Std. Error t value Pr(>|t|)
(Intercept)  0.1982  0.6302      0.3145  0.7535
              sp  1.0457  0.1453      7.1962  0.0000
```

Regression Diagnostics:

```

R-Squared 0.2378
Adjusted R-Squared 0.2332
Durbin-Watson Stat 2.0290

```

```

Residual Diagnostics:
      Stat P-Value
Jarque-Bera  2.5348  0.2816
Ljung-Box  24.2132  0.3362

```

Residual standard error: 8.13 on 166 degrees of freedom

Thus, the fitted model is

$$r_t = 0.20 + 1.0457r_{M,t} + e_t, \quad \hat{\sigma}_e = 8.13.$$

Based on the residual diagnostics, the model appears to be adequate for the GM stock returns with adjusted $R^2 = 23.3\%$.

As shown in Section 11.3, model (11.86) is a special case of the state-space model. We then estimate the model using SsfPack. The result is as follows:

```

> reg.m=function(parm,mX=NULL){
+   parm=exp(parm) % log(sigma.e) is used to ensure
positiveness.
+   ssf.reg=GetSsfReg(mX)
+   ssf.reg$mOmega[3,3]=parm[1]
+   CheckSsf(ssf.reg)
+ }
> c.start=c(10)
> X.mtx=cbind(rep(1,168),sp)
> reg.fit=SsfFit(c.start,gm,"reg.m",mX=X.mtx)
RELATIVE FUNCTION CONVERGENCE
> names(reg.fit)
[1] "parameters" "objective" "message" "grad.norm"
"iterations"
[6] "f.evals"      "g.evals"      "hessian"      "scale"        "aux"
[11] "call"         "vcov"
> sqrt(exp(reg.fit$parameters))
[1] 8.130114
> ssf.reg$mOmega[3,3]=exp(reg.fit$parameters)
> reg.s=SsfMomentEst(gm,ssf.reg,task="STSMO")
> reg.s$state.moment[10,]
state.1 state.2
0.1982025 1.045702
> sqrt(reg.s$state.variance[10,])
state.1 state.2
0.6302091 0.1453139

```

As expected, the result is in total agreement with that of the OLS method.

Finally, we entertain the time-varying CAPM of Section 11.3.1. The estimation result, including time plot of the smoothed response variable, is given below. The command `SsfCondDens` is used to compute the smoothed estimates of the state vector and observation without variance estimation.

```
> tv.capm = function(parm, mX=NULL) { %setup model for estimation
+   parm = exp(parm) %parameterize in log for positiveness.
+   Phi.t = rbind(diag(2), rep(0, 2))
+   Omega = diag(parm)
+   JPhi = matrix(-1, 3, 2)
+   JPhi[3, 1] = 1
+   JPhi[3, 2] = 2
+   Sigma = -Phi.t
+   ssf.tv = list(mPhi = Phi.t,
+   mOmega = Omega,
+   mJPhi = JPhi,
+   mSigma = Sigma,
+   mX = mX)
+   CheckSsf(ssf.tv)
+ }
> tv.start = c(0, 0, 0) % starting values
> tv.mle = SsfFit(tv.start, gm, "tv.capm", mX = X.mtx) % estimation
> sigma.mle = sqrt(exp(tv.mle$parameters))
> sigma.mle
[1] 4.907845e-05 1.219885e-02 8.125213e+00
% Smoothing
> smoEst.tv = SsfCondDens(gm, tv.capm(tv.mle$parameters, mX = X.
mtx),
+ task = "STSMO")
> names(smoEst.tv)
[1] "state" "response" "task"
> par(mfcol = c(2, 2)) % plotting
> plot(gm, type = 'l', ylab = 'excess return')
> title(main = ' (a) Monthly simple excess returns ')
> plot(smoEst.tv$response, type = 'l', ylab = 'rtn')
> title(main = ' (b) Expected returns ')
> plot(smoEst.tv$state[, 1], type = 'l', ylab = 'value')
> title(main = ' (c) Alpha(t) ')
> plot(smoEst.tv$state[, 2], type = 'l', ylab = 'value')
> title(main = ' (d) Beta(t) ')
```

Note that estimates of σ_η and σ_ε are 4.91×10^{-5} and 1.22×10^{-2} , respectively. These estimates are close to zero, indicating that α_t and β_t of the time-varying market model are essentially constant for the GM stock returns. This is in agreement with the fact that the fixed-coefficient market model fits the data well. Figure 11.5 shows some plots for the time-varying CAPM fit. Part (a) is the monthly simple excess returns of GM stock from January 1990 to December 2003. Part (b) is the expected returns of GM stock, that is, $r_{t|T}$, where $T = 168$ is the sample size. Parts

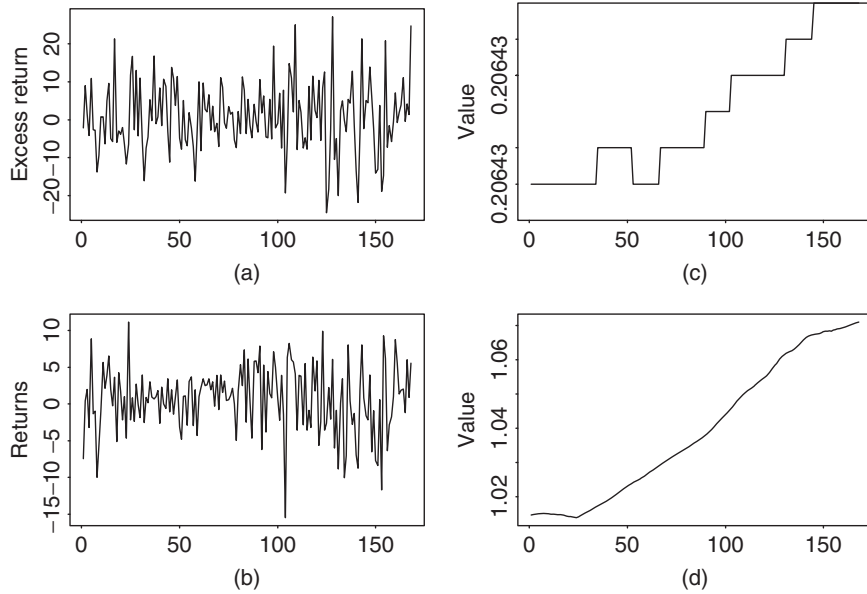


Figure 11.5 Time plots of some statistics for time-varying CAPM applied to monthly simple excess returns of General Motors stock. S&P 500 composite index return is used as market return: (a) monthly simple excess return, (b) expected returns $r_{t|T}$, (c) α_t estimate, and (d) β_t estimate.

(c) and (d) are the time plots of the estimates of α_t and β_t . Given the tightness in the vertical scale, these two time plots confirm the assertion that a fixed-coefficient market model is adequate for the monthly GM stock return.

Example 11.3. In this example we reanalyze the series of quarterly earnings per share of Johnson & Johnson from 1960 to 1980 using the unobserved component model; see Chapter 2 for details of the data. The model considered is

$$y_t = \mu_t + \gamma_t + e_t, \quad e_t \sim N(0, \sigma_e^2), \quad (11.87)$$

where y_t is the logarithm of the observed earnings per share, μ_t is the local trend component satisfying

$$\mu_{t+1} = \mu_t + \eta_t, \quad \eta_t \sim N(0, \sigma_\eta^2),$$

and γ_t is the seasonal component that satisfies

$$(1 + B + B^2 + B^3)\gamma_t = \omega_t, \quad \omega_t \sim N(0, \sigma_\omega^2),$$

that is, $\gamma_t = -\sum_{j=1}^3 \gamma_{t-j} + \omega_t$. This model has three parameters— σ_e , σ_η , and σ_ω —and is a simple unobserved component model. It can be put in a state-space

form as

$$\begin{bmatrix} \mu_{t+1} \\ \gamma_{t+1} \\ \gamma_t \\ \gamma_{t-1} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & -1 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \mu_t \\ \gamma_t \\ \gamma_{t-1} \\ \gamma_{t-2} \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \eta_t \\ \omega_t \end{bmatrix},$$

where the covariance matrix of $(\eta_t, \omega_t)'$ is $\text{diag}\{\sigma_\eta^2, \sigma_\omega^2\}$, and $y_t = [1, 1, 0, 0]s_t + e_t$; see Section 11.3. This is a special case of the structural time series in `SsfPack` and can easily be specified using the command `GetSsfStsm`. Performing the maximum-likelihood estimation, we obtain $(\hat{\sigma}_e, \hat{\sigma}_\eta, \hat{\sigma}_\omega) = (2.04 \times 10^{-6}, 7.27 \times 10^{-2}, 2.93 \times 10^{-2})$.

```
> jnj=scan(file='q-jnj.txt')
> y=log(jnj)
% Estimation
> jnj.m=function(parm){
+   parm=exp(parm)
+   jnj.sea=GetSsfStsm(irregular=parm[1],level=parm[2],
+   seasonalDummy=c(parm[3],4))
+   CheckSsf(jnj.sea)
+ }
>
> c.start=c(0,0,0) % Starting values
> jnj.est=SsfFit(c.start,y,"jnj.m")
> names(jnj.est)
[1] "parameters" "objective" "message" "grad.norm" "itera-
tions"
[6] "f.evals" "g.evals" "hessian" "scale" "aux"
[11] "call"
> jnjest=exp(jnj.est$parameters)
> jnjest % estimates
[1] 2.044516e-06 7.269655e-02 2.931691e-02
> jnj.ssf=GetSsfStsm(irregular=jnjest[1],level=jnjest[2],
+ seasonalDummy=c(jnjest[3],4)) % specify the model with esti-
mates
> CheckSsf(jnj.ssf)
$mPhi:
      [,1] [,2] [,3] [,4]
[1,]    1    0    0    0
[2,]    0   -1   -1   -1
[3,]    0    1    0    0
[4,]    0    0    1    0
[5,]    1    1    0    0
```



```

$mOmega:
      [,1]      [,2] [,3] [,4]      [,5]
[1,] 0.005284788 0.000000000 0 0 0.000000e+00
[2,] 0.000000000 0.000859481 0 0 0.000000e+00
[3,] 0.000000000 0.000000000 0 0 0.000000e+00
[4,] 0.000000000 0.000000000 0 0 0.000000e+00
[5,] 0.000000000 0.000000000 0 0 4.180047e-12

$mSigma:
      [,1] [,2] [,3] [,4]
[1,] -1 0 0 0
[2,] 0 -1 0 0
[3,] 0 0 -1 0
[4,] 0 0 0 -1
[5,] 0 0 0 0

$mDelta:
      [,1]
[1,] 0
[2,] 0
[3,] 0
[4,] 0
[5,] 0

$mJPhi:
[1] 0

$mJOmega:
[1] 0

$mJDelta:
[1] 0

$mX:
[1] 0

$cT:
[1] 0

$cX:
[1] 0

$cY:
[1] 1

$cSt:
[1] 4

attr(,"class"):
[1] "ssf" % below: smoothed components
> jnj.smo=SsfMomentEst(y,jnj.ssf,task="STSMO")
> up1=jnj.smo$state.moment[,1]+
+ 2*sqrt(jnj.smo$state.variance[,1])
> lw1=jnj.smo$state.moment[,1]-
+ 2*sqrt(jnj.smo$state.variance[,1])
> max(up1) % obtain the range for plotting

```

```

[1] 2.795702
> min(lw1)
[1] -0.5948943
> up=jnj.smo$state.moment[,2]+
+ 2*sqrt(jnj.smo$state.variance[,2])
> lw=jnj.smo$state.moment[,2]-
+ 2*sqrt(jnj.smo$state.variance[,2])
> max(up)
[1] 0.3788652
> min(lw)
[1] -0.3552441
> par(mfcol=c(2,1)) % plotting
> plot(tdx,jnj.smo$state.moment[,1],type='l',xlab='year',
+ ylab='value',ylim=c(-1,3))
> lines(tdx,up1,lty=2)
> lines(tdx,lw1,lty=2)
> title(main='(a) Trend component')
> plot(tdx,jnj.smo$state.moment[,2],type='l',xlab='year',
+ ylab='value',ylim=c(-.5,.5))
> lines(tdx,up,lty=2)
> lines(tdx,lw,lty=2)
> title(main='(b) Seasonal component')
% Filtering and smoothing
> jnj.fil=KalmanFil(y,jnj.ssf,task="STFIL")
> jnj.smo1=KalmanSmo(jnj.fil,jnj.ssf)
> plot(tdx,jnj.fil$mOut[,1],type='l',xlab='year',ylab='resi')
> title(main='(a) 1-Step forecast error')
> plot(tdx,jnj.smo1$response.residuals[2:85],type='l',
+ xlab='year',ylab='resi')
> title(main='(b) Smoothing residual')

```

Figure 11.6 shows the smoothed estimates of the trend and seasonal components, that is, $\mu_{t|T}$ and $\gamma_{t|T}$ with $T = 84$, of the data. Of particular interest is that the seasonal pattern seems to evolve over time. Also shown are 95% pointwise confidence regions of the unobserved components. Figure 11.7 shows the residual plots, where part (a) gives the 1-step-ahead forecast errors computed by Kalman filter and part (b) is the smoothed response residuals of the fitted model. Thus, state-space modeling provides an alternative approach for analyzing seasonal time series. It should be noted that the estimated components in Figure 11.6 are not unique. They depend on the model specified and constraints used. In fact, there are infinitely many ways to decompose an observed time series into unobserved components. For instance, one can use a different specification for the seasonal component, for example, `sesonalTrig` in `SsfPack`, to obtain another decomposition for the earnings series of Johnson & Johnson. Thus, care must be exercised in interpreting the estimated components. However, for forecasting purposes, the choice of decomposition does not matter provided that the chosen one is a valid decomposition.

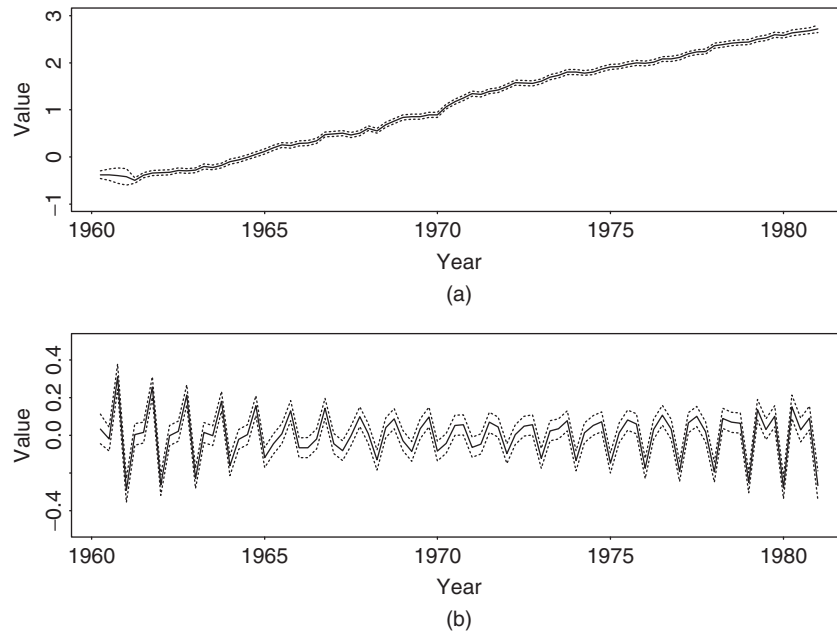


Figure 11.6 Smoothed components of fitting model (11.87) to logarithm of quarterly earnings per share of Johnson & Johnson from 1960 to 1980: (a) trend component and (b) seasonal component. Dotted lines indicate pointwise 95% confidence regions.

EXERCISES

- 11.1. Consider the ARMA(1,1) model $y_t - 0.8y_{t-1} = a_t + 0.4a_{t-1}$ with $a_t \sim N(0, 0.49)$. Convert the model into a state-space form using (a) Akaike's method, (b) Harvey's approach, and (c) Aoki's approach.
- 11.2. The file `aa-rv-20m.txt` contains the realized daily volatility series of Alcoa stock returns from January 2, 2003, to May 7, 2004; see the example in Section 11.1. The volatility series is constructed using 20-minute intradaily log returns.
 - (a) Fit an ARIMA(0,1,1) model to the log volatility series and write down the model.
 - (b) Estimate the local trend model in Eqs. (11.1) and (11.2) for the log volatility series. What are the estimates of σ_e and σ_η ? Obtain time plots for the filtered and smoothed state variables with pointwise 95% confidence interval.
- 11.3. Consider the monthly simple excess returns of Pfizer stock and the S&P 500 composite index from January 1990 to December 2003. The excess returns are in `m-pfesp-ex9003.txt` with Pfizer stock returns in the first column.

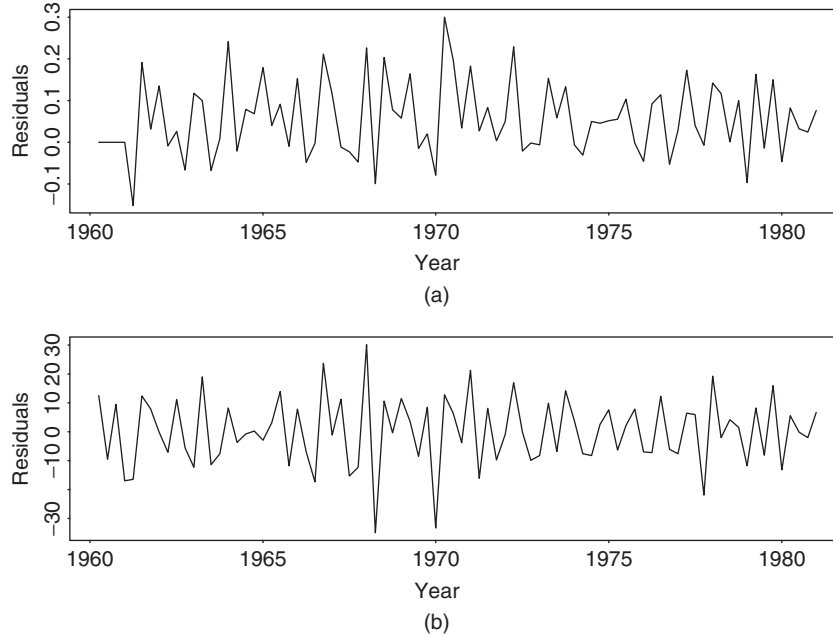


Figure 11.7 Residual series of fitting model (11.87) to logarithm of quarterly earnings per share of Johnson & Johnson from 1960 to 1980: (a) 1-step-ahead forecast error v_t and (b) smoothed residuals of response variable.

- (a) Fit a fixed-coefficient market model to the Pfizer stock return. Write down the fitted model.
- (b) Fit a time-varying CAPM to the Pfizer stock return. What are the estimated standard errors of the innovations to the α_t and β_t series? Obtain time plots of the smoothed estimates of α_t and β_t .

11.4. Consider the AR(3) model

$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \phi_3 x_{t-3} + a_t, \quad a_t \sim N(0, \sigma_a^2),$$

and suppose that the observed data are

$$y_t = x_t + e_t, \quad e_t \sim N(0, \sigma_e^2),$$

where $\{e_t\}$ and $\{a_t\}$ are independent and the initial values of x_j with $j \leq 0$ are independent of e_t and a_t for $t > 0$.

- (a) Convert the model into a state-space form.
- (b) If $E(e_t) = c$, which is not zero, what is the corresponding state-space form for the system?

- 11.5. The file `m-ppiaco4709.txt` contains year, month, day, and U.S. producer price index (PPI) from January 1947 to November 2009. The index is for all commodities and not seasonally adjusted. Let $z_t = \ln(Z_t) - \ln(Z_{t-1})$, where Z_t is the observed monthly PPI. It turns out that an AR(3) model is adequate for z_t if the minor seasonal dependence is ignored. Let y_t be the sample mean-corrected series of z_t .
- Fit an AR(3) model to y_t and write down the fitted model.
 - Suppose that y_t has independent measurement errors so that $y_t = x_t + e_t$, where x_t is a zero-mean AR(3) process and $\text{Var}(e_t) = \sigma_e^2$. Use a state-space form to estimate parameters, including the innovational variances to the state and σ_e^2 . Write down the fitted model and obtain a time plot of the smoothed estimate of x_t . Also, show the time plot of filtered response residuals of the fitted state-space model.

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